

NHC External Curve Catalog — All Rounds

Contents

1	Round 1: NHC External (2131 bases)	5
1.1	Unimodular ($ \det = 1$) — 474 bases	7
1.1.1	NHC $(-2)(-2)(-3)$	7
1.1.2	NHC $(-2)(-3)$	8
1.1.3	NHC $(-2)(-3)(-2)$	50
1.2	Non-unimodular ($ \det > 1$) — 1657 bases	51
1.2.1	NHC $(-2)(-2)(-3)$	51
1.2.2	NHC $(-2)(-3)$	61
1.2.3	NHC $(-2)(-3)(-2)$	167
2	Round 2: NHC External 2-ext (1536 bases)	174
2.1	Unimodular ($ \det = 1$) — 59 bases	174
2.1.1	NHC $(-2)(-3) + \mathfrak{su}_2$	174
2.1.2	NHC $(-2)(-3) + \mathfrak{su}_3$	179
2.1.3	NHC $(-2)(-3)(-2) + \mathfrak{su}_2$	180
2.2	Non-unimodular ($ \det > 1$) — 1477 bases	181
2.2.1	NHC $(-2)(-2)(-3) + \mathfrak{so}_8$	181
2.2.2	NHC $(-2)(-2)(-3) + \mathfrak{su}_2$	181
2.2.3	NHC $(-2)(-2)(-3) + \mathfrak{su}_3$	189
2.2.4	NHC $(-2)(-3) + \mathfrak{f}_4$	190
2.2.5	NHC $(-2)(-3) + \text{NHC}(-2)(-2)(-3)$	190
2.2.6	NHC $(-2)(-3) + \text{NHC}(-2)(-3)$	190
2.2.7	NHC $(-2)(-3) + \mathfrak{so}_8$	191
2.2.8	NHC $(-2)(-3) + \mathfrak{su}_2$	193
2.2.9	NHC $(-2)(-3) + \mathfrak{su}_3$	284
2.2.10	NHC $(-2)(-3)(-2) + \mathfrak{so}_8$	301
2.2.11	NHC $(-2)(-3)(-2) + \mathfrak{su}_2$	302
3	Round 3: NHC External 3-ext (1750 bases)	315
3.1	Unimodular ($ \det = 1$) — 78 bases	315
3.1.1	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	315
3.1.2	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	315
3.1.3	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	318
3.1.4	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	321
3.1.5	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	321
3.1.6	NHC $(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	324
3.2	Non-unimodular ($ \det > 1$) — 1672 bases	324
3.2.1	NHC $(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	324
3.2.2	NHC $(-2)(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix}$	325
3.2.3	NHC $(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	325
3.2.4	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	325
3.2.5	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	328
3.2.6	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \text{su2n3mix}$	328
3.2.7	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$	328

3.2.8	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	328
3.2.9	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	339
3.2.10	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix}$	339
3.2.11	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	343
3.2.12	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$	343
3.2.13	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	345
3.2.14	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	399
3.2.15	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix}$	410
3.2.16	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	442
3.2.17	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	445
3.2.18	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	476
3.2.19	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix}$	479
3.2.20	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3$	484
3.2.21	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_2$	485
3.2.22	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{so}_8$	485
3.2.23	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2$	485
3.2.24	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	490
3.2.25	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \text{su2n3mix}$	493

4 Round 4: NHC External 4-ext (1161 bases) 507

4.1	Unimodular ($ \det = 1$) — 46 bases	507
4.1.1	$\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	507
4.1.2	$\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	508
4.1.3	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	509
4.1.4	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	510
4.1.5	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	511
4.1.6	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	511
4.1.7	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	512
4.1.8	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	512
4.1.9	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \text{su2n3mix}$	513
4.1.10	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	513
4.1.11	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	513
4.1.12	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	513
4.2	Non-unimodular ($ \det > 1$) — 1115 bases	513
4.2.1	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	513
4.2.2	$\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	514
4.2.3	$\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	514
4.2.4	$\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	515
4.2.5	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix}$	515
4.2.6	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	516
4.2.7	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	519
4.2.8	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	519
4.2.9	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2$	520
4.2.10	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	523
4.2.11	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix}$	523
4.2.12	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	523
4.2.13	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	524
4.2.14	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	526
4.2.15	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{su2n3mix}$	528
4.2.16	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	530
4.2.17	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	530
4.2.18	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	530
4.2.19	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	544
4.2.20	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	545

4.2.21	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	547
4.2.22	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	581
4.2.23	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \text{su2n3mix}$	583
4.2.24	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	584
4.2.25	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	586
4.2.26	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{su2n3mix}$	587
4.2.27	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	588
4.2.28	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	601
4.2.29	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{su2n3mix}$	608
4.2.30	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	616
4.2.31	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	624
4.2.32	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	633
4.2.33	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	633
4.2.34	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	643

5 Round 5: NHC External 5-ext (269 bases) 648

5.1	Unimodular ($ \det = 1$) — 12 bases	648
5.1.1	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	648
5.1.2	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	648
5.1.3	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	649
5.1.4	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	649
5.1.5	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	649
5.1.6	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	649
5.1.7	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	649
5.1.8	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	649
5.2	Non-unimodular ($ \det > 1$) — 257 bases	650
5.2.1	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	650
5.2.2	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \text{su2n3mix}$	650
5.2.3	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	650
5.2.4	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2$	651
5.2.5	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	651
5.2.6	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 + \text{su2n3mix}$	652
5.2.7	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	652
5.2.8	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix} + \mathfrak{su}_2$	653
5.2.9	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	653
5.2.10	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	654
5.2.11	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	654
5.2.12	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	654
5.2.13	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	654
5.2.14	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \text{su2n3mix}$	654
5.2.15	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	655
5.2.16	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 + \text{su2n3mix}$	656
5.2.17	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \text{su2n3mix} + \mathfrak{su}_2$	657
5.2.18	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	657
5.2.19	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	658
5.2.20	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	659
5.2.21	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	661
5.2.22	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	666
5.2.23	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	667
5.2.24	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	672
5.2.25	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	673
5.2.26	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2$	677
5.2.27	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	678
5.2.28	$\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2$	679

6	Round 6: NHC External 6-ext (25 bases)	680
6.1	Non-unimodular ($ \det > 1$) — 25 bases	680
6.1.1	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	680
6.1.2	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	680
6.1.3	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	680
6.1.4	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	680
6.1.5	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	681
6.1.6	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	681
6.1.7	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	682
6.1.8	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	682
6.1.9	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	683
6.1.10	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	683
7	Round 7: NHC External 7-ext (1 base)	684
7.1	Non-unimodular ($ \det > 1$) — 1 bases	684
7.1.1	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	684

1 Round 1: NHC External (2131 bases)

Colored curves in quiver show attachment points. Matching color in external label: **ext1** **ext2** **ext3**

Total: 2131 bases (474 uni, 1657 non-uni).

T	Count
5	4
6	23
7	39
8	61
9	60
10	67
11	77
12	86
13	92
14	95
15	84
16	82
17	75
18	91
19	82
20	83
21	69
22	76
23	65
24	63
25	57
26	54
27	51
28	43
29	38
30	39
31	37
32	39
33	29
34	22
35	25
36	29
37	19
38	24
39	10
40	14
41	15
42	20
43	19
44	16
45	9
46	14
47	16
48	12
49	16
50	7
51	12
52	9
53	12
54	8
55	6
56	4
57	8
58	8
59	7
60	5
61	2
62	2

1.1 Unimodular ($|\det| = 1$) — 474 bases

1.1.1 NHC(-2)(-2)(-3)

$$T = 12 \quad [72] \quad 1^{\overline{1}} 31^{\overline{1}} 513221^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 2 & 5 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 5 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[73] \quad 1^{\overline{1}} 231^{\overline{1}} 51321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [89] \quad 1^{\overline{1}} 7^{\overline{1}} 12321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 15 \quad [96] \quad 1^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[97] \quad 1^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [98] \quad 1^{\overline{1}} 6^{\overline{1}} 131612321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 17 \quad [102] \quad 1^{\overline{1}} 8^{\overline{1}} 1231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[103] \quad 1^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[104] \quad 1^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [107] \quad 1^{\mathbf{1}^{131}} 6 \quad 13161315131^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, \quad T_{\max} = 18, \quad \Delta = 0, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[108] \quad 1^{\mathbf{1}^{12321}} 8 \quad 123216131^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, \quad T_{\max} = 18, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[109] \quad 1^{\mathbf{1}^{12321}} 8 \quad 123216131^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, \quad T_{\max} = 18, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [112] \quad 1^{\mathbf{1}^{1321}} 8 \quad 12321712321^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 19, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[113] \quad 1^{\mathbf{1}^{12321}} 8 \quad 1232171231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 19, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[114] \quad 1^{\mathbf{1}^{12321}} 8 \quad 1232171231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 19, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

1.1.2 NHC(-2)(-3)

$$T = 9 \quad [71] \quad 1^{\mathbf{1}^{2231321}} 1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -39, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[72] \quad 1^{\mathbf{1}^{2231321}} 1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -39, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 10 \quad [108] \quad 1^{\mathbf{1}^{31413221}} 1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -37, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, \quad T_{\max} = 11, \quad T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[109] \quad 1^{\mathbf{1}^{31413221}} 1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -37, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, \quad T_{\max} = 11, \quad T \in [10, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8 \\
[110] & 1^1 23141321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
\\
T = 11 & [167] 1^1 314141321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[168] & 1^1 141413221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[169] & 1^1 5^{1231} 1321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[170] & 1^1 231512321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[171] & 1^1 223151231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[172] & 1^1 314141321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
[173] & 1^1 223151231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[174] & 1^1 141413221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[175] & 1^1 5^{12231} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[176] & 1^1 5^{12231} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[177] & 1^1 231512321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9
\end{aligned}$$

$$\begin{aligned}
& T = 12 \quad [225] \quad 1^1 4131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [226] \quad 1^1 4131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
& [227] \quad 1^1 2321612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [228] \quad 1^1 3141414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
& [229] \quad 1^1 1414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [230] \quad 1^1 1414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
& T = 13 \quad [301] \quad 1^1 22315132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [302] \quad 1^1 2321 1^1 7 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
& [303] \quad 1^1 15131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \quad \begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
& [304] \quad 1^1 15131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [305] \quad 1^1 23151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [306] \quad 1^1 14141414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
& [307] \quad 1^1 22315132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[308] & 1^1 23151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[309] & 1^1 14141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
T = 14 & [381] 21^1 8^{12321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[379] & 1^1 316123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[380] & 1^1 316123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[378] & 1^1 321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[382] & 1^1 232161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[383] & 1^1 232161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[384] & 1^1 14141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[385] & 1^1 612231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[386] & 1^1 8^{12321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[387] & 1^1 161231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}
\end{aligned}$$

[388] $21\overset{1}{6}131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 14, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[389] $1\overset{1}{6}131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 14, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[390] $1^{\overset{1}{2}315131}6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[391] $1^{\overset{1}{2}315131}6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[392] $1^1321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

$T = 15$ [460] $131\overset{1}{6}131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[461] $1712231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[462] $171231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[463] $1^13216131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[464] $1^12321712315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[465] $1^{\overset{1}{3}31}6131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

[466] $1^{\overset{1}{3}31}6131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[562] \quad & 1^1 31612321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[544] \quad & 1^1 31513161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
\\
& T = 17 \quad [616] \quad 1^1 6^{131} 13161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[617] \quad & 1^1 321712321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[618] \quad & 1^1 15^{131} 1316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[619] \quad & 1^1 13218^{1231} 1231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[620] \quad & 1^1 8^{12321} 123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[621] \quad & 1^1 216^{131} 131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[622] \quad & 1^1 8^{12321} 123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
[623] \quad & 1^1 15^{131} 1316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[624] \quad & 1^1 321712321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[625] \quad & 1^1 321613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}
\end{aligned}$$

[626] $1^1 413218123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[627] $1^1 413218123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[628] $1^1 321613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[629] $1^1 6^{131} 13161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

$T = 18$ [689] $1^1 2321712321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[688] $1^1 8^{12321} 1232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[687] $1^1 2321712321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[690] $1^1 6^{131} 13161316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[691] $1^1 15^{13218123151321^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[692] $216^{131} 1316131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[693] $1^1 15^{13218123151321^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[694] $1^1 2315132181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[695] $1^1 2315132181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[696] & 1^1 3216131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[697] & 1^1 3151321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[698] & 1^1 1316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[699] & 1^1 6^1 131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[700] & 1^1 4131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[701] & 1^1 8^1 1232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[702] & 1^1 4131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[703] & 1^1 8^1 12321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[704] & 1^1 6^1 131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
[705] & 1^1 8^1 12321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16} \\
T = 19 & [770] 1^1 8^1 12321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[766] & 1^1 41321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}
\end{aligned}$$

[767] $1^{1^{12321}} 1^8 12321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[768] $1^{1^{131}} 1^6 1316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[769] $1^{1^{131}} 1^6 1316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[765] $1^{1^4} 1321812321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[771] $1^{1^1} 16123218123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[772] $1^{1^{1321}} 1^8 12321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[773] $1^{1^1} 131613151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 20, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[774] $1^{1^{1321}} 1^8 12321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[775] $1^{1^1} 13161316131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[776] $1^{1^1} 15131613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[777] $1^{1^1} 131612321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 20, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[778] $1^{1^1} 141316131613161231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 19, T_{\max} = 19, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[779] \ 1^1 41316131613161231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 19, T_{\max} = 19, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[780] \ 1^1 31613151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 19, T_{\max} = 20, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[781] \ 1^1 15131613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 19, T_{\max} = 19, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[782] \ 1^1 31612321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 19, T_{\max} = 20, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 20 \quad [856] \ 1^1 6^{1^1 31} 1316131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[844] \ 1^1 316123218123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[845] \ 1^{21} (12)_{1^1 231513221} 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[846] \ 1^1 321712321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[847] \ 1^1 321712321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[848] \ 1^{21} (12)_{1^1 231513221} 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[849] \ 1^1 413161316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 20, T_{\max} = 20, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

[850] $1^1 316131612321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[851] $1^1 151321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[852] $1^1 2321 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[853] $1^1 151316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[854] $1^1 151316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[855] $1^1 6^{131} 1316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[843] $1^1 2321 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[857] $1^1 316131612321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[858] $1^1 216131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[859] $1^1 231513221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[860] $1^1 231513221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[861] $1^1 217123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[862] $1^1 232171232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

[863] $1^1 232171232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[864] $1^1 151321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$

$T = 21$ [943] $1^1 3218123151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[929] $1^1 1612321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[930] $1^1 2161316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[931] $1^1 8^{12321} 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[932] $1^1 8^{12321} 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[933] $1^1 8^{12321} 123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$

[934] $1^1 1513161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[935] $1^1 2218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[936] \ 1^1 3221812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[937] \ 21 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[938] \ 1^1 321 \quad 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[939] \ 21 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[940] \ 1^1 3221812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[941] \ 1^1 321 \quad 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[942] \ 1^1 2321 \quad 123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -22, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[928] \ 1^1 3218123151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 22, \ \Delta = -29, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 22, \ T \in [21, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[944] \ 216131613161316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -17, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[945] \ 1^1 513161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -19, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[946] \ 1^1 3161232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[947] \ 1^1 3161232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, \ T_{\max} = 21, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 21, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, \ T_{\max} = 21, \ T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 22 \quad [999] \ 21 \quad \begin{matrix} 1^1 \\ 15 \end{matrix} 13221 \quad 1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -23, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1000] \ 16 \ 12321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -21, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1001] \ 1^1 5 \ 131613161316131^1 5 \ 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -14, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1002] \ 1^1 32171232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -27, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1003] \ 1321 \ 8 \ 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1004] \ 21 \ 6 \ 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -16, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1005] \ 21 \quad \begin{matrix} 1^1 \\ 15 \end{matrix} 13221 \quad 1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -23, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1006] \ 1 \quad \begin{matrix} 1 \\ 1^1 \end{matrix} 131513221 \quad 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 23, \ \Delta = -30, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 23, \ T \in [22, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1007] \ 1 \ 8 \ 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 22, \ T_{\max} = 22, \ \Delta = -27, \ \det = 1, \ (n_+, n_-, n_0) = (1, 22, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 22, \ T_{\max} = 22, \ T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

[1008] $1^{1321} 1^8 123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1009] $1^{1513221} 1^{12} 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1010] $1^{1231513221} 1^{12} 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1011] $1^{1231513221} 1^{12} 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1012] $1^1 31612321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1013] $1^1 31612321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1014] $1^{12} 21712321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1015] $1^{1513221} 1^{12} 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1016] $1^1 41321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1017] $1^1 41321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1018] $1^{131513221} 21^{12} 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -30, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 23 \quad [1068] \quad 1^1 321712321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1069] \quad 1^1 413218123218123216131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[1070] \quad 1^1 315132181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1071] \quad 21^1 6131613161316131^1 51 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1072] \quad 221^1 812321812321712321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[1073] \quad 1^1 4141413221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1074] \quad 1^1 413218123218123216131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1075] \quad 1^1 321712321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[1076] \quad 15^1 1321812321812315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1077] \quad 1^1 4141413221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$[1078] \quad 1^1 8^1 1232181232171231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1079] & 1^1 321812321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1080] & 21^1 7^1 123218123218^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1081] & 1^1 321^1 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21} \\
[1082] & 1^1 321812321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1083] & 1^1 315132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1084] & 1^1 151321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
T = 24 & [1128] 1^1 2321812321^1 8^1 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1129] & 221^1 8^1 123218123218^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[1130] & 1^1 1612321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[1131] & 1^1 5131513221^1 (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$[1143] \quad 1^{15131513221} (12) \quad 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$T = 25 \quad [1190] \quad 1^{15132181232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[1191] \quad 1^{16131513221} (12) \quad 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[1192] \quad 21^{1^1 3221} (12) \quad 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[1193] \quad 1^{132171232181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[1194] \quad 1^{1231513221} (12) \quad 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[1195] \quad 21^1 712321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[1196] \quad 1^{132171232181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[1197] \quad 1^{15132181232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$$

$$[1198] \quad 1^{131513221} (12) \quad 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 26, T \in [25, 26]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

[1199] $1^1 41321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1200] $1^1 41321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[1201] $1^1 31513221 (12) 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 26, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 26, T \in [25, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[1202] $1^1 321 8 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[1203] $1^1 321 8 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[1204] $21 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1205] $1^1 16 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23}$

[1245] $1^1 231513221 (12) 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1246] $1^1 232181231513221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1247] $21 7 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1248] $2218^{11}12321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[1249] $1^{11}322181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[1250] $15^{11}1321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1251] $15^{11}1321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[1252] $1^{11}321123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[1253] $17^{11}1231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1254] $1^{11}322181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1255] $1^{11}413218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1256] $1^{11}3211232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$

[1257] $1^{11}3211232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$

[1258] $1^{11}321123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[1259] \quad 1^{1232181231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -27, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[1260] \quad 1^{161232181232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -19, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$T = 27 \quad [1300] \quad 218^{11231513221} (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1301] \quad 1^{1812231513221} (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1302] \quad 1^{1513221} (12) \quad 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -22, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1303] \quad 1^{1513221} (12) \quad 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -22, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1304] \quad 1^{161232181232181232181231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -18, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1305] \quad 2218^{1123218123218123216131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -17, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1306] \quad 1^{131513221} (12) \quad 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -28, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1307] \quad 1^{1513218123218123218123141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -18, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1308] \quad 1^{131513221} (12) 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1309] \quad 1^{141413221} (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1310] \quad 1^{151316131513221} (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1311] \quad 1^{141413221} (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1312] \quad 21^1 71232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1313] \quad 1^{1612321812321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1314] \quad 1^{1513218123218123218123141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1315] \quad 1^{14132181232181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1316] \quad 1^{14132181232181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[1317] \quad 1^{151316131513221} (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$\begin{aligned}
& T = 28 \quad [1349] \quad 1^1 5 132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1350] \quad 21^1 7 1232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1351] \quad 1^1 5 132181232181232181231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -13, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1352] \quad 221^1 8 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1353] \quad 1^1 6 1316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1354] \quad 1^1 6 123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1355] \quad 1^1 5 132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
& [1356] \quad 21^1 7 12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1357] \quad 1^1 3161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
& [1358] \quad 1^1 3161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
& [1359] \quad 1^1 316131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[1360] \quad 1^1 51232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$T = 29 \quad [1391] \quad 1^1 6123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1392] \quad 1^1 5132181231513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1393] \quad 21^1 7123218123218123218123141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1394] \quad 1^1 5132181231513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1395] \quad 1^1 232181231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1396] \quad 221^1 812321812321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1397] \quad 1^1 232181231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1398] \quad 221^1 81232181232181232181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[1399] \quad 1^1 61232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -14, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$T = 30 \quad [1428] \quad 1^1 61231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -22, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[1429] \quad 221^1 8123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -14, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1430] \quad 1^1 316131513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -27, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[1431] \quad 1^1 316131513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -27, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1432] \quad 1^1 3221 (12) 1223151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -27, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[1433] \quad 1^1 6131513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -21, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[1434] \quad 1^1 61232181231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -22, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1435] \quad 1^1 3221 (12) 1223151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -27, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1436] \quad 21^1 71232181232181232181231^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$[1437] \quad 21^1 71232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -12, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$\begin{aligned}
& T = 31 \quad [1469] \quad 1^{1^1 31513221} (12)^{21} 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1470] \quad 1^{1^1 413221} (12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{30} \\
& [1471] \quad 1^{1^1 413221} (12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1472] \quad 1^{1^1 71231513221} (12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{30} \\
& [1473] \quad 1^{1^1 71232181231513221} (12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1474] \quad 1^{1^1 31513221} (12)^{21} 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1475] \quad 22181232181232181232181231^{1^1} 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1476] \quad 22181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{29} \\
& [1477] \quad 1^{1^1 712231513221} (12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, T_{\max} = 31, T \in [31, 31] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{30} \\
& T = 32 \quad [1508] \quad 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{30} \\
& [1509] \quad 2181231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1510] \quad 1 \quad (12) \quad 12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[1511] \quad 1^1 1513221(12)1223151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[1512] \quad 1 \quad (12) \quad 12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[1513] \quad 1^1 1513221(12)1223151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1514] \quad 1^1 32181232181231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[1515] \quad 1^1 32181231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[1516] \quad 1^1 1812231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1517] \quad 1^1 32181232181231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$T = 33 \quad [1544] \quad 21 \quad (12) \quad 12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1545] \quad 1 \quad (12) \quad 12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1546] \quad 1 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1547] \quad 1 \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1548] \quad 1 \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[1549] \quad 21 \quad (12) \quad 12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$T = 34 \quad [1564] \quad 21 \quad (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[1565] \quad 21 \quad (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{32}$$

$$T = 35 \quad [1586] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[1587] \quad 1 \quad (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1588] \quad 1 \quad (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[1589] \quad 1^1 232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[1590] \ 1^1 232181232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1591] \ 1^1 316131513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$T = 36 \quad [1615] \ 1^1 1513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1616] \ 1^1 41413221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1617] \ 21^1 13221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1618] \ 21^1 13221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1619] \ 1^1 41413221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1620] \ 1^1 231513221(12)12231513221(12)12231513161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1621] \ 1^1 16131513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1622] \ 1^1 1513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[1623] \ 1 \quad (12) \quad {}^{1^1 31513221} 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \ T_{\max} = 36, \ \Delta = -21, \ \det = 1, \ (n_+, n_-, n_0) = (1, 36, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \ T_{\max} = 36, \ T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$T = 37 \quad [1640] \ 1 \quad (12) \quad {}^{1^1 32181231513221(12)12231513221} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 37, \ \Delta = -24, \ \det = -1, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, \ T_{\max} = 37, \ T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1641] \ 1 \quad (12) \quad {}^{1^1 32181231513221(12)12231513221} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 37, \ \Delta = -24, \ \det = -1, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, \ T_{\max} = 37, \ T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1642] \ 1 \quad (12) \quad {}^{1^1 231513221(12)12231513221} 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 37, \ \Delta = -20, \ \det = -1, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, \ T_{\max} = 37, \ T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1643] \ 1 \ 7 \ 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 37, \ \Delta = -19, \ \det = -1, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, \ T_{\max} = 37, \ T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1644] \ 1 \quad (12) \quad {}^{1^1 31513221} 12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, \ T_{\max} = 37, \ \Delta = -19, \ \det = -1, \ (n_+, n_-, n_0) = (1, 37, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 37, \ T_{\max} = 37, \ T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$T = 38 \quad [1664] \ 1 \ 8 \ 1231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 38, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36}$$

$$[1665] \ 21 \ 8 \ 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -18, \ \det = 1, \ (n_+, n_-, n_0) = (1, 38, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36}$$

$$[1666] \ 1 \quad (12) \quad {}^{1^1 231513221(12)12231513221} 1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, \ T_{\max} = 38, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 38, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, \ T_{\max} = 38, \ T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36}$$

$$\begin{aligned}
& \text{[1667]} \quad 1 \quad \overset{1}{5} \overset{1}{13221} \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \text{[1668]} \quad 1 \quad \overset{1}{5} \overset{1}{13221} \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \text{[1669]} \quad 1 \quad 8 \overset{1}{12231513221(12)12231513221(12)122315131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -18, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \text{[1670]} \quad 1 \quad \overset{1}{41413221(12)12231513221(12)122315132181231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = \\
& 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \text{[1671]} \quad 1 \quad \overset{1}{316131513221(12)12231513221(12)12231513151}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -22, \det = \\
& 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \text{[1672]} \quad 1 \quad \overset{1}{316131513221(12)12231513221(12)12231513151}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -22, \det = \\
& 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \text{[1673]} \quad 1 \quad \overset{1}{41413221(12)12231513221(12)122315132181231}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = \\
& 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& T = 39 \quad \text{[1680]} \quad 1 \quad \overset{1}{316131513221(12)12231513221(12)122315132161}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = \\
& -20, \det = -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \text{[1681]} \quad 1 \quad \overset{1}{6131513221(12)12231513221(12)1223151316131}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -19, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \text{[1682]} \quad 1 \quad \overset{1}{61316131513221(12)12231513221(12)1223151321}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37}
\end{aligned}$$

$$\begin{aligned}
& T = 40 \quad [1690] \quad 21 \overset{1}{(12)} 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -24, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{38} \\
& [1691] \quad 1^1 32181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1692] \quad 1^1 71231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{38} \\
& [1693] \quad 1^1 32181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{38} \\
& [1694] \quad 1^1 316131513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{38} \\
& T = 41 \quad [1706] \quad 1^1 61231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = \\
& -19, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1707] \quad 1^1 6131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -18, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1708] \quad 21 \overset{1}{(12)} 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1709] \quad 21^1 81231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1710] \quad 1^1 812231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = \\
& -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{39} \\
& [1711] \quad 21 \overset{1}{(12)} 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det =
\end{aligned}$$

$$-1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39}$$

$$T = 42 \quad [1724] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1725] \quad 1 \overset{1}{4} 13221(12)12231513221(12)12231513221 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1726] \quad 1 \overset{1}{3} 3221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1727] \quad 1 \overset{1}{3} 3221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1728] \quad 1 \overset{1}{7} 1232181231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1729] \quad 21(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1730] \quad 21(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1731] \quad 1 \overset{1}{7} 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$[1732] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 42, T_{\max} = 42, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}$$

$$\begin{aligned}
& \text{[1733]} \quad 1 \overset{1}{\overset{1}{5}} 13221 (12) \overset{21}{12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 42, \quad T_{\max} = 42, \quad \Delta = -18, \quad \det = \\
& 1, \quad (n_+, n_-, n_0) = (1, 42, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{40}
\end{aligned}$$

$$\begin{aligned}
& T = 43 \quad \text{[1745]} \quad 1 \overset{1}{\overset{1}{41413221(12)12231513221(12)12231513221}} (12) \overset{1}{1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -21, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1746]} \quad 1 \overset{1}{\overset{1}{413221(12)12231513221}} (12) \overset{1}{122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -25, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1747]} \quad 1 \overset{1}{\overset{1}{5}} 13221 (12) \overset{1}{12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -17, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1748]} \quad 1 \overset{1}{\overset{1}{5}} 13221 (12) \overset{1}{12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -17, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1749]} \quad 1 \overset{1}{\overset{1}{413221(12)12231513221}} (12) \overset{1}{122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -25, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{42}
\end{aligned}$$

$$\begin{aligned}
& \text{[1750]} \quad 1 \overset{1}{\overset{1}{41413221(12)12231513221(12)12231513221}} (12) \overset{1}{1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -21, \quad \det = -1, \quad (n_+, n_-, n_0) = \\
& (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1751]} \quad 1 \overset{1}{181232181231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -18, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1752]} \quad 1 \overset{1}{1812231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -16, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{smallmatrix} \right)_{b_0} \rightarrow \text{bd}_{41}
\end{aligned}$$

$$\begin{aligned}
& \text{[1753]} \quad 1 \overset{1}{2181231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 43, \quad T_{\max} = 43, \quad \Delta = -16, \quad \det = \\
& -1, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$T = 44 \quad [1763] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[1764] \quad 1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$$

$$[1765] \quad 21 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$$

$$[1766] \quad 21 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[1767] \quad 1 \overset{1}{(12)} 12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[1768] \quad 1 \overset{1}{(12)} 12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[1769] \quad 1 \overset{1}{(12)} 12231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[1770] \quad 1 \overset{1}{4} 1413221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$T = 45 \quad [1774] \quad 1 \overset{1}{(12)} 12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$$

[1775] $^{1\bar{1}3221}$ 21 (12) 12231513221(12)12231513221(12)122315132161¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 45, T_{\max} = 45, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$

[1776] $^{1\bar{1}5131513221(12)12231513221(12)12231513221}$ 1 (12) 1223141¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$

[1777] $^{1\bar{1}16131513221(12)12231513221(12)12231513221(12)12231^{21}}$ 1 + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 45, T_{\max} = 45, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$

$T = 46$ [1785] $^{1\bar{1}1513221}$ 1 (12) 12231513221(12)12231513221(12)12231513151¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1786] $^{1\bar{1}61231513221(12)12231513221(12)12231513221}$ 1 (12) 1223141¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

[1787] $^{1\bar{1}1513221}$ 1 (12) 12231513221(12)12231513221(12)12231513151¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

[1788] $^{1\bar{1}3221}$ 21 (12) 12231513221(12)12231513221(12)1223151322171¹ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

[1789] $^{1\bar{1}41413221(12)12231513221(12)12231513221(12)12231513151^{1\bar{1}}}$ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

[1790] $^{1\bar{1}16131513221(12)12231513221(12)12231513221(12)1223141^{1\bar{1}}}$ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

[1791] $^{1\bar{1}41413221(12)12231513221(12)12231513221(12)12231513151^{1\bar{1}}}$ + [$\mathfrak{su}_2 \cdot \mathfrak{g}_2$] $T = 46, T_{\max} = 46, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1792] $1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$

$T = 47$ [1802] $1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1803] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1804] $1^1 41413221(12)12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1805] $21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1806] $1 \overset{1}{7} 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1807] $1 \overset{1}{8} 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1808] $1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

[1809] $1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$

$T = 48$ [1815] $1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 48, T_{\max} = 48, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1816] \quad 1^1 8 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1817] \quad 1^1 21 8 1231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1818] \quad 1^1 15 13221(12)12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1819] \quad 1^1 17 1231513221(12)12231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1820] \quad 1^1 51 31513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$[1821] \quad 1^1 41 413221(12)12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46}$$

$$T = 49 \quad [1832] \quad 1^1 16 131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$$[1833] \quad 1^1 61 231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -14, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$$[1834] \quad 1^1 21 8 1231513221(12)12231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$$[1835] \quad 1^1 18 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

[1836] $218^{11}1231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

[1837] $18^{11}12231513221(12)12231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

[1838] $1^{11}231513221(12)12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$T = 50$ [1841] $1^{11}712231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48}$$

[1842] $17^{11}1231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48}$$

$T = 51$ [1851] $218^{11}1231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49}$$

[1852] $18^{11}12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49}$$

$T = 53$ [1876] $1^{11}31513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51}$$

[1877] $1^{11}31513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51}$$

$$T = 54 \quad [1886] \quad 1^1 81232181231513221(12)12231513221(12)12231513221(12)1223151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{52}$$

1.1.3 NHC(-2)(-3)(-2)

$$T = 12 \quad [39] \quad 1^1 41413221^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[40] \quad 1^1 31513221^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [45] \quad 1^1 414141321^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 \\ 3 & 5 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [57] \quad 1^1 2315131^1 6^{131} 131^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 15, T_{\max} = 15, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [61] \quad 1^1 6^{131} 131612321^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 17 \quad [64] \quad 1^1 23151321^1 8^{131} 1231^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [67] \quad 1^1 6^{131} 13161315131^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 18, T_{\max} = 18, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [70] \quad 1^1 8^{1321} 12321712321^1 + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[71] \ 1^{\overset{1}{1}31} \ 6 \ 131613161231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [72] \ 1^{\overset{1}{1}2321} \ 8 \ 1232181231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 3 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 22 \quad [75] \ 1^{\overset{1}{1}321} \ 8 \ 12321812315131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 22, T_{\max} = 22, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 23 \quad [76] \ 1^{\overset{1}{1}321} \ 8 \ 123218123216131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 23, T_{\max} = 23, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

1.2 Non-unimodular ($|\det| > 1$) — 1657 bases

1.2.1 NHC(-2)(-2)(-3)

$$T = 6 \quad [0] \ 21^{\overset{1}{1}31^{\overset{1}{1}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{1}{1}31^{\overset{1}{1}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{\overset{1}{1}31^{\overset{1}{1}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 21^{\overset{1}{1}31^{\overset{1}{1}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overset{1}{1}31^{\overset{1}{1}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 47, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 221^{\overset{1}{1}3} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 6, T_{\max} = 10, \Delta = -116, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [6] \quad 21^1 321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 31^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 2^1 2^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 2^1 2^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 2^1 3^1 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 2^1 3^1 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 2^1 3^1 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 8 & 11 & 10 & 6 \\ 4 & 11 & 8 & 11 & 6 \\ 4 & 10 & 11 & 7 & 5 \\ 2 & 6 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

- [16] $1^{11}2321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$
- [17] $1^{11}3221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$
- [18] $1^{12}321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -61, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$
- [19] $1^{12}321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [20] $1^{11}3221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$
- [21] $41^12231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$
- [22] $1^141^1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$
- [23] $1^141^1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^141^1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$
- [25] $2^13141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [26] $2^13141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 4 & 3 & 3 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [27] $2^13141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 2 \quad 3 \quad 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 8 & 8 & 6 \\ 4 & 8 & 5 & 5 \\ 3 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 41^{11} 2 \quad 3 \quad 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 41^{11} 3 \quad 22 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 5 & 8 & 5 \\ 3 & 8 & 8 & 4 & 4 \\ 2 & 6 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 41^{11} 2231^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 41^{11} 232 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 41^{11} 3 \quad 22 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 4 & 3 & 2 \\ 3 & 5 & 9 & 7 & 5 \\ 4 & 9 & 10 & 9 & 6 \\ 3 & 7 & 9 & 4 & 4 \\ 2 & 5 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 41^{11} 322 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 3 & 2 \\ 4 & 10 & 13 & 8 & 6 \\ 4 & 13 & 10 & 9 & 6 \\ 3 & 8 & 9 & 4 & 4 \\ 2 & 6 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 41^{111} 3 \quad 22 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [36] \quad 1^{11} 41321^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 41321^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[38] \quad 31^{11} 3151^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 8 & 11 & 4 & 6 \\ 5 & 11 & 8 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $21^{11}4132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 12 & 15 & 14 & 8 \\ 4 & 15 & 12 & 15 & 8 \\ 4 & 14 & 15 & 11 & 7 \\ 2 & 8 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [51] $1^{11}413221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

[52] $1^{11}315131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 12$, $\Delta = -60$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{11}315131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 12$, $\Delta = -60$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{11}315131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 12$, $\Delta = -60$, $\det = 5$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[55] $31^{11}41^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -88$, $\det = 52$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 5 & 1 & 2 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[56] $41^{11}31513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -88$, $\det = 56$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[57] $31^{11}41^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -88$, $\det = 56$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 3 & 1 & 2 \\ 5 & 8 & 7 & 0 & 6 \\ 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[58] $41^{11}31513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -88$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[59] $3141^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -88$, $\det = 248$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[60] $2231^{11}3161^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 251$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 7 & 1 & 7 & 2 \\ 7 & 33 & 6 & 35 & 11 \\ 1 & 6 & 0 & 6 & 2 \\ 7 & 35 & 6 & 32 & 10 \\ 2 & 11 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [61] \quad 1^{\overline{1}1} 31^{\overline{1}} 51321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{\overline{1}1} 31^{\overline{1}} 51321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{\overline{1}1} 31^{\overline{1}} 51321^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 2^{\overline{1}1} 3^{\overline{1}} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[69] \quad 2^{\overline{1}1} 3^{\overline{1}} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[70] \quad 21^{\overline{1}1} 414132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -416, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 16 & 19 & 18 & 10 \\ 4 & 19 & 16 & 19 & 10 \\ 4 & 18 & 19 & 15 & 9 \\ 2 & 10 & 10 & 9 & 5 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
T = 12 \quad [71] \quad 1^1 231^1 5 13221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0 \\
[74] \quad 1^1 31^1 6 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[75] \quad 1^1 1^{12321} 6 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[76] \quad 1^1 1^{12321} 6 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
T = 13 \quad [77] \quad 1^1 1^{12321} 7 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[78] \quad 1^1 1^{12321} 7 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[79] \quad 1^1 2321^1 8 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[80] \quad 1^1 2321^1 8 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[81] \quad 1^1 2321^1 8 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[82] \quad 1^1 5131^1 31^1 6 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[83] \quad 1^1 5131^1 31^1 6 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -35, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[84] \quad 2^1 1^1 4^1 14141^1 3^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 5 & 6 \\ 2 & 7 & 4 & 6 & 6 \\ 2 & 5 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[85] \quad 31^{1^{11}5^{131}} 6^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[86] \quad 2^{1^{11}4^{14141}} 3^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 4 & 6 & 6 \\ 2 & 6 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[87] \quad 31^{1^{11}5^{131}} 6^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[88] \quad 21^{1^{11}4^{14141}3^2} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -416, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 20 & 23 & 22 & 12 \\ 4 & 23 & 20 & 23 & 12 \\ 4 & 22 & 23 & 19 & 11 \\ 2 & 12 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [90] \quad 1^{1^{11}2^{321}} 8^{1231^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1^{11}3^{15131}} 6^{131^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1^{11}3^{221}} 8^{12321^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1^{11}3^{15131}} 6^{131^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1^{11}2^{321}} 8^{1231^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1^{11}3^{221}} 8^{12321^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [99] \quad 1^{\mathbf{1}} 1^{\mathbf{1}} 3151321 \quad 8 \quad 1231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, \quad T_{\max} = 17, \quad \Delta = -29, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 16, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{\mathbf{1}} 1^{\mathbf{1}} 3151321 \quad 8 \quad 1231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, \quad T_{\max} = 17, \quad \Delta = -29, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 16, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 16.67 \rightarrow 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [101] \quad 1^{\mathbf{1}} 1^{\mathbf{1}} 23151321 \quad 8 \quad 12321^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, \quad T_{\max} = 17, \quad \Delta = -3, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 17, 1), \quad c_{\text{ext}} = 4.0$$

$$[105] \quad 31^{\mathbf{1}} 6131613151^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, \quad T_{\max} = 18, \quad \Delta = -29, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 17.93 \rightarrow 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[106] \quad 31^{\mathbf{1}} 6131613151^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, \quad T_{\max} = 18, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [110] \quad 1^{\mathbf{1}} 3221 \quad (12) \quad 12231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, \quad T_{\max} = 19, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [111] \quad 1^{\mathbf{1}} 1^{\mathbf{1}} 2321 \quad 8 \quad 12321712321^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 4.0$$

$$[115] \quad 3221 \quad (12) \quad 12231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 21, \quad \Delta = -58, \quad \det = -26, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 20.92 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[116] \quad 3221 \quad (12) \quad 12231^{\mathbf{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 21, \quad \Delta = -58, \quad \det = -35, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 20.89 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[117] \quad 3221 \quad (12) \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 21, \quad \Delta = -58, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 20.93 \rightarrow 21, \quad T_{\max} = 21, \quad T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 3221 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 5 & 1 \end{pmatrix} 13221 \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, \quad T_{\max} = 21, \quad \Delta = -58, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, \quad T_{\max} = 21, \quad T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [119] \quad 1 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 8 & 1 \end{pmatrix} 12321 \quad 8 \quad 12321 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 20, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 22 \quad [120] \quad 1 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 8 & 1 \end{pmatrix} 12321 \quad 8 \quad 12321 \quad 8 \quad 12315131 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 23 \quad [121] \quad 1 \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 8 & 1 \end{pmatrix} 12321 \quad 8 \quad 12321 \quad 8 \quad 12321 \quad 6131 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 23, \quad T_{\max} = 23, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 23, 1), \quad c_{\text{ext}} = 4.0$$

1.2.2 NHC(-2)(-3)

$$T = 5 \quad [0] \quad 131 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 5, \quad T_{\max} = 10, \quad \Delta = -145, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 21 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 5, \quad T_{\max} = 10, \quad \Delta = -145, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9.44 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 21 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 5, \quad T_{\max} = 10, \quad \Delta = -145, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9.33 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 221 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 5, \quad T_{\max} = 10, \quad \Delta = -145, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9.73 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1321 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 6, \quad T_{\max} = 10, \quad \Delta = -117, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 13131 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 6, \quad T_{\max} = 10, \quad \Delta = -124, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1321 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 6, \quad T_{\max} = 10, \quad \Delta = -117, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 2141 \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 6, \quad T_{\max} = 10, \quad \Delta = -136, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

- [8] $1^1 1^1 4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -136, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [9] $21^1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [10] $12 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [11] $2 2 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [12] $21^1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [13] $2 2 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -117, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.45 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [14] $2^1 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -119, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [15] $31^1 2 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -125, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [16] $2^1 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 6, T_{\max} = 10, \Delta = -119, \det = 112, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 8 & 11 & 6 \\ 4 & 11 & 7 & 5 \\ 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$
- $T = 7$ [17] $1^1 2 3 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 10, \Delta = -96, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [18] $1^1 1 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 10, \Delta = -88, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [19] $1^1 1 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 10, \Delta = -88, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [20] $1^1 1^2 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 10, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$

- [21] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$ $T = 7, T_{\max} = 10, \Delta = -96, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [22] $21^1 3221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 10, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [23] $1^1 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 10, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [24] $1^1 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 10, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [25] $1^1 41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [26] $1^1 41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [27] $1^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 10, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [28] $1^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 10, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [29] $1^1 1512 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [30] $1^1 5^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [31] $221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [32] $1^1 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -116, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [33] $1^1 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -116, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[34] \quad & 31^1 3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -123, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[35] \quad & 141^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -116, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[36] \quad & 41^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -116, \quad \det = -30, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.80 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[37] \quad & 41^1 232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -118, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[38] \quad & 41^1 322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -116, \quad \det = -62, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 9 & 6 \\ 3 & 9 & 4 & 4 \\ 2 & 6 & 4 & 3 \end{pmatrix}_{b_0} \\
[39] \quad & 41^1 322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, \quad T_{\max} = 11, \quad \Delta = -116, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
T = 8 \quad [40] \quad & 1^1 223131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -67, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[41] \quad & 1^1 223131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -67, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[42] \quad & 1^1 231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -68, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, \quad T_{\max} = 10, \quad T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[43] \quad & 1^1 412321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -89, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, \quad T_{\max} = 11, \quad T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[44] \quad & 1^1 314131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -94, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, \quad T_{\max} = 11, \quad T \in [10, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[45] \quad & 1^1 41321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -87, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, \quad T_{\max} = 11, \quad T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[46] \quad & 1^1 41321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -87, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[60] $3131\bar{5}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$

[61] $2^1\bar{3}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[62] $2^1\bar{3}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -89, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 6 \\ 3 & 6 & 5 \end{pmatrix}_{b_0}$

[63] $2^1\bar{3}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -89, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[64] $31\bar{5}13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[65] $1\bar{5}131^1\bar{3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$

[66] $31^1\bar{3}2151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -119, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$

[67] $231^1\bar{3}151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -119, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$

[68] $31^1\bar{5}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -119, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 8 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[69] $21^{11}\bar{4}132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -89, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 12 & 15 & 8 \\ 4 & 15 & 11 & 7 \\ 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$

$T = 9$ [70] $1^1\bar{2}2313221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 10, \Delta = -39, \det = 0, (n_+, n_-, n_0) = (1, 9, 1), c_{\text{ext}} = 3.0$

[73] $1^1\bar{4}13221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[74] $1^1\bar{4}13221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

- $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [88] $161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [89] $16131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [90] $1315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [91] $1315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [92] $1^1 4132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [93] $1612231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [94] $1^1 4132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [95] $2161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [96] $2315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [97] $1513141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [98] $1513141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [99] $1^1 41 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[100] $221\overset{1}{6}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[101] $23\overset{1}{15}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -90, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[102] $31\overset{1}{41}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[103] $41\overset{1}{31}513 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[104] $2231\overset{1}{3161}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -116, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 7 & 2 \\ 1 & 0 & 6 & 2 \\ 7 & 6 & 32 & 10 \\ 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[105] $3141\overset{1}{3151}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -117, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [106] $1^1 231413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$

[107] $1^1 231413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$

[111] $1^1 23215131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

[112] $1^1 23151231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

[113] $1^1 31414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -64, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[114] $1\overset{1}{5}13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

[115] $1^1 23215131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

[116] $1^1 \overset{1}{5} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[117] & 1 \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[118] & 1 \overset{1}{5} 13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[119] & 1 \overset{1}{4} 141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[120] & 1 \overset{1}{4} 141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[121] & 1 \overset{1}{2} 1513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0} \\
[122] & 1 \overset{1}{2} 1513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[123] & 1 \overset{1}{2} 1513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[135] & 1 \overset{1}{3} 151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[125] & 1 \overset{1}{3} 1612231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[126] & 1 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -86, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[127] & 1 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[128] & 1 \overset{1}{3} 151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

- [129] $1^1 21612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [130] $1^1 131612231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [131] $1^1 1^3 321 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [132] $1^1 1612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [133] $1^1 141315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [134] $1^1 141315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [124] $21^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -86, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [136] $1^1 23151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$
- [137] $1^1 1414141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [138] $1^1 1^3 32161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [151] $1^1 141322161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [140] $1^1 16123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [141] $1^1 141413151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$

$$[154] \quad 21\overset{1}{6}13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^1\overset{1}{5}1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[156] \quad 23216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[157] \quad 31\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -94, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[158] \quad 2^1\overset{1}{4}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[159] \quad 21\overset{1}{4}1414132^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 16 & 19 & 10 \\ 4 & 19 & 15 & 9 \\ 2 & 10 & 9 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [160] \quad 1^1\overset{1}{2}231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[161] \quad 1^1\overset{1}{5}1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[162] \quad 1^1\overset{1}{5}1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[163] \quad 1^1\overset{1}{3}141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[164] \quad 1^1\overset{1}{3}141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[165] \quad 1^1\overset{1}{2}231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[166] \quad 1^1\overset{1}{2}314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -37, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[184] \quad 1^1\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[179] \quad 131\overset{1}{5}13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^1\overset{1}{4}13151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[181] \quad 131\overset{1}{5}13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

- [182] $1^1 413151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [183] $1^1 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [178] $1^1 321612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [185] $1^1 141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [186] $1^1 141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [187] $1^1 321612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [192] $1^1 315132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [189] $1^1 315132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [190] $1^1 413161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [191] $1^1 413161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [188] $1^1 1712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [193] $1^1 413216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [194] $1^1 131^1 6^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [195] $1^1 413216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[196] & 1^1 321712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[197] & 1^1 321712231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[198] & 1^1 151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[199] & 1^1 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[200] & 1^1 221712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[201] & 1^1 1^1 31 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[202] & 1^1 1^1 31 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[203] & 1^1 1^1 321 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[204] & 1^1 1^1 151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[211] & 1^1 161232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[206] & 1^1 1^1 414132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[207] & 1^1 1^1 161314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[208] $\overset{1}{1}6\overset{1}{1}231\overset{1}{5}1 + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \, T_{\max} = 14, \, \Delta = -102, \, \det = -4, \, (n_+, n_-, n_0) = (1, 11, 0), \, c_{\text{ext}} = 3.0$
 $T_{\min} = 14, \, T_{\max} = 14, \, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[209] $221\bar{8}12231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2]$ $T = 11$, $T_{\max} = 14$, $\Delta = -105$, $\det = -4$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

$$[210] \text{ } 222181231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 14, \ \Delta = -105, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^1 15 1322161^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -102, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[212] \overset{1}{1}7\overset{1}{1}223141\overset{1}{1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -105, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[213] \ 21\overset{1}{7}123141\overset{1}{4} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 14, \ \Delta = -105, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[214] \begin{smallmatrix} 1 & 1 \\ 16 & 131 \\ 1 & 51 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -101, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[215] \begin{array}{c} 221 \\ 1^1 \end{array} 1^8 1231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \, T_{\max} = 14, \, \Delta = -105, \, \det = -4, \, (n_+, n_-, n_0) = (1, 11, 0), \, c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[216] \begin{array}{c} 21 \\ 1^4 \end{array} 1812231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -105, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[217] \overset{1}{2}3\overset{1}{2}171231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[218] \begin{array}{c} 21 \\ 1^1 \end{array} 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \, T_{\max} = 14, \, \Delta = -105, \, \det = -4, \, (n_+, n_-, n_0) = (1, 11, 0), \, c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, \, T_{\max} = 14, \, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[219] $\overset{1}{2}16\overset{1}{1}3151 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[220] $1^1 232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 12$ [224] $1^1 31414141321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[222] $1^1 31414141321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[223] $1^1 14141413221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[221] $1^1 14141413221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[231] $1^1 141414141 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

[232] $1^1 4131612321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[233] $1^1 1513151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[234] $1^1 3151315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

[235] $1^1 2315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

[236] $1^1 7^{12321} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[237] $1^1 7^{12321} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[238] $1^1 2321712231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[239] $1^1 4131612321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[240] $1^1 2315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

- [241] $1^1 2321712231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [242] $1^1 1513151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$
- [259] $1^1 6^{15} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [269] $1^1 161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [245] $21^1 8^{1321} 1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [246] $1^1 8^{13221} 1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [247] $1^1 8^{13221} 1231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [248] $1^1 4141316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [249] $21^1 8^{12321} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [250] $1^1 3221812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [251] $21^1 61315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [252] $1^1 3161232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [253] $1^1 3161232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

- [254] $1^1 4132171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [255] $1^1 2321712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$
- [256] $1^1 1513216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [257] $1^1 5131 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [258] $1^1 1513216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [243] $1^1 4131613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [260] $1^1 4141316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [261] $1^1 1513161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [262] $1^1 4132171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [263] $1^1 1321712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [264] $1^1 812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [265] $1^1 1513161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [266] $1^1 1612315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

- $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
 [267] $1 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
 [268] $1^1 6122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
 [244] $2221^1 812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
 [277] $31321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -93, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
 [271] $1^1 1712231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
 [272] $1^1 171232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
 [273] $1^1 4141413161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
 [274] $1^1 5131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
 [275] $1^1 171231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
 [276] $1^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
 [270] $21^1 71231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
 [278] $221^1 8123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

- $$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$
- [279] $1^1 17 12314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$
- [280] $1^1 21 8 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$
- [281] $1^1 31 5 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [282] $1^1 23 21812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [283] $1^1 14141322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$
- [284] $1^1 23 2181232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -89, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$
- [285] $1^1 21 7 1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$
- [286] $1^1 321^1 8^{12} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$
- [290] $1^1 23 21812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [288] $2^1 3^{21} 1^1 8^{12} 1231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [289] $1^1 31 5 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$
- [287] $1^1 5131^1 31^1 6^{12} 131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 15, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[291] \quad 31^{\overset{1}{1}5131} \quad 6 \quad 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -88, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[292] \quad 31^{\overset{1}{1}5131} \quad 6 \quad 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[293] \quad 2^{\overset{1}{1}414141} \quad 3 \quad 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[294] \quad 21^{\overset{2}{1}141414132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 112, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 20 & 23 & 12 \\ 4 & 23 & 19 & 11 \\ 2 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [295] \quad 1^1 231513151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[296] \quad 1^1 223151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[297] \quad 1^1 314141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[298] \quad 1^1 41414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[299] \quad 1^1 41414141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[300] \quad 1^1 223151315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[318] \quad 1^1 15131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^1 8 \quad 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^1 6 \quad 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^1 8 \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[314] \quad 1^1 8 \quad 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^1 61223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

- [328] $16123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [329] $1^1 32161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [352] $16123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [331] $15132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [332] $17122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [333] $13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [334] $31321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -66, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [335] $1^1 32171232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [336] $2161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [337] $15131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [338] $15131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [339] $1^1 712315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [340] $1^1 32171232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

- $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [341] $1 \overset{1}{8} \overset{2}{3} \overset{2}{1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$
- [342] $1 \overset{1}{5} 132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [343] $2 \overset{1}{3} \overset{1}{2} \overset{1}{3} \overset{1}{8} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [344] $1 \overset{1}{8} \overset{1}{3} 21^1 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [345] $1 \overset{1}{8} \overset{1}{3} 21^1 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [346] $21 \overset{1}{7} 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [347] $1 \overset{1}{4} 1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [348] $1 \overset{1}{4} 1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [349] $1 \overset{1}{4} 1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [350] $1 \overset{1}{4} 1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [351] $21 \overset{1}{6} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [330] $1321 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

- [353] $13221^1 812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [362] $1^1 6^1 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [355] $18122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [356] $16131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [357] $181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [358] $2218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [359] $22181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [360] $2181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [361] $18123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [354] $21812314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [363] $181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [364] $1^1 51321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[365] & 1^1 51321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[366] & 1^1 41414132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[367] & 1^1 61231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
T = 14 & [374] 1^1 414141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[369] & 1^1 2321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[370] & 1^1 2321613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[371] & 1^1 3216131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[372] & 1^1 3216131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[373] & 1^1 22315131^1 6^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[368] & 1^1 3161231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[375] & 1^1 22315131^1 6^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[376] & 1^1 414141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[377] & 1^1 3161231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0 \\
[403] & 21^1 7123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[394] & 1^1 2321^1 8^1 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[395] & 21^1 6131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[396] & 1^1 31^1 6^1 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[397] & 1^1 413221812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[398] & 1^1 316123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

- [399] $1^{131} 1^6 13161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [400] $1^{14} 13161315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [401] $1^{131} 1^6 13161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [402] $1^{1232171232151} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [393] $1^{14} 13161315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [404] $1^{17} 1223151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [405] $1^{17} 123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$
- [406] $1^{14} 13221812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [407] $1^{131} 1^6 13151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [408] $1^{1321712315131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [409] $1^{1321712315131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [410] $1^{15} 1321712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [411] $1^{1321613161231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

- [412] $1^{1\bar{1}2321} 1^{\bar{1}} 8^{123141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
- [413] $1^{1\bar{1}1321712321} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [414] $1^{1\bar{1}232171232151} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [423] $21^{\bar{1}} 7^{123216131} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [439] $1^{1\bar{1}1322181231} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [417] $1^{1\bar{1}414132181231} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [418] $1^{1\bar{1}1316131} 5^{\bar{1}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [419] $412321^{\bar{1}} 8^{12321} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [420] $1^{21} 8^{12315131} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [421] $1^{1\bar{1}5^{1321}} 8^{1231} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [422] $1^{1\bar{1}5^{1321}} 8^{1231} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [415] $1^{1\bar{1}1322181231} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [424] $1^{1\bar{1}413218123141} 1^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 16, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[425] $1^1 218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[426] $1^1 221812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[427] $1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[428] $1^1 216131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[429] $1^1 512321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[430] $1^1 18122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[431] $1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[432] $1^1 151321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[433] $1^1 151321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[434] $1^1 512321 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[435] $1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[436] $1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[437] $1^1 31 6 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[438] & 1^1 161232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[416] & 1^1 1414132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[440] & 1^1 315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[441] & 1^1 181231414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[442] & 1^1 161316123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[443] & 1^1 1712231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[444] & 1^1 1414141322181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[445] & 1^1 513217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[446] & 1^1 1612321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[447] & 1^1 171231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[448] & 1^1 16131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[449] & 1^1 31^1 61316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 6 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

- [450] $31\overset{1}{5}132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 14, T_{\max} = 17, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [451] $31\overset{31}{6}131\overset{1}{6}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 14, T_{\max} = 16, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 2 & 4 \\ 2 & 4 & 4 & 5 \end{pmatrix}_{b_0}$
- $T = 15$ [457] $1^1 23217123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [453] $1^1 32171231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [454] $1^1 41316131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [455] $1^1 14141414141\overset{1}{4}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 15, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [456] $1^1 23217123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [452] $1^1 41316131513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [458] $1^1 23216131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [459] $1^1 32171231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$
- [486] $1^1 4131613161231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [501] $1^1 \overset{31}{6}131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
- [476] $1^1 131\overset{1}{6}131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [477] $1^1 \overset{31}{6}131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
- [478] $1^1 1513221812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [479] $1^1 1513221812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [480] $1^1 1513161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [481] $1^1 13161232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [482] $1^1 4131613161231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$

- $$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$
- [496] $1 \overset{1}{\overset{15}{8}} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$
- [497] $1 \overset{1}{\overset{15}{8}} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$
- [498] $1 \overset{1}{\overset{15}{8}} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$
- [499] $16 \overset{1}{16} 12321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$
- [500] $15 \overset{1}{15} 13161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$
- [475] $18 \overset{21}{18} 123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$
- [502] $1 \overset{1}{1} 3161316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$
- [503] $1 \overset{1}{1} 6 \overset{1}{15} 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$
- [504] $23 \overset{1}{23} 21812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$
- [505] $1 \overset{1}{1} 3161316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$
- [506] $221 \overset{1}{221} 8123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$
- [507] $15 \overset{1}{15} 13218123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

- [508] $21\overset{1}{7}1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [509] $1^1\overset{1}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [510] $1613151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [511] $1513218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [512] $21\overset{1}{6}1316131^1\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [513] $1^1\overset{1}{6}123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [514] $1^1\overset{1}{4}132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [515] $31\overset{1}{5}1321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$
- [516] $161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [517] $1321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [518] $1^1\overset{1}{4}132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [519] $1612321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [522] $1^1\overset{1}{6}123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 18, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$

- [542] $1^1 315131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$
- [564] $21 6^1 13161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [565] $1^1 15 131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [566] $1^1 1^1 31612321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [567] $21 7^1 12321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [568] $1^1 8^1 123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [569] $1^1 16 12321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [570] $1^1 6^1 1316131^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [571] $1^1 6^1 1316131^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [572] $1^1 8^1 123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [573] $1^1 1^1 31612321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [574] $1^1 1^1 41321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [575] $1^1 1^1 32171232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [576] $1^1 1^1 31613151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[577] \quad & 1^1 41321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[578] \quad & 1^1 13218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[579] \quad & 1^1 41316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[580] \quad & 1^1 15131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[585] \quad & 1^1 22181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[582] \quad & 1^1 16123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[583] \quad & 1^1 21712321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[584] \quad & 1^1 131613161232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[581] \quad & 1^1 23218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[586] \quad & 1^1 17123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[587] \quad & 1^1 1513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -73, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[588] \quad & 1^{221} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& 1^1 3221 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[589] \quad & 1^1 31613151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[590] \quad & 1^1 15132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[591] \quad & 1^1 31612321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[592] \quad & 1^1 1613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[593] \quad & 1^1 15132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[594] \quad & 1^1 32181231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[595] \quad & 1^1 32181231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[596] \quad & 1^1 51232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[597] \quad & 21^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& 1^1 3221 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[598] \quad & 1^1 2171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[599] \quad & 1^1 3221^{21} 2221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[600] \quad & 1^1 61315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[601] \quad & 1^1 17123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[602] \quad 1^{1^1 51321} 8^{1232161^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[603] \quad 1^{1^1 61232181223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[604] \quad 21^1 8^{12321613151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[605] \quad 31^{31} 6^{131613151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17.80 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [606] \quad 1^{1^1 2321613161315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[607] \quad 1^{1^1 2321712321712321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[608] \quad 1^{1^1 31} 6^{131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[609] \quad 21^1 6^{1316131513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[610] \quad 1^{1^1 31} 6^{131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[611] \quad 1^{1^1 4132181231513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[612] \quad 1^{1^1 3216131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[613] \quad 1^{1^1 4132181231513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[614] \quad 1^{1^1 2321613161315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[615] \quad 1^{1^1 3216131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0$$

$$[630] \quad 21^1 6^{131613161231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[631] \quad 1^{1^1} 5^{1321812315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[632] \quad 221^1 8^{12321712321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[633] \quad 1^{1^1 232181231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[634] \quad 1^{1^1 232181231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[648] & 1^1 315132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[649] & 21^1 712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[661] & 221^1 812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[651] & 221^1 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[652] & 21^1 7123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[653] & 1^1 81223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[654] & 21^1 8123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[655] & 1^1 513161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[656] & 1^1 513161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[657] & 221^1 81232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[658] & 1^1 61315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[659] & 1^1 3221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -64, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[660] $17^{11}1232161316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[650] $16^{11}123218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[662] $3217^{11}12321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$

[663] $1^{11}612315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[664] $16^{11}1232181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[665] $23^{11}2181232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[666] $221^{13221}(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 19, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[667] $218^{11}123217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[668] $17^{11}123218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[669] $1^{11}513161315132171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[670] $17^{11}1232181223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[671] $1^{11}51316131^{16}13161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 17, T_{\max} = 20, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[740] & \begin{matrix} 1 \\ 2218123218123141 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[741] & \begin{matrix} 1 \\ 221 \end{matrix} \begin{matrix} 1513221 \\ (12) \end{matrix} 12231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[742] & \begin{matrix} 1 \\ 2171232181232151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[743] & \begin{matrix} 1 \\ 17122315132181231 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[745] & \begin{matrix} 1 \\ 16131513218123151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[744] & \begin{matrix} 1 \\ 1812315131613151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[746] & \begin{matrix} 1 \\ 15131613161232171 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[747] & \begin{matrix} 1 \\ 2181232181223151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[748] & \begin{matrix} 1 \\ 218123218123151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[749] & \begin{matrix} 1 \\ 15131613151322181 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[750] & \begin{matrix} 1 \\ 3221 \end{matrix} \begin{matrix} 1513221 \\ (12) \end{matrix} 12231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[751] & \begin{matrix} 1 \\ 3221 \end{matrix} \begin{matrix} 1513221 \\ (12) \end{matrix} 12231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 10, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.60 \rightarrow 21, T_{\max} = 21, T \in [21, 21]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[752] $1^{231} \overset{1}{5} 13221 (12) 12231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 13, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.77 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[753] $31 \overset{1}{6} 1316131 \overset{1}{6} 131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -34, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 3 & 5 \\ 2 & 5 & 5 & 6 \end{pmatrix}_{b_0}$$

[754] $3221^{11} \overset{1}{5} 13221 (12) 1223 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.80 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[755] $2231^{11} \overset{1}{5} 13221 (12) 12231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 21, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 20.86 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 4 \end{pmatrix}_{b_0}$$

$T = 19$ [764] $1^{15} 1316131613151321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[757] $1^{13} 16123218123151321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[758] $1^{14} 13161316131612321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[759] $1^{14} 13161316131612321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[760] $1^{13} 15131613161315131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[761] $1^{12321} \overset{1}{8} 12321812321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[762] $1^{16} 1232181231513221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[763] $1^{13} 16123218123151321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[756] $1^{15} 1316131613151321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[789] $1^{13} 2171232181232151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[784] $1^{13} 1613161232181231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[785] $1^{16} 123218123216131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[786] \quad & 1^1 15132181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[787] \quad & 1^1 15132181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[788] \quad & 1^1 32171232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[783] \quad & 1^1 51321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[790] \quad & 1^1 31513221 \quad 221 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[791] \quad & 1^1 321 \quad 1^1 8 \quad 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[792] \quad & 1^1 41321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[793] \quad & 1^1 21 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[794] \quad & 1^1 21 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[795] \quad & 1^1 321 \quad 1^1 8 \quad 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[796] \quad & 1^1 2161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[797] \quad & 1^1 21712321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

[798] $1\bar{7}123151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[799] $221 \quad (12) \quad 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[800] $1^1 51321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[801] $1^1 41321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[802] $1^1 31613161232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[803] $1\bar{6}131612321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[804] $1^1 71223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[813] $1^1 8 \quad 1232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[806] $1^1 812315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[807] $1^1 3161316131 \quad 6 \quad 13161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[808] $1^1 41413221 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[809] $1^1 8122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[842] $1^1 2321712321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[886] $1^1 8^{1321} 1232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[866] $1^1 232181232181223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[867] $1^1 612321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[868] $1^1 161232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[869] $1^1 413218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[870] $1^1 151321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[871] $1^1 217123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[872] $1^1 151321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[873] $1^1 322181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[874] $1^1 181223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[875] $1^1 322181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[876] $1^1 218123151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[877] & 1^{1^1} 221812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[878] & 1^{1^1} 321812315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[879] & 1^{1^1} 32181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[880] & 1^{1^1} 31513221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[881] & 1^{1^1} 31513221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[882] & 1^{1^1} 51321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[883] & 1^{1^1} 51321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[884] & 1^{1^1} 3218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[885] & 1^{1^1} 232181232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[865] & 1^{1^1} 3218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[897] & 1^{1^1} 414141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[888] & 1^{1^1} 5131513221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[889] $1 \overset{1}{7} 1232171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[890] $1 \overset{1}{4} 141413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[891] $1 \overset{1}{(12)} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[892] $1 \overset{1}{4} 141413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[893] $1 \overset{1}{(12)} 12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[894] $1 \overset{1}{4} 14141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[895] $1 \overset{1}{3} 16131613161232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[896] $1 \overset{1}{3} 321812321 \overset{1}{8} 1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[887] $1 \overset{1}{4} 1413221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[898] $1 \overset{1}{(12)} 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[899] $1 \overset{1}{(12)} 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[900] $1^1 316131613151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[901] $21^{1^1} (12)^{1^1} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[902] $1^1 181231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[903] $21^{1^1} (12)^{1^1} 12231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[904] $21^{1^1} (12)^{1^1} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[905] $31^1 513221(12)12231^1 513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[906] $1^1 812231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[907] $1^1 181231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 21$ [908] $1^1 6^{1^1} 1316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[909] $21^1 613161316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[910] $1^1 41316131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[911] $221^1 81232181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[912] $1^1 6^{1^1} 131613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[913] $1^1 31612321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[914] $1^1 31612321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

3.0

[915] $1^{\overline{1^1 2321}} 1^{\overline{8}} 12321812315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[916] $1^{\overline{1^1 2321}} 1^{\overline{8}} 12321812315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[917] $1^{\overline{1^1 321}} 1^{\overline{8}} 123218123151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[918] $1^{\overline{1^1 321}} 1^{\overline{8}} 123218123151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[919] $13161316131613151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[920] $1^{\overline{1^1 3221}} 123218123151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[921] $1^{\overline{1^1 3221}} 123218123151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[922] $1^{\overline{1^1 41316131613161315131}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[923] $21^{\overline{1^1 2231513221}} (12) 1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[924] $1^{\overline{1^1 23218123151321812321}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[925] $1^{\overline{1^1 31}} 1^{\overline{6}} 1316131613161231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[926] $21^{\overline{1^1 2231513221}} (12) 1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[927] $1^{\overline{1^1 32161316131613161231}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[948] $1^{\overline{1^1 31513221}} (12) 122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[949] $1^{\overline{1^1 4132181232181232151}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[950] $161232181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[951] $1513218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[952] $21^{\overline{1^1 513221}} (12) 122315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[953] $1^{\overline{1^1}} (12) 122315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[954] & 1^{11} (12) \quad 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -51, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& \begin{pmatrix} 1 & 5 & 1 & 3 & 2 & 2 & 1 \end{pmatrix} \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[955] & 1^{11} 16 \quad 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -48, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[956] & 1^{11} 2218 \quad 123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[957] & 1^{11} 31513221 (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -55, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[958] & 1^{11} 17 \quad 12321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[959] & 1^{11} 23218123218 \quad 1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -50, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[960] & 1^{11} 15 \quad 13218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -48, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[961] & 1^{11} 217 \quad 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[962] & 1^{11} 513221 (12) \quad 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -51, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[963] & 1^{11} 4132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -50, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[964] & 1^{11} 3221 (12) \quad 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, \quad T_{\max} = 23, \quad \Delta = -78, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} =
\end{aligned}$$

3.0
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[965] $16^{1^1} 131513221^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[966] $15^{1^1} 13221(12) 12231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[967] $15^{1^1} 13221(12) 12231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[968] $17^{1^1} 1232181^{1^1} 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[969] $1^{1^1 5131513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[970] $1^{1^1 5131513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[971] $1^{31^1 5^{1^1} 13221} (12) 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[972] $21^{1^1} 81232171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[973] $18^{1^1} 12321613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[974] $1^{1^1 5131613161315132181^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 21, T_{\max} = 24, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[975] $1^1 7123217123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21$, $T_{\max} = 24$, $\Delta = -99$, $\det = -4$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 3.0$

$T_{\min} = 24$, $T_{\max} = 24$, $T \in [24, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 22$ [976] $1^1 151316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -22$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[977] $1^1 6^{131} 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -24$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[978] $1^1 8^{12321} 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[979] $21^{15} 12^{13221} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[980] $1^{1231513221} 1^{12} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[981] $1^1 5^{1231513221} 151316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -22$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[982] $1^{12231513221} 1^{12} 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -27$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[983] $1^{12231513221} 1^{12} 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -27$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[984] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[985] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[986] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[987] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[988] $1^{15} 1^{12} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[989] $1^{15} 1^{12} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[990] $21^{1231513221} 1^{12} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[991] $21^{1231513221} 1^{12} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[992] $1321^1 81232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[993] $1^1 321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[994] $21^{15} 1^{12} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[995] $1^1 321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[996] $1^1 6^{131} 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 22$, $\Delta = -24$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[997] $1^{1231513221} 1^{12} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[998] $1^1 8^{12321} 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 3.0$

[1019] $21 \overset{1}{7} 1232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1020] $221 \overset{1}{8} 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1021] $21 \overset{1}{(12)} 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1022] $1 \overset{1}{6} 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1023] $21 \overset{1}{(12)} 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1024] $21 \overset{1}{7} 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1025] $21 \overset{1}{8} 12321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1026] $1 \overset{1}{5} 132181232181231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1027] $1 \overset{1}{5} 132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1028] $1 \overset{1}{4} 141413221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1029] $1 \overset{1}{5} 1232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1030] \quad 1^1 15 132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1031] \quad 1^1 32181232171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1032] \quad 1^1 17 12321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1033] \quad 1^1 4141413221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1034] \quad 1^1 17 1231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1035] \quad 1^1 5131513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1036] \quad 1^1 5131513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1037] \quad 1^1 41413221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1038] \quad 1^1 41413221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1039] \quad 1^1 16 131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1040] \quad 1^1 21 8 1232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} =$$

3.0
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
[1041] $1^1 61315132181232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
 $1^1 3221$
[1042] $21 (12) 1223151322171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
 $1^1 61231513221$
[1043] $1 (12) 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
 $1^1 513221$
[1044] $21 (12) 12231513161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 25, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
[1045] $1^1 51316131613161232181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
 1^1
[1046] $18 123217123218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
 $1^1 51321812321$
[1047] $1 8 1232171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
 31
[1048] $31 6 131613161316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 22, T_{\max} = 22, \Delta = -4, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 4 & 6 \\ 2 & 6 & 6 & 7 \end{pmatrix}_{b_0}$

 $T = 23$ [1049] $1^1 31 6 131613161316131^1 51 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$
[1050] $1^1 4141413221(12)12231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$
[1051] $1^1 2315132181232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 23, T_{\max} = 23, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$
 $1^1 31$
[1052] $1 6 131613161316131^1 51 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1053] $1^1 2315132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1054] $1^1 4131613161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1055] $1^1 1513161316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1056] $1^1 2321712321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1057] $1^1 1513161316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1058] $1^1 2321712321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1059] $1^1 8^{1321} 12321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1060] $1^1 8^{1321} 12321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1061] $1^1 8^{12321} 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1062] $1^1 8^{12321} 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1063] $1^1 2321812321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1064] $1^1 1513218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1065] $1^1 4141413221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1066] $1^1 1513218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1067] $1^1 3151321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 3.0$

[1085] $1^1 217123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1086] $1^1 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1087] $1^1 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1088] $1^1 613151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1089] $1^1 21812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} =$

3.0

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1090] \ 2218^{1^1}1232181232181231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1091] \ 1^{1^1}5131513221(12) \ 122315131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1092] \ 1^{1^1}5131513221(12) \ 122315131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1093] \ 1^{1^1}3218123218 \ 1232181231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1094] \ 2218^{1^1}12321812321812231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1095] \ 16^{1^1}1232181232181232151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1096] \ 16^{1^1}1232181232181231^{1^1}51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1097] \ 1^{1^1}613161232181232181231^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1098] \ 1^{1^1}41413221(12)122315132161^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1099] \ 1^{1^1}3221(12)122315131613151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1100] & 1^{1712231513221} (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1101] & 1^{1316131613161315132181^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1102] & 1^{1513221} (12) 122315132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1103] & 21^1 81231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1104] & 1^{13221} (12) 122315131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1105] & 1^1 812231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1106] & 1^1 6131513221(12)122315^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1107] & 1^1 6131513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1108] & 1^{1712315132181232181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1109] & 1^1 71231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1110] \ 21 \quad \begin{matrix} 1 \\ 1513221 \end{matrix} (12) \quad 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, \ T_{\max} = 26, \ \Delta = -99, \ \det = -4, \ (n_+, n_-, n_0) = (1, 23, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, \ T_{\max} = 26, \ T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1111] \ 18 \ 123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, \ T_{\max} = 26, \ \Delta = -97, \ \det = -4, \ (n_+, n_-, n_0) = (1, 23, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, \ T_{\max} = 26, \ T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [1112] \ 1 \quad \begin{matrix} 1 \\ 12321 \end{matrix} 8 \ 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1113] \ 16 \ 123218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -22, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1114] \ 1^1 41321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1115] \ 1^1 41321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1116] \ 15 \ 131613161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -18, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1117] \ 15 \ 131613161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -18, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1118] \ 1 \quad \begin{matrix} 1 \\ 15131513221 \end{matrix} (12) \quad 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -25, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1119] \ 1 \quad \begin{matrix} 1 \\ 1321 \end{matrix} 8 \ 12321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1120] \ 1 \quad \begin{matrix} 1 \\ 1321 \end{matrix} 8 \ 12321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1121] \ 1 \quad \begin{matrix} 1 \\ 15131513221 \end{matrix} (12) \quad 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -25, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1122] \ 216 \ 13161316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -17, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1123] \ 216 \ 1316131613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -16, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1124] \ 1^1 31612321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 25, \ \Delta = -29, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1125] \ 1 \quad \begin{matrix} 1 \\ 2321812321 \end{matrix} 8 \ 12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 25, \ \Delta = -29, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1126] \ 1 \quad \begin{matrix} 1 \\ 12321 \end{matrix} 8 \ 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 24, \ \Delta = -27, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1127] \ 1^1 31612321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 25, \ \Delta = -29, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 3.0$$

$$[1144] \ 1 \quad \begin{matrix} 1 \\ 131513221 \end{matrix} (12) \quad 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \ T_{\max} = 25, \ \Delta = -51, \ \det = 2, \ (n_+, n_-, n_0) = (1, 24, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, \ T_{\max} = 25, \ T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

[1145] $1^1 6131612321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -49, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1146] $1^1 7123151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -49, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1147] $1^1 6131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -50, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1148] $1^1 41413221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -54, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1149] $1^1 41413221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -54, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1150] $221^1 8123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1151] $21^1 71232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1152] $1^1 513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -50, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1153] $21^1 3221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -55, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1154] $21^1 3221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -55, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$
 $T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1155] $1^1 513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, \quad T_{\max} = 25, \quad \Delta = -50, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 3.0$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1156] \quad 1^1 21 7 1232181232181231^1 5 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1157] \quad 1^1 18 12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1158] \quad 1^1 31 6 131513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1159] \quad 1^1 41 3221(12)122315131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1160] \quad 1^1 21 8 1231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1161] \quad 1^1 41 413221(12)1223151322171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1162] \quad 1^1 41 3221(12)122315131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1163] \quad 1^1 51 31513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1164] \quad 1^1 18 12315132181232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1165] \quad 1^1 31 61316131613161232181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[1166] $1^1 17 1231513221(12)1223151 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1167] $1^1 17 1231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[1168] $1^1 3218123218123151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1169] $1^1 15 13221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1170] $1^1 31513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[1171] $1^1 161316131513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1172] $1^1 181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 27, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1173] $1^1 21(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 27, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 25$ [1174] $1^1 21613161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1175] $1^1 232171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1176] $1^1 2231513221(12)122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1177] $1^1 151321812321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1178] $21 \overset{1}{7} 123218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1179] $1 \overset{1}{8} \overset{1}{2321} 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1180] $1 \overset{1}{5} 1321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1181] $1 \overset{1}{3} 21712321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1182] $1 \overset{1}{3} 21712321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1183] $1 \overset{1}{5} 1316131613161316131 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1184] $1 \overset{1}{3} 16123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1185] $1 \overset{1}{6} 131513221 \overset{1}{(12)} 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1186] $1 \overset{1}{231513221} \overset{1}{(12)} 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1187] $1 \overset{1}{231513221} \overset{1}{(12)} 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1188] $1 \overset{1}{232171232181232181232151}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1189] $1 \overset{1}{8} \overset{1}{2321} 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$

[1206] $1 \overset{1}{32181231513221} \overset{1}{(12)} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1207] $1 \overset{1}{32181231513221} \overset{1}{(12)} 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1208] $221 \overset{1}{8} 1232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1209] $221 \overset{1}{8} 1232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1210] $1 \overset{1}{3221} \overset{1}{(12)} 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[1211] \quad & 1^1 8 123151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -48, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1212] \quad & 1^1 3221 (12) 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -55, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1213] \quad & 1^1 23218123218123151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -48, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1214] \quad & 1^1 7 1231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1215] \quad & 1^1 31513221 (12) 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1216] \quad & 1^1 6 131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1217] \quad & 1^1 6 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -75, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1218] \quad & 1^1 8 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -69, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1219] \quad & 1^1 6 1231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -74, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1220] \quad & 1^1 5 13221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1221] \quad & 1^1 5 13221(12)122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = \\
& (1, 25, 0), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1222] \quad 1^1 2181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1223] \quad 1^1 (12) 12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1224] \quad 1^1 (12) 12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1225] \quad 1^1 2181231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[1226] \quad 1^1 1812231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[1227] \quad 1^1 171232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1228] \quad 1^1 151321812321812315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 28, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1229] \quad 1^1 (12) 122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 28, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [1230] \quad 1^1 216131613161316131613151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$$

$$[1231] \quad 1^1 3161232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$$

$$[1232] \quad 1^1 812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$$

$$[1233] \quad 1^1 (12) 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) =$$

[1267] $1^1 316131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1268] $1^1 413221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1269] $1^1 413221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1270] $1^1 712321712321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1271] $1^1 812231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1272] $1^1 513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1273] $1^1 81231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1274] $1^1 3221(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1275] $1^1 513161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 28, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1276] $1^1 81232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 28, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1277] $1^1 61316131513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 28, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1278] & 1^{1\bar{1}}712231513221(12)12231513151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -72, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1279] & 1^{1\bar{1}}413221(12)12231513218123151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1280] & 1^{1\bar{1}}413221(12)12231513218123151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -77, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1281] & 1^{1\bar{1}}513161316131513221(12)12231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -78, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1282] & 1^{1\bar{1}}321812321812321 \quad 8 \quad 1232171^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1283] & 1^{1\bar{1}}71231513221(12)12231513151^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 26, T_{\max} = 28, \Delta = -72, \det = 3, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
\\
T = 27 & [1284] 21^{\bar{1}}712321812321812321712321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
& 1^{1\bar{1}}31513221 \\
[1285] & 1^{1\bar{1}}(12) 1223151321812321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[1286] & 1^{1\bar{1}}51316131513221(12)12231513221^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
& 1^{1\bar{1}}31513221 \\
[1287] & 1^{1\bar{1}}(12) 1223151321812321^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[1288] & 1^{1\bar{1}}31612321812321812321812231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[1289] & 1^{1\bar{1}}41321812321812321812315131^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
& 1^{1\bar{1}}231513221 \\
[1290] & 1^{1\bar{1}}(12) 122315132181231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[1291] & 1^{1\bar{1}}31612321812321812321812231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
& 1^{1\bar{1}}231513221 \\
[1292] & 1^{1\bar{1}}(12) 122315132181231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0 \\
[1293] & 1^{1\bar{1}}32171232181232181232171231^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[1294] $1^1 8 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1295] $1^1 6 12321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1296] $1^1 8^{1321} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1297] $1^1 8^{1321} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1298] $21^1 8 1231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1299] $1^1 51316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[1318] $1^1 6 1316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1319] $1^1 5 13221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1320] $1^1 2321812321812321^1 8 1232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1321] $1^1 8 12321712321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1322] $1^1 316131513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[1323] $1^1 6 131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1324] $1^1 5 13221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1325] $1^1 5 13221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -72, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1326] & 1^1 413221(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -76, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1327] & 1^1 81232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1328] & 1^1 2181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -71, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1329] & 1^1 812231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -71, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1330] & 1^1 1513221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -72, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1331] & 1^1 171232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1332] & 1^1 1613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1333] & 1^1 8123217123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 30, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
\\
T = 28 & [1334] 1^1 413218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = \\
& (1, 28, 1), c_{\text{ext}} = 3.0 \\
[1335] & 1^1 321712321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = \\
& (1, 28, 1), c_{\text{ext}} = 3.0 \\
[1336] & 1^1 221812321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = \\
& (1, 28, 1), c_{\text{ext}} = 3.0 \\
[1337] & 1^1 151321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = \\
& (1, 28, 1), c_{\text{ext}} = 3.0 \\
[1338] & 1^1 3161316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 28, 1), c_{\text{ext}} = 3.0$
 $[1339] \ 1^1 3161316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1340] \ 1^1 51321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1341] \ 1^1 315132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1342] \ 1^1 315132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1343] \ 1^1 61316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1344] \ 1^1 8^{1321} 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1345] \ 1^1 8^{1321} 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1346] \ 1^1 321712321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1347] \ 21^1 712321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1348] \ 1^1 413218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

$[1361] \ 21^{13221} (12)^{21} 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

$[1362] \ 1^1 5132181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$[1363] \ 1^1 812321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$[1364] \ 1^1 316131513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

$[1365] \ 1^1 32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

$[1366] \ 1^1 32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\text{max}} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$[1367] \ 1^1 31613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\text{max}} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1368] \quad 1^1 17 1231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1369] \quad 1^1 5132181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1370] \quad 1^1 31613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1371] \quad 1^1 17 1232181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[1372] \quad 1^1 15 13221(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1373] \quad 21^1 8 1232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1374] \quad 1^1 513221 (12) 122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1375] \quad 1^1 613161316131513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1376] \quad 1^1 51321812321812321 8 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 31, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 29 \quad [1377] \quad 1^1 3221812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$$

$$[1378] \quad 1^1 3161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$$

[1379] $1^1 4132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1380] $1^1 4132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1381] $1^1 3161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1382] $1^1 513218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1383] $1^1 1612321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1384] $1^1 5132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1385] $221^1 812321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1386] $1^1 81232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1387] $1^1 5132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1388] $1^1 812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1389] $1^1 513218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1390] $1^1 812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[1400] $21^1 81231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1401] $1^1 16131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1402] $1^1 61232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1403] $1^1 1(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1404] $1^1 1(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1405] $21^1 1(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 29, T_{\max} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1406] \quad 1^{13221} 21 (12) 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1407] \quad 1^{161231513221}(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1408] \quad 1^{1812231513221}(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1409] \quad 1^{171232181231513221}(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1410] \quad 1^{2181232181231513221}(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[1411] \quad 1^{1321812321812321812315132181^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1412] \quad 1^{13221} 1 (12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1413] \quad 1^{13221} 1 (12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 31, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1414] \quad 1^{181232181231513221}(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 32, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [1415] \quad 1^{1316131513221}(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$$

$$[1416] \quad 1^{1316131513221}(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) =$$

$(1, 30, 1), c_{\text{ext}} = 3.0$
 $[1417] \ 1^{\overline{1}32171232181232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1418] \ 1^{\overline{1}32171232181232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1419] \ 21 \overline{1}712321812321812321812315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1420] \ 1 \overline{1}5132181232181232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1421] \ 1 \overline{1}5132181232181232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1422] \ 1 \overline{1}6123218123218123218123216131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1423] \ 1^{\overline{1}41321812321812321812321812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1424] \ 1^{\overline{1}61232181231513221(12)12231513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 30, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1425] \ 1^{\overline{1}41321812321812321812321812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1426] \ 1^{\overline{1}321} \overline{1}8 \ 123218123218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1427] \ 1^{\overline{1}321} \overline{1}8 \ 123218123218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[1438] \ 1^{\overline{1}712231513221(12)122315132181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 $[1439] \ 21 \overline{1}^{13221} (12) 12231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
 $[1440] \ 21 \overline{1}^{13221} (12) 12231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
 $[1441] \ 1^{\overline{1}413221(12)12231513221} \overline{1}^{1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
 $[1442] \ 1^{\overline{1}2321812321812321812315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
 $[1443] \ 1^{\overline{1}71232181231513221(12)122315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]} \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1444] \quad 1^1 171231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1445] \quad 1^1 413221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1446] \quad 1^1 513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1447] \quad 1^1 413221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1448] \quad 1^1 513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1449] \quad 1^1 5132181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1450] \quad 21^1 513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[1451] \quad 21^1 81232181231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1452] \quad 1^1 5132181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1453] \quad 21^1 513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 32, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 31 \quad [1454] \quad 1^{1231513221} (12) \quad 12231513221^{21} (12) 12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1455] \quad 1^{1321} 8^{1232181232181232181231} 5^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1456] \quad 1^{1322181232181232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1457] \quad 21^{17123218123218123218123216131} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1458] \quad 1^{161232181232181232181232171231} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1459] \quad 1^{1231513221} (12) \quad 12231513221^{21} (12) 12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1460] \quad 1^{1413218123218123218123218123141} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1461] \quad 1^{1322181232181232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1462] \quad 1^{151321812321812321812321812231} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1463] \quad 1^{151321812321812321812321812231} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1464] \quad 1^{1321} 8^{1232181232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1465] \quad 1^{181232181232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1466] \quad 1^{171232181231513221} (12) 12231513221^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1467] \quad 221^{1812321812321812321812315131} + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1468] \quad 1^{1321} 8^{1232181232181232181231} 5^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$$

$$[1478] \quad 1^{141413221} (12) 12231513221^{21} 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1479] \quad 1^{1513221} (12) 122315132181232181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1480] \quad 1^{141413221} (12) 12231513221^{21} 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1481] \quad & \overset{1}{1} \overset{1}{5} 13221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1482] \quad & \overset{1}{1} \overset{1}{5} 13221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1483] \quad & \overset{1}{1} 8 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -46, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1484] \quad & \overset{1}{2} 1 \overset{1}{8} 1232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1485] \quad & \overset{1}{2} 1 \overset{1}{8} 1231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -46, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1486] \quad & \overset{1}{1} 5 13221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -48, \det = -2, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1487] \quad & \overset{1}{1} \overset{1}{5} 13221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -71, \det = -3, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1488] \quad & \overset{1}{1} 6 1232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -76, \det = -3, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1489] \quad & \overset{1}{1} 8 1232171232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1490] \quad & \overset{1}{1} \overset{1}{5} 13221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, T_{\max} = 33, \Delta = -71, \det = -3, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& T = 32 \quad [1491] \quad 1 \overset{1}{6} 1232181232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1492] \quad 1 \overset{1}{3} 2181231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1493] \quad 1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1494] \quad 1 \overset{1}{3} 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1495] \quad 1 \overset{1}{3} 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1496] \quad 1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1497] \quad 21 \overset{1}{8} 1232181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1498] \quad 1 \overset{1}{5} 13218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1499] \quad 1 \overset{1}{231513221} (12) \overset{1}{12231513221} (12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1500] \quad 1 \overset{1}{231513221} (12) \overset{1}{12231513221} (12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1501] \quad 1 \overset{1}{5} 13218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1502] \quad 1 \overset{1}{3} 2181231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1503] \quad 1 \overset{1}{2} 32181232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1504] \quad 1 \overset{1}{2} 32181232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1505] \quad 1 \overset{1}{6} 12321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1506] \quad 21 \overset{1}{7} 1232181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -16, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1507] \quad 221 \overset{1}{8} 123218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -16, \det = 0, (n_+, n_-, n_0) = \\
& (1, 32, 1), c_{\text{ext}} = 3.0 \\
& [1518] \quad 1 \overset{1}{5} 13221 (12) \overset{1}{12231513221} (12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [1519] \quad 21 \overset{1}{13221} (12) \overset{1}{12231513221} (12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [1520] \quad 21 \overset{1}{1513221} (12) \overset{1}{12231513221} (12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [1521] \quad 1 \overset{1}{5} 13221 (12) \overset{1}{12231513221} (12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1522] \quad 1 \quad 1^1 5 13221 \quad (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1523] \quad 1 \quad 1^1 5 13221 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[1524] \quad 1^1 41413221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1525] \quad 21 \quad 1^1 3221 \quad (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1526] \quad 1^1 812321712321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1527] \quad 1^1 171232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1528] \quad 1 \quad 1^1 3221 \quad (12) \quad 12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1529] \quad 1 \quad 1^1 321812321812321812321 \quad 8 \quad 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 34, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [1530] \quad 1^1 5 132181232181232181232181231^1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -12, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[1531] \quad 21 \quad 1^1 7 12321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$[1532] \quad 221 \quad 1^1 8 1232181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
[1533] & 1^{1^1 31513221} (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1534] & 1^{1^1 231513221} (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1535] & 1^{1^1} 16123218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1536] & 1^{1^1 231513221} (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1537] & 21^{1^1 513221} (12) \quad 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1538] & 21^{1^1 513221} (12) \quad 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1539] & 1^{1^1 231513221} (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1540] & 1^{1^1 231513221} (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1541] & 1^{1^1} 51232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1542] & 1^{1^1} 5132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1543] & 1^{1^1} 5132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[1550] & 1^{1^1 5131513221(12)12231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1551] & 1^{1^1 5131513221(12)12231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1552] & 1^{1^1 2321812321812321812321} 8 \quad 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1553] & 21^{1^1 3221} (12) \quad 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1554] & 1^{1^1} 16131513221(12)12231513221(12)12231^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& \text{[1555]} \quad 1 \quad \begin{matrix} 1^1 413221(12)12231513221 \\ (12) \end{matrix} \quad 12231513161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& \text{[1556]} \quad 21^1 8 \quad \begin{matrix} 1232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \\ (12) \end{matrix} \quad T = 33, T_{\max} = 35, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{smallmatrix} \right)_{b_0} \\
& \text{[1557]} \quad 1 \quad \begin{matrix} 1^1 3221 \\ (12) \end{matrix} \quad 12231513221(12)122315132171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& \text{[1558]} \quad 1^1 71232181232181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 33, T_{\max} = 36, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{smallmatrix} \right)_{b_0} \\
& T = 34 \quad \text{[1559]} \quad 221^1 8 \quad \begin{matrix} 12321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \\ (12) \end{matrix} \quad T = 34, T_{\max} = 34, \Delta = -14, \det = \\
& 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0 \\
& \text{[1560]} \quad 16^1 1232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -13, \det = 0, (n_+, n_-, n_0) = \\
& (1, 34, 1), c_{\text{ext}} = 3.0 \\
& \text{[1561]} \quad 16^1 1232181232181232181232181231^1 5^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -11, \det = 0, (n_+, n_-, n_0) = \\
& (1, 34, 1), c_{\text{ext}} = 3.0 \\
& \text{[1562]} \quad 221^1 8 \quad \begin{matrix} 1232181232181232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \\ (12) \end{matrix} \quad T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = \\
& (1, 34, 1), c_{\text{ext}} = 3.0 \\
& \text{[1563]} \quad 21^1 7123218123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = \\
& (1, 34, 1), c_{\text{ext}} = 3.0 \\
& \text{[1566]} \quad 1 \quad \begin{matrix} 1^1 513221 \\ (12) \end{matrix} \quad 12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \\
& \text{[1567]} \quad 1 \quad \begin{matrix} 1^1 61231513221(12)12231513221 \\ (12) \end{matrix} \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -47, \det = 2, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{smallmatrix} \right)_{b_0} \\
& \text{[1568]} \quad 21 \quad \begin{matrix} 1^1 3221 \\ (12) \end{matrix} \quad 12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{smallmatrix} \right)_{b_0} \\
& \text{[1569]} \quad 1^1 32181232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{smallmatrix} \right)_{b_0}
\end{aligned}$$

[1570] $1^1 32181232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1571] $1^1 \overset{1}{5} 13221 (12) 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1572] $1^1 16131513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1573] $1^1 171231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1574] $1^1 41413221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1575] $1^1 41413221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1576] $1^1 181231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1577] $1^1 \overset{1}{3} 221 (12) 12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1578] $1^1 413221(12)12231513221 (12) 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1579] $1^1 \overset{1}{5} 13221(12)12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 36, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1580] $1^1 181232181232181231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 34, T_{\max} = 37, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix} b_0$$

$$T = 35 \quad [1581] \quad 1 \quad (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -24, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$\begin{array}{l} \text{[1582]} \quad 21 \overset{1}{7} 1232181232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -11, \det = 0, (n_+, n_-, n_0) = \\ (1, 35, 1), c_{\text{ext}} = 3.0 \end{array}$$

$$\begin{array}{l} \text{[1583]} \quad 1 \quad (12) \quad 12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -24, \det = 0, (n_+, n_-, n_0) = \\ (1, 35, 1), c_{\text{ext}} = 3.0 \end{array}$$

[1584] $221\overset{1}{8}123218123218123218123218123141^1 + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 35, 1), c_{\text{ext}} = 3.0$

$$\begin{array}{l} \text{[1585]} \quad 21\overset{1}{7}1232181232181232181232181231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -9, \det = 0, (n_+, n_-, n_0) = \\ (1, 35, 1), c_{\text{ext}} = 3.0 \end{array}$$

$$[1592] \overset{1}{1} 8 12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{\mathfrak{su}}_2 : \textcolor{blue}{\mathfrak{g}}_2] \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix} b_0$$

$$\begin{array}{l} \text{[1593] } 1\overset{1}{7}1231513221(12)12231513221(12)1223141\overset{1}{4}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2] \quad T=35, T_{\max}=36, \Delta=-45, \det=-2, (n_+, n_-, n_0)= \\ (1, 35, 0), c_{\text{ext}}=3.0 \end{array}$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix} b_0$$

$$[1594] \begin{array}{c} 1^1 \\ 16131513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \\ (1, 35, 0), c_{\text{ext}} = 3.0 \end{array} \quad T = 35, T_{\text{max}} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) =$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1595] \quad 1^{1712231513221(12)12231513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[1596] $21\overset{1}{8}1231513221(12)12231513221(12)12231^{\overset{21}{1}}+[\textcolor{red}{\mathfrak{su}}_2\textcolor{blue}{\mathfrak{g}}_2]$ $T = 35$, $T_{\max} = 36$, $\Delta = -45$, $\det = -2$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 3.0$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$\begin{array}{l} \text{[1597] } 1 \quad \text{(12)} \quad 12231513161^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -49, \det = -2, (n_+, n_-, n_0) = \\ (1, 35, 0), c_{\text{ext}} = 3.0 \end{array}$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[1598] $1^1 41413221(12)12231513221(12)122315132161^1 + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [1599] \quad 1 \quad \overset{1}{1} \overset{1}{5} 13221 (12) 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -42, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [1600] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 36, \Delta = -41, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [1601] \quad 1 \overset{1}{6} 1231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [1602] \quad 1 \quad \overset{1}{4} 13221(12)12231513221 \quad 1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [1603] \quad 1 \overset{1}{5} 13221(12)12231513221(12)122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -69, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [1604] \quad 1 \quad \overset{1}{1} 3221 (12) 12231513221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 37, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [1605] \quad 1 \quad \overset{1}{1} 513221 (12) 12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 38, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& T = 36 \quad [1606] \quad 1 \quad \overset{1}{1} 231513221 (12) 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -22, \det = \\
& 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1607] \quad 21 \quad \overset{1}{1} 3221 (12) 12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1608] \quad 221 \overset{1}{8} 1232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -10, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1609] \quad 1 \quad \overset{1}{1} 31513221 (12) 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1610] \quad 1 \quad \overset{1}{1} 31513221 (12) 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1611] \quad 221 \overset{1}{8} 1232181232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -8, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [1612] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 36, 1), c_{\text{ext}} = 3.0$$

$$[1613] \quad 1 \quad \begin{matrix} 1^1 2231513221(12)12231513221 \\ (12) \end{matrix} \quad 12231513161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[1614] \quad 21 \quad \begin{matrix} 1^1 3221 \\ (12) \end{matrix} \quad 12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0$$

$$[1624] \quad 1 \quad \begin{matrix} 1^1 513221 \\ (12) \end{matrix} \quad 12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[1625] \quad 1^1 41413221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[1626] \quad 21^1 81231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1627] \quad 1^1 71231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[1628] \quad 1^1 812231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1629] \quad 1 \quad \begin{matrix} 1^1 31513221(12)12231513221 \\ (12) \end{matrix} \quad 122315132171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1630] \quad 1^1 5131513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1631] \quad 1^1 71231513221(12)12231513221(12)12231^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 37, T_{\text{max}} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1632] \quad 1^1 513221(12)12231513221(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\text{max}} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$$

$$T_{\text{min}} = 38, T_{\text{max}} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[1633] $1^1 8 1231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 36, T_{\max} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1634] $1^1 413221(12)12231513221(12)122315131613161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 36, T_{\max} = 38, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$T = 37$ [1635] $1^1 2231513221(12)12231513221 122315132171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[1636] $1^1 232181231513221(12)12231513221 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[1637] $1^1 232181231513221(12)12231513221 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[1638] $1^1 231513221 12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[1639] $1^1 7 1231513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[1645] $1^1 8 1231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1646] $1^1 21 8 1231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1647] $1^1 8 12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1648] $1^1 61231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1649] $1^1 21 8 1231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1650] $1^1 16 131513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1651] \quad 1^1 18 12231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1652] \quad 1^1 131513221(12)12231513221(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1653] \quad 1^1 15 13221(12)12231513221(12)122315131613161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 37, T_{\max} = 39, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 38 \quad [1654] \quad 1^1 41413221(12)12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1655] \quad 1^1 18 12231513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1656] \quad 1^1 2231513221(12)12231513221(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1657] \quad 1^1 31513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1658] \quad 1^1 15 13221(12)12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1659] \quad 1^1 15 13221(12)12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1660] \quad 1^1 31513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1661] \quad 1^1 218 1231513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1662] \quad 1^1 18 1231513221(12)12231513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1663] \quad 1^1 41413221(12)12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$$

$$[1674] \quad 1^1 17 1231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 39, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1675] \quad 1^1 712231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 39, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1676] \quad 1^1 61316131513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 38, T_{\max} = 39, \Delta = -48, \det =$$

$$2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$1^{13221}$$

$$[1677] \ 1 \ (12) \ 12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 40, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 40, T_{\max} = 40, T \in [40, 40]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 39 \quad [1678] \ 1^1 316131513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0$$

$$[1679] \ 1^1 61316131513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0$$

$$[1683] \ 1^1 812231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 40, T_{\max} = 40, T \in [40, 40]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1684] \ 21^1 81231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 40, T_{\max} = 40, T \in [40, 40]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1685] \ 1^1 (12) \ 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 40, T_{\max} = 40, T \in [40, 40]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1686] \ 1^1 413221(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1687] \ 1^1 81232181231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$T = 40 \quad [1688] \ 1^1 232181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0$$

$$[1689] \ 1^1 232181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0$$

$$[1695] \ 1^1 (12) \ 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1696] \ 1^1 (12) \ 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 513221 & 1 \\ [1697] & 1 & (12) & 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 40, \quad T_{\max} = 42, \quad \Delta = -76, \quad \det = \\ & 3, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 \\ [1698] & 18 & 1232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 40, \quad T_{\max} = 42, \quad \Delta = -72, \quad \det = \\ & 3, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 \\ [1699] & 15 & 13221(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 40, \quad T_{\max} = 42, \quad \Delta = -68, \quad \det = \\ & 3, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 513221 & 1 \\ [1700] & 1 & (12) & 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 40, \quad T_{\max} = 42, \quad \Delta = -76, \quad \det = \\ & 3, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 513221 \\ [1701] & 1 & (12) & 12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 40, \quad T_{\max} = 43, \quad \Delta = -96, \quad \det = 4, \quad (n_+, n_-, n_0) = \\ & (1, 40, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 41 \quad \begin{matrix} 1^1 \\ [1702] & 316131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 41, \quad \Delta = -26, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 41, 1), \quad c_{\text{ext}} = 3.0$$

$$\begin{aligned} & \begin{matrix} 1^1 \\ [1703] & 16 & 1231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 41, \quad \Delta = -21, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 41, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & \begin{matrix} 1^1 \\ [1704] & 16 & 131513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 41, \quad \Delta = -20, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 41, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & \begin{matrix} 1^1 \\ [1705] & 1316131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 41, \quad \Delta = -26, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 41, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 3221 & 1 \\ [1712] & 1 & (12) & 12231513221(12)12231513221(12)122315^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 42, \quad \Delta = -47, \quad \det = \\ & -2, \quad (n_+, n_-, n_0) = (1, 41, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 413221 & 12231513221 \\ [1713] & 1 & (12) & 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 42, \quad \Delta = -51, \quad \det = -2, \quad (n_+, n_-, n_0) = \\ & (1, 41, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & \begin{matrix} 1^1 & 3221 & 1 \\ [1714] & 1 & (12) & 12231513221(12)12231513221(12)122315^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 41, \quad T_{\max} = 42, \quad \Delta = -47, \quad \det = \\ & -2, \quad (n_+, n_-, n_0) = (1, 41, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42] \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1715] \quad 1^1 71232181231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, \quad T_{\max} = 42, \quad \Delta = -47, \quad \det =$$

$$-2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1716] \quad 1^1 18 1232181231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 43, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [1717] \quad 1^1 71232181231513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1718] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1719] \quad 1^1 712321513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1720] \quad 1^1 71231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1721] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1722] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1723] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0$$

$$[1734] \quad 1^1 15 13221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1735] \quad 1^1 18 1232181231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1736] \quad 1^1 15 13221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$T = 43 \quad [1737] \quad 21^1 8 1231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$$

$$[1738] \quad 1^1 413221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$$

$$[1739] \quad 1^1 18 1232181231513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$$

$$[1740] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$$

$$[1741] \quad 1^1 413221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$$

[1742] $1^{1\bar{1}31513221} (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$

[1743] $1^{1\bar{1}32181231513221} (12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$

[1744] $1^{1\bar{1}1812231513221} (12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$

[1754] $1^{1\bar{1}1513221} (12)12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 44, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$
 $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0}$

[1755] $1^{1\bar{1}6131513221} (12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 45, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$
 $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0}$

$T = 44$ [1756] $1^{1\bar{1}31513221} (12) \quad 12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1757] $1^{1\bar{1}31513221} (12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1758] $1^{1\bar{1}31513221} (12) \quad 12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1759] $1^{1\bar{1}31513221} (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1760] $1^{1\bar{1}31513221} (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1761] $1^{1\bar{1}1513221} (12)12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1762] $1^{1\bar{1}1513221} (12)12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[1771] $1^{1\bar{1}71231513221} (12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 46, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0}$

$T = 45$ [1772] $21^{1\bar{1}3221} (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0$

[1773] $21^{1\bar{1}3221} (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0$

[1778] $1^{1\bar{1}3221} (12) \quad 12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0}$

[1779] $1^{1\bar{1}181231513221} (12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned} & 1^1 812231513221(12)12231513221 \\ [1780] \quad & 1 \quad (12) \quad 12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, \quad T_{\max} = 47, \quad \Delta = -73, \quad \det = \\ & -3, \quad (n_+, n_-, n_0) = (1, 45, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 47, \quad T_{\max} = 47, \quad T \in [47, 47] \\ & \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & 1^1 316131513221(12)12231513221(12)12231513221 \\ T = 46 \quad [1781] \quad & 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -24, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 31513221 \\ [1782] \quad & 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -22, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 31513221 \\ [1783] \quad & 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -22, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 316131513221(12)12231513221(12)12231513221 \\ [1784] \quad & 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -24, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 413221(12)12231513221(12)12231513221 \\ [1793] \quad & 1 \quad (12) \quad 122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 48, \quad \Delta = -72, \quad \det = \\ & 3, \quad (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 48, \quad T_{\max} = 48, \quad T \in [48, 48] \\ & \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & 1^1 81231513221(12)12231513221 \\ [1794] \quad & 1 \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 49, \quad \Delta = -96, \quad \det = \\ & 4, \quad (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 49, \quad T_{\max} = 49, \quad T \in [49, 49] \\ & \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} & 1^1 \\ T = 47 \quad [1795] \quad & 1 \quad 6 \quad 131513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = \\ & -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 41413221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = \\ [1796] \quad & 1 \quad 41413221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = \\ & -23, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 513221 \\ [1797] \quad & 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -19, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 3221 \\ [1798] \quad & 21 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -24, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 3221 \\ [1799] \quad & 21 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -24, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 513221 \\ [1800] \quad & 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -19, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 31513221 \\ [1801] \quad & 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -20, \quad \det = \\ & 0, \quad (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \end{aligned}$$

$$\begin{aligned} & 1^1 \\ [1810] \quad & 1 \quad 5 \quad 13221(12)12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 49, \quad \Delta = -67, \quad \det = \\ & -3, \quad (n_+, n_-, n_0) = (1, 47, 0), \quad c_{\text{ext}} = 3.0 \\ & T_{\min} = 49, \quad T_{\max} = 49, \quad T \in [49, 49] \\ & \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned}
& T = 48 \quad [1811] \quad 1 \quad (12) \quad 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 3.0 \\
& [1812] \quad 1 \quad (12) \quad 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 3.0 \\
& [1813] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151322171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 3.0 \\
& [1814] \quad 17 \quad 1231513221(12)12231513221(12)12231513221(12)122315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 3.0 \\
& [1822] \quad 1 \quad (12) \quad 122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 48, \quad T_{\max} = 49, \quad \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 48, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49, \quad T_{\max} = 49, \quad T \in [49, 49] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
& T = 49 \quad [1823] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1824] \quad 1 \quad (12) \quad 122315132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1825] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1826] \quad 18 \quad 12231513221(12)12231513221(12)12231513221(12)122315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1827] \quad 218 \quad 1231513221(12)12231513221(12)12231513221(12)122315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1828] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1829] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1830] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& [1831] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 49, \quad T_{\max} = 49, \quad \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), \quad c_{\text{ext}} = 3.0 \\
& T = 50 \quad [1839] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)122315132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, \quad T_{\max} = 50, \quad \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), \quad c_{\text{ext}} = 3.0 \\
& [1840] \quad 16 \quad 131513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, \quad T_{\max} = 50, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), \quad c_{\text{ext}} = 3.0 \\
& [1843] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, \quad T_{\max} = 51, \quad \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 51, \quad T_{\max} = 51, \quad T \in [51, 51] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
& [1844] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, \quad T_{\max} = 51, \quad \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 51, \quad T_{\max} = 51, \quad T \in [51, 51] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [1845] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181232181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 50, \quad T_{\max} = 52, \quad \Delta = -72, \det =
\end{aligned}$$

$$3, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 52, T_{\max} = 52, T \in [52, 52]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 51 \quad [1846] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0$$

$$[1847] \quad 1^1 32181231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0$$

$$[1848] \quad 1^1 32181231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0$$

$$[1849] \quad 21 \begin{matrix} 1^1 3221 \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0$$

$$[1850] \quad 1^1 171231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0$$

$$[1853] \quad 1 \begin{matrix} 1^1 413221(12)12231513221(12)12231513221 \\ (12) \end{matrix} 12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 52, T_{\max} = 52, T \in [52, 52]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1854] \quad 1 \begin{matrix} 1^1 413221(12)12231513221(12)12231513221 \\ (12) \end{matrix} 12231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 52, T_{\max} = 52, T \in [52, 52]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[1855] \quad 1 \begin{matrix} 1^1 3221(12)12231513221(12)12231513221 \\ (12) \end{matrix} 12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[1856] \quad 1 \begin{matrix} 1^1 3221(12)12231513221(12)12231513221 \\ (12) \end{matrix} 12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1857] \quad 1^1 413221(12)12231513221(12)12231513221(12)122315132181232181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 52 \quad [1858] \quad 1^1 1812231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$$

$$[1859] \quad 1^1 16131513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$$

$$[1860] \quad 1^1 61231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$$

$$[1861] \quad 21 \begin{matrix} 1^1 3221 \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$$

$$[1862] \quad 21 \begin{matrix} 1^1 3221 \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$$

[1863] $21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0$

[1864] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1865] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1866] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)122315132181232181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 52, T_{\max} = 54, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

$T = 53$ [1867] $21 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1868] $1 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1869] $1 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1870] $1 \overset{1}{413221(12)12231513221(12)12231513221(12)12231513221} (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1871] $1 \overset{1}{71231513221(12)12231513221(12)12231513221(12)122315132181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1872] $21 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1873] $1 \overset{1}{71231513221(12)12231513221(12)12231513221(12)122315132181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1874] $1 \overset{1}{231513221(12)12231513221(12)12231513221} (12) 12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1875] $1 \overset{1}{231513221(12)12231513221(12)12231513221} (12) 12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$

[1878] $1 \overset{1}{81232181231513221(12)12231513221(12)12231513221(12)122315131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 53, T_{\max} = 54, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$T = 54$ [1879] $1 \overset{1}{(12)} 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$

[1880] $1 \overset{1}{41413221(12)12231513221(12)12231513221(12)12231513221} (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$

[1881] $1 \overset{1}{812231513221(12)12231513221(12)12231513221(12)122315132181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 54, T_{\max} = 54, \Delta =$

$$\begin{array}{c} 1^1 \\ 15 \ 13221 \end{array} \quad [1882] \quad 1 \ (12) \ 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, \ T_{\max} = 54, \ \Delta = -16, \ \det = 0, \ (n_+, n_-, n_0) = (1, 54, 1), \ c_{\text{ext}} = 3.0$$

$$\begin{array}{l}
1^1 41413221(12)12231513221(12)12231513221(12)12231513221 \\
[1884] \ 1 \qquad \qquad \qquad (12) \qquad \qquad \qquad 1223141^1 + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, \ T_{\max} = 54, \ \Delta = -20, \ \det = \\
0, \ (n_+, n_-, n_0) = (1, 54, 1), \ c_{\text{ext}} = 3.0
\end{array}$$

$$T = 55 \quad [1887] \ 1^4 141413221(12)12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2\text{-}\mathfrak{g}_2] \quad T = 55, \ T_{\max} = 55, \ \Delta = -19, \ \det = 0, \ (n_+, n_-, n_0) = (1, 55, 1), \ c_{\text{ext}} = 3.0$$

$$\begin{array}{l} \text{[1889] } 1 \text{ } \overset{1}{15} \text{ (12) } 12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 55, T_{\max} = 55, \Delta = \\ -15, \det = 0, (n_+, n_-, n_0) = (1, 55, 1), c_{\text{ext}} = 3.0 \end{array}$$

$$\begin{array}{c} \textcolor{red}{1} \\ 1\ 5\ 13221 \end{array} \\ [1890] \ 1 \ (12) \ 12231513221(12)12231513221(12)12231513221(12)122314141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 55, \ T_{\max} = 55, \ \Delta = \\ -15, \ \det = 0, \ (n_+, n_-, n_0) = (1, 55, 1), \ c_{\text{ext}} = 3.0$$

$$\begin{array}{c} 1^1 \\ 15\ 13221 \end{array} \quad [1891] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)1223151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 55, \quad T_{\max} = 55, \quad \Delta = -11, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 55, 1), \quad c_{\text{ext}} = 3.0$$

[1892] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 55$, $T_{\max} = 55$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 55, 1)$, $c_{\text{ext}} = 3.0$

$$T = 56 \quad [1893] \quad 1^{15131513221(12)12231513221(12)12231513221(12)12231513221} \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 56, \quad T_{\max} = 56, \quad \Delta = -18, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 56, 1), \quad c_{\text{ext}} = 3.0$$

$$[1894] \ 1 \quad (12) \quad 1223141^{1+[\text{su}_2 \cdot \text{g}_2]} \quad T = 56, \ T_{\max} = 56, \ \Delta = -18, \ \det = 0, \ (n_+, n_-, n_0) = (1, 56, 1), \ c_{\text{ext}} = 3.0$$

[1895] $1^1 16 131513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 56$, $T_{\max} = 56$, $\Delta = -16$, $\det = 0$, $(n_+, n_-, n_0) = (1, 56, 1)$, $c_{\text{ext}} = 3.0$

$$\begin{array}{l} \text{[1896] } 21 \text{ (12) } 12231513221(12)12231513221(12)12231513221(12)122315132161^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}_2} \textcolor{blue}{\cdot} \textcolor{blue}{\mathfrak{g}_2}] \quad T = 56, T_{\max} = 56, \Delta = \\ -17, \det = 0, (n_+, n_-, n_0) = (1, 56, 1), c_{\text{ext}} = 3.0 \end{array}$$

$$T = 57 \quad [1897] \quad 1^1 41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 57, T_{\max} = 57, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0$$

$$[1898] \ 1^{1^1 61231513221(12)12231513221(12)12231513221(12)12231513221} \quad (12) \quad 1223141^{1+[\text{su}_2 \cdot \text{g}_2]} \quad T = 57, \ T_{\max} = 57, \ \Delta = -16, \ \det = 0, \ (n_+, n_-, n_0) = (1, 57, 1), \ c_{\text{ext}} = 3.0$$

[1899] $16^1 13 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 15 13 22 1(12) 12 23 14 1^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 57, T_{\max} = 57, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0$

[1900] $1 \overset{1}{\underset{15}{(12)}} 12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{s}u_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 57, T_{\max} = 57, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0$

$$\begin{array}{c} 1^1 \\ 1 \ 5 \ 13221 \end{array} [1901] \ 1 \ (12) \ 12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 : \textcolor{blue}{g}_2] \quad T = 57, \ T_{\max} = 57, \ \Delta = -13, \ \det = 0, \ (n_+, n_-, n_0) = (1, 57, 1), \ c_{\text{ext}} = 3.0$$

[1902] $1^1 41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 57$, $T_{\max} = 57$, $\Delta = -17$, $\det = 0$, $(n_+, n_-, n_0) = (1, 57, 1)$, $c_{\text{ext}} = 3.0$

$$T = 61 \quad [1925] \quad 1^1 7 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 61, T_{\max} = 61, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 61, 1), c_{\text{ext}} = 3.0$$

$$[1926] \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 61, T_{\max} = 61, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 61, 1), c_{\text{ext}} = 3.0$$

$$T = 62 \quad [1927] \quad 1^1 8 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 62, T_{\max} = 62, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 62, 1), c_{\text{ext}} = 3.0$$

$$[1928] \quad 21^1 8 1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 62, T_{\max} = 62, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 62, 1), c_{\text{ext}} = 3.0$$

1.2.3 NHC(-2)(-3)(-2)

$$T = 6 \quad [0] \quad 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 10, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 8 & 6 \\ 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^1 31^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 10, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 21^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 10, \Delta = -118, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 221^{11} 3 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 10, \Delta = -118, \det = 216, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 3 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 21^1 321^{11} + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3 21^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 3 21^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 2231^{11} + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 2^1 2 31^1 + [\text{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -90, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 2 \overset{1}{\underset{11}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 10, \Delta = -92, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [10] \quad 1 \overset{1}{\underset{11}{2}} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1 \overset{1}{\underset{11}{4}} 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 2 1 \overset{1}{\underset{11}{4}} 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1 \overset{1}{\underset{11}{3}} 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1 \overset{1}{\underset{11}{4}} 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 2 \overset{1}{\underset{11}{3}} 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1 \overset{1}{\underset{11}{4}} 1 \overset{1}{\underset{11}{2}} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 2 \overset{1}{\underset{11}{3}} 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 4 1 \overset{1}{\underset{11}{3}} 2 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 8 & 4 \\ 3 & 8 & 8 & 5 & 5 \\ 2 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 4 1 \overset{1}{\underset{11}{2}} 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 11, \Delta = -89, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[20] $41^{11}232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 11, \Delta = -91, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 5 & 2 \\ 4 & 12 & 5 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

$T = 9$ [21] $1^{11}41321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[22] $314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[23] $1^{11}41^{11}3141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 5 & 1 & 1 & 3 \\ 2 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

[24] $31^{11}4141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 3 & 9 & 1 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[25] $2^13^{11}21 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 6 & 8 & 6 \\ 4 & 8 & 6 & 6 \\ 3 & 6 & 6 & 5 \end{pmatrix}_{b_0}$

[26] $31^14^{11}141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 2 & 7 & 1 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[27] $2^{11}3^{11}41 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 4 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[28] $41^{11}23141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[29] $321^{11}4141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 9, T_{\max} = 12, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 8 & 8 & 3 \\ 8 & 17 & 19 & 9 \\ 8 & 19 & 17 & 9 \\ 3 & 9 & 9 & 4 \end{pmatrix}_{b_0}$

$T = 10$ [30] $1^{11}413221 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 11, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[31] $1^{\overline{1}1} 1^{\overline{1}1} 5 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{\overline{1}1} 1^{\overline{1}1} 4 14 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [33] $1^{\overline{1}1} 1^{\overline{1}1} 31 5 1321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[34] $1^{\overline{1}1} 1^{\overline{1}1} 4 14 1321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 2 & 4 \\ 3 & 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[35] $1^{\overline{1}1} 1^{\overline{1}1} 6 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[36] $1^{\overline{1}1} 1^{\overline{1}1} 6 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{\overline{1}1} 1^{\overline{1}1} 6 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[38] $2^{\overline{1}1} 3^{\overline{1}1} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 \\ 2 & 5 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$T = 12$ [41] $1^{\overline{1}1} 1^{\overline{1}1} 6 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

[42] $1^{\overline{1}1} 1^{\overline{1}1} 6 13141^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{\overline{1}1321} 7^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 12, T_{\max} = 14, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[44] $1^{\overline{1}131} 6^{11} 13141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & -1 & 4 & 4 \end{pmatrix}_{b_0}$

$T = 13$ [46] $1^{\overline{1}12321} 7^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 14, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$

[47] $2^{\overline{1}1} 3^{\overline{1}1} 218^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[48] $1^{\overline{1}13221} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[49] $1^{\overline{1}13221} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[50] $1^{\overline{1}1321} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[51] $2^{\overline{1}11} 3^{\overline{1}1} 4^{11} 14141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 13, T_{\max} = 13, \Delta = -2, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 3 & 6 & 5 \\ 2 & 6 & 6 & 4 & 6 \\ 2 & 6 & 5 & 6 & 6 \end{pmatrix}_{b_0}$

$T = 14$ [52] $1^{\overline{1}12321} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[53] $1^{\overline{1}13221} 8^{11} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{131} \overset{1}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, \ T_{\max} = 15, \ \Delta = -32, \ \det = 2, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[55] \ 31 \overset{1}{6} \overset{1}{131} \overset{1}{61} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, \ T_{\max} = 16, \ \Delta = -58, \ \det = 10, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 15.80 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 5 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [56] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{131} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, \ T_{\max} = 15, \ \Delta = -4, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.0$$

$$[58] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{131} \overset{1}{61} \overset{1}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, \ T_{\max} = 16, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{1231} \overset{1}{41} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, \ T_{\max} = 16, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{131} \overset{1}{61} \overset{1}{231} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, \ T_{\max} = 16, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [62] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{131} \overset{1}{51} \overset{1}{321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 16, \ T_{\max} = 17, \ \Delta = -31, \ \det = 2, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [63] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{1231} \overset{1}{51} \overset{1}{321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 17, \ T_{\max} = 17, \ \Delta = -3, \ \det = 0, \ (n_+, n_-, n_0) = (1, 17, 1), \ c_{\text{ext}} = 4.0$$

$$[65] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{12321} \overset{1}{61} \overset{1}{31} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 17, \ T_{\max} = 18, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, \ T_{\max} = 18, \ T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [66] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{131} \overset{1}{61} \overset{1}{31} \overset{1}{51} \overset{1}{321} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 18, \ T_{\max} = 18, \ \Delta = -3, \ \det = 0, \ (n_+, n_-, n_0) = (1, 18, 1), \ c_{\text{ext}} = 4.0$$

$$[68] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{12321} \overset{1}{71} \overset{1}{231} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 18, \ T_{\max} = 19, \ \Delta = -29, \ \det = 2, \ (n_+, n_-, n_0) = (1, 18, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, \ T_{\max} = 19, \ T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [69] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}6} \quad 1316131612321^{\overset{1}{\color{blue}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 21 \quad [73] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}6} \quad 131613161315131^{\overset{1}{\color{blue}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 21, \quad T_{\max} = 21, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 21, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 22 \quad [74] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}8} \quad 123218123151321^{\overset{1}{\color{blue}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 24 \quad [77] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}8} \quad 12321812321712321^{\overset{1}{\color{blue}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 24, \quad T_{\max} = 24, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 24, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 25 \quad [78] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}8} \quad 12321812321 \overset{1}{\color{blue}8} 1231^{\overset{1}{\color{green}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 25, 1), \quad c_{\text{ext}} = 4.0$$

$$T = 27 \quad [79] \quad 1^{\overset{1}{\color{red}1}} \overset{1}{\color{blue}8} \quad 12321812321812315131^{\overset{1}{\color{blue}1}} + [\color{red}{\mathfrak{su}}_2 \cdot \color{blue}{\mathfrak{so}}_7 \cdot \color{green}{\mathfrak{su}}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 27, 1), \quad c_{\text{ext}} = 4.0$$

2 Round 2: NHC External 2-ext (1536 bases)

Colored curves in quiver show attachment points. Matching color in external label: ext1 ext2 ext3 ext4
 Total: 1536 bases (59 uni, 1477 non-uni).

T	Count
6	10
7	39
8	180
9	208
10	137
11	145
12	247
13	74
14	94
15	101
16	51
17	43
18	31
19	30
20	13
21	18
22	13
23	15
24	16
25	10
26	13
27	5
28	7
29	3
30	4
31	3
32	2
33	2
34	1
35	2
36	3
37	3
38	2
40	1
41	1
42	2
44	2
46	1
47	1
48	1
49	2

2.1 Unimodular ($|\det| = 1$) — 59 bases

2.1.1 NHC(-2)(-3) + $\mathfrak{su}_2$

$$T = 11 \quad [321] \quad 1^{\overset{1}{\textcolor{red}{1}}31\overset{1}{5}1321^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [436] \quad 1^{\overset{1}{\textcolor{green}{4}}31\overset{1}{5}13221^{\textcolor{blue}{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 19 & 4 \\ 8 & 0 & 19 & 115 & 27 \\ 2 & 0 & 4 & 27 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[437] \ 1^{\overline{1}} \overline{6} \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[438] \ 1^{\overline{2}} 31^{\overline{1}} 5 \ 13221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 7 & 4 \\ 4 & 0 & 7 & 19 & 11 \\ 2 & 0 & 4 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[439] \ 1^{\overline{1}} 31^{\overline{1}} 5 \ 13221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 5 & 1 & 3 & 5 \\ 2 & 5 & 2 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [626] \ 1^{\overline{1}} \overline{7} \ 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[627] \ 1^{\overline{1}} \overline{4} \ 14141^{\overline{1}} \overline{4} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 6 & 7 \\ 3 & 6 & 4 & 6 \\ 3 & 7 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [677] \ 1^{\overline{1}} \overline{8} \ 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[678] \ 1^{\overline{1}} \overline{8} \ 12321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[679] \ 1^{\overline{1}} \overline{6} \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 15 \quad [756] \ 1^{\overline{1}} \overline{6} \ 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 3 & 1 & 3 & 5 \\ 2 & 3 & 2 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[759] \ 1^{\overline{1}} \overline{6} \ 131^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 4 & 3 & 19 & 11 \\ 2 & 2 & 2 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[760] $1^{\overline{11}} \overset{1}{6} \overset{13}{2315131} 131^{\overline{4}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 83 & 23 \\ 2 & 2 & 2 & 23 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[758] $1^{\overline{131}} \overset{1}{6} 13161231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[755] $1^{\overline{12321}} \overset{1}{8} 123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[757] $1^{\overline{131}} \overset{1}{6} 1316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 4 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$T = 16$ [836] $1^{\overline{131}} \overset{1}{6} 131613141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 5 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

[837] $1^{\overline{13151321}} \overset{1}{8} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

[838] $1^{\overline{1512321}} \overset{1}{8} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

[839] $1^{\overline{1}} \overset{1}{15} 1321812321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$T = 17$ [874] $1^{\overline{1321}} \overset{1}{8} 123216131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 17, T_{\max} = 17, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

[875] $1^{\overline{1}} \overset{1}{15} 13161316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 17, T_{\max} = 17, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

[876] $1 \overset{1}{6} 12321 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 17, T_{\max} = 17, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$T = 18$ [913] $1 \overset{1}{1} \overset{1}{3} 21 \overset{1}{8} 12321^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 18, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

[914] $21 \overset{1}{7} 12321 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 18, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$T = 19$ [940] $1 \overset{1}{1} \overset{1}{5} 13161316131 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

[939] $221 \overset{1}{8} 12321 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

[938] $1 \overset{1}{1} \overset{1}{3} 21 \overset{1}{8} 12321 \overset{1}{8} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

[941] $1 \overset{1}{1} \overset{1}{3} 21 \overset{1}{8} 12321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 4 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$T = 20$ [957] $1 \overset{1}{1} \overset{1}{3} 21 \overset{1}{8} 12321 \overset{1}{8} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 5 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

[958] $21 \overset{1}{1} \overset{1}{6} 13161316131 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 20, T_{\max} = 20, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 21 \quad [968] \quad 1^{\underline{1}} 5 1321812321^{\underline{1}} 8 1231^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[969] \quad 1^{\underline{1}} 1^{\underline{1}} 321 8 1232181232151^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 23 \quad [990] \quad 1^{\underline{1}} 1^{\underline{1}} 5 132181232181231^{\underline{1}} 5 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 24 \quad [1001] \quad 1^{\underline{1}} 1^{\underline{1}} 6 1232181232181231^{\underline{1}} 5 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 25 \quad [1012] \quad 21^{\underline{1}} 7 1232181232181231^{\underline{1}} 5 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$T = 33 \quad [1055] \quad 1^{\underline{1}} 1^{\underline{1}} 5 13221 (12) 12231513221 (12) 12231^{\underline{1}} 5 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 33, T_{\max} = 33, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$T = 36 \quad [1060] \quad 1^{\underline{1}} 1^{\underline{1}} 5 13221 (12) 12231513221 (12) 122315131^{\underline{1}} 6 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 36, T_{\max} = 36, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$T = 37 \quad [1062] \quad 1^{\underline{1}} 1^{\underline{1}} 5 13221 (12) 12231513221 (12) 1223151321^{\underline{1}} 7 1^{\underline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 37, T_{\max} = 37, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$T = 38 \quad [1065] \quad 21^{1^1} 8^{1^1} 1231513221(12)12231513221(12)1223151^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, T_{\max} = 38, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 7 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 7 & 4 & 19 & 11 \\ 2 & 4 & 2 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[1066] \quad 1^{1^1} 5^{1^1} 13221(12)12231513221(12)1223151322181^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, T_{\max} = 38, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$T = 44 \quad [1072] \quad 1^{1^1} 5^{1^1} 13221(12)12231513221(12)12231513221(12)1223151^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 44, T_{\max} = 44, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 \\ 3 & 3 & 4 & 6 \\ 3 & 4 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

2.1.2 NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 12 \quad [85] \quad 1^{1^1} 2315^{1^1} 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 12, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[86] \quad 1^{1^1} 315^{1^1} 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 12, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 4 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [140] \quad 1^{1^1} 6^{1^1} 2315131^{1^1} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 16 \quad [150] \quad 1^{1^1} 6^{1^1} 131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [167] \quad 1^{1^1} 6^{1^1} 13161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 18, \Delta = -7, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [175] \quad 1^{131} 6^{131} 13161316131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 3 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 5 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[174] \quad 1^{131} 6^{131} 131613161231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [182] \quad 1^{131} 6^{131} 1316131613141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 20, T_{\max} = 20, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 6 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

2.1.3 NHC(-2)(-3)(-2) + su₂

$$T = 12 \quad [116] \quad 1^{11} 31^1 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[117] \quad 1^{12} 31^1 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 13 \quad [122] \quad 1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[123] \quad 1^{11} 6^{11} 131^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[124] \quad 1^{12} 6^{12} 131^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 2 & 9 & 3 \\ 2 & 2 & 2 & 15 & 4 \\ 12 & 9 & 15 & 67 & 21 \\ 3 & 3 & 4 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[125] \ 1^{11} \overset{1^{12}2321^1}{6} \ 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 2 & 12 & 3 \\ 2 & 2 & 2 & 19 & 4 \\ 14 & 12 & 19 & 106 & 26 \\ 3 & 3 & 4 & 26 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [126] \ 1^{11} \overset{1^{11}2321}{7} \ 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [128] \ 1^{11} \overset{1^{11}315131}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[129] \ 1^{11} \overset{1^{11}315131}{6} \ 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

2.2 Non-unimodular ($|\det| > 1$) — 1477 bases

2.2.1 $\text{NHC}(-2)(-2)(-3) + \textcolor{brown}{so}_8$

$$T = 9 \quad [0] \ 1^{11} \overset{1^1}{2} 321^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{11} \overset{1^1}{3} 151321^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 5 & 5 \\ 2 & 3 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

2.2.2 $\text{NHC}(-2)(-2)(-3) + \textcolor{blue}{su}_2$

$$T = 8 \quad [0] \ 2 \overset{1^{11}}{3} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1^1}{3} 1^{11} 31^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

- [2] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$
- [3] $2^{11}3^{11}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 5 & 3 \\ 2 & 4 & 1 & 3 & 5 & 3 \\ 2 & 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 8 & 6 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$
- [4] $2^{112}3^{112}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 10 & 3 \\ 2 & 4 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [5] $2^{111}3^{111}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$
- [6] $2^{1112}3^{1112}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 6 & 3 \\ 2 & 4 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 6 & 4 & 10 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$
- [7] $2^{1113}3^{1113}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 9 & 3 \\ 2 & 4 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 9 & 6 & 25 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [8] $2^{1114}3^{1114}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 12 & 3 \\ 2 & 4 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 12 & 8 & 46 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [9] $2^{1115}3^{1115}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 15 & 3 \\ 2 & 4 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 15 & 10 & 73 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [10] $2^{1112}3^{1112}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 8 & 3 \\ 2 & 4 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 8 & 7 & 21 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 2 \overset{1}{\underset{116}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 256, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 18 & 3 \\ 2 & 4 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 18 & 12 & 106 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 2 \overset{1}{\underset{117}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 308, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 4 & 2 & 21 & 3 \\ 2 & 4 & 1 & 3 & 21 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 21 & 14 & 145 & 21 \\ 2 & 3 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 2 \overset{12}{\underset{113}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 340, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 13 & 3 \\ 2 & 4 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 13 & 12 & 61 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 2 \overset{11}{\underset{113}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 352, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 11 & 3 \\ 2 & 4 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 11 & 9 & 40 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 2 \overset{11}{\underset{114}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 14 & 3 \\ 2 & 4 & 1 & 3 & 14 & 3 \\ 2 & 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 14 & 11 & 65 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 2 \overset{12}{\underset{114}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 576, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 16 & 3 \\ 2 & 4 & 1 & 3 & 16 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 16 & 14 & 90 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [19] \ 1 \overset{12}{\underset{114}{2}} 3 \overset{11}{\underset{11}{21}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 6 & 2 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1 \overset{12}{\underset{114}{2}} 3 \overset{11}{\underset{11}{21}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1 \overset{11}{\underset{114}{2}} 3 \overset{11}{\underset{11}{21}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \ 1 \overset{11}{\underset{114}{2}} 3 \overset{11}{\underset{11}{21}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 2^3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 2^3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{12} 2^3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13} 2^3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 9 & 2 \\ 6 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 2^3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 2^3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{13} 2^3 21^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{14} 2^3 21^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 2^3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{13} 2^3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{14} 2^3 3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 10 & 2 \\ 8 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{12} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 5 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{14} 2^3 3 21^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 1 & 13 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{13} 2^3 3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{12} 2^3 3 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 6 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{12} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{13} 2^3 3 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 10 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{14} 2^3 3 21^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{14} 2^3 3 21^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 10 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{14} 2^3 3 21^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 16 & 3 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 12 & 0 & 16 & 16 & 126 & 24 \\ 2 & 0 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{14} 2^3 3 21^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 10 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{14} 2^3 3 21^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 11 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [56] \ 231^{11} 3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 566, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 5 & 2 \\ 5 & 33 & 36 & 0 & 35 & 11 \\ 5 & 36 & 33 & 1 & 35 & 11 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 35 & 35 & 0 & 33 & 11 \\ 2 & 11 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [64] \ 1^{11} 31^5 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[57] \ 1^{11} 31^5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[58] \ 1^{14} 231^5 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[59] \ 1^{11} 231^5 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[60] \ 1^{13} 231^5 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[61] \ 1^{11} 31^5 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[62] \ 1^{12} 231^5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 2), c_{\text{ext}} = 5.5$$

$$[63] \ 1^{12} 31^5 13221^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$$

$$[66] \ 1^{11} 31^5 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 1 & 5 & 3 \\ 3 & 0 & 4 & 5 & 7 & 7 \\ 2 & 0 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 31 \bar{5} 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 1 & 10 & 3 \\ 6 & 0 & 8 & 10 & 34 & 14 \\ 2 & 0 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{13} 31 \bar{5} 1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 1 & 15 & 3 \\ 9 & 0 & 12 & 15 & 79 & 21 \\ 2 & 0 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 31 \bar{5} 1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 1 & 16 & 3 \\ 9 & 0 & 15 & 16 & 106 & 24 \\ 2 & 0 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{13} 31 \bar{5} 1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 2 & 1 & 15 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} 31 \bar{5} 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 2 & 1 & 10 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{11} 31 \bar{5} 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 31 \bar{5} 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{14} 31 \bar{5} 1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 2 & 1 & 20 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 20 & 0 & 142 & 28 \\ 2 & 3 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{11} 31 \bar{5} 1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 5 & 4 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 5 & 4 & 7 & 7 \\ 2 & 0 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11} 31 \bar{5} 1321^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{12} 31^1 5 \ 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{12} 31^1 5 \ 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 13 & 4 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 6 & 0 & 13 & 6 & 43 & 15 \\ 2 & 0 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{12} 31^1 5 \ 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{12} 31^1 5 \ 1321^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 8 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{12} 31^1 5 \ 1321^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 7 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{13} 31^1 5 \ 1321^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 19 & 4 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 9 & 0 & 19 & 13 & 113 & 25 \\ 2 & 0 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{13} 31^1 5 \ 1321^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

2.2.3 NHC(-2)(-2)(-3) + su₃

$$T = 7 \quad [0] \ 21^{12} 31^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 5 & 21 & 4 \\ 3 & 1 & 5 & 15 & 3 \\ 5 & 5 & 4 & 22 & 4 \\ 21 & 15 & 22 & 84 & 16 \\ 4 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 21^{113}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 22 & 4 \\ 15 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{14}3^{12}3^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{14}3^{12}3^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

2.2.4 NHC(-2)(-3) + $\mathfrak{f}_4$

$$T = 12 \quad [0] \quad 1^{131}31^6 \quad 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

2.2.5 NHC(-2)(-3) + NHC(-2)(-2)(-3)

$$T = 11 \quad [0] \quad 1^{15}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 5 & 0 & 5 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 5 & 3 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

2.2.6 NHC(-2)(-3) + NHC(-2)(-3)

$$T = 7 \quad [0] \quad 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 10, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{13}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{14}4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -64, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} \overline{5} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \quad 1^{\overline{1}} \overline{1} 4131^{\overline{1}} \overline{5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [7] \quad 1^{\overline{1}} \overline{1} \overline{5} 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

2.2.7 NHC(-2)(-3) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{\overline{1}} \overline{1} 2321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} \overline{1} 2321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} \overline{1} 2321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} \overline{1} 2321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} \overline{1} 2321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 2 \quad 3 \quad 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{11} 31^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 31^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 31^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [9] \quad 1^{11} 31^{11} 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 31^{11} 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 2 & 4 \\ 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 31^{11} 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 5 & 5 \\ 2 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 2 \quad 3 \quad 151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [13] \quad 1^{11} 31^{11} 513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 5 \\ 3 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^{11} 6 \quad 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 12, T_{\max} = 14, \Delta = -63, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{1}12321} 6^{11} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{1}131513221} 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 6 & 6 \\ 2 & 5 & -1 & 6 & 5 \end{pmatrix}_{b_0}$$

$T = 14$ [17] $23218^{11} 12321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$T = 15$ [18] $1^{\overline{1}12315131} 6^{11} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$T = 18$ [19] $1^{\overline{1}13221} (12) 12231^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 18, T_{\max} = 20, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 19.60 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

2.2.8 NHC(-2)(-3) + $\mathfrak{su}_2$

$T = 6$ [0] $1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 9, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 7$ [1] $1^{\overline{1}1} 321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 9, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1}1} 321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 9, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1}1} 3131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad \begin{array}{c} 1^1 \\ 1^1 4^1 1^1 \end{array} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -105, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\left(\begin{array}{cccc} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{array} \right)_{b_0}$$

[5] $21^{\overline{1}}41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -105, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$
 $T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$
 $\left(\begin{smallmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{smallmatrix} \right)_{b_0}$

$$[6] \begin{array}{c} 1 \\ 1 \end{array} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \, T_{\max} = 10, \, \Delta = -105, \, \det = -6, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\left(\begin{array}{cccc} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{array} \right)_{b_0}$$

[7] $1^{11}3131^1 + [\textcolor{red}{su}_2, \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -93, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

$$[8] \ 1^1 1^1 1^1 31^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -93, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\textcolor{red}{1}}1^{\textcolor{blue}{1}}3131^{\textcolor{blue}{1}} + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{green}{u}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}31^131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -93, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \begin{array}{c} 1^{11} \\ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{u}}_2 \\ 1^1 \end{array} \quad T = 7, \ T_{\max} = 10, \ \Delta = -88, \ \det = -36, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\left(\begin{array}{c} 0 \ 2 \ 2 \ 2 \ 2 \\ 2 \ 1 \ 3 \ 2 \ 3 \\ 2 \ 3 \ 0 \ 3 \ 2 \\ 2 \ 2 \ 3 \ 1 \ 3 \\ 2 \ 3 \ 2 \ 3 \ 3 \end{array} \right)_{b_0}$$

[12] $1^{\textcolor{red}{1}}1^{\textcolor{blue}{1}}31^{\textcolor{green}{1}}1^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

$$[13] \begin{array}{c} 1 \\ 2 \ 3 \ 2 + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}u_2 \\ 1 \ 1 \end{array} \quad T = 7, \ T_{\max} = 10, \ \Delta = -88, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 2 \begin{smallmatrix} 1 \\ 2 \end{smallmatrix} \begin{smallmatrix} 3 \\ 1 \end{smallmatrix} \begin{smallmatrix} 2 \\ 2 \end{smallmatrix} + [\textcolor{red}{\mathfrak{s}u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}u}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\left(\begin{smallmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{smallmatrix} \right)_{b_0}$$

[15] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -67$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 2 \overset{1}{\underset{1^{13}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 2 \overset{1^{11}}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1 \overset{12}{\underset{1^{11}}{3}} 31 \overset{2}{\underset{1^{11}}{3}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 2 \overset{1}{\underset{1^{14}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 46 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 2 \overset{1}{\underset{1^{15}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 10 & 73 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 2 \overset{1^{12}}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 6 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 2 \overset{1^{11}}{\underset{1^{12}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 2 \overset{1}{\underset{1^{16}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 106 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 2 \overset{1}{\underset{1^{17}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 21 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 14 & 145 & 21 \\ 2 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 2 \overset{1^{11}}{\underset{1^{13}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 9 & 40 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 2 \overset{1^{12}}{\underset{1^{12}}{3}} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 2 \overset{1^{11}}{\underset{1^{14}}{3}} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 65 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 2 \overset{1^{12}}{\underset{1^{13}}{3}} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 12 & 61 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 2 \overset{1^{12}}{\underset{1^{14}}{3}} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 10, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 14 & 90 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [30] \quad 1 \overset{1^1}{\underset{1^1}{3}} 2 2 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 9, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[31] \quad 1 \overset{1^1}{\underset{1^1}{2}} 3 2 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[32] \quad 1 \overset{1^{12}}{\underset{1^{11}}{3}} 2 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1 \overset{1^{11}}{\underset{1^{11}}{2}} 3 2 1^2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 17 & 9 \\ 2 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1 \overset{1^{11}}{\underset{1^{13}}{2}} 3 2 1^4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1 \overset{1^{11}}{\underset{1^{12}}{2}} 3 2 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1 \overset{1^{11}}{\underset{1^{11}}{2}} 3 1 3 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[37] $1^1 1^1 4 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[38] $1^1 2 3 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[39] $1^1 2 3 1^1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 4 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[40] $1^1 1^1 2 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[41] $1^1 1^1 4 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[42] $4 1^1 2^1 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[43] $2 2 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[44] $1^1 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[45] $1^1 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[46] $2 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[47] $1^1 1^1 5 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{\overline{11}} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{\overline{1}} \begin{smallmatrix} 5 \\ 1^{\overline{1}}, 1^{\overline{1}} \end{smallmatrix} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{\overline{1}} \begin{smallmatrix} 5 \\ 1^{\overline{1}} \end{smallmatrix} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{\overline{11}} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{\overline{14}} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{4}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{\overline{13}} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{\overline{12}} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{\overline{11}} 2^{\overline{12}} 3^{\overline{1}} 21^{\overline{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{\overline{12}} 2^{\overline{11}} 3^{\overline{1}} 21^{\overline{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 49 & 15 \\ 2 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{\overline{11}} 2^{\overline{11}} 3^{\overline{1}} 21^{\overline{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{\overline{12}} 2^{\overline{11}} 3^{\overline{1}} 21^{\overline{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{11}23131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{11}231^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[63] $31^131^141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{13}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 13 & 0 & 97 & 21 \\ 2 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{12}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{11}231^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 7 & 3 \\ 4 & 0 & 7 & 14 & 8 \\ 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{12}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

- [70] $1^{12}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$
- [71] $1^{12}231^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -65$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 11 & 0 & 32 & 12 \\ 2 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0}$
- [72] $1^{13}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [73] $1^{12}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [74] $1^{12}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [75] $1^{13}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [76] $1^{12}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [77] $1^{13}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 77$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$
- [78] $41^{11}232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 20 & 6 \\ 2 & 5 & 0 & 10 & 2 \\ 8 & 20 & 10 & 38 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [79] $1^{13}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$
- [80] $1^{13}2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 87$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

- [81] $1^{13}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$
- [82] $41^{11}232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$
- [83] $1^{13}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [84] $1^{13}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [85] $1^{14}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [86] $1^{13}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [87] $1^{14}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [88] $1^{14}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$
- [89] $41^{12}232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 24 & 6 \\ 2 & 5 & 0 & 8 & 2 \\ 8 & 24 & 8 & 46 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [90] $1^{14}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$
- [91] $41^{11}232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 6 & 2 \\ 4 & 10 & 5 & 16 & 6 \\ 2 & 5 & 0 & 7 & 2 \\ 6 & 16 & 7 & 21 & 9 \\ 2 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{14}2^321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{14}2^321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[94] $1^{14}2^321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{14}2^321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[96] $41^{13}2^32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 36 & 6 \\ 2 & 5 & 0 & 12 & 2 \\ 12 & 36 & 12 & 106 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[97] $41^{14}2^32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 48 & 6 \\ 2 & 5 & 0 & 16 & 2 \\ 16 & 48 & 16 & 190 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[98] $41^{12}2^332 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 10 & 2 \\ 4 & 10 & 5 & 28 & 6 \\ 2 & 5 & 0 & 11 & 2 \\ 10 & 28 & 11 & 65 & 15 \\ 2 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[99] $41^{12}2^332 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 32 & 6 \\ 2 & 5 & 0 & 14 & 2 \\ 12 & 32 & 14 & 90 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[100] $41^{13}2^332 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 14 & 2 \\ 4 & 10 & 5 & 40 & 6 \\ 2 & 5 & 0 & 15 & 2 \\ 14 & 40 & 15 & 133 & 21 \\ 2 & 6 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[101] $41^{13}2^332 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 44 & 6 \\ 2 & 5 & 0 & 18 & 2 \\ 16 & 44 & 18 & 166 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[102] $41^{14}2^332 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 18 & 2 \\ 4 & 10 & 5 & 52 & 6 \\ 2 & 5 & 0 & 19 & 2 \\ 18 & 52 & 19 & 225 & 27 \\ 2 & 6 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 41^{14} 2^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 1184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 20 & 2 \\ 4 & 10 & 5 & 56 & 6 \\ 2 & 5 & 0 & 22 & 2 \\ 20 & 56 & 22 & 266 & 30 \\ 2 & 6 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [104] \ 1^1 4 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[105] \ 1^1 4 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[106] \ 1^1 4 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[107] \ 1^1 4 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[108] \ 1^1 4 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[109] \ 1^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 \\ 2 & 3 & 1 & 3 \\ 3 & 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[110] \ 2^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[111] \ 1^1 3 1^1 4 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^1 5 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[113] \ 1^1 3 14 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

- [114] $1^1 1^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$
- [115] $1^1 1^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [116] $1^1 1^1 5 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [117] $1^1 1^1 2231^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 3 & 1 & 8 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 8 & 0 & 15 & 9 \\ 2 & 5 & -1 & 9 & 5 \end{pmatrix}_{b_0}$
- [118] $1^1 1^1 223131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$
- [119] $1^1 1^1 231321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$
- [120] $1^1 1^1 223131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$
- [121] $2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 4 & 2 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [125] $1^1 1^1 3141^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$
- [126] $31^1 31^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [127] $1^1 1^1 41^1 2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0}$
- [128] $1^1 1^1 412321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$

- [123] $1^{141}1^{2321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0}$
- [124] $1^{11}1^{314131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$
- [122] $1^{11}1^{412321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$
- [129] $1^{11}1^{2231}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -36$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 8 & 4 \\ 4 & 0 & 8 & 15 & 9 \\ 2 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0}$
- [130] $2^{141}2^{321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 3 & 5 \\ 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$
- [131] $31^{231}1^{51}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$
- [132] $1^{11}1^{31}1^{4131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 6 & 1 & 14 & 8 \\ 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$
- [133] $2^{11}2^{321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 6 & 6 \\ 2 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$
- [134] $1^{141}1^{2321}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$
- [135] $1^{11}1^{41}1^{2321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$
- [136] $2^{141}2^{321}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 12 & 6 \\ 6 & 12 & 18 & 10 \\ 3 & 6 & 10 & 5 \end{pmatrix}_{b_0}$
- [137] $2^{11}2^{321}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

- [138] $2 \overset{1^{12}41}{3} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 14 & 8 \\ 2 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [139] $2 \overset{1^{11}41}{3} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 11 & 6 \\ 6 & 11 & 15 & 9 \\ 3 & 6 & 9 & 5 \end{pmatrix}_{b_0}$
- [140] $31 \overset{1^1}{\underset{1^{11}}{5}} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [141] $2 \overset{1^{11}41}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [142] $31 \overset{1^1}{3} 31 \overset{1^1}{5} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 21 & 4 \\ 18 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$
- [143] $1 \overset{1^1}{2} 3141 \overset{1^1}{31} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 7 & 1 & 17 & 9 \\ 2 & 3 & 0 & 9 & 4 \end{pmatrix}_{b_0}$
- [144] $1 \overset{1^1}{4} 1^{12} 2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0}$
- [145] $1 \overset{1^1}{2} 41 \overset{1^1}{3} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0}$
- [146] $1 \overset{1^1}{4} 1^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$
- [147] $1 \overset{1^1}{4} 1^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$
- [148] $2 \overset{1^{12}41}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 14 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [149] $2 \overset{1^1}{3} 3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 18 & 6 \\ 9 & 18 & 43 & 15 \\ 3 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

[150] $31^{11}31\bar{5}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[151] $31\bar{5}^{11}13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[152] $2\bar{3}^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 1 & 9 & 3 \\ 5 & 4 & 9 & 20 & 10 \\ 2 & 2 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[153] $2\bar{3}^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 13 & 3 \\ 7 & 5 & 13 & 42 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{12}31^{14}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 10 & 1 & 39 & 13 \\ 2 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[155] $31\bar{5}^{11}13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[156] $2\bar{3}^{13}4^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 6 & 12 & 9 & 34 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[157] $31^{11}321^{15}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

[158] $1^{11}41^{23}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$$

[159] $1^{14}1^{23}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$$

[160] $2\bar{3}^{14}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 24 & 6 \\ 12 & 24 & 78 & 20 \\ 3 & 6 & 20 & 5 \end{pmatrix}_{b_0}$$

[161] $1^{12}41^{23}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

- $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$
- [162] $1^{11}41^{12}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$
- [163] $2^{11}3^{11}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 17 & 6 \\ 9 & 17 & 36 & 14 \\ 3 & 6 & 14 & 5 \end{pmatrix}_{b_0}$
- [164] $2^{11}3^{11}41^{11}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 7 & 4 \\ 2 & 4 & 1 & 7 & 3 \\ 4 & 7 & 7 & 12 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [165] $2^{11}3^{12}41^{12}21 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 1 & 10 & 3 \\ 6 & 6 & 10 & 30 & 12 \\ 2 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$
- [166] $2^{11}3^{14}41^{11}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 8 & 16 & 12 & 62 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$
- [167] $31^{12}31^{12}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 16 & 5 \\ 12 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$
- [168] $1^{11}41^{12}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$
- [169] $1^{11}41^{12}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$
- [170] $1^{12}41^{12}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 74 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$
- [171] $1^{11}41^{12}2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$
- [172] $1^{13}41^{12}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$
- [173] $2^{11}3^{13}41^{12}21 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 11 & 3 \\ 7 & 8 & 11 & 44 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[174] \quad 2^3 3^{15} 21^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 30 & 6 \\ 15 & 30 & 123 & 25 \\ 3 & 6 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{141} 2^{2321} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{1241} 2^{2321} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[177] \quad 2^3 3^{15} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 20 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 10 & 20 & 15 & 98 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[178] \quad 2^3 3^{13} 21^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 23 & 6 \\ 12 & 23 & 67 & 19 \\ 3 & 6 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{141} 3^{2321} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{1141} 3^{2321} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[181] \quad 2^3 3^{1241} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 1 & 14 & 3 \\ 8 & 7 & 14 & 54 & 16 \\ 2 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{1341} 2^{1321} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{1341} 2^{1321} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[184] \quad 231^{11} 3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 35 & 0 & 33 & 11 \\ 2 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 2^3 3^{16} 21^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

- $$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 6 & 36 & 6 \\ 18 & 36 & 178 & 30 \\ 3 & 6 & 30 & 5 \end{pmatrix}_{b_0}$$
- [186] $2^{1^{12}4^1}3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 11 & 4 \\ 2 & 4 & 1 & 10 & 3 \\ 6 & 11 & 10 & 30 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$
- [187] $1^{12}4^{23}2^{321}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$
- [188] $1^{11}4^{23}2^{321}1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$
- [189] $2^{1^{16}4^1}3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 4 & 24 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 12 & 24 & 18 & 142 & 24 \\ 2 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$
- [190] $31^{13}31^51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 24 & 5 \\ 18 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$
- [191] $2^{1^{12}4^1}3^{21}1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 18 & 3 \\ 10 & 8 & 18 & 86 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$
- [192] $1^{14}4^{12}3^{2321}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 16 & 3 \\ 16 & 16 & 106 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$
- [193] $1^{12}4^{23}2^{321}1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 142 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$
- [194] $31^{12}3^{21}5^{21}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 30 & 5 \\ 22 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$
- [195] $2^{1^{13}4^1}3^{21}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 1 & 15 & 3 \\ 9 & 9 & 15 & 70 & 18 \\ 2 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$
- [196] $2^{1^{11}4^1}3^{21}1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 29 & 6 \\ 15 & 29 & 108 & 24 \\ 3 & 6 & 24 & 5 \end{pmatrix}_{b_0}$$
- [197] $1^{12}4^{32}3^{2321}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

- $$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 138 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$
- [198] $2^{\overline{1^1 41}} 2^{\overline{1^7}} 21^{\overline{1^7}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 6 & 42 & 6 \\ 21 & 42 & 243 & 35 \\ 3 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$
- [199] $1^{\overline{1^3 41}} 2^{\overline{2 3 21}} 13^{\overline{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 1 & 16 & 3 \\ 18 & 16 & 118 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$
- [200] $2^{\overline{1^4 41}} 2^{\overline{1^3}} 21^{\overline{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 1 & 16 & 3 \\ 10 & 11 & 16 & 90 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$
- [201] $2^{\overline{1^7 41}} 2^{\overline{1^7}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 2 & 4 & 28 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 14 & 28 & 21 & 194 & 28 \\ 2 & 4 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$
- [202] $2^{\overline{1^3 41}} 2^{\overline{1^3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 15 & 4 \\ 2 & 4 & 1 & 13 & 3 \\ 8 & 15 & 13 & 56 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$
- [203] $1^{\overline{1^2 41}} 2^{\overline{3 2 3 21}} 14^{\overline{1^4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 1 & 18 & 3 \\ 23 & 18 & 180 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$$
- [204] $1^{\overline{1^4 41}} 2^{\overline{2 3 2 3 21}} 13^{\overline{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 18 & 3 \\ 19 & 18 & 140 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$
- [205] $1^{\overline{1^3 41}} 2^{\overline{2 3 2 3 21}} 14^{\overline{1^4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$$
- [206] $1^{\overline{1^3 41}} 2^{\overline{3 2 3 21}} 13^{\overline{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$$
- [207] $2^{\overline{1^3 41}} 2^{\overline{1^4}} 21^{\overline{1^4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 1 & 19 & 3 \\ 11 & 10 & 19 & 104 & 22 \\ 2 & 2 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$
- [208] $31^{\overline{1^4}} 31^{\overline{1^4}} 51^{\overline{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 32 & 5 \\ 24 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$
- [209] $1^{\overline{1^3 41}} 2^{\overline{4 2 3 2 3 21}} 13^{\overline{1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

- [210] $\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 214 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$ $1^{12}41^{42}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 26 & 3 \\ 2 & 1 & 20 & 3 \\ 26 & 20 & 230 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0}$
- [211] $1^{14}41^{32}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -318$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 186 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$
- [212] $231^{12}3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -88$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 1 & 10 & 2 \\ 5 & 33 & 1 & 70 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 70 & 0 & 138 & 22 \\ 2 & 11 & -1 & 22 & 3 \end{pmatrix}_{b_0}$
- [213] $2^{1^{14}41^{11}}3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 19 & 4 \\ 2 & 4 & 1 & 16 & 3 \\ 10 & 19 & 16 & 90 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$
- [214] $1^{14}41^{22}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -330$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 182 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$
- [215] $1^{13}41^{32}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -338$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 202 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$
- [216] $2^{1^{14}41}3^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -350$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 1 & 20 & 3 \\ 12 & 12 & 20 & 126 & 24 \\ 2 & 2 & 3 & 24 & 4 \end{pmatrix}_{b_0}$
- [217] $1^{13}41^{42}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -395$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 27 & 3 \\ 2 & 1 & 22 & 3 \\ 27 & 22 & 256 & 38 \\ 3 & 3 & 38 & 5 \end{pmatrix}_{b_0}$
- [218] $31^{13}321^{35}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -88$, $\det = -400$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 33 & 3 \\ 5 & 5 & 45 & 5 \\ 33 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$
- [219] $1^{14}41^{32}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -407$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 25 & 3 \\ 2 & 1 & 22 & 3 \\ 25 & 22 & 228 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0}$
- [220] $1^{14}41^{42}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -470$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 28 & 3 \\ 2 & 1 & 24 & 3 \\ 28 & 24 & 286 & 40 \\ 3 & 3 & 40 & 5 \end{pmatrix}_{b_0}$
- [221] $231^{13}3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -88$, $\det = -599$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 1 & 15 & 2 \\ 5 & 33 & 1 & 105 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 105 & 0 & 313 & 33 \\ 2 & 11 & -1 & 33 & 3 \end{pmatrix}_{b_0}$
- [222] $231^{14}3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -88$, $\det = -984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 20 & 2 \\ 5 & 33 & 1 & 140 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 140 & 0 & 558 & 44 \\ 2 & 11 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 231^{1\bar{5}}3151^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -1479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 25 & 2 \\ 5 & 33 & 1 & 175 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 175 & 0 & 873 & 55 \\ 2 & 11 & -1 & 55 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [224] \ 1^{\bar{1}}\bar{5}12321^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{\bar{1}}\bar{5}1321^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{\bar{1}}\bar{4}14131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{\bar{1}}\bar{3}1\bar{5}131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{\bar{1}}\bar{3}1\bar{5}131^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{\bar{1}}\bar{4}131\bar{5}1^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{\bar{1}}\bar{4}14141^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{\bar{1}}\bar{3}1\bar{5}1231^{\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{\bar{1}}\bar{3}31\bar{5}131^{\bar{4}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

- [233] $1^1 3151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [234] $1^1 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [235] $1^{11} 2231321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$
- [236] $1^1 161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [237] $216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -76, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [238] $1^{11} 315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [239] $1^1 413151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -78, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 6 \end{pmatrix}_{b_0}$
- [240] $1^1 612231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [241] $1^{11} 2231321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$
- [242] $1^{11} 315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [243] $1^{11} 315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

- [244] $1^{11}315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$
- [245] $1^{12}315131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$
- [246] $1^{13}315131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$
- [247] $1^{14}315131^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$
- [248] $1^131^141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$
- [249] $1^{11}3141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$
- [250] $1^{11}3141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$
- [251] $1^{11}315131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$
- [252] $1^{12}315131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$
- [253] $2^315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$
- [254] $2^315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [255] $1^{11}3151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$

- $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [256] $1^{11}3151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$
- $1^{12}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [258] $1^{11}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [259] $1^{11}3151^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 27 & 11 \\ 2 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$
- [260] $1^{11}31^1 41321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 8 & 4 \\ 6 & 8 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}$
- $1^{11}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- $1^{12}2315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- $1^{12}31^1 5131^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- $1^{11}2315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- $1^{12}3151^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$
- $1^{13}1^{12}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[268] $1^{12}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[267] $1^{11}3151^{1231^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[269] $1^{12}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[271] $1^{13}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[272] $1^{13}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[273] $1^{12}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[270] $1^{12}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[274] $1^{11}3151^{1231^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[275] $1^{12}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[276] $1^{12}3151^{1231^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

[277] $1^{12}2315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[278] $1^{12} 2^3 3 15 13 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[279] $1^{13} 3 15 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[282] $1^{13} 3 15 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[281] $1^{12} 3 15 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 7 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[280] $1^{13} 3 15 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0}$$

[283] $1^{13} 3 15 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[284] $1^{13} 3 15 13 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3 15 13 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[286] $1^{13} 3 15 13 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[287] $1^{13} 3 15 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0}$$

[288] $1^{12} 3 15 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 10 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[289] $1^{14}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[290] $1^{13}315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[291] $1^{13}3151^2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 1 & 22 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 22 & 0 & 154 & 26 \\ 2 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[292] $1^{12}3151^1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 8 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[295] $1^{14}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[294] $2^3315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[293] $1^{14}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[296] $2^3315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{13}3151^1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 17 & 3 \\ 9 & 0 & 17 & 79 & 19 \\ 2 & 0 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[298] $1^{14}3151^1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 1 & 21 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 21 & 0 & 117 & 23 \\ 2 & 4 & -1 & 23 & 4 \end{pmatrix}_{b_0}$$

[299] $1^{14}315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

- [311] $1^{13}3151^2231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 206$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 12 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$
- [312] $1^{14}2315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 234$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$
- [313] $1^{13}3151^3231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 249$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 15 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$
- [314] $1^{13}3151^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 13 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$
- [315] $1^{14}3151^2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 314$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 26 & 3 \\ 14 & 0 & 26 & 202 & 30 \\ 2 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$
- [316] $1^{13}3151^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 318$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 16 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$
- [317] $1^{14}3151^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 383$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 31 & 3 \\ 17 & 0 & 31 & 317 & 37 \\ 2 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$
- $T = 11$ [318] $1^1231513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 11, 2)$, $c_{\text{ext}} = 4.0$
- [319] $1^123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = 0$, $(n_+, n_-, n_0) = (1, 11, 1)$, $c_{\text{ext}} = 4.0$
- [320] $1^131513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 11, 1)$, $c_{\text{ext}} = 4.0$
- [322] $1^16131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -56$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$
- [325] $1^16131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -56$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$
- [327] $1^1414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -45$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 5 & 6 \\ 2 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0}$
- [328] $1^43151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 16 & 3 \\ 8 & 0 & 16 & 106 & 24 \\ 2 & 0 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^1 16 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^1 31^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 5 & 3 \\ 2 & 0 & 5 & 7 & 7 \\ 2 & 0 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^1 4 141 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 5 & 6 \\ 3 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^1 6 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^1 5 132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^1 5 131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^1 31^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 2 & 4 \\ 2 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[333] \quad 1^1 5 131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[341] \quad 1^{11} 31413221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[338] \quad 1^1 5 132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -74, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[337] \quad 21^1 7 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

- [340] $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
 1^1
 $[340] \ 1712231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [336] $31^1 1316131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [342] $1^{11} 31413221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$
- [335] $1^1 6123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -76, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$
- [334] $1^{11} 23141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -7, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$
- [339] $1^1 1513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$
- [343] $1^{12} 43151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 78 & 20 \\ 2 & 0 & 3 & 20 & 5 \end{pmatrix}_{b_0}$
- [344] $1^1 13151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 7 & 7 \\ 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0}$
- [345] $1^{11} 123151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 18 & 10 \\ 2 & 0 & 3 & 10 & 5 \end{pmatrix}_{b_0}$
- [346] $1^1 43151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 142 & 28 \\ 2 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0}$
- [347] $1^{14} 23151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

- [348] $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$
- [349] $1^{13} 2^3 15 13 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 10 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 10 & 0 & 34 & 14 \\ 2 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0}$
- [350] $1^{31} 5 13 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 79 & 21 \\ 2 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0}$
- [351] $1^{12} 2^3 15 13 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$
- [352] $1^{11} 2^3 15 13 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$
- [355] $1^{11} 2^3 21^1 6 13 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 9 & 3 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 9 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [353] $1^{31} 5 13 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 11 & 3 \\ 6 & 0 & 11 & 53 & 17 \\ 2 & 0 & 3 & 17 & 5 \end{pmatrix}_{b_0}$
- [354] $1^{11} 3 14 14 1^{13} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & 6 & 6 \\ 2 & 4 & 0 & 6 & 5 \end{pmatrix}_{b_0}$
- [356] $1^{11} 2^3 21 15 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$
- [358] $1^{12} 1^{11} 3 15 13 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 4 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$
- [359] $1^{11} 2^3 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

- [360] $1^{11}23215131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$
- [361] $1^{11}5^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$
- [362] $1^{11}31414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 4 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0}$
- [363] $1^{11}23151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$
- [364] $1^{11}315^{11}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$
- [357] $1^{11}23151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$
- [365] $2^13^1216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$
- [366] $2^13^1216131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [367] $1^{11}315^{11}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$
- [368] $31^16131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 2 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [369] $1^{12}315^{12}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

- [370] $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$ $1^{11}31^1414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 9 & 1 & 23 & 11 \\ 2 & 4 & 0 & 11 & 5 \end{pmatrix}_{b_0}$
- [373] $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$ $1^{12}31^151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
- [372] $1^{12}23151^1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 15 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 53 & 17 \\ 2 & 5 & -1 & 17 & 5 \end{pmatrix}_{b_0}$
- [371] $1^{11}23151^1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 10 & 4 \\ 5 & 0 & 10 & 28 & 12 \\ 2 & 0 & 4 & 12 & 5 \end{pmatrix}_{b_0}$
- [374] $1^{12}23216131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [375] $1^{11}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$
- [376] $1^{12}31^151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$
- [377] $1^{11}23216131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [378] $1^{11}2321^15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 12 & 4 \\ 6 & 0 & 12 & 38 & 14 \\ 2 & 0 & 4 & 14 & 5 \end{pmatrix}_{b_0}$
- [379] $1^{11}31^151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [380] $1^{11}31^151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[381] $1^{1^3}2^31^51321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 7 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[382] $1^{1^2}2^321^15131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 17 & 0 & 67 & 19 \\ 2 & 5 & -1 & 19 & 5 \end{pmatrix}_{b_0}$$

[383] $1^{1^1}2^31^51321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 5 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[384] $2^321^{1^{11}}6131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 16 & 11 & 101 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[385] $1^{1^2}2^31^51321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[386] $2^321^{1^{11}}6131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[388] $1^{1^1}2^31^51321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[387] $1^{1^2}2^31^4141^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 14 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 14 & 1 & 55 & 17 \\ 2 & 4 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

[390] $1^{1^3}2^31^51321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[389] $1^{1^1}2^31^51321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[393] $1^{1^2}2^31^51^{12}2^31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 15 & 4 \\ 7 & 0 & 15 & 53 & 17 \\ 2 & 0 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[392] $1^{13}3151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[391] $1^{13}23151^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 20 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 20 & 0 & 88 & 22 \\ 2 & 5 & -1 & 22 & 5 \end{pmatrix}_{b_0}$$

[394] $1^{12}3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[395] $1^{13}3151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[396] $1^{12}23216131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[397] $1^{11}2321^{16}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 8 & 7 & 32 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[398] $1^{12}23216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[399] $1^{13}2321^15131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 22 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 22 & 0 & 106 & 24 \\ 2 & 5 & -1 & 24 & 5 \end{pmatrix}_{b_0}$$

[400] $1^{12}2321^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 17 & 4 \\ 8 & 0 & 17 & 67 & 19 \\ 2 & 0 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

[401] $1^{13}3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 8 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[402] $1^{14}3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 10 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[403] $1^{12}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[404] $1^{14}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[405] $1^{13}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[406] $1^{12}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[407] $1^{13}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

[408] $1^{13}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

[410] $1^{14}23151^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 25 & 0 & 133 & 27 \\ 2 & 5 & -1 & 27 & 5 \end{pmatrix}_{b_0}$$

[409] $1^{13}23151^{12}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 20 & 4 \\ 9 & 0 & 20 & 88 & 22 \\ 2 & 0 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[411] $1^{14}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[412] $1^{14}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 11 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[413] $1^{14}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[414] $1^{12} 2 3 21^1 6131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 13 & 14 & 95 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[415] $1^{13} 2 3 216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[416] $1^{12} 2 3 21^1 6131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 11 & 8 & 51 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[417] $1^{13} 23151^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -129, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 25 & 4 \\ 12 & 0 & 25 & 163 & 29 \\ 2 & 0 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[418] $1^{14} 31^1 51321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -129, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 11 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[419] $1^{14} 31^1 51321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -132, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[420] $1^{13} 2 3 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -136, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[421] $1^{12} 2 3 21^1 6131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 12 & 11 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[422] $1^{14} 23151^2 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 30 & 4 \\ 14 & 0 & 30 & 218 & 34 \\ 2 & 0 & 4 & 34 & 5 \end{pmatrix}_{b_0}$$

[424] $1^{14} 2 3 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[423] $1^{14} 2 3 216131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[425] \quad 1^{13} 2^3 21^1 6131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -220, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 17 & 18 & 157 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[426] \quad 1^{14} 2^3 216131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[427] \quad 1^{13} 2^3 21^1 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -248, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 15 & 12 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[428] \quad 1^{13} 2^3 21^1 6131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -262, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 16 & 15 & 124 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[429] \quad 1^{14} 2^3 21^1 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -342, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 18 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 14 & 18 & 13 & 130 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[431] \quad 1^{14} 2^3 21^1 6131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 1 & 2 & 19 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 15 & 19 & 16 & 159 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[430] \quad 1^{14} 2^3 21^1 6131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 1 & 2 & 20 & 3 \\ 1 & 2 & 0 & 19 & 2 \\ 16 & 20 & 19 & 194 & 30 \\ 2 & 3 & 2 & 30 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [432] \quad 1^{14} 231^1 5^2 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[433] \quad 1^{14} 231^1 5^1 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[434] \quad 1^{11} 231^1 5^2 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0$$

$$[435] \quad 1^{12} 231^1 5^1 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.5$$

$$[442] \quad 1^{11} 1^{13} 2321^1 6^1 131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 110 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[444] \quad 1^{11} 5^1 1315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[445] \quad & 1^{\overline{1}^{1321}} \overline{7}^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[446] \quad & 1^{\overline{1}^{131}} \overline{6}^{13141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & -1 & 3 & 5 \end{pmatrix}_{b_0} \\
[447] \quad & 1^{\overline{1}^{131}} \overline{6}^{13141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[448] \quad & 1^{\overline{1}^{231}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 26 & 12 \\ 2 & 3 & 2 & 12 & 5 \end{pmatrix}_{b_0} \\
[449] \quad & 21 \overline{7}^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[450] \quad & 1^{\overline{1}^{131}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0} \\
[451] \quad & 1^{\overline{1}^{131}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 8 \\ 2 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0} \\
[443] \quad & 1^{\overline{1}^{112321}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 6 & 3 \\ 8 & 6 & 30 & 14 \\ 3 & 3 & 14 & 6 \end{pmatrix}_{b_0} \\
[441] \quad & 1^{\overline{1}^{112321}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 7 & 7 \\ 3 & 3 & 7 & 6 \end{pmatrix}_{b_0} \\
[440] \quad & 1^{\overline{1}^{132321}} \overline{6}^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 122 & 28 \\ 3 & 3 & 28 & 6 \end{pmatrix}_{b_0} \\
[459] \quad & 21 \overline{6}^{131} \overline{5}^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[456] \quad & 1^1 6 131 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[457] \quad & 21^1 6 131 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
[455] \quad & 1^1 6 12231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[454] \quad & 1^1 6 131 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[453] \quad & 1^1 6^{11} 2321 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 2 & 1 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0} \\
[460] \quad & 1^1 6 1231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[452] \quad & 1^1 6 131 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[458] \quad & 1^1 6 1231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[471] \quad & 1^1 5^{11} 2231 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 2 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[465] \quad & 221^1 8 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[466] \quad & 1^{11} 231512321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$$

[467] $1^{\overline{1}1}71223141^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[468] $1^{\overline{1}1}223151231^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

[469] $1^{\overline{1}1}314141321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[470] $1^{\overline{1}1}223151231^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$$

[464] $1^{\overline{1}}1513161^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[472] $1^{\overline{1}1}31513221^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 4 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0}$$

[473] $1^{\overline{1}1}5^{\overline{1}1}2231^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 6 & 5 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 6 & 4 & 8 & 8 \\ 2 & 5 & 2 & 8 & 6 \end{pmatrix}_{b_0}$$

[474] $1^{\overline{1}1}231512321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

[475] $1^{\overline{1}}31513221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 8 & 8 \\ 2 & 5 & -1 & 8 & 6 \end{pmatrix}_{b_0}$$

[476] $1^{\overline{2}3}1513221^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 1 & 12 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 12 & 0 & 38 & 16 \\ 2 & 5 & -1 & 16 & 6 \end{pmatrix}_{b_0}$$

[477] $1^{\overline{3}}31513221^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 18 & 0 & 88 & 24 \\ 2 & 5 & -1 & 24 & 6 \end{pmatrix}_{b_0}$$

[478] $1^{11} 31 5 13221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 24 & 0 & 158 & 32 \\ 2 & 5 & -1 & 32 & 6 \end{pmatrix}_{b_0}$$

[479] $1^{11} 6 1314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[480] $1^{11} 15 132161^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -71, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[481] $1^{11} 314141321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[463] $1^{11} 231 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

[462] $1^{11} 231 1^{11} 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

[461] $21 8 12231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[482] $1^{11} 6 1232151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 14, \Delta = -73, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[483] $1^{11} 31 5 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 13 & 4 \\ 6 & 0 & 13 & 57 & 19 \\ 2 & 0 & 4 & 19 & 6 \end{pmatrix}_{b_0}$$

[484] $1^{11} 31 5 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0}$$

[485] $1^{11} 1^{11} 2321 6 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 6 & 3 \\ 6 & 6 & 22 & 12 \\ 3 & 3 & 12 & 6 \end{pmatrix}_{b_0} \\
[486] & 1^{1^2} 6^{1^2} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 9 & 3 \\ 12 & 9 & 67 & 21 \\ 3 & 3 & 21 & 6 \end{pmatrix}_{b_0} \\
[487] & 1^{1^2} 6^{1^2} 131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 9 & 3 \\ 10 & 9 & 57 & 19 \\ 3 & 3 & 19 & 6 \end{pmatrix}_{b_0} \\
[491] & 1^{1^2} 31^2 5^{1^2} 13221^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0} \\
[492] & 1^{1^2} 223151^{1^2} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 4 & 1 & 17 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 17 & 0 & 57 & 19 \\ 2 & 6 & -1 & 19 & 6 \end{pmatrix}_{b_0} \\
[493] & 1^{1^2} 6^{1^2} 131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \\
[494] & 1^{1^2} 6^{1^2} 131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[495] & 1^{1^2} 231^2 5^{1^2} 1321^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0} \\
[496] & 1^{1^2} 41^{1^2} 3151321^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 2 & 3 & 7 & 7 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0} \\
[497] & 1^{1^2} 413151321^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[498] & 1^{1^2} 231^2 5^{1^2} 1321^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0} \\
[499] & 1^{1^2} 6^{1^2} 131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 7 & 4 \\ 6 & 7 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

- [500] $1^{11} 1^{11} 2321^{11} 6^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$
- [490] $1^{11} 321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$
- [488] $1^{11} 321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$
- [489] $1^{11} 413151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$
- [501] $1^{11} 31513221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 11$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0}$
- [502] $1^{11} 1^{11} 2321^{11} 6^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 12$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 11 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$
- [503] $1^{11} 1^{12} 2321^{11} 6^{11} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 14$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 106 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$
- [504] $1^{11} 31513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$
- [506] $1^{12} 31513221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 5 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
- [507] $1^{11} 31513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$
- [505] $1^{12} 31513221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 7 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
- [508] $1^{11} 41^1 3151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 18$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 4 & 3 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0} \\
[509] & 1^{1^{11}2321} 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 10 & 4 \\ 7 & 10 & 32 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0} \\
[510] & 1^{12} 31^1 5^{12} 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 6 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0} \\
[511] & 1^{11} 321^1 6^{12} 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0} \\
[512] & 1^{1^{11}2321} 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 9 & 4 \\ 6 & 9 & 22 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \\
[513] & 1^{12} 321^1 6^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0} \\
[514] & 1^{14} 1^{13} 15^{12} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 8 & 4 \\ 7 & 8 & 20 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \\
[515] & 1^{1^{12}2321} 6^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 4 & 4 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0} \\
[516] & 1^{1^{12}2321} 6^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 4 & 5 & 21 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0} \\
[520] & 1^{13} 231^1 5^{12} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0} \\
[521] & 1^{12} 31^1 5^{12} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 17 & 5 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 17 & 9 & 67 & 21 \\ 2 & 5 & 2 & 21 & 6 \end{pmatrix}_{b_0} \\
[519] & 1^{13} 231^1 5^{12} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

- [517] $1^{12}223151^{1231^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 17 & 5 \\ 7 & 0 & 17 & 57 & 19 \\ 2 & 0 & 5 & 19 & 6 \end{pmatrix}_{b_0}$
- [518] $1^{13}223151^{1231^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 4 & 1 & 23 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 23 & 0 & 97 & 25 \\ 2 & 6 & -1 & 25 & 6 \end{pmatrix}_{b_0}$
- [522] $1^{13}2315^{12}1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$
- [523] $1^{12}32161^{12321^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$
- [524] $1^{11}2321^{11}6^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$
- [525] $1^{12}2321^{11}6^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$
- [527] $1^{12}2321^{11}6^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 6 & 5 & 31 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$
- [528] $1^{11}2321^{11}6^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
- [529] $1^{12}2321^{11}6^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
- [530] $1^{141}13151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 11 & 4 \\ 8 & 11 & 35 & 15 \\ 3 & 4 & 15 & 6 \end{pmatrix}_{b_0}$
- [531] $1^{13}2321^{11}6^{131^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 64 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [532] $1^{11}32161^{12321^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$

- [533] $1^{\overline{13}2321} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 43 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [526] $1^{\overline{12}2321^1} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$
- [537] $1^{\overline{13}31^4} 5^{13} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 10 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$
- [536] $1^{\overline{13}31^2} 5^{13} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 8 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$
- [535] $1^{\overline{12}2321} 6^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 12 & 4 \\ 10 & 12 & 50 & 18 \\ 3 & 4 & 18 & 6 \end{pmatrix}_{b_0}$
- [534] $1^{\overline{11}41^1} 3^{15} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 4 & 6 & 22 & 12 \\ 2 & 2 & 2 & 12 & 5 \end{pmatrix}_{b_0}$
- [538] $1^{\overline{12}31^2} 5^{13} 13221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$
- [539] $1^{\overline{13}31^3} 5^{13} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 9 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$
- [540] $1^{\overline{12}31^3} 5^{13} 13221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$
- [541] $1^{\overline{13}223151^1} 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 23 & 5 \\ 9 & 0 & 23 & 97 & 25 \\ 2 & 0 & 5 & 25 & 6 \end{pmatrix}_{b_0}$
- [542] $1^{\overline{14}231^3} 5^{13} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$

- [543] $1^{14}31^151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$
- [544] $1^{14}31^1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 9 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
- [545] $1^{14}31^1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 10 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
- [546] $1^21^{12}2321^16131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 14 & 4 \\ 13 & 14 & 72 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$
- [547] $1^11^{12}23216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 14 & 4 \\ 10 & 14 & 57 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0}$
- [548] $1^11^{12}2321^16131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [549] $1^11^{13}23216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 6 & 7 & 34 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$
- [550] $1^31^{13}23216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$
- [552] $1^{12}321^1612321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
- [553] $1^{11}321^1612321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
- [555] $1^21^{13}2321^16131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 9 & 6 & 7 & 49 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0}$
- [551] $1^31^{12}23216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 85 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$$

[554] $1^{1^{13}2321^1} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[556] $1^{12} 41^{13} 51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 32 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$$

[557] $1^{1^{13}2321} 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 7 & 6 & 8 & 44 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[558] $1^{12} 6^{1^{12}2321} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 15 & 4 \\ 11 & 15 & 69 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$$

[559] $1^{11} 41^{13} 51321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 53 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0}$$

[560] $1^{14} 31^3 513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 11 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

[561] $1^{14} 31^4 513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 12 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

[562] $1^{1^{13}2321^1} 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 6 & 10 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[563] $1^{12} 32161^{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[564] $1^{12} 321^{16} 1^{12} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 88 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

[565] $1^{1^{13}2321^1} 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

- [568] $1^2 \overset{13}{2} \overset{2321}{6} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 99 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0}$
- [570] $1^2 \overset{14}{2} \overset{2321}{6} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 10 & 8 & 7 & 63 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$
- [571] $1^{11} 41^1 3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 6 & 9 & 59 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$
- [572] $1^{12} 41^1 3151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 47 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0}$
- [573] $1^3 \overset{14}{2} \overset{2321}{6} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 11 & 8 & 8 & 79 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$
- [569] $1^{12} \overset{2321}{6} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$
- [567] $1^4 \overset{13}{2} \overset{2321}{6} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 19 & 4 \\ 16 & 19 & 135 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0}$
- [566] $1^{13} 32161^1 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$
- [577] $1^2 \overset{14}{2} \overset{2321}{6} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 8 & 8 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$
- [578] $1^{11} 41^1 3151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 74 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$
- [579] $1^3 \overset{12}{2} \overset{2321}{6} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
- [580] $1^4 \overset{14}{2} \overset{2321}{6} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

- [574] $1^1 41^2 3151321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 8 & 10 & 90 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0}$
- [575] $1^1 1^3 2321^1 6^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 86 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$
- [576] $1^1 1^3 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
- [581] $1^1 1^2 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$
- [582] $1^1 1^4 2321^1 6^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 72 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$
- [583] $1^2 1^2 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 81$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 95 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0}$
- [586] $1^2 1^4 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 98$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 9 & 8 & 11 & 77 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$
- [585] $1^3 1^4 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 98$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 10 & 8 & 12 & 93 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$
- [584] $1^1 1^3 2321^1 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 98$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
- [588] $1^1 1^2 2321^1 6^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 99$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0}$
- [589] $1^1 2321^1 612321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 99$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$
- [587] $1^1 321^1 61^1 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 99$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[591] \ 1^{13}321^{16}12321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[590] \ 1^3 \overset{1^{12}2321^1}{6} \ 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 146 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[592] \ 1^{13}41^{13}151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 61 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[593] \ 1^2 \overset{1^{12}2321^1}{6} \ 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{11}321^{16}1^{12}321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 75 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[595] \ 1^3 \overset{1^{13}2321^1}{6} \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 20 & 4 \\ 18 & 20 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

$$[596] \ 1^2 \overset{1^{13}2321^1}{6} \ 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[597] \ 1^2 \overset{1^{13}2321^1}{6} \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 103, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 19 & 4 \\ 17 & 19 & 125 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0}$$

$$[598] \ 1^4 \overset{1^{13}2321^1}{6} \ 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 22 & 4 \\ 17 & 22 & 164 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

$$[599] \ 1^{12}41^{23}151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 73 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[602] \ 1^1 \overset{1^{14}2321^1}{6} \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0} \\
[600] & 1^3 \overset{1^{13}2321}{6} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 142 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0} \\
[601] & 1^3 \overset{1^{14}2321^1}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 12 & 8 & 11 & 98 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0} \\
[603] & 1^2 \overset{1^{14}2321^1}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 11 & 8 & 10 & 80 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0} \\
[604] & 1^{12}321^161^12321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 96 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0} \\
[605] & 1^3 \overset{1^{14}2321}{6} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 184 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0} \\
[606] & 1^{13}41^23151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 10 & 9 & 91 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0} \\
[607] & 1^2 \overset{1^{14}2321^1}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 13 & 2 \\ 12 & 8 & 13 & 103 & 25 \\ 2 & 2 & 2 & 25 & 5 \end{pmatrix}_{b_0} \\
[608] & 1^1 \overset{1^{14}2321^1}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0} \\
[609] & 1^4 \overset{1^{14}2321}{6} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 206 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0} \\
[610] & 1^{12}41^23151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 94 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0} \\
[611] & 1^{13}321^161^12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 133, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 21 & 4 \\ 18 & 21 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

$$[612] \ 1^{\overline{1}^{13}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 137, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[613] \ 1^{\overline{1}^{13}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 140, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 202 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[614] \ 1^{\overline{1}^{13}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 144, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 22 & 4 \\ 18 & 22 & 154 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

$$[615] \ 1^{\overline{1}^{13}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 176 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

$$[616] \ 1^{\overline{1}^{14}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 2 & 24 & 4 \\ 21 & 24 & 196 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[617] \ 1^{\overline{1}^{12}41^{\overline{3}}3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 167, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 10 & 12 & 132 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[618] \ 1^{\overline{1}^{14}321^{\overline{1}}61^{\overline{1}}2321^{\overline{13}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 170, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 24 & 4 \\ 19 & 24 & 178 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

$$[619] \ 1^{\overline{1}^{14}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 170, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 2 & 26 & 4 \\ 23 & 26 & 244 & 40 \\ 3 & 4 & 40 & 6 \end{pmatrix}_{b_0}$$

$$[620] \ 1^{\overline{1}^{14}2321^{\overline{1}}} \ 6 \ 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 25 & 4 \\ 22 & 25 & 218 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$$

$$[621] \ 1^{\overline{1}^{13}41^{\overline{2}}3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 175, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 13 & 10 & 12 & 112 & 26 \\ 2 & 2 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[622] \ 1^{\overline{1}^{13}41^{\overline{3}}3151321^{\overline{14}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 204, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 16 & 12 & 12 & 154 & 30 \\ 2 & 2 & 2 & 30 & 5 \end{pmatrix}_{b_0}$$

[623] $1^{14}321^161^12321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 204$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 26 & 4 \\ 22 & 26 & 220 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$$

[624] $1^{14}41^23151321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 208$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 14 & 12 & 12 & 134 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [625] $1^{12321}7^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 13, 1)$, $c_{\text{ext}} = 4.0$

[630] $221^18^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -47$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[629] $1^15^{1131}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -49$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[632] $1^{1321}8^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -54$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[633] $1^{115131}6^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -49$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[634] $1^{11612315131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -50$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[635] $1^15^{1131}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -49$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[636] $1^{13221}8^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -54$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[637] $1^1316123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13$, $T_{\max} = 14$, $\Delta = -50$, $\det = -2$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[638] $1^1 1^1 3 2 1 6 1 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[628] $1^1 1^1 6 1 3 1 5 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[631] $1^1 1^1 3 2 2 1^1 8^{1^1} 1 2 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[639] $1^1 1^1 7^{1^1} 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$

[640] $2 1^1 7^{1^1} 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[641] $2 1^1 7^{1^1} 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[642] $1^1 1^1 7^{1^1} 1 2 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[643] $1^1 1^1 7^{1^1} 1 2 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[644] $1^1 1^1 7^{1^1} 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[645] $1^1 1^1 7^{1^1} 1 2 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$

[651] $1^1 1^1 1 5^{1^1} 1 3 2 2 1^1 7 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

- [650] $1^{11}4131513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$
- [649] $1^{11}3141414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 5 & 1 & 4 & 6 \\ 2 & 5 & 0 & 6 & 6 \end{pmatrix}_{b_0}$
- [652] $1^{11}71232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [653] $1^{11}5132171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [654] $1^{11}712314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$
- [655] $31^{11}32181231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 15, \Delta = -62, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$
- [646] $1^{11}41^{11}31513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 4 & 2 & 4 & 8 & 8 \\ 2 & 2 & 3 & 8 & 6 \end{pmatrix}_{b_0}$
- [648] $1^{11}2321612321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$
- [647] $1^{11}4131513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [658] $2^3 3^1 21712321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$
- [656] $2^3 3^1 21712321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$
- [659] $2^3 3^1 21712321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[657] $1^{13} 2^3 21712321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[660] $1^{14} 1^{13} 151513221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 10 & 5 \\ 7 & 10 & 24 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0}$$

[661] $1^{11} 1^{13} 15151315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 14, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

[662] $1^{14} 1^{13} 151513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 3 & 14 & 5 \\ 8 & 14 & 44 & 18 \\ 3 & 5 & 18 & 7 \end{pmatrix}_{b_0}$$

[663] $1^{11} 1^{14} 1^{13} 151513221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 6 & 4 & 8 & 26 & 14 \\ 2 & 2 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

[664] $1^{12} 2^3 21812^3 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 3 & 16 & 5 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 16 & 7 & 37 & 13 \\ 2 & 5 & 2 & 13 & 4 \end{pmatrix}_{b_0}$$

[665] $1^{12} 31^1 414141^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -3, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 18 & 1 & 71 & 21 \\ 2 & 5 & 0 & 21 & 6 \end{pmatrix}_{b_0}$$

[667] $1^{11} 1^{14} 2^3 151513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0}$$

[666] $1^{12} 1^{14} 1^{13} 151513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 6 & 12 & 56 & 20 \\ 2 & 2 & 3 & 20 & 6 \end{pmatrix}_{b_0}$$

[668] $1^{11} 1^{14} 1^{13} 151513221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 20 & 5 \\ 12 & 20 & 90 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0}$$

[669] $1^{13} 1^{14} 1^{13} 151513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 9 & 8 & 12 & 70 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0}$$

[670] $1^{14} 2^3 21812^3 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 5 & 3 & 32 & 5 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 32 & 14 & 154 & 26 \\ 2 & 5 & 2 & 26 & 4 \end{pmatrix}_{b_0} \\
[671] & 1^{12}41^231513221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0} \\
[672] & 1^{13}41^231513221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -83, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 13 & 10 & 16 & 128 & 30 \\ 2 & 2 & 3 & 30 & 6 \end{pmatrix}_{b_0} \\
\\
T = 14 & [673] 1^{12}31^141414141^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.5 \\
[674] & 1^{11}4141414141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[675] & 1^{12321}8^{11}12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[676] & 1^{12315131}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 4.0 \\
[687] & 1^{11}7122315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
[684] & 1^{11}4131613151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[685] & 216^{11}1316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[683] & 1^{11}712315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[682] & 23218^{11}12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[688] & 1^{11}3217123151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[689] & 1^{11}8^{11}123141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[681] & 1^{11}6123216131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

[698] $1\overset{1}{8}12231\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$

[699] $21\overset{1}{8}12231\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[700] $1\overset{1}{8}12231\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$

[701] $21\overset{1}{8}12231\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[708] $1\overset{1}{6}131513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[705] $1\overset{1}{5}13221\overset{1}{8}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[707] $21\overset{1}{8}1232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[704] $1\overset{1}{1}23151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$

[709] $1\overset{1}{8}122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[710] $1\overset{1}{7}12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$

[702] $1\overset{1}{1}23151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$

[703] $1 \overset{1}{\overset{11}{7}} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 3 & 7 & 5 \\ 5 & 7 & 12 & 10 \\ 3 & 5 & 10 & 7 \end{pmatrix}_{b_0}$

[706] $1 \overset{1}{\overset{1}{15}} 1321 \overset{1}{8} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[714] $1 \overset{2}{\overset{12}{8}} 12321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[715] $1 \overset{2}{\overset{13}{8}} 12321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$

[716] $1 \overset{2}{\overset{14}{8}} 12321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$

[717] $1 \overset{1}{\overset{1}{23}} 21812321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 7 & 7 \\ 2 & 2 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[718] $1 \overset{2}{\overset{1}{23}} 21812321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 4 & 10 & 34 & 14 \\ 2 & 2 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[711] $1 \overset{4}{\overset{1}{23}} 21812321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 8 & 20 & 142 & 28 \\ 2 & 2 & 3 & 28 & 5 \end{pmatrix}_{b_0}$

[713] $1 \overset{2}{\overset{11}{8}} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[712] $1 \overset{3}{\overset{1}{23}} 21812321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 6 & 15 & 79 & 21 \\ 2 & 2 & 3 & 21 & 5 \end{pmatrix}_{b_0}$

[719] $1 \overset{11}{\overset{1}{23}} 21812321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 1 & 2 & 5 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 13 & 9 \\ 2 & 3 & 3 & 9 & 5 \end{pmatrix}_{b_0} \\
[720] & 1^{11}31612315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[721] & 1^{11}31612315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \\
[722] & 1^{11}32161315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[723] & 1^{11}32161315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \\
[724] & 1^{11}2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 3 & 11 & 5 \\ 6 & 11 & 26 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0} \\
[725] & 2^1 3^1 21812321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 9 & 11 & 44 & 16 \\ 2 & 3 & 3 & 16 & 5 \end{pmatrix}_{b_0} \\
[727] & 1^{11}321^{12}61315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0} \\
[726] & 1^{11}321^{12}61315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0} \\
[728] & 2^1 3^1 21812321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 13 & 16 & 93 & 23 \\ 2 & 3 & 3 & 23 & 5 \end{pmatrix}_{b_0} \\
[729] & 1^{11}2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 3 & 17 & 5 \\ 10 & 17 & 66 & 22 \\ 3 & 5 & 22 & 7 \end{pmatrix}_{b_0} \\
[730] & 2^1 3^1 21812321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 26, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 1 & 21 & 3 \\ 13 & 17 & 21 & 160 & 30 \\ 2 & 3 & 3 & 30 & 5 \end{pmatrix}_{b_0} \\
[732] & 1^{11}3161^{12}2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} =
\end{aligned}$$

4.0
 $T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$
[731] $1^{12}3161^12315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$
[733] $1^{12}3161^12315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
[734] $1^{12}321^161315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$
[735] $1^{11}321^161315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
[736] $1^{11}3161^12315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$
[737] $1^{12}3161^12315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 6 & 9 & 52 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$
[738] $1^{13}3161^12315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$
[739] $1^{13}3161^12315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$
[740] $1^{14}3161^12315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0}$
[741] $1^{13}321^161315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$

$$[742] \ 1^{12}321^1 61315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 99, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 14.99 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[743] \ 1^{12}3161^1 2315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 107, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 14.92 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[744] \ 1^{13}3161^1 2315131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 129, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 14.99 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[745] \ 1^{14}3161^1 2315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 137, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.93 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [753] \ 1^{13}3161^1 231513221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.8$$

$$[749] \ 1^{13}22315131^1 6^{13} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.8$$

$$[748] \ 1^{14}22315131^1 6^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 5.0$$

$$[747] \ 1^{12}22315131^1 6^{12} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.0$$

$$[746] \ 1^{14}22315131^1 6^{14} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.5$$

$$[754] \ 1^{11}22315131^1 6^{11} 131^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.0$$

$$[750] \ 1^{12}22315131^1 6^{12} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.5$$

$$[752] \ 1^{131} 6^{131} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -28, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.0$$

$$[751] \ 1^{11}22315131^1 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 4.0$$

$$[761] \ 1^1 15 13218 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 16, \ \Delta = -48, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[762] \ 1^1 15 1316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 16, \ \Delta = -43, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[763] \ 1^1 5 1316131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 16, \ \Delta = -43, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[764] \ 21^1 8 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, \ T_{\max} = 16, \ \Delta = -47, \ \det = -2, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[765] & 1 \overset{1}{5} 1321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[766] & 1 \overset{1}{5} \overset{1}{8} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[767] & 1 \overset{1}{5} \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
[768] & 1 \overset{1}{8} 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[769] & 1 \overset{1}{3} 22181231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[770] & 1 \overset{1}{5} \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[771] & 31 \overset{1}{6} 1316131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 6 & 10 & 6 \\ 6 & 10 & 14 & 10 \\ 3 & 6 & 10 & 6 \end{pmatrix}_{b_0} \\
[772] & 1 \overset{1}{6} 1232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[773] & 1 \overset{1}{3} 21812231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[774] & 1 \overset{1}{6} 131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[775] & 1 \overset{1}{6} 131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[776] $1 \overset{1}{\overset{11}{2321}} 8 \overset{11}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$

[777] $1 \overset{1}{\overset{11}{5131}} 6 \overset{11}{13161} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[779] $1 \overset{11}{11} 316123151321 \overset{11}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$

[780] $21 \overset{1}{\overset{11}{2321}} 8 \overset{11}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$

[781] $1 \overset{11}{11} 232161315131 \overset{11}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$

[782] $1 \overset{11}{11} 232161315131 \overset{11}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$

[783] $1 \overset{1}{\overset{11}{2321}} 8 \overset{11}{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 3 & 6 & 6 & 14 & 10 \\ 2 & 4 & 3 & 10 & 6 \end{pmatrix}_{b_0}$

[784] $1 \overset{1}{\overset{11}{2315131}} 6 \overset{11}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$

[785] $1 \overset{1}{\overset{11}{2315131}} 6 \overset{11}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 5 & 3 & 16 & 10 \\ 2 & 3 & 2 & 10 & 6 \end{pmatrix}_{b_0}$

[786] $1 \overset{1}{\overset{11}{2315131}} 6 \overset{11}{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$

[787] $1 \overset{11}{11} 321613151321 \overset{11}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[788] & 1^{\overline{1}} 8^{1231414141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[789] & 1^{\overline{1}} 6^{1316123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[790] & 1^{\overline{1}} 7^{1231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[791] & 1^{\overline{11}} 321613151321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[778] & 1^{\overline{11}} 316123151321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[792] & 1^{\overline{11}} 1^{12} 2315131^{\overline{12}} 6^{131^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 5 & 6 & 45 & 17 \\ 2 & 2 & 2 & 17 & 6 \end{pmatrix}_{b_0} \\
[793] & 1^{\overline{11}} 2315131^{\overline{11}} 6^{131^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -6, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 8 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0} \\
[794] & 31^{\overline{11}} 6^{1316131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 8 & 0 & 6 & 6 \\ 2 & 6 & 0 & 6 & 5 \end{pmatrix}_{b_0} \\
[796] & 1^{\overline{11}} 8^{12321^{\overline{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 4 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0} \\
[797] & 1^{\overline{12}} 2315131^{\overline{12}} 6^{131^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 6 & 5 & 25 & 13 \\ 2 & 3 & 2 & 13 & 6 \end{pmatrix}_{b_0} \\
[795] & 1^{\overline{1}} 321^{\overline{1}} 613151321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =
\end{aligned}$$

4.0

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 6 & 4 & 22 & 12 \\ 2 & 3 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[800] \ 1^{\overline{1}} \begin{smallmatrix} 1 \\ 1^{\overline{1}} \end{smallmatrix} \begin{smallmatrix} 8 \\ 2321 \end{smallmatrix} \ 12321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 5 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

$$[799] \ 21^{\overline{1}} \begin{smallmatrix} 1 \\ 1^{\overline{1}} \end{smallmatrix} \begin{smallmatrix} 8 \\ 2321 \end{smallmatrix} \ 12321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 12 & 5 \\ 7 & 12 & 34 & 16 \\ 3 & 5 & 16 & 7 \end{pmatrix}_{b_0}$$

$$[798] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{12}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{11}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 6 & 4 & 17 & 11 \\ 2 & 3 & 2 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[801] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{13}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{11}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[803] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{12}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{12}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 5 & 7 & 6 & 28 & 14 \\ 2 & 3 & 2 & 14 & 6 \end{pmatrix}_{b_0}$$

$$[804] \ 31^{\overline{1}} \begin{smallmatrix} 31 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1316131^{\overline{11}} \\ 131^{\overline{11}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 2 & 5 & 4 \\ 2 & 3 & 5 & 3 & 5 \\ 2 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[805] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{12}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{12}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[802] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{13}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{11}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 9 & 5 & 40 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0}$$

$$[806] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{12}} \\ 6 \end{smallmatrix} \begin{smallmatrix} 2315131 \\ 131^{\overline{12}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 8 & 6 & 38 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{\overline{1}} \begin{smallmatrix} 1^{\overline{11}} \\ 31 \end{smallmatrix} \begin{smallmatrix} 1316131^{\overline{11}} \\ 131^{\overline{11}} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 8 & 1 & 7 & 7 \\ 2 & 6 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

[808] $1^{1^{13}2315131}6^{131}12 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 9 & 7 & 43 & 17 \\ 2 & 3 & 2 & 17 & 6 \end{pmatrix}_{b_0}$$

[809] $1^{1^{12}2321}8^{12321}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 6 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0}$$

[810] $1^{1^{12}3161^123151321}12 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[811] $1^{1^{12}321^1613151321}12 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 10 & 6 & 50 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[812] $1^{1^{11}3161^123151321}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[813] $1^{1^{11}321^1613151321}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 9 & 7 & 48 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[814] $1^{1^{12}2321}8^{12321}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 7 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0}$$

[815] $21^{1^{12}2321}8^{12321}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 18 & 5 \\ 11 & 18 & 76 & 24 \\ 3 & 5 & 24 & 7 \end{pmatrix}_{b_0}$$

[816] $1^{1^{13}2315131}4^{131}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 13 & 9 & 108 & 26 \\ 2 & 3 & 2 & 26 & 6 \end{pmatrix}_{b_0}$$

[817] $1^{1^{13}2315131}12^{131}13 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 11 & 9 & 72 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0}$$

- [818] $1^2 \overset{14}{6} \overset{2315131}{131} \overset{12}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 12 & 8 & 74 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0}$
- [819] $1^3 \overset{13}{6} \overset{2315131}{131} \overset{13}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$
- [820] $1^3 \overset{14}{6} \overset{2315131}{131} \overset{12}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$
- [822] $1^{\textcolor{red}{12}} 3161^{\textcolor{red}{1}} 23151321^{\textcolor{red}{14}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$
- [825] $1^{\textcolor{red}{13}} 3161^{\textcolor{red}{1}} 23151321^{\textcolor{red}{13}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$
- [824] $1^3 \overset{13}{6} \overset{2315131}{131} \overset{14}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 15 & 9 & 115 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0}$
- [823] $1^{\textcolor{red}{13}} 321^{\textcolor{red}{1}} 613151321^{\textcolor{red}{13}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 13 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$
- [821] $1^2 \overset{13}{6} \overset{2315131}{131} \overset{14}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 14 & 9 & 97 & 25 \\ 2 & 3 & 2 & 25 & 6 \end{pmatrix}_{b_0}$
- [826] $1^{\textcolor{red}{1}} \overset{13}{8} \overset{2321}{12321} \overset{14}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 20 & 4 \\ 1 & 2 & 1 & 17 & 3 \\ 9 & 20 & 17 & 138 & 30 \\ 2 & 4 & 3 & 30 & 6 \end{pmatrix}_{b_0}$
- [827] $21^{\textcolor{red}{1}} \overset{13}{8} \overset{2321}{12321} \overset{14}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0}$
- [828] $31^{\textcolor{red}{12}} \overset{31}{6} \overset{1316131}{131} \overset{11}{+} [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -53, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15.98 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 6 & 1 & 16 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 16 & 1 & 31 & 13 \\ 2 & 6 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[829] \ 1^{\overline{14}} 3^{14} 6^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 15 & 12 & 136 & 30 \\ 2 & 3 & 2 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[830] \ 1^{\overline{14}} 3161^{\overline{1}} 23151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 15 & 9 & 109 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[831] \ 1^{\overline{14}} 3^{14} 6^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 16 & 12 & 158 & 32 \\ 2 & 3 & 2 & 32 & 6 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [832] \ 1^{\overline{14}} 8^{14} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 2), c_{\text{ext}} = 4.0$$

$$[833] \ 1^{\overline{14}} 8^{14} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$$

$$[834] \ 1^{\overline{14}} 8^{14} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$$

$$[835] \ 1^{\overline{14}} 8^{14} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$$

$$[843] \ 1^{\overline{14}} 8^{14} 1232161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[845] \ 1^{\overline{14}} 413218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[842] \ 161232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[840] \ 1^{\overline{14}} 613151316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[844] \ 21^{\overline{14}} 6131613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[841] \ 1^{\overline{14}} 612321812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

- [846] $1^{\overline{1}151321} 8^{123151} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 18, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$
- [847] $1^{\overline{1}1} 3217123151321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$
- [848] $1^{\overline{1}1} 712321613151^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [852] $1^{\overline{1}1} 3216131612321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [853] $1^{\overline{1}1} 2321712315131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [854] $1^{\overline{1}1} 812231513151^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [855] $1^{\overline{1}1} 3216131612321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$
- [850] $1^{\overline{1}1} 3217123151321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$
- [851] $1^{\overline{1}1} 2321712315131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 16, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$
- [849] $1^{\overline{1}1} 181231513151^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 18, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [857] $2^{\overline{1}12} 3^{\overline{1}12} 21812315131^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 6 & 18 & 6 \\ 10 & 18 & 50 & 18 \\ 3 & 6 & 18 & 6 \end{pmatrix}_{b_0}$
- [860] $31^{\overline{1}13} 5^{\overline{1}13} 1321812321^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[861] \quad 1^{11} 31 \ 5 \ 1321812321^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[862] \quad 1^{12} 31 \ 5 \ 1321812321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[859] \quad 1^{14} 2 \ 3 \ 21812315131^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 6 & 36 & 6 \\ 20 & 36 & 206 & 36 \\ 3 & 6 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[856] \quad 1^{11} 2 \ 3 \ 21812315131^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 11 & 9 \\ 3 & 6 & 9 & 6 \end{pmatrix}_{b_0}$$

$$[858] \quad 1^{13} 2 \ 3 \ 21812315131^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 6 & 27 & 6 \\ 15 & 27 & 115 & 27 \\ 3 & 6 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[863] \quad 1^{13} 321^1 6131612321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 5 & 4 & 7 & 27 & 13 \\ 2 & 2 & 3 & 13 & 6 \end{pmatrix}_{b_0}$$

$$[864] \quad 1^{11} 321^1 6131612321^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 7 & 6 & 11 & 55 & 19 \\ 2 & 2 & 3 & 19 & 6 \end{pmatrix}_{b_0}$$

$$[865] \quad 1^{12} 321^1 6131612321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 19 & 5 \\ 11 & 19 & 87 & 25 \\ 3 & 5 & 25 & 7 \end{pmatrix}_{b_0}$$

$$[866] \quad 1^{11} 321^1 613161^{12} 2321^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 11 & 8 & 13 & 91 & 25 \\ 2 & 2 & 3 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[867] \quad 1^{13} 321^1 613161^{12} 2321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0}$$

$$[868] \quad 1^{13} 321^1 6131612321^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0}$$

$$[869] \ 1^{14} 321^1 613161^1 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 85, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 3 & 31 & 5 \\ 19 & 31 & 227 & 41 \\ 3 & 5 & 41 & 7 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [870] \ 1^{11} 8^{12321} 123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[871] \ 1^{11} 6^{131} 13161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[872] \ 1^{113} 8^{23151321} 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[873] \ 1^{11} 8^{12321} 123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[877] \ 1^{11} 6^{123218123141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[879] \ 1^{11} 5^{132181231} 5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[880] \ 1^{11} 5^{132181231} 5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[881] \ 21^1 7^{12321} 8^{1231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[882] \ 1^{11} 7^{123151316131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[883] \ 1^{11} 6^{1316131} 13161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[878] \ 1^{11} 5^{132181232151} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 18, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[884] $1^{\overline{1}^{11}3151321} 8^{11} 1232161^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 19, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 5 & 5 \\ 2 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[885] $1^{\overline{1}^{11}31513161315131^{\overline{1}^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[886] $1^{\overline{1}^{11}3151321} 8^{11} 12321^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[887] $1^{\overline{1}^{11}3151321} 8^{11} 12321^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 3 & 7 & 5 \\ 5 & 7 & 12 & 10 \\ 3 & 5 & 10 & 7 \end{pmatrix}_{b_0}$$

[888] $1^{\overline{1}^{11}3151321} 8^{11} 12321^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[889] $1^{\overline{1}^{11}23218122315131^{\overline{1}^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[890] $1^{\overline{1}^{11}3151321} 8^{11} 12321^{\overline{1}^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 3 & 2 & 5 & 8 & 8 \\ 2 & 2 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

[891] $1^{\overline{1}^7} 123217123151^{\overline{1}^7} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 19, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[892] $1^{\overline{1}^{11}31612321712321^{\overline{1}^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[893] $1^{\overline{1}^{11}23218122315131^{\overline{1}^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[894] $1^{\overline{1}^{11}31612321712321^{\overline{1}^{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} =$

4.0
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$

[896] $1^{1^{11}3151321}8^{12321^{12}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 3 & 11 & 5 \\ 6 & 11 & 26 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0}$

[895] $1^{1^{12}3151321}8^{12321^{11}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 4 & 2 \\ 1 & 1 & 5 & 3 \\ 4 & 4 & 5 & 14 \\ 2 & 2 & 3 & 10 & 6 \end{pmatrix}_{b_0}$

[900] $1^{1^{13}3151321}8^{12321^{12}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 6 & 2 \\ 1 & 0 & 6 & 2 \\ 1 & 1 & 9 & 3 \\ 6 & 6 & 9 & 38 & 16 \\ 2 & 2 & 3 & 16 & 6 \end{pmatrix}_{b_0}$

[899] $1^{1^{12}3151321}8^{12321^{13}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 3 & 17 & 5 \\ 10 & 17 & 66 & 22 \\ 3 & 5 & 22 & 7 \end{pmatrix}_{b_0}$

[898] $1^{1^{11}3161^12321712321^{13}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 8 & 2 \\ 1 & 0 & 6 & 2 \\ 1 & 1 & 11 & 3 \\ 8 & 6 & 11 & 62 & 20 \\ 2 & 2 & 3 & 20 & 6 \end{pmatrix}_{b_0}$

[897] $1^{1^{12}3161^12321712321^{12}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 19 & 5 \\ 12 & 19 & 94 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0}$

[901] $1^{1^{12}232181^122315131^{13}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 10 & 2 \\ 1 & 0 & 6 & 2 \\ 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0}$

[902] $1^{1^{13}3161^12321712321^{13}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 3 & 25 & 5 \\ 16 & 25 & 158 & 34 \\ 3 & 5 & 34 & 7 \end{pmatrix}_{b_0}$

[903] $1^{1^{12}3161^12321712321^{14}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 10 & 2 \\ 1 & 0 & 8 & 2 \\ 1 & 1 & 15 & 3 \\ 10 & 8 & 15 & 102 & 26 \\ 2 & 2 & 3 & 26 & 6 \end{pmatrix}_{b_0}$

[904] $1^{1^{14}232181^122315131^{12}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2} T = 17, T_{\max} = 17, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.96 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0}$$

[905] $1^{14}3161^12321712321^{13}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 3 & 29 & 5 \\ 17 & 29 & 200 & 38 \\ 3 & 5 & 38 & 7 \end{pmatrix}_{b_0}$$

[906] $1^{13}232181^122315131^{14}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 17, T_{\max} = 17, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.8$

$T_{\min} = 16.97 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

$T = 18$ [907] $1^{11}6^{131}131613161231^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 18, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$

[908] $1^{11}8^{1321}12321712321^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$

[909] $1^{11}6^{131}13161316131^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$

[910] $1^{11}8^{12321}1232171231^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$

[911] $1^{11}8^{12321}12321712321^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 2), c_{\text{ext}} = 4.0$

[912] $1^{11}8^{12321}1232171231^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 18, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$

[921] $1^{11}6^{13}1315132181231^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[919] $1^{11}8^{1223151316131^1}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[918] $1^{11}6^{1232181231^1}51^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[917] $1^1151321812321^{11}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[922] $1^{11}3221(12)12231^1+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[923] $1^{11}3221$ (12) $12231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 8 \\ 2 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$$

[924] $1^{12}3221$ (12) $12231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 26 & 12 \\ 2 & 3 & 2 & 12 & 5 \end{pmatrix}_{b_0}$$

[916] $22181232181231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

[915] $1^{11}71232161316131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[920] $1^{11}61232181232151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 19, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[925] $1^{11}61316131613151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[926] $1^{11}7123218123151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[927] $1^{11}7123218123151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[928] $1^{11}71232181223151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[929] $1^{1^1 51321} 8^{1^1} 12321 7 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 19$ [930] $1^{1^1 321} 8^{1^1} 12321 8 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[931] $1^{1^1 2321} 8^{1^1} 12321 8 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[932] $1^{1^1 2321} 8^{1^1} 12321 8 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[933] $1^{1^1 2321} 8^{1^1} 12321 8 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 4.0$

[934] $1^{1^1 2321} 8^{1^1} 12321 8 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[935] $1^{1^1 31} 6^{1^1} 1316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[936] $1^{1^1 321} 8^{1^1} 12321 8 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[937] $1^{1^1 2321} 8^{1^1} 12321 7 12321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 19, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[943] $1^{21} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[945] $21^{1^1} 7^{1^1} 12321 8 1231^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 20, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$

[944] $1^{1^1} 7^{1^1} 12315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 20, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[942] $1^{1^1} 6^{1^1} 13161232181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 20, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[946] $221^{1^1} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[947] $21^{1^1} 8^{1^1} 12321 8 123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 19, T_{\max} = 21, \Delta = -67, \det = -3, (n_+, n_-, n_0) =$

$$(1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[948] \quad 1^1 8^{1^1} 123218^{1^1} 123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 19, T_{\max} = 21, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[949] \quad 1^1 8^{1^1} 12315131613151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 19, T_{\max} = 21, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[950] \quad 1^1 3^{1^1} 151321812315131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[951] \quad 31^{31} 1613161316^{1^1} 131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 4 & 5 \\ 2 & 5 & 3 & 6 & 5 \\ 2 & 4 & 6 & 4 & 6 \\ 2 & 5 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [952] \quad 1^1 5^{1^1} 131613161316^{1^1} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[953] \quad 1^1 6^{1^1} 131613161315^{1^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[954] \quad 1^{1^3} 3161^{1^1} 23218123151321^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[955] \quad 1^1 8^{1^1} 123218^{1^1} 123141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[956] \quad 1^1 8^{1^1} 123218^{1^1} 123141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[959] \quad 1^1 8^{1^1} 122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 21, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[960] \quad 21^{1^1} (12)^{1^1} 1223141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 21, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[961] \quad 1^1 8^{1^1} 123216131613151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 20, T_{\max} = 22, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [967] \quad 1^1 15 1321812321^1 8 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$[966] \quad 1^1 1^1 8^1 12321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$[965] \quad 1^1 15 1321812321^1 8 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$[964] \quad 1^1 1^1 2321^1 8^1 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$[963] \quad 1^1 1^1 8^1 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$[962] \quad 1^1 1^1 2321^1 8^1 12321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 2), c_{\text{ext}} = 4.0$$

$$[970] \quad 21 \quad (12) \quad 1^1 15 13221^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[971] \quad 1^1 1^1 8^1 1231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[972] \quad 1^1 1^1 5^1 13221^1 1^1 (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[973] \quad 1^1 (12) \quad 12231^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$1^1 1^1 5^1 13221^1$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[974] \quad 1^1 (12) \quad 12231^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$1^1 1^1 5^1 13221^1$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[975] \quad 1^1 1^1 7^1 1232171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[976] $1^1 5^{1^1} 13221^{1^1} 1223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21$, $T_{\max} = 22$, $\Delta = -41$, $\det = -2$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[977] $1^1 8^{1^1} 1231513218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21$, $T_{\max} = 23$, $\Delta = -68$, $\det = -4$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$T = 22$ [978] $1^1 5^{1^1} 13161316131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 22$, $\Delta = -12$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 4.0$

[979] $1^1 8^{1^1} 123218123216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 22$, $\Delta = -27$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 2)$, $c_{\text{ext}} = 4.0$

[980] $1^1 8^{1^1} 123218123216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 22$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 4.0$

[981] $1^1 16^{1^1} 12321812321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 22$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 22, 1)$, $c_{\text{ext}} = 4.0$

[982] $1^1 4141413221(12)1223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 23$, $\Delta = -44$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[983] $1^1 321812321^{1^1} 1232171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 23$, $\Delta = -44$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[984] $1^1 8^{1^1} 12321613161316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 23$, $\Delta = -44$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[985] $21 (12) 12231513161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 22$, $T_{\max} = 23$, $\Delta = -46$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$T = 23$ [989] $1^1 8^{1^1} 1232181232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 23$, $T_{\max} = 23$, $\Delta = -24$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 4.0$

[988] $217^{1^1} 12321812321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 23$, $T_{\max} = 23$, $\Delta = -17$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 4.0$

[986] $1^1 8^{1^1} 1232181232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 23$, $T_{\max} = 23$, $\Delta = -26$, $\det = 0$, $(n_+, n_-, n_0) =$

$$(1, 23, 2), c_{\text{ext}} = 4.0$$

$$[987] \quad 21 \overset{1^1}{6} 131613161316131 \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 23, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

$$[991] \quad 21 \overset{1^1}{(12)} 1223151321 \overset{1^1}{7} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 24, T_{\text{max}} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[993] \quad 1 \overset{1^1}{(12)} 122315131 \overset{1^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 24, T_{\text{max}} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[992] \quad 1 \overset{1^1}{(12)} 12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 24, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 24, T_{\text{max}} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[994] \quad 1 \overset{1^1}{7} 12321 \overset{1^1}{8} 123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 25, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 25, T_{\text{max}} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[995] \quad 1 \overset{1^1}{8} 123217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\text{max}} = 25, \Delta = -67, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 25, T_{\text{max}} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [996] \quad 1 \overset{1^1}{8} \overset{1^1}{1321} 12321812321 \overset{1^1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$[997] \quad 1 \overset{1^1}{8} \overset{1^1}{1321} 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$[998] \quad 1 \overset{1^1}{8} \overset{1^1}{1321} 12321812321 \overset{1^1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0$$

$$[999] \quad 221 \overset{1^1}{8} 12321812321 \overset{1^1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 24, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$[1000] \quad 1 \overset{1^1}{8} \overset{1^1}{12321} 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0$$

$$[1002] \quad 1 \overset{1^1}{(12)} 1223151321 \overset{1^1}{7} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\text{min}} = 25, T_{\text{max}} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1003] \quad 1 \overset{1^1}{(12)} 122315131 \overset{1^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\text{max}} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1004] \quad 1^{1^1} 21^{1^1} (12) 12231513221^{1^1} 81^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1005] \quad 1^{1^1} 15^{1^1} 13221^{1^1} (12) 122315131^{1^1} 61^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1006] \quad 1^{1^1} 6^{1^1} 131513221^{1^1} (12) 122314141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1007] \quad 1^{1^1} 18^{1^1} 12321^{1^1} 8^{1^1} 123218123151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1008] \quad 1^{1^1} 8^{1^1} 12321^{1^1} 8^{1^1} 123218123151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1009] \quad 1^{1^1} 5^{1^1} 1321812321^{1^1} 8^{1^1} 12321^{1^1} 81^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [1010] \quad 1^{1^1} 8^{1^1} 12321^{1^1} 123218123218123141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 2), c_{\text{ext}} = 4.0$$

$$[1011] \quad 1^{1^1} 8^{1^1} 12321^{1^1} 123218123218123141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 4.0$$

$$[1015] \quad 1^{1^1} 5^{1^1} 13221^{1^1} (12) 1223151321^{1^1} 71^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1017] \quad 1^{1^1} 1^{1^1} (12) 1223151316131^{1^1} 61^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1018] \quad 1^{1^1} 12^{1^1} 1223151322181^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -43, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1016] \quad 1^{1^1} 7^{1^1} 1231513221(12)122314141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -43, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1014] \quad 1^{1^1} 5^{1^1} 13221(12)122315132171^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -39, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1013] \quad 1^{1^1} 8^{1^1} 12315132181232181231^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -44, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1019] \quad 21^{1^1} 12^{1^1} 122315132181^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 25, \quad T_{\max} = 27, \quad \Delta = -67, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [1020] \quad 1^{1^1} 5^{1^1} 13218123218123218^{1^1} 12321^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 2), \quad c_{\text{ext}} = 4.0$$

$$[1021] \quad 1^{1^1} 5^{1^1} 13218123218123218^{1^1} 1231^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -17, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 1), \quad c_{\text{ext}} = 4.0$$

$$[1022] \quad 1^{1^1} 8^{1^1} 1232181232181232151^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 1), \quad c_{\text{ext}} = 4.0$$

$$[1023] \quad 1^{1^1} 8^{1^1} 1232181232181232151^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -21, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 2), \quad c_{\text{ext}} = 4.0$$

$$[1024] \quad 1^{1^1} 8^{1^1} 123218123218123151^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -17, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 26, 1), \quad c_{\text{ext}} = 4.0$$

$$[1025] \quad 1^{1^1} 6^{1^1} 131513221(12)12231513151^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1026] \quad 1^{1^1} 6^{1^1} 1316131513221(12)1223141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1027] \quad 1^1 5 13221(12)12231513221 8 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -38, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1028] \quad 21^1 8 1231513221(12)12231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -38, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 5 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 15 & 9 \\ 2 & 3 & 2 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[1029] \quad 1^1 5 13221(12)12231513221 8 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -38, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1030] \quad 1^1 5 1316131513221(12)12231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[1031] \quad 1^1 8 12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 27 \quad [1032] \quad 1^1 6 12321812321812321 8 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -16, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 27, 1), \quad c_{\text{ext}} = 4.0$$

$$[1034] \quad 1^1 6 1316131513221(12)12231 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -39, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1033] \quad 1^1 8 1232171232181232181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -43, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1036] \quad 1^1 7 1231513221(12)12231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -41, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1035] \quad 1^1 5 13221(12)122315131613161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -39, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 28 \quad [1037] \quad 1^1 \bar{5} 132181232181232181231 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 28, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 4.0$$

$$[1038] \quad 1^1 \bar{5} 132181231513221(12)12231 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 29, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[1039] \quad 1^1 \bar{3} 21812321812321 \quad 8 \quad 12321 \bar{8} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 29, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1040] \quad 1^1 \bar{6} 13161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1041] \quad 1^1 \bar{3} 221 \quad 1(12) \quad 1223151321812321 \bar{7} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1042] \quad 1^1 \bar{8} 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 28, T_{\max} = 29, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 29 \quad [1043] \quad 1^1 \bar{6} 1232181232181232181231 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 29, T_{\max} = 29, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 4.0$$

$$[1044] \quad 1^1 \bar{7} 1232181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 29, T_{\max} = 30, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1045] \quad 1^1 \bar{8} 12321 \bar{8} 123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 29, T_{\max} = 31, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [1046] \quad 21^1 \bar{7} 1232181232181232181231 \bar{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 30, T_{\max} = 30, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 4.0$$

[1047] $1^{\overline{1}} 7^{1^1} 1232181231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[1048] $1^{\overline{1}} 5^{1^1} 13221(12)122315132181232171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1049] $1^{\overline{1}} 5^{1^1} 13221(12)122315132181232181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 30, T_{\max} = 32, \Delta = -66, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 31$ [1050] $221^{\overline{1}} 8^{1^1} 123218123218123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 31, T_{\max} = 31, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 4.0$

[1051] $1^{\overline{1}} 8^{1^1} 13221(12)123218123218123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 31, T_{\max} = 31, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 31, 2), c_{\text{ext}} = 4.0$

[1052] $21^{\overline{1}} 8^{1^1} 1232181231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 31, T_{\max} = 32, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$

$T = 32$ [1053] $1^{\overline{1}} 6^{1^1} 1232181232181232181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 32, T_{\max} = 32, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 32, 2), c_{\text{ext}} = 4.0$

[1054] $21^{\overline{1}} 5^{1^1} 13221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 32, T_{\max} = 33, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

$T = 33$ [1056] $1^{\overline{1}} 7^{1^1} 1232181232181231513221(12)12231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 33, T_{\max} = 34, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$T = 35$ [1057] $1^{\overline{1}} 3221(12)12231513221(12)122315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 35, T_{\max} = 36, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[1058] $1^{\overline{1}} 8^{1^1} 1232181232181231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 35, T_{\max} = 37, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 36 \quad [1059] \quad 221 \overset{1^1}{8} 1232181232181232181232181231 \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 36, T_{\max} = 36, \Delta = -6, \det = 0, (n_+, n_-, n_0) = (1, 36, 2), c_{\text{ext}} = 4.0$$

$$[1061] \quad 1 \overset{1^1}{413221(12)12231513221} 122315132181 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 36, T_{\max} = 37, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 37 \quad [1063] \quad 1 \overset{1^1}{5} 13221(12)12231513221(12)122315132181 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 37, T_{\max} = 38, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[1064] \quad 1 \overset{1^1}{5} 13221(12)12231513221 122315132181 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 37, T_{\max} = 38, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 40 \quad [1067] \quad 1 \overset{1^1}{3221} (12)12231513221(12)1223151321812321 \overset{1^1}{8} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 40, T_{\max} = 41, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [1068] \quad 1 \overset{1^1}{8} 1232181231513221(12)12231513221(12)1223141 \overset{1^1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 41, T_{\max} = 42, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [1069] \quad 1 \overset{1^1}{5} 13221(12)12231513221(12)1223151321812321 \overset{1^1}{8} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 42, T_{\max} = 43, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1070] \quad 1 \overset{1^1}{8} 1232181231513221(12)12231513221(12)12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 42, T_{\max} = 43, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 44 \quad [1071] \quad 1 \quad \overset{1^1}{5} \overset{1^1}{13221} \quad (12) \quad 12231513221(12)12231513221(12)12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 44, T_{\max} = 44, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 4.0$$

$$T = 46 \quad [1073] \quad 1 \quad \overset{1^1}{3221}(12)12231513221 \quad (12) \quad 12231513221(12)1223151321 \overset{1^1}{8} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 46, T_{\max} = 47, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 47 \quad [1074] \quad 1 \quad \overset{1^1}{5} \overset{1^1}{13221} \quad (12) \quad 12231513221(12)12231513221(12)122315131 \overset{1^1}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 47, T_{\max} = 47, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 4.0$$

$$T = 48 \quad [1075] \quad 1 \quad \overset{1^1}{5} \overset{1^1}{13221} \quad (12) \quad 12231513221(12)12231513221(12)1223151321 \overset{1^1}{7} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 48, T_{\max} = 48, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 4.0$$

$$T = 49 \quad [1076] \quad 21 \overset{1^1}{8} \overset{1^1}{1231513221}(12)12231513221(12)12231513221(12)12231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 49, T_{\max} = 49, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 4.0$$

$$[1077] \quad 1 \quad \overset{1^1}{5} \overset{1^1}{13221} \quad (12) \quad 12231513221(12)12231513221(12)12231513221 \overset{1^1}{8} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 49, T_{\max} = 49, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 4.0$$

2.2.9 NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 6 \quad [0] \quad 1 \overset{1^1}{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 21 \overset{1^1}{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \overset{1^1}{1^1} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 21 \overset{1^1}{1^2} 31 \overset{1^4}{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 164, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 84 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 21 \overset{1^1}{1^2} 31 \overset{1^4}{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 117 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1 \overset{1^1}{1^3} 31 \overset{1^4}{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 184, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 21 \overset{1^1}{1^3} 31 \overset{1^4}{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 248, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 21^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 6, T_{\max} = 10, \Delta = -123, \det = 368, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [9] \quad 1^{11}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -95, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -95, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [13] \quad 1^{11}3^1221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}3^1221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[18] $1^1 4 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -93, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 4 & 3 & 5 \end{pmatrix}_{b_0}$

[19] $1^1 4 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[20] $1^1 3 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$

[21] $1^1 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -93, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 1 & 3 \\ 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

$T = 9$ [22] $1^1 4 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[23] $21^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[24] $1^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[25] $1^1 4 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$

[26] $1^1 5 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$

[27] $1^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 3 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[28] $1^1 5 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

$$[29] \quad 1\overset{1}{5}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1\overset{1}{4}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[31] \quad 1\overset{1}{5}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1\overset{1}{5}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 \\ 4 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 21\overset{1}{5}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1\overset{1}{5}1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1\overset{1}{5}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 21\overset{1}{5}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 21\overset{1}{5}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11}3221^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 30 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1\overset{1}{5}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{12}3221^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -89, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [41] \quad 1^1 1^1 4 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^1 1^1 5 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 3 & 6 \\ 2 & 3 & 0 & 2 \\ 3 & 6 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^1 1^1 3 1^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^1 1^1 4 14131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^1 1^1 5 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^1 1^1 3 1^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^1 1^1 4 14131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^1 1^1 3 1^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^1 1^1 5 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^1 1^1 5 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^1 1^1 4 131^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^1 4 1 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^1 4 1 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 6 & 9 & 6 \\ 4 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^1 5 1 2 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^1 4 1 4 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[59] \quad 2 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 2 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^1 4 1 3 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^1 5 1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[63] $21^{1\bar{1}}5^{1\bar{1}}1321^{1\bar{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[64] $1^{1\bar{1}}41^{1\bar{1}}3151^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [65] $1^{1\bar{1}}315^{1\bar{1}}1321^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 5 \end{pmatrix}_{b_0}$$

[66] $15^{1\bar{1}}13221^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

[67] $1^{1\bar{1}}4141321^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[68] $1^{1\bar{1}}315^{1\bar{1}}1321^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[69] $1^{1\bar{1}}315^{1\bar{1}}1321^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 5 \end{pmatrix}_{b_0}$$

[70] $1^{1\bar{1}}6^{1\bar{1}}131^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{1\bar{1}}6^{1\bar{1}}131^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{1\bar{1}}6^{1\bar{1}}131^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

[73] $16^{1\bar{1}}12321^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{\overset{1}{\underset{1}{1}}} 6^{1321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{\overset{1}{\underset{1}{1}}} 6^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -64, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{\overset{1}{\underset{1}{1}}} 5^{13221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{\overset{1}{\underset{1}{1}}} 4^{141321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[78] \ 21^{\overset{1}{\underset{1}{1}}} 5^{13221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{\overset{1}{\underset{1}{1}}} 5^{13221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 7 & 10 & 7 \\ 4 & 10 & 10 & 8 \\ 3 & 7 & 8 & 6 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{\overset{1}{\underset{1}{1}}} 6^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[81] \ 21^{\overset{1}{\underset{1}{1}}} 6^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{\overset{1}{\underset{1}{1}}} 5^{13221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$$

$$[83] \ 21^{\overset{1}{\underset{1}{1}}} 5^{13221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$$

- [84] $1^{11} 231^1 513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 12$, $\Delta = -11$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 5.0$
- [87] $1^{11} 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -38$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$
- [88] $1^{11} 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -38$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$
- [89] $1^{11} 31^1 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -43$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$
- [90] $1^{11} 41414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -33$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [91] $1^{11} 51315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 14$, $\Delta = -61$, $\det = 3$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$
- [92] $1^{11} 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -38$, $\det = 3$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0}$
- [93] $1^{11} 41413221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 12$, $\Delta = -6$, $\det = 4$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 3 & 2 & 3 & 5 \\ 2 & 3 & 2 & 5 & 6 \end{pmatrix}_{b_0}$
- [94] $1^{11} 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 14$, $\Delta = -63$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [95] $1^{11} 51315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 14$, $\Delta = -61$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [96] $1^{11} 6^{131} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 14$, $\Delta = -63$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 4 \end{pmatrix}_{b_0}$
- [97] $1^{11} 41414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 12$, $T_{\max} = 13$, $\Delta = -33$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$

- [98] $1^{\overline{1}} 1^{\overline{1}} 5 1 3 1 5 1 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 14, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$
- $T = 13$ [99] $1^{\overline{1}} 4 1 4 1 4 1 3 2 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 13, \Delta = -5, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$
- [100] $1^{\overline{1}} 1^{\overline{1}} 7 1 2 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 14, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$
- [101] $1^{\overline{1}} 5 1 3 1 5 1 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [102] $1^{\overline{1}} 3 2 1 6 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$
- [103] $1^{\overline{1}} 1^{\overline{1}} 5 1 3 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [104] $1^{\overline{1}} 1^{\overline{1}} 5 1 3 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$
- [105] $1^{\overline{1}} 1^{\overline{1}} 3 1 6 1 2 3 1^{\overline{1}} 5 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$
- [106] $1^{\overline{1}} 1^{\overline{1}} 4 1 4 1 4 1 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 13, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$
- [107] $1^{\overline{1}} 1^{\overline{1}} 3 2 1 6 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [108] $1^{\overline{1}} 1^{\overline{1}} 5 1 3 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [109] $1^{\overline{1}} 1^{\overline{1}} 5 1 3 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[110] \ 1^1 316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[111] \ 1^1 151316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[112] \ 1^1 612315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[113] \ 1^1 161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[114] \ 1^1 513151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 7 & 7 \\ 3 & 7 & 3 & 5 \\ 3 & 7 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{11} 321613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[116] \ 1^1 61315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{11} 5131 \ 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{11} 316123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[119] \ 21^1 61315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{11} 5131 \ 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [121] \quad 1^1 4 14 14 14 13 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -4, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 14, 1), \quad c_{\text{ext}} = 5.0$$

$$[122] \quad 1^1 16 13 15 13 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -30, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^1 1^1 23 21 6 13 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -32, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^1 1^1 31 5 13 1^1 6 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -37, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^1 1^1 32 21^1 8 12 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -36, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^1 1^1 31 5 13 1^1 6 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -37, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^1 1^1 23 21^1 8 12 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -36, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^1 1^1 6 12 3 15 13 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -31, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^1 1^1 31 5 13 1^1 6 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -37, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.67 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^1 1^1 32 17 12 3 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 16, \quad \Delta = -58, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^1 1^1 4 14 14 14 14 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -3, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 5 \\ 3 & 5 & 4 & 6 \\ 3 & 5 & 6 & 7 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^1 1^1 5 13 15 13 22 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -4, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 8 & 8 & 8 \\ 3 & 8 & 4 & 6 \\ 3 & 8 & 6 & 7 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{1}} 6 123216131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{1}} 32171231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{\overline{1}} 23216131^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{\overline{1}} 6 13151321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[137] \ 21^{\overline{1}} 6 13151321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{\overline{1}} 32171231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [139] \ 1^{\overline{1}} 414141414131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$$

$$[141] \ 1^{\overline{1}} 6 1316131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{\overline{1}} 6 13161231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 16, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{\overline{1}} 232171231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{\overline{1}} 6 1316131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 4 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[145] $1^{1^{131}} 6^{13161231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -35, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[146] $1^{1^{11}} 6^{131513221^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 3 & 7 & 5 \\ 3 & 2 & 7 & 7 & 7 \\ 2 & 2 & 5 & 7 & 6 \end{pmatrix}_{b_0}$$

[147] $21^{1^{11}} 6^{131513221^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 3 & 7 & 6 \end{pmatrix}_{b_0}$$

[148] $21^{1^{11}} 6^{131612321^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 13 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$$

[149] $1^{11} 232171231^{11} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$T = 16$ [152] $1^{1^{11}} 8^{1^{11} 5^{1321}} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[153] $1^{1^{11}} 6^{12321712321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[154] $1^{1^{11}} 3^{151316131^{11}} 5^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[155] $1^{1^{131}} 6^{131613141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 3 & 4 \\ 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0}$$

[156] $1^{1^{13151321}} 8^{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[151] $1^{1^1}232181223151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[157] $1^{1^1}6^{1^1}131613141^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & -1 & 3 & 5 \end{pmatrix}_{b_0}$

[158] $1^{1^1}232181223151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[159] $1^{1^1}1^{1^1}315131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[160] $1^{1^1}5^{1^1}1321^{1^1}12321^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

$T = 17$ [161] $1^{1^1}22315131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$

[162] $1^{1^1}2321^{1^1}12321^{1^1}61^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[163] $1^{1^1}3216131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[164] $1^{1^1}1^{1^1}2315131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 17, T_{\max} = 17, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

[165] $1^{1^1}1^{1^1}3216131613151^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

$T = 18$ [166] $21^{1^1}61316131513221^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 18, T_{\max} = 18, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 5.0$

[168] $1^{1\bar{1}3221} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 19, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[169] $1^{1\bar{1}315132181231} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 19, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[170] $21^{1\bar{1}6131613151321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 18, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 10 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0}$$

[171] $1^{1\bar{1}232161316131} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 18, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

[172] $1^{1\bar{1}315132181231} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 18, T_{\max} = 19, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$T = 19$ [173] $1^{1\bar{1}22315132181231} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 5.0$

[176] $1^{1\bar{1}612321812315131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 19, T_{\max} = 20, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[177] $1^{1\bar{1}3161232181231} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 19, T_{\max} = 20, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[178] $1^{1\bar{1}613151321812321} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 19, T_{\max} = 20, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[179] $1^{1\bar{1}2315132181231} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 19, T_{\max} = 19, \Delta = -2, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

[180] $1^{11}316123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 19$, $T_{\max} = 20$, $\Delta = -29$, $\det = -8$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

$T = 20$ [181] $1^{11}23151316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 20$, $T_{\max} = 20$, $\Delta = -2$, $\det = 0$, $(n_+, n_-, n_0) = (1, 20, 1)$, $c_{\text{ext}} = 5.0$

[183] $1^{11}3151316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 20$, $T_{\max} = 20$, $\Delta = -1$, $\det = 4$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

$T = 21$ [184] $1^{11}232161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 21$, $T_{\max} = 21$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 21, 1)$, $c_{\text{ext}} = 5.0$

[185] $1^{11}23217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 21$, $T_{\max} = 21$, $\Delta = -1$, $\det = -4$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

$T = 22$ [186] $1^{11}513221$ $122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[187] $21^{11}513221$ $122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[188] $1^{11}2321$ $123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 22$, $T_{\max} = 22$, $\Delta = 0$, $\det = 4$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 22$, $T_{\max} = 22$, $T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

[189] $21^{11}513221$ $122315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 8$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[190] $1^{11}513221$ $122315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 22$, $T_{\max} = 23$, $\Delta = -29$, $\det = 8$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 23$, $T_{\max} = 23$, $T \in [23, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 5 & 3 & 6 & 6 \\ 2 & 4 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [191] \quad 1^{11}3151316131613161315^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$[192] \quad 1 \quad (12) \quad 12231513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$[193] \quad 21 \quad (12) \quad 12231513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$[194] \quad 21 \quad (12) \quad 1223151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[195] \quad 1 \quad (12) \quad 1223151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 23, T_{\max} = 23, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [196] \quad 1^{11}23151321812321812315^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 24, T_{\max} = 24, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 5.0$$

$$[197] \quad 1^{11}3151321812321812315^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 24, T_{\max} = 24, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [198] \quad 1^{11}2321712321812321812315^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 26, T_{\max} = 26, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 5.0$$

$$T = 28 \quad [199] \quad 1 \quad (12) \quad 1223151321812321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 28, T_{\max} = 28, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 34 \quad [200] \quad 1 \quad (12) \quad 12231513221^{11} (12) 122315^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 34, T_{\max} = 34, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 5.0$$

2.2.10 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[1] $2^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 5 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

$T = 9 \quad [2] 1^{11} 3 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

$T = 10 \quad [3] 2^3 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 3 \\ 3 & 6 & 6 & 3 & 4 \end{pmatrix}_{b_0}$

2.2.11 NHC(-2)(-3)(-2) + su_2

$T = 8 \quad [0] 1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[1] $1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[2] $1^{11} 3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 6 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$

[3] $2^2 3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 4 & 4 \\ 3 & 5 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[4] $1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

[5] $1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{14} 3^{21} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 16 & 3 \\ 8 & 12 & 0 & 16 & 110 & 20 \\ 2 & 2 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{12} 3^{21} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 4 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{13} 3^{21} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 6 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^{21} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 4 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[12] $21^{11} 3^{21} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 5 & 7 & 15 & 5 \\ 5 & 7 & 5 & 15 & 5 \\ 12 & 15 & 15 & 32 & 12 \\ 4 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{12} 3^{21} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 6 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{12} 3^{21} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 8 & 8 & 0 & 14 & 70 & 16 \\ 2 & 2 & -1 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13} 3^{21} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 8 & 10 & 0 & 15 & 88 & 18 \\ 2 & 2 & -1 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3^{21} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[18] $21^{12} 321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 5 & 7 & 30 & 5 \\ 5 & 7 & 5 & 30 & 5 \\ 24 & 30 & 30 & 134 & 24 \\ 4 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[20] $2^2 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 2 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 2 & 0 & 3 & 2 \\ 3 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{12} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{12} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3^{21} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[24] $2^3 3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^{21} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{13} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[27] $2 2 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 2 & 4 \\ 3 & 5 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[28] $2 2 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[29] $2 2 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 8 & 6 & 26 & 10 \\ 2 & 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[30] $2 2 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{12} 3 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[33] $2 2 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 34 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{12} 3 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 2 \overset{1}{\underset{1^{115}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 3 & 10 & 2 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 2 \overset{1}{\underset{1^{116}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 3 & 12 & 2 \\ 2 & 3 & 3 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 2 \overset{1}{\underset{1^{117}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 3 & 14 & 2 \\ 2 & 3 & 3 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 2 \overset{1}{\underset{1^{114}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 8 & 2 \\ 2 & 3 & 3 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 2 \overset{1}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 3 & 6 & 2 \\ 2 & 3 & 3 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 2 \overset{1}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 4 & 2 \\ 2 & 3 & 3 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 2 \overset{1}{\underset{1^{111}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 2 \overset{1}{\underset{1^{114}}{2}} 2 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 2 \overset{1}{\underset{1^{112}}{2}} 2 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1 \overset{1}{\underset{1^{13}}{3}} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12} 3^{13} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{13} 3^{14} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12} 3^{14} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{12} 3^{14} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[50] \ 2^{13} 2^2 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 2 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 2 & 0 & 9 & 2 \\ 9 & 9 & 12 & 9 & 61 & 15 \\ 2 & 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[51] \ 2^{115} 2^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 98 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 2^{111} 2^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 5 & 2 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{14} 3^{14} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[54] \ 2^{112} 2^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 32 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{13} 3^{13} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{12} 3^{14} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 2^2 31^{16} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 142 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{13} 3^{14} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{13} 3^{14} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 2^2 31^{17} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 5 & 7 & 35 & 5 \\ 4 & 7 & 5 & 35 & 5 \\ 21 & 35 & 35 & 194 & 28 \\ 3 & 5 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{14} 3^{14} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{13} 3^{14} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[63] \ 2^2 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 2 & 0 & 12 & 2 \\ 12 & 12 & 16 & 12 & 110 & 20 \\ 2 & 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[64] \ 2^2 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 58 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 2^3 3^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 3 & 7 & 2 \\ 2 & 3 & 3 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[66] $2^2 2^1 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 94 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{14} 3^1 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[68] $2^2 2^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 92 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[69] $2^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 3 & 9 & 2 \\ 2 & 3 & 3 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[70] $2^2 2^1 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 134 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[71] $2^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 3 & 11 & 2 \\ 2 & 3 & 3 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [72] $1^{11} 2^1 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{13} 2^1 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3^1 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{14} 2^1 3^1 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{14} 3^{221} 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 20 & 0 & 126 & 24 \\ 2 & 2 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[97] \ 41^{111} 232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -59, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 4 & 2 \\ 4 & 10 & 5 & 12 & 12 & 6 \\ 2 & 5 & 0 & 5 & 4 & 2 \\ 4 & 12 & 5 & 10 & 12 & 6 \\ 4 & 12 & 4 & 12 & 10 & 6 \\ 2 & 6 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [98] \ 2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 \\ 3 & 6 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[99] \ 2^3 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 6 & 8 & 12 & 6 \\ 4 & 8 & 6 & 12 & 6 \\ 6 & 12 & 12 & 18 & 10 \\ 3 & 6 & 6 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[100] \ 2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 6 & 8 & 18 & 6 \\ 4 & 8 & 6 & 18 & 6 \\ 9 & 18 & 18 & 43 & 15 \\ 3 & 6 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[101] \ 2^3 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 6 & 8 & 24 & 6 \\ 4 & 8 & 6 & 24 & 6 \\ 12 & 24 & 24 & 78 & 20 \\ 3 & 6 & 6 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[102] \ 2^3 21^{115} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 6 & 8 & 30 & 6 \\ 4 & 8 & 6 & 30 & 6 \\ 15 & 30 & 30 & 123 & 25 \\ 3 & 6 & 6 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[103] \ 2^3 21^{116} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 6 & 8 & 36 & 6 \\ 4 & 8 & 6 & 36 & 6 \\ 18 & 36 & 36 & 178 & 30 \\ 3 & 6 & 6 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[104] \ 2^3 21^{117} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 6 & 8 & 42 & 6 \\ 4 & 8 & 6 & 42 & 6 \\ 21 & 42 & 42 & 243 & 35 \\ 3 & 6 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[105] \ 2^{111} 3^{141} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[106] $2 \overset{1^{112}1^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 4 & 6 & 3 \\ 2 & 4 & 4 & 2 & 8 & 4 \\ 4 & 8 & 6 & 8 & 14 & 8 \\ 2 & 4 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[107] $2 \overset{1^{113}1^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 4 & 9 & 3 \\ 2 & 4 & 4 & 2 & 12 & 4 \\ 6 & 12 & 9 & 12 & 34 & 12 \\ 2 & 4 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[108] $2 \overset{1^{114}1^1}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 4 & 12 & 3 \\ 2 & 4 & 4 & 2 & 16 & 4 \\ 8 & 16 & 12 & 16 & 62 & 16 \\ 2 & 4 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [109] $1 \overset{1^{11}}{1^3} 31 \overset{1^{11}}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[110] $1 \overset{1^{11}}{1^3} 31 \overset{1^{11}}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[111] $1 \overset{1^{11}}{1^3} 31 \overset{1^{11}}{5} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[112] $1 \overset{1^{12}}{1^3} 31 \overset{1^{12}}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[113] $1 \overset{1^{12}}{1^4} 31 \overset{1^{12}}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[114] $1 \overset{1^{11}}{1^2} 31 \overset{1^{11}}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

5.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [115] \quad 1^{11} 31^1 5^{13} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[118] \quad 2^{1^{11} 1^1 4^{14} 141} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 & 4 \\ 2 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[119] \quad 2^{1^{11} 1^1 2^1 4^{14} 141} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 2 & 5 & 8 & 4 \\ 2 & 5 & 5 & 3 & 10 & 5 \\ 4 & 10 & 8 & 10 & 18 & 10 \\ 2 & 5 & 4 & 5 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[120] \quad 2^{1^{11} 1^1 3^1 4^{14} 141} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 3 & 5 & 5 & 15 & 5 \\ 2 & 5 & 2 & 5 & 12 & 4 \\ 2 & 5 & 5 & 3 & 15 & 5 \\ 6 & 15 & 12 & 15 & 43 & 15 \\ 2 & 5 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[121] \quad 2^{1^{11} 1^1 4^1 4^{14} 141} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 3 & 5 & 5 & 20 & 5 \\ 2 & 5 & 2 & 5 & 16 & 4 \\ 2 & 5 & 5 & 3 & 20 & 5 \\ 8 & 20 & 16 & 20 & 78 & 20 \\ 2 & 5 & 4 & 5 & 20 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [127] \quad 1^{11} 6^{1^{11} 2315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$$

$$T = 16 \quad [130] \quad 1^{11} 6^{1^{11} 31} 131612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$[131] \quad 1^{11} 6^{1^{14} 31} 13161^{12} 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$T = 17 \quad [132] \quad 1^{11} 8^{1^{11} 23151321} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

3 Round 3: NHC External 3-ext (1750 bases)

Colored curves in quiver show attachment points. Matching color in external label: ext1 ext2 ext3 ext4 ext5
 Total: 1750 bases (78 uni, 1672 non-uni).

T	Count
7	26
8	116
9	484
10	197
11	425
12	278
13	58
14	53
15	44
16	18
17	24
18	2
19	13
20	2
22	2
23	1
24	1
25	2
28	1
33	1
36	1
44	1

3.1 Unimodular ($|\det| = 1$) — 78 bases

3.1.1 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 13 \quad [90] \quad 1^{12} 31^1 5^{11} 13221^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 26 & 5 \\ 15 & 0 & 26 & 148 & 30 \\ 3 & 0 & 5 & 30 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[91] \quad 1^{11} 31^1 5^{11} 13221^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

3.1.2 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [304] \quad 1^{11} 5^{11} 1321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 11, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 12 \quad [351] \quad 1^{11} 1^{11} 31^1 5^{11} 1321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[353] $1^{11} 6^{11} 12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[354] $1^{11} 4^{11} 14141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 6 & 5 & 5 & 6 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 3 & 5 \\ 3 & 6 & 5 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[352] $1^{11} 6^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[355] $1^{11} 12^{11} 31^1 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 4 & 5 & 7 & 13 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[349] $1^{11} 11^{11} 31^1 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[350] $1^{11} 12^{11} 31^1 5^{11} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 6 & 3 \\ 2 & 0 & 5 & 7 & 13 & 7 \\ 4 & 0 & 6 & 13 & 18 & 10 \\ 2 & 0 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$T = 14$ [469] $1^{11} 8^{11} 1231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

[470] $1^{11} 6^{11} 1315131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

[471] $1^{11} 6^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$\begin{aligned}
T = 16 \quad [501] \quad 1^{1^{1^1}5^{1^1}1321} 8^{1^1} 1231^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[503] \quad 1^{1^{1^1}5^{1^1}13161315^{1^1}1^{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -13, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[502] \quad 1^{1^1} 8^{1^1} 122315131^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
T = 17 \quad [508] \quad 1^{1^1} 6^{1^1} 123218^{1^1} 1231^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
T = 18 \quad [510] \quad 1^{1^1} 5^{1^1} 1321812315^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
T = 19 \quad [515] \quad 1^{1^1} 6^{1^1} 12321812315^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15} \\
T = 22 \quad [517] \quad 1^{1^1} (12) \quad 122315^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 22, T_{\max} = 22, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 6 & 4 & 6 \\ 1 & 2 & 1 & 4 & 1 & 3 \\ 2 & 3 & 2 & 6 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[518] \quad 21^{1^1} (12) \quad 122315^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 22, T_{\max} = 22, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0 \\
T_{\min} = 22, T_{\max} = 22, T \in [22, 22]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 25 \quad [521] \quad 1^{11} 5^{13} 221 \quad 122315131 \quad 6 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -10, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 6 & 4 & 6 \\ 1 & 2 & 1 & 4 & 1 & 3 \\ 2 & 3 & 2 & 6 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 28 \quad [523] \quad 1^{11} 6^{13} 16131513221(12)1223151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 28, T_{\max} = 28, \Delta = -9, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

3.1.3 NHC(-2)(-3) + su₂ + su₂ → (-3)

$$T = 10 \quad [17] \quad 1^{11} 22313^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 11 \quad [33] \quad 1^{11} 3^{11} 141321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -7, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [99] \quad 1^{11} 3^{11} 15^{13} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 6 & 1 & 3 \\ 4 & 0 & 6 & 18 & 5 & 10 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 3 & 10 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[95] \quad 1^{11} 3^{11} 15^{13} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 1 & 3 \\ 2 & 0 & 5 & 7 & 3 & 7 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[100] \quad 1^{11} 3^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 2 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[93] \quad 1^{11} 3^{11} 1414131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 0 & 5 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[94] $1^{13}3^11^451321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 10 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[102] $1^{12}3^11^151321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 5 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[103] $1^{11}2321513^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

[104] $1^{43}1^151321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 8 & 0 & 16 & 106 & 11 & 24 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 3 & 24 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[105] $1^{13}1^151321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[106] $1^{11}3^11^251321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 4 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[101] $1^{12}3^11^351321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 7 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[98] $1^{14}3^11^351321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 11 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

[97] $1^{13}3^11^251321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 8 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[96] \ 1^{11}2315123^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [114] \ 1^{11}3^121612321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [116] \ 1^{11}3^1151315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 4 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 4 & 5 \\ 1 & 4 & 0 & 4 & 2 & 4 \\ 2 & 5 & -1 & 5 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[115] \ 1^{11}2321^17 \ 123^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [119] \ 1^{11}3^12161315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[120] \ 1^{11}3216131513^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[121] \ 1^{11}3161231513^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[122] \ 1^{11}8^1123^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[118] \ 1^{11}3161^1231513^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[117] \ 1^{13} 3161^{12} 31513^{11} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 12 & 12 & 103 & 27 \\ 3 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [124] \ 1^{13151321} 8^{12311} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

3.1.4 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 16 \quad [23] \ 1^{131} 6^{131} 1316131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & 3 & -1 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [24] \ 1^{1151321} 8^{12321} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 0 & 4 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

3.1.5 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [148] \ 1^{111} 4^{113221} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -6, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [196] \ 1^{111} 4^{1141321} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -5, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[197] \ 1^{114} 5^{13221} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[198] \ 1^{111} 5^{13221} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [255] \quad 1^{11} 4^{11} 1414131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 2 & 4 & 0 & 3 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [265] \quad 1^{11} 5^{11} 13151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -3, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 7 & 7 \\ 2 & 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 15 \quad [291] \quad 1^{11} 6^{14} 13151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[292] \quad 1^{11} 6^{12} 23151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[293] \quad 1^{11} 6^{11} 13151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[294] \quad 1^{11} 6^{11} 123151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[295] \quad 1^{11} 8^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -6, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 5 & 4 & 7 & 7 \\ 2 & 2 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[290] \quad 1^{11} 23216131^{11} 5^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -2, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[289] \ 1^{\overline{11}} \begin{smallmatrix} 1 \\ 8 \end{smallmatrix} \begin{smallmatrix} 1231 \\ 1^{12321} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = -6, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 16 \quad [302] \ 1^{\overline{11}} \begin{smallmatrix} 1^{131} \\ 6 \end{smallmatrix} \begin{smallmatrix} 1316131 \\ 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 4 & 2 & -1 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[303] \ 1^{\overline{11}} \begin{smallmatrix} 1^{12} \\ 6 \end{smallmatrix} \begin{smallmatrix} 1316131 \\ 1^{11} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 0 & 11 & 4 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 11 & 6 & 0 & 34 & 14 \\ 2 & 4 & 2 & -1 & 14 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[304] \ 1^{\overline{11}} \begin{smallmatrix} 1^{11} \\ 232171231 \end{smallmatrix} \begin{smallmatrix} 5 \\ 1 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 1^{11} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 17 \quad [316] \ 1^{\overline{11}} \begin{smallmatrix} 1^{12} \\ 8 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 5 \end{smallmatrix} \begin{smallmatrix} 1321 \\ 1321 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 11 & 0 & 28 & 14 \\ 3 & 4 & 0 & 14 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[314] \ 1^{\overline{11}} \begin{smallmatrix} 1^{11} \\ 8 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 5 \end{smallmatrix} \begin{smallmatrix} 1321 \\ 1321 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[315] \ 1^{\overline{11}} \begin{smallmatrix} 1^{11} \\ 8 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 5 \end{smallmatrix} \begin{smallmatrix} 1321 \\ 1321 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[313] \ 1^{\overline{12}} \begin{smallmatrix} 1^{12} \\ 232181 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 2231 \end{smallmatrix} \begin{smallmatrix} 5 \\ 1 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 1^{11} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 52 & 18 \\ 3 & 4 & 0 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[317] \ 1^{\overline{11}} \begin{smallmatrix} 1^{11} \\ 6 \end{smallmatrix} \begin{smallmatrix} 1^{11} \\ 1 \end{smallmatrix} \begin{smallmatrix} 12321712321 \\ 1^{12} \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 2 & 11 & 4 \\ 1 & 2 & 0 & 7 & 2 \\ 10 & 11 & 7 & 50 & 18 \\ 3 & 4 & 2 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[318] \ 1^{11} 6^{11} 12321712321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[312] \ 1^{11} 3151316131^{11} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[311] \ 1^{11} 2321812231^{11} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 19 \quad [331] \ 1^{11} 315132181231^{11} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

3.1.6 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [22] \ 1^{12} 3^{11} 1^5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 0 & 7 & 2 & 3 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 5 & 9 \\ 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[21] \ 1^{11} 3^{11} 1^5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 & 3 \\ 1 & 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 6 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

3.2 Non-unimodular ($|\det| > 1$) — 1672 bases

3.2.1 NHC(-2)(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 31^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

3.2.2 NHC(-2)(-2)(-3) + su₂ + su_{2n3mix}

$$\begin{aligned}
T = 12 \quad [3] \quad 1^{13} 3^1 1^5 13221^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 2), c_{\text{ext}} = 6.0 \\
[2] \quad 1^{13} 3^1 1^5 1321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[1] \quad 1^{11} 3^1 1^5 1321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[0] \quad 1^{11} 3^1 1^5 1321^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0
\end{aligned}$$

3.2.3 NHC(-2)(-2)(-3) + su₃ + su₂

$$\begin{aligned}
T = 8 \quad [0] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0} \\
[2] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[3] \quad 1^{14} 3^1 1^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.6 \\
T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 10 & 13 \\ 1 & 1 & 2 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -1 & 13 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

3.2.4 NHC(-2)(-3) + NHC(-2)(-3) + su₂

$$\begin{aligned}
T = 8 \quad [0] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0} \\
T = 11 \quad [1] \quad 1^{11} 5^1 131^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2] $1^{11} \overset{1}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 13 & 10 & 0 & 75 & 17 \\ 2 & 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{12} \overset{1}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 5 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} \overset{1}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{13} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 11 & 9 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} \overset{1}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{12} \overset{1}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 14 & 11 & 0 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{\frac{1}{1^4}} 5^{\frac{1}{1^4}} 131^{\frac{1}{1^4}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 3 & 16 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^2}} 1^{\frac{1}{2}} 231^{\frac{1}{1^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 1 & 22 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 16 & 30 & 22 & 0 & 154 & 26 \\ 3 & 5 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{\frac{1}{1^3}} 5^{\frac{1}{1^3}} 131^{\frac{1}{1^3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 3 & 12 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{\frac{1}{1^2}} 5^{\frac{1}{1^2}} 131^{\frac{1}{1^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 3 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 131^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 3 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^3}} 1^{\frac{1}{3}} 231^{\frac{1}{1^3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 5 & 5 & 0 & 45 & 5 \\ 2 & 5 & 2 & 1 & 33 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 24 & 45 & 33 & 0 & 349 & 39 \\ 3 & 5 & 4 & -1 & 39 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{\frac{1}{1^1}} 5^{\frac{1}{1^1}} 1^{\frac{1}{1^1}} 231^{\frac{1}{1^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 1 & 11 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 13 \\ 3 & 5 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{\frac{1}{1^3}} 5^{\frac{1}{1^3}} 131^{\frac{1}{1^3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

3.2.5 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{11} \bar{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 \\ 1 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

3.2.6 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 11 \quad [1] \quad 1^{12} \bar{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 16 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 16 & 12 & 0 & 117 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{12} \bar{5} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 12 & 9 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} \bar{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 1 & 0 & 17 & 2 \\ 1 & 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 10 & 13 & 17 & 13 & 0 & 141 & 23 \\ 2 & 2 & 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

3.2.7 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} \bar{2} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

3.2.8 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} \bar{2} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 6 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \bar{2} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 2^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 2^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 2^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 2^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 2^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 2^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 2^1 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 3 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 2^1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 2^1 3 2 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 1^{14} 2^3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 8 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 1^{13} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 6 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 1^{12} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 4 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 1^{11} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{13} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -38, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{12} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -45, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & -2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 2^3 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -45, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 7 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{12} 2^3 3 \ 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{11} 1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 3 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{12} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 9 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{12} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{12} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 8 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{13} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -85$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 15 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{13} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -103$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{14} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{13} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{13} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & -2 & 15 & 3 \end{pmatrix}_{b_0}$$

- [32] $1^{11} 1^{12} 2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 5 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [35] $1^{12} 2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$
- [33] $1^{14} 2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{12} 2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 10 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [36] $1^{14} 2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 2 & 1 & 28 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 28 & 0 & 266 & 34 \\ 3 & 4 & -1 & 34 & 4 \end{pmatrix}_{b_0}$
- [37] $1^{13} 2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$
- [39] $1^{13} 2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$
- [40] $1^{13} 2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 12 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$
- [38] $1^{11} 2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 6 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{13} 2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$
- [42] $1^{14} 2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{12}2^13^121^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 11 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{13}2^13^121^{12}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 16 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{11}2^13^121^{13}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 7 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{12}2^13^121^{12}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 15 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{12}2^13^121^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{14}2^13^121^{13}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{14}2^13^121^{13}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 22 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{12}2^13^121^{12}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 18 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{13}2^13^121^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{11}2^13^121^{14}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 9 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

- [53] $1^{14}2^321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$
- [54] $1^{12}2^321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 16 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$
- [55] $1^{14}2^321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & -2 & 24 & 3 \end{pmatrix}_{b_0}$
- [56] $1^{13}2^321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 21 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$
- [57] $1^{11}2^321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 8 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [58] $1^{13}2^321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 17 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$
- [59] $1^{13}2^321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 26 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 26 & 0 & 28 & 322 & 36 \\ 3 & 0 & 3 & 36 & 4 \end{pmatrix}_{b_0}$
- [60] $1^{13}2^321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$
- [61] $1^{14}2^321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 27 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0}$
- [62] $1^{14}2^321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -357$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 19 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$
- [63] $1^{14}2^321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 23 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{13}2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 22 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{13}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 18 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{11}4^2 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 10 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{14}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -480$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{14}2^3 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -480$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 28 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 28 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{14}2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 24 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [70] $1^{11}31^5 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[72] $1^{12}31^5 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 6$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 3 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 190 & 32 \\ 3 & 5 & -1 & 32 & 5 \end{pmatrix}_{b_0}$$

[71] $1^{11}31^5 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 6$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 3 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 14 & 0 & 46 & 16 \\ 3 & 5 & -1 & 16 & 5 \end{pmatrix}_{b_0}$$

[73] $1^{11}31^5 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 15$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[74] $1^{13}3151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 27 & 5 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{14}3151321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 36 & 5 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{12}3151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 18 & 5 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{11}3151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 9 & 5 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[78] $1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 7 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[79] $1^{11}3151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 9 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[80] $1^{12}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 15 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[81] $1^{11}3151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 10 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[82] $1^{11}3151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 12 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

- [83] $1^{12}31^15^1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 16 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$
- [84] $1^{12}31^15^1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 18 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$
- [85] $1^{13}31^15^1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 22 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$
- [86] $1^{11}31^15^1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 147, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 13 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$
- [87] $1^{12}31^15^1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 183, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 17 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$
- [88] $1^{12}31^15^1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 213, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 19 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$
- [89] $1^{13}31^15^1321^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 233, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 23 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$
- $T = 13$ [92] $1^{11}31^16^131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$
- [93] $1^{11}31^15^13221^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 22 & 5 \\ 12 & 0 & 22 & 88 & 24 \\ 3 & 0 & 5 & 24 & 6 \end{pmatrix}_{b_0}$$
- [94] $1^{11}31^15^13221^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{orange}{o}_8 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 13, T_{\max} = 13, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 16 & 5 \\ 9 & 0 & 16 & 50 & 18 \\ 3 & 0 & 5 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{12} 31^1 5^{12} 13221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 32 & 5 \\ 18 & 0 & 32 & 206 & 36 \\ 3 & 0 & 5 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{14} 31^1 5^{14} 13221^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 3 & 1 & 6 & 24 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 40 & 6 \\ 8 & 24 & 0 & 40 & 158 & 32 \\ 2 & 5 & -1 & 6 & 32 & 5 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{11} 31^1 5^{11} 13221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 10 & 6 \\ 2 & 6 & 0 & 10 & 8 & 8 \\ 2 & 5 & -1 & 6 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{12} 31^1 5^{12} 13221^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 3 & 1 & 6 & 12 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 20 & 6 \\ 4 & 12 & 0 & 20 & 38 & 16 \\ 2 & 5 & -1 & 6 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{13} 31^1 5^{13} 13221^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 6 & 18 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 30 & 6 \\ 6 & 18 & 0 & 30 & 88 & 24 \\ 2 & 5 & -1 & 6 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11} 31^1 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{12} 31^1 6^{12} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [102] \ 1^{11} 3221^{11} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 20, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{11}3221 \quad (12) \quad 12231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -32, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{11}3221 \quad (12) \quad 12231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -32, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

3.2.9 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{11}31513^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 4 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31513^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 4 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 \\ 3 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

3.2.10 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2\mathfrak{n}3\text{mix}$

$$T = 11 \quad [0] \quad 1^{11}31513^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 11 & 4 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

- [2] $1^{11}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 5 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [3] $1^{11}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$
- [5] $1^{11}3^{11}15131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$
- [4] $1^{12}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 11 & 0 & 32 & 12 \\ 3 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0}$
- [6] $1^{12}3^{11}15131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 65 & 17 \\ 3 & 4 & -1 & 17 & 4 \end{pmatrix}_{b_0}$
- [7] $1^{11}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 6 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{13}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 19 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 19 & 0 & 110 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{11}3^{11}15131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 9 & 0 & 11 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{13}3^{11}15131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 23 & 0 & 167 & 27 \\ 3 & 4 & -1 & 27 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{14}31513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 23 & 0 & 152 & 26 \\ 3 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{14}3^1 1^1 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -101, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 27 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 27 & 0 & 217 & 31 \\ 3 & 4 & -1 & 31 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 10 & 0 & 11 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 13 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{12}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 65 & 17 \\ 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{12}3^1 1^1 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -189, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{13}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -212, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 15 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{13}3^1 1^1 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 22 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{12}3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 14 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{13}3^{11}5^{12}13^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -245, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{13}3^{11}5^{12}13^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 17 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{11}5^{13}13^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -293, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 24 & 0 & 31 & 317 & 37 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{13}3^{11}5^{13}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -309, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{13}3^{11}5^{12}13^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 19 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{13}3^{11}5^{13}13^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 23 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^{11}5^{13}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -365, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 152 & 26 \\ 3 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{14}3^{11}5^{12}13^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 21 & 0 & 27 & 217 & 31 \\ 3 & 0 & 3 & 31 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{14}3^{11}5^{14}13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 29 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 35 & 3 \\ 29 & 0 & 35 & 441 & 43 \\ 3 & 0 & 3 & 43 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{14}3^{11}5^{13}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -456, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 25 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 25 & 0 & 31 & 317 & 37 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [31] $1^{13}3^{11}5^{13}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 8 & 4 \\ 5 & 0 & 8 & 15 & 9 \\ 3 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}5^{12}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 22 & 4 \\ 15 & 0 & 22 & 119 & 25 \\ 3 & 0 & 4 & 25 & 5 \end{pmatrix}_{b_0}$$

3.2.11 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$T = 8$ [0] $1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

3.2.12 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 6 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{11}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 4 & 4 \\ 2 & 3 & -1 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{121}2321^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 16 & 12 \\ 2 & 4 & 0 & 16 & 4 & 4 \\ 2 & 3 & -1 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{131}2321^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 24 & 18 \\ 2 & 4 & 0 & 24 & 4 & 4 \\ 2 & 3 & -1 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{141}2321^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 32 & 24 \\ 2 & 4 & 0 & 32 & 4 & 4 \\ 2 & 3 & -1 & 24 & 4 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [8] $1^{11}3151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[10] $1^{21}3151321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 10 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 18 & 14 \\ 2 & 5 & 0 & 18 & 5 & 5 \\ 2 & 4 & -1 & 14 & 5 & 4 \end{pmatrix}_{b_0}$$

[9] $1^{41}3151321^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 2 & 1 & 20 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 36 & 28 \\ 2 & 5 & 0 & 36 & 5 & 5 \\ 2 & 4 & -1 & 28 & 5 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{31}3151321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 2 & 1 & 15 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 27 & 21 \\ 2 & 5 & 0 & 27 & 5 & 5 \\ 2 & 4 & -1 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{11}3151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 5 & 5 \\ 2 & 4 & -1 & 7 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [13] \quad 1^{11} 31^1 5 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 8 & 10 & 8 \\ 2 & 6 & 0 & 10 & 6 & 6 \\ 2 & 5 & -1 & 8 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{21} 31^1 5 13221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 3 & 1 & 12 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 38 & 20 & 16 \\ 2 & 6 & 0 & 20 & 6 & 6 \\ 2 & 5 & -1 & 16 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{31} 31^1 5 13221^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 3 & 1 & 18 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 18 & 0 & 88 & 30 & 24 \\ 2 & 6 & 0 & 30 & 6 & 6 \\ 2 & 5 & -1 & 24 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{41} 31^1 5 13221^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 3 & 1 & 24 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 24 & 0 & 158 & 40 & 32 \\ 2 & 6 & 0 & 40 & 6 & 6 \\ 2 & 5 & -1 & 32 & 6 & 5 \end{pmatrix}_{b_0}$$

3.2.13 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_2$

$$T = 7 \quad [0] \quad 1^{11} 3^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 9, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [3] \quad 1^{11} 31^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 \\ 2 & 3 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 31^{131} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 28 & 10 \\ 2 & 3 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad \begin{matrix} 1 & 1 \\ 1 & 4 \end{matrix} \begin{matrix} 1 \\ 1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -75, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\left(\begin{matrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{matrix} \right)_{b_0}$$

$$[6] \begin{array}{c} 1^1 \\ 1^1 \end{array} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -75, \, \det = 7, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\left(\begin{array}{ccccc} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{array} \right)_{b_0}$$

$$[7] \quad 1^{\textcolor{teal}{1}\textcolor{teal}{1}\textcolor{teal}{1}} 31^{\textcolor{teal}{1}\textcolor{teal}{1}} 31^{\textcolor{teal}{1}\textcolor{teal}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{red}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{\textcolor{teal}{1}\textcolor{teal}{1}\textcolor{teal}{1}}3131^{\textcolor{teal}{1}\textcolor{teal}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

$$[9] \quad 1^{\textcolor{brown}{1}1^{\textcolor{brown}{1}}1^{\textcolor{brown}{1}}31^{\textcolor{brown}{1}}31^{\textcolor{brown}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -63, \, \det = 21, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\textcolor{blue}{1}}1^{\textcolor{blue}{1}}31^{\textcolor{blue}{1}}31^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \text{ } 1^{11} 31^{11} 31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 4 \\ 3 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[12] $2 \overset{11}{\underset{11}{3} \ 2} + [\textcolor{red}{\mathfrak{s}u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}u}_2 + \textcolor{orange}{\mathfrak{s}u}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[13] $2 \begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix} 2 + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}u_2 + \textcolor{orange}{5}u_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\left(\begin{smallmatrix} 0 & 2 & 2 & 6 & 6 & 2 \\ 2 & 1 & 3 & 9 & 8 & 3 \\ 2 & 3 & 0 & 6 & 7 & 2 \\ 6 & 9 & 6 & 25 & 24 & 9 \\ 6 & 8 & 7 & 24 & 21 & 9 \\ 2 & 3 & 2 & 9 & 9 & 3 \end{smallmatrix} \right)_{b_0}$

[14] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 4 \\ 4 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[15] $1^{12}31^{11}31^{11} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

- [16] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 12 & 4 \\ 5 & 9 & 0 & 12 & 28 & 10 \\ 2 & 3 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [17] $2^{11}3^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 7 \\ 3 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{112}31^231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{112}31^231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{112}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [23] $2^{11}3^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{112}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [25] $1^{112}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

- [26] $2 \overset{1}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [27] $2 \overset{12}{\underset{1^{111}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 7 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 7 & 3 \\ 6 & 7 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$
- [28] $1 \overset{112}{\underset{1^{111}}{3}} 31 \overset{11}{\underset{1^{111}}{3}} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [29] $2 \overset{1}{\underset{1^{122}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 6 & 6 & 3 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 4 & 6 & 4 & 10 & 12 & 6 \\ 4 & 6 & 4 & 12 & 10 & 6 \\ 2 & 3 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$
- [30] $1 \overset{111}{\underset{1^{111}}{3}} 31 \overset{11}{\underset{1^{111}}{3}} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [31] $2 \overset{1}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$
- [32] $2 \overset{11}{\underset{1^{111}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 4 & 5 & 5 & 8 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [33] $2 \overset{1}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 9 & 3 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 4 & 6 & 4 & 10 & 18 & 6 \\ 6 & 9 & 6 & 18 & 25 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [34] $2 \overset{12}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 13 & 3 \\ 2 & 3 & 0 & 4 & 12 & 2 \\ 4 & 6 & 4 & 10 & 26 & 6 \\ 10 & 13 & 12 & 26 & 61 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0}$
- [35] $2 \overset{11}{\underset{1^{122}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 8 & 3 \\ 2 & 3 & 0 & 4 & 7 & 2 \\ 4 & 6 & 4 & 10 & 16 & 6 \\ 6 & 8 & 7 & 16 & 21 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 2 \overset{1}{\underset{1^{114}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 2 \overset{1}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 12 & 3 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 4 & 6 & 4 & 10 & 24 & 6 \\ 8 & 12 & 8 & 24 & 46 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 2 \overset{1}{\underset{1^{115}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 2 \overset{1^{11}}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 2 \overset{1^{11}}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 11 & 3 \\ 2 & 3 & 0 & 4 & 9 & 2 \\ 4 & 6 & 4 & 10 & 22 & 6 \\ 8 & 11 & 9 & 22 & 40 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 2 \overset{1}{\underset{1^{116}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 2 \overset{1^{12}}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 2 \overset{1^{11}}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 2 \overset{1^{12}}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 12 & 2 \\ 2 & 1 & 3 & 6 & 16 & 3 \\ 2 & 3 & 0 & 4 & 14 & 2 \\ 4 & 6 & 4 & 10 & 32 & 6 \\ 12 & 16 & 14 & 32 & 90 & 18 \\ 2 & 3 & 2 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 2 \overset{1}{\underset{1^{117}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 2 \overset{11}{\underset{1^{124}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 14 & 3 \\ 2 & 3 & 0 & 4 & 11 & 2 \\ 4 & 6 & 4 & 10 & 28 & 6 \\ 10 & 14 & 11 & 28 & 65 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 2 \overset{11}{\underset{1^{114}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 2 \overset{112}{\underset{1^{113}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 10 & 13 & 12 & 13 & 61 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 2 \overset{112}{\underset{1^{114}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 16 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 16 & 3 \\ 12 & 16 & 14 & 16 & 90 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [50] \quad 1 \overset{112}{\underset{1^{133}}{2}} 3 21 \overset{123}{\underset{1^{123}}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 6 & 11 & 9 & 49 & 60 & 15 \\ 8 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1 \overset{112}{\underset{1^{111}}{2}} 3 21 \overset{11}{\underset{1^{11}}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1 \overset{112}{\underset{1^{112}}{3}} 21 \overset{111}{\underset{1^{111}}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1 \overset{111}{\underset{1^{111}}{2}} 3 21 \overset{21}{\underset{1^{11}}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 17 & 12 & 9 \\ 2 & 4 & 0 & 12 & 6 & 6 \\ 2 & 3 & -1 & 9 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1 \overset{112}{\underset{1^{112}}{2}} 3 21 \overset{13}{\underset{1^{13}}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 2 & 0 & 1 & 1 & 10 & 3 \\ 6 & 0 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{114}2^321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{11}231^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 7 & 0 & 14 & 8 \\ 2 & 0 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{112}2^321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{133}2^321^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 84 & 18 \\ 8 & 13 & 0 & 84 & 97 & 21 \\ 2 & 3 & -1 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{11}23131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{111}231^{131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 \\ 3 & 7 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{11}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{111,1^1}5^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

[63] $1^15^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{111}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0} \\
[66] & 1^{\overset{1^1}{1^1,1^1}} 5^{\overset{1^1}{1^1}} 1^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0} \\
[65] & 1^{\overset{1^1}{1^1,1^1}} 5^{\overset{1^1}{1^1}} 1^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[69] & 1^{\overset{1^1}{1^1,1^1}} 5^{\overset{1^1}{1^1}} 1^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[68] & 21^{\overset{1^1}{1^1}} 5^{\overset{1^1}{1^1}} 1^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0} \\
[67] & 21^{\overset{1^1}{1^1}} 5^{\overset{1^1}{1^1}} 1^{\overset{1^1}{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[71] & 1^{\overset{1^{12}}{1^{12}}} 2^{\overset{1^{12}}{1^{12}}} 3^{\overset{1^{12}}{1^{12}}} 21^{\overset{1^{12}}{1^{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0} \\
[70] & 1^{\overset{1^{12}}{1^{12}}} 2^{\overset{1^{12}}{1^{12}}} 3^{\overset{1^{12}}{1^{12}}} 21^{\overset{1^{12}}{1^{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 40 & 12 \\ 6 & 9 & 0 & 40 & 49 & 15 \\ 2 & 3 & -1 & 12 & 15 & 4 \end{pmatrix}_{b_0} \\
[72] & 1^{\overset{1^1}{1^1,1^1}} 5^{\overset{1^1}{1^1,1^1}} 1^{\overset{1^1}{1^1,1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[73] & 1^{\overset{1^{11}}{1^{11}}} 2^{\overset{1^{11}}{1^{11}}} 3^{\overset{1^{11}}{1^{11}}} 21^{\overset{1^{11}}{1^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 2 & 0 & 1 & 1 & 7 & 3 \\ 4 & 0 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0} \\
[74] & 1^{\overset{1^{13}}{1^{13}}} 2^{\overset{1^{13}}{1^{13}}} 3^{\overset{1^{13}}{1^{13}}} 21^{\overset{1^{13}}{1^{13}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 9 & 2 \\ 8 & 0 & 13 & 97 & 71 & 21 \\ 6 & 0 & 9 & 71 & 49 & 15 \\ 2 & 0 & 2 & 21 & 15 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{112}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{114}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 9 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{111}231^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 2 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

[78] $1^{112}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 9 & 2 \\ 2 & 0 & 4 & 6 & 20 & 6 \\ 6 & 0 & 9 & 20 & 49 & 15 \\ 2 & 0 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[79] $1^{111}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 5 & 2 \\ 2 & 0 & 4 & 6 & 12 & 6 \\ 4 & 0 & 5 & 12 & 17 & 9 \\ 2 & 0 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{113}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[83] $1^{114}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[82] $1^{12}2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 2 & 0 & 1 & 1 & 14 & 3 \\ 8 & 0 & 10 & 14 & 74 & 18 \\ 2 & 0 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{143}2321^{43} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 96 & 24 \\ 6 & 12 & 0 & 96 & 70 & 18 \\ 2 & 3 & -1 & 24 & 18 & 4 \end{pmatrix}_{b_0}$$

[86] $1^{112}2^321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 \\ 2 & 0 & 4 & 6 & 24 & 6 \\ 8 & 0 & 10 & 24 & 74 & 18 \\ 2 & 0 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[87] $1^{122}2^321^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 32 & 12 \\ 4 & 8 & 0 & 32 & 30 & 12 \\ 2 & 3 & -1 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[88] $1^{144}2^321^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 128 & 24 \\ 8 & 16 & 0 & 128 & 126 & 24 \\ 2 & 3 & -1 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[89] $1^{132}2^321^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 48 & 18 \\ 4 & 8 & 0 & 48 & 30 & 12 \\ 2 & 3 & -1 & 18 & 12 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{142}2^321^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 64 & 24 \\ 4 & 8 & 0 & 64 & 30 & 12 \\ 2 & 3 & -1 & 24 & 12 & 4 \end{pmatrix}_{b_0}$$

[94] $1^{111}2^321^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 6 & 2 \\ 6 & 0 & 6 & 34 & 36 & 12 \\ 6 & 0 & 6 & 36 & 34 & 12 \\ 2 & 0 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{133}2^321^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 72 & 18 \\ 6 & 12 & 0 & 72 & 70 & 18 \\ 2 & 3 & -1 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{113}2^321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 24 & 6 \\ 6 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[92] $1^{114}2^321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 32 & 6 \\ 8 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{111}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{112}2^321^{12}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 16 & 6 \\ 4 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{112}2^321^{13}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 20 & 6 \\ 6 & 9 & 0 & 20 & 49 & 15 \\ 2 & 3 & -1 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{111}2^321^{32}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 5 & 2 \\ 6 & 0 & 6 & 34 & 26 & 12 \\ 4 & 0 & 5 & 26 & 17 & 9 \\ 2 & 0 & 2 & 12 & 9 & 4 \end{pmatrix}_{b_0}$$

[98] $1^{12}2^321^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[99] $1^{123}2^321^{24}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -22$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 56 & 12 \\ 8 & 13 & 0 & 56 & 97 & 21 \\ 2 & 3 & -1 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[100] $1^{12}2^321^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -23$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[101] $1^{112}2^321^{34}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 10 & 2 \\ 6 & 0 & 6 & 34 & 52 & 12 \\ 8 & 0 & 10 & 52 & 74 & 18 \\ 2 & 0 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[102] $1^{112}2^321^{23}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 9 & 2 \\ 4 & 0 & 5 & 17 & 31 & 9 \\ 6 & 0 & 9 & 31 & 49 & 15 \\ 2 & 0 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[103] $1^{122}2^321^{12}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 6 & 7 & 5 & 17 & 24 & 9 \\ 7 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{112}2^321^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[106] $1^{111}2^3 21^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 5 & 2 \\ 4 & 0 & 5 & 17 & 19 & 9 \\ 4 & 0 & 5 & 19 & 17 & 9 \\ 2 & 0 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[107] $1^{122}2^3 21^{33} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 9 & 2 \\ 6 & 0 & 9 & 49 & 51 & 15 \\ 6 & 0 & 9 & 51 & 49 & 15 \\ 2 & 0 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[105] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 4 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[108] $1^{113}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[109] $1^{122}2^3 21^{34} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 10 & 2 \\ 6 & 0 & 9 & 49 & 62 & 15 \\ 8 & 0 & 10 & 62 & 74 & 18 \\ 2 & 0 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[111] $1^{114}2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[112] $1^{13}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[113] $1^{13}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[114] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 6 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[115] $1^{113}2^321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 28 & 6 \\ 8 & 13 & 0 & 28 & 97 & 21 \\ 2 & 3 & -1 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[116] $1^{111}2^321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 4 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[117] $1^{114}2^321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[118] $1^{114}2^321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[119] $1^{12}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[120] $1^{111}2^321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[121] $1^{112}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[122] $1^{113}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[123] $1^{111}231^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -35$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 2 & 0 & 4 & 2 & 7 & 4 \\ 4 & 0 & 7 & 7 & 14 & 8 \\ 2 & 0 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[125] $1^{122}2^321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 5 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[124] $1^{123}2^13^121^{122}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[127] $1^{122}2^13^121^{112}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 5 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[126] $1^{123}2^13^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[129] $1^{112}2^13^121^{24}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 10 & 2 \\ 4 & 0 & 5 & 17 & 38 & 9 \\ 8 & 0 & 10 & 38 & 74 & 18 \\ 2 & 0 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[128] $1^{112}2^13^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[130] $1^{112}2^13^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[131] $1^{112}2^13^121^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[132] $1^{113}2^13^121^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[133] $1^{113}2^13^121^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 9 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[134] $1^{114}2^13^121^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 11 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[135] $1^{13}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[136] $1^{13}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[137] $1^{123}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{112}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[139] $1^{111}2^321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[140] $1^{122}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[141] $1^{112}2^321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[142] $1^{122}2^321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 10 & 2 \\ 8 & 0 & 10 & 74 & 76 & 18 \\ 8 & 0 & 10 & 76 & 74 & 18 \\ 2 & 0 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{134}2^321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{144}2^321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 10 & 14 & 10 & 74 & 86 & 18 \\ 11 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{114}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{113}2^321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{123}2^321^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{123}2^321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 6 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[149] $1^{13}2^321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[150] $1^{122}2^321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{14}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{14}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{144}2^321^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 8 & 14 & 10 & 74 & 86 & 18 \\ 9 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{114}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 8 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[155] $1^{144}2^321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 9 & 14 & 10 & 74 & 86 & 18 \\ 10 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[156] $1^{122}2^321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{124}2^321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[158] $1^{114}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[159] $1^{133}2^321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 9 & 11 & 9 & 49 & 60 & 15 \\ 10 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[160] $1^{113}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 6 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[161] $1^{113}2^321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[162] $1^{113}2^321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[163] $1^{114}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

- [164] $1^{123}2^13^221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -102$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 5 & 17 & 26 & 9 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$
- [165] $1^{113}2^13^221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$
- [166] $1^{114}2^13^221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 9 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$
- [167] $1^{14}2^13^221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$
- [170] $1^{134}2^13^221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$
- [168] $1^{112}2^13^221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 5 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$
- [171] $1^{134}2^13^221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$
- [169] $1^{112}2^13^221^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 7 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$
- [172] $1^{124}2^13^221^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[173] $1^{14}2^321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[174] $1^{13}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[175] $1^{22}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[176] $1^{33}2^321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 8 & 11 & 9 & 49 & 60 & 15 \\ 9 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{12}2^321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{14}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{11}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[180] $1^{11}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 6 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{24}2^321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{123} 2^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{123} 2^3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{133} 2^3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 1 & 2 & 10 & 10 & 3 \\ 1 & 2 & 0 & 6 & 6 & 2 \\ 7 & 10 & 6 & 34 & 36 & 12 \\ 6 & 10 & 6 & 36 & 34 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{134} 2^3 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{122} 2^3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{124} 2^3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{122} 2^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 5 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{112} 2^3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

- [188] $1^{122}2^13^{32}21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$
- [191] $1^{14}2^13^{14}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$
- [193] $1^{124}2^13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$
- [192] $1^{114}2^13^{13}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$
- [194] $1^{113}2^13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -135$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$
- [195] $1^{124}2^13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$
- [196] $1^{123}2^13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -143$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$
- [197] $1^{124}2^13^{123}21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -143$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$
- [198] $1^{114}2^13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[199] $1^{114}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[200] $1^{123}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[201] $1^{123}2^13^121^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[202] $1^{114}2^13^121^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 4 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[203] $1^{114}2^13^121^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[204] $1^{123}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[205] $1^{113}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[206] $1^{123}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[207] $1^{113}2^13^121^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

5.8

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{123} 2^1 3^{33} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{123} 2^1 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 6 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[209] \ 1^{123} 2^1 3^{13} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{134} 2^1 3^{22} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 8 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{134} 2^1 3^{44} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 9 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{114} 2^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{123} 2^1 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{134} 2^1 3^{33} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[216] $1^{124}2^13^121^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[217] $1^{122}2^13^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[218] $1^{133}2^13^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.6$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[219] $1^{123}2^13^121^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$

$T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[220] $1^{134}2^13^121^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[221] $1^{134}2^13^121^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[222] $1^{134}2^13^121^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 9 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[223] $1^{134}2^13^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[224] $1^{133}2^13^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

6.6

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[225] $1^{134}2^13^221^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 9 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[226] $1^{134}2^13^221^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{124}2^13^221^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{124}2^13^221^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[228] $1^{113}2^13^221^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[229] $1^{124}2^13^221^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 7 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{124}2^13^221^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[231] $1^{113}2^13^221^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[233] $1^{133}2^321^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.6$

$T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[234] $1^{134}2^321^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[236] $1^{134}2^321^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[235] $1^{114}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[239] $1^{123}2^321^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 8 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{113}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[238] $1^{123}2^321^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 10 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[240] $1^{124}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -253$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 9 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[241] $1^{114}2^321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

6.0

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[242] $1^{124}2^1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[243] $1^{124}2^1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{124}2^1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{124}2^1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[246] $1^{134}2^1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[247] $1^{134}2^1321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 11 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[248] $1^{134}2^1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[249] $1^{124}2^1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[250] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[251] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[252] $1^{133}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 8 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[253] $1^{133}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[254] $1^{124}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[255] $1^{144}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[256] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[257] $1^{114}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[258] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.6

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 9 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{144} 2^3 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{123} 2^3 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{114} 2^3 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{134} 2^3 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{133} 2^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{133} 2^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{134} 2^3 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 10 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{134} 2^3 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 8 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[267] $1^{134}2^13^121^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[268] $1^{134}2^13^121^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[269] $1^{114}2^13^121^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -396$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 11 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[270] $1^{144}2^13^121^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -413$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[272] $1^{114}2^13^121^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[271] $1^{124}2^13^121^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -420$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 11 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[273] $1^{144}2^13^121^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -437$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[275] $1^{144}2^13^121^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 8 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[274] $1^{133}2^13^121^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

6.6

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 9 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[276] $1^{134}2^1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -453, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[277] $1^{134}2^1321^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 11 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[278] $1^{134}2^1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[279] $1^{144}2^1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 9 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[280] $1^{144}2^1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[281] $1^{124}2^1321^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[282] $1^{144}2^1321^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 11 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[283] $1^{144}2^1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 9 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[284] $1^{134}2^13^421^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -612$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[285] $1^{144}2^13^321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -648$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[286] $1^{144}2^13^421^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 11 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[287] $1^{144}2^13^421^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 12 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [290] $1^{111}3141^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[289] $1^{111}31^14131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 1 & 6 & 14 & 8 \\ 2 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[293] $1^{11}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -45$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[292] $1^15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -49$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[288] $1^{11}31^141^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 3 & 4 & 0 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[291] & 1^{111} 314131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[294] & 1^{11} 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 4 & 5 \\ 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 \\ 3 & 5 & 4 & 4 & 5 \end{pmatrix}_{b_0} \\
[296] & 1^{11} 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -49, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[295] & 1^{111} 2231^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 3 & 1 & 8 & 5 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 8 & 0 & 15 & 8 & 9 \\ 2 & 5 & 0 & 8 & 3 & 5 \\ 2 & 5 & -1 & 9 & 5 & 5 \end{pmatrix}_{b_0} \\
[298] & 1^{111} 2231^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 8 & 4 \\ 2 & 0 & 5 & 3 & 8 & 5 \\ 4 & 0 & 8 & 8 & 15 & 9 \\ 2 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[297] & 1^{111} 3141^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[299] & 1^{111} 31^{141} 1^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 4 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 6 & 1 & 9 & 14 & 8 \\ 2 & 3 & 0 & 5 & 8 & 4 \end{pmatrix}_{b_0} \\
[300] & 1^{112} 31^{141} 1^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 7 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 6 & 1 & 14 & 16 & 8 \\ 5 & 7 & 1 & 16 & 17 & 9 \\ 2 & 3 & 0 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[301] & 1^{111} 31^{141} 1^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 7 & 4 \\ 4 & 6 & 1 & 7 & 14 & 8 \\ 2 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0} \\
[302] & 1^{112} 31^{141} 1^{131} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 11 & 4 \\ 7 & 10 & 1 & 11 & 39 & 13 \\ 2 & 3 & 0 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

[303] $231^{111}3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 428$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 5 & 2 \\ 5 & 33 & 1 & 35 & 35 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 35 & 11 \\ 5 & 35 & 0 & 35 & 33 & 11 \\ 2 & 11 & -1 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [305] $1^{111}3151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[306] $1^{11}3151231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[307] $1^{11}61231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -47$, $\det = -4$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[308] $1^{11}31^141321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 11$, $\Delta = -5$, $\det = -4$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[309] $1^{111}3151^1231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -7$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 5 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[310] $1^{111}31^141321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 11$, $\Delta = -5$, $\det = -9$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 3 \\ 2 & 2 & 5 & 8 & 4 \\ 4 & 5 & 7 & 12 & 7 \\ 6 & 8 & 12 & 19 & 11 \\ 3 & 4 & 7 & 11 & 6 \end{pmatrix}_{b_0}$$

[311] $1^{111}3151^1231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -9$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 6 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[312] $1^{114}3151^2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 2 & 1 & 4 & 26 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 26 & 4 \\ 11 & 26 & 0 & 26 & 202 & 30 \\ 2 & 4 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

[313] $1^{12}3151^{2231^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 0 & 18 & 0 & 114 & 22 \\ 2 & 0 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[314] $1^{13}3151^{2231^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -20$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 0 & 22 & 0 & 154 & 26 \\ 2 & 0 & 3 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

[315] $1^{111}2315131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -28$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 2 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[316] $1^{113}2315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 8 & 2 \\ 2 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[317] $1^{12}3151^{1231^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -53$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 4 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 4 \\ 6 & 13 & 0 & 13 & 49 & 15 \\ 2 & 4 & -1 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[318] $1^{113}3151^{1231^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 17 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 17 & 4 \\ 7 & 17 & 0 & 17 & 79 & 19 \\ 2 & 4 & -1 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

[319] $1^{111}2315131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[320] $1^{112}2315131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -93$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[322] $1^{112}3151^{1231^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -107$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 7 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{114}3151^3231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -107, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 31 & 3 \\ 2 & 0 & 4 & 2 & 31 & 4 \\ 17 & 0 & 31 & 31 & 317 & 37 \\ 2 & 0 & 3 & 4 & 37 & 4 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{113}3151^2231^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 12 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{112}3151^1231^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -133, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 8 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{112}2 \ 3 \ 15131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -148, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{123}2 \ 3 \ 15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -148, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 8 & 2 \\ 2 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{113}3151^2231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -182, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 13 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{114}2 \ 3 \ 15131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -216, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 10 & 2 \\ 2 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{122}2 \ 3 \ 15131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -220, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 5 & 2 \\ 2 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[330] \ 1^{122}2 \ 3 \ 15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -220, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[331] $1^{113} 2 \ 3 \ 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -241, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 8 & 2 \\ 2 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[332] $1^{113} 2 \ 3 \ 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[333] $1^{123} 2 \ 3 \ 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -313, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 8 & 2 \\ 2 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{114} 3151^2 231^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 26 & 3 \\ 2 & 0 & 4 & 2 & 26 & 4 \\ 14 & 0 & 26 & 26 & 202 & 30 \\ 2 & 0 & 3 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

[335] $1^{124} 2 \ 3 \ 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 11 & 2 \\ 2 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[337] $1^{134} 2 \ 3 \ 15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 10 & 2 \\ 2 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[336] $1^{114} 2 \ 3 \ 15131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[338] $1^{114} 2 \ 3 \ 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -437, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 11 & 2 \\ 2 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[339] $1^{124} 2 \ 3 \ 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -508, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 10 & 2 \\ 2 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[340] \ 2 \ 3 \ 15131^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 8 & 2 \\ 2 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[341] \ 2 \ 3 \ 15131^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -796, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 10 & 2 \\ 2 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [348] \ 1^{111} 31 \ 5 \ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[347] \ 1^{13} 31 \ 5 \ 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[346] \ 1^{12} 31 \ 5 \ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$$

$$[345] \ 1^{11} 31 \ 5 \ 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[344] \ 1^{13} 231 \ 5 \ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[343] \ 1^{11} 31 \ 5 \ 13221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$$

$$[342] \ 1^{114} 31 \ 5 \ 13221^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[356] \ 1^{11} 5 \ 131 \ 5 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 4 & 5 \\ 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 \\ 3 & 5 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{11} 5 \ 131 \ 5 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{112} 2321^{15131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 5 & 17 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 17 & 5 \\ 6 & 17 & 0 & 17 & 67 & 19 \\ 2 & 5 & -1 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{11} 7 \ 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -45, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{111} 314141^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & 6 & 5 & 6 \\ 2 & 4 & 1 & 5 & 3 & 5 \\ 2 & 4 & 0 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{13}31 \ 5 \ 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 11 & 3 \\ 2 & 0 & 5 & 7 & 22 & 7 \\ 6 & 0 & 11 & 22 & 53 & 17 \\ 2 & 0 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{11}31 \ 5 \ 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{112}31 \ 5 \ 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 8 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{111}31 \ 414131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 10 & 5 \\ 4 & 9 & 1 & 10 & 23 & 11 \\ 2 & 4 & 0 & 5 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{12}31 \ 5 \ 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{112}31 \ 5 \ 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 6 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[385] \ 2 \ 3 \ 151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{33}31 \ 5 \ 1321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 2 & 1 & 15 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 81 & 21 \\ 6 & 15 & 0 & 81 & 79 & 21 \\ 2 & 4 & -1 & 21 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[376] \ 2 \ 3 \ 151321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 1 & 15 & 5 & 3 \\ 9 & 12 & 15 & 79 & 27 & 21 \\ 3 & 4 & 5 & 27 & 7 & 7 \\ 2 & 3 & 3 & 21 & 7 & 5 \end{pmatrix}_{b_0}$$

[386] $1^{14}3151321^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 36 & 7 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

[377] $1^{132}23151321^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 79 & 54 & 21 \\ 6 & 8 & 10 & 54 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0}$$

[378] $1^{133}23151321^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 9 & 2 \\ 2 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 1 & 15 & 15 & 3 \\ 9 & 12 & 15 & 79 & 81 & 21 \\ 9 & 12 & 15 & 81 & 79 & 21 \\ 2 & 3 & 3 & 21 & 21 & 5 \end{pmatrix}_{b_0}$$

[384] $1^{121}23151321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 1 & 10 & 5 & 3 \\ 6 & 8 & 10 & 34 & 18 & 14 \\ 3 & 4 & 5 & 18 & 7 & 7 \\ 2 & 3 & 3 & 14 & 7 & 5 \end{pmatrix}_{b_0}$$

[374] $1^{143}23151321^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 9 & 2 \\ 2 & 1 & 2 & 16 & 12 & 3 \\ 1 & 2 & 1 & 20 & 15 & 3 \\ 12 & 16 & 20 & 142 & 108 & 28 \\ 9 & 12 & 15 & 108 & 79 & 21 \\ 2 & 3 & 3 & 28 & 21 & 5 \end{pmatrix}_{b_0}$$

[373] $1^{142}23151321^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 6 & 2 \\ 2 & 1 & 2 & 16 & 8 & 3 \\ 1 & 2 & 1 & 20 & 10 & 3 \\ 12 & 16 & 20 & 142 & 72 & 28 \\ 6 & 8 & 10 & 72 & 34 & 14 \\ 2 & 3 & 3 & 28 & 14 & 5 \end{pmatrix}_{b_0}$$

[379] $1^{22}3151321^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 1 & 10 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 36 & 14 \\ 4 & 10 & 0 & 36 & 34 & 14 \\ 2 & 4 & -1 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[383] $1^{11}3151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[380] $1^{23}3151321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 1 & 10 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 54 & 14 \\ 6 & 15 & 0 & 54 & 79 & 21 \\ 2 & 4 & -1 & 14 & 21 & 5 \end{pmatrix}_{b_0}$$

[372] $1^{141}23151321^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 1 & 20 & 5 & 3 \\ 12 & 16 & 20 & 142 & 36 & 28 \\ 3 & 4 & 5 & 36 & 7 & 7 \\ 2 & 3 & 3 & 28 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{13}3151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 27 & 7 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{44}3151321^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 2 & 1 & 20 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 144 & 28 \\ 8 & 20 & 0 & 144 & 142 & 28 \\ 2 & 4 & -1 & 28 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{43}3151321^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 2 & 1 & 20 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 108 & 28 \\ 6 & 15 & 0 & 108 & 79 & 21 \\ 2 & 4 & -1 & 28 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[368] \ 1^{42}3151321^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 2 & 1 & 20 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 72 & 28 \\ 4 & 10 & 0 & 72 & 34 & 14 \\ 2 & 4 & -1 & 28 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[367] \ 2^3 3 151321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{12}3151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 18 & 7 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[375] \ 2^3 3 151321^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 12 & 2 \\ 2 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 1 & 20 & 20 & 3 \\ 12 & 16 & 20 & 142 & 144 & 28 \\ 12 & 16 & 20 & 144 & 142 & 28 \\ 2 & 3 & 3 & 28 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{11}3151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 5 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{11}3151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 4 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[390] $1^{111}31\bar{5}1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 5 & 4 & 7 & 9 & 7 \\ 3 & 5 & 4 & 9 & 7 & 7 \\ 2 & 4 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[389] $1^{112}31\bar{5}1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 13 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0}$$

[391] $1^{12}31\bar{5}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$$

[395] $1^{111}31\bar{5}1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 2 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[392] $1^{111}31\bar{5}1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 4 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[393] $1^{112}31\bar{5}1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

[394] $1^{112}31\bar{5}1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

[400] $1^{13}31\bar{5}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

[396] $1^{111}31\bar{5}1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 7 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{124}31 \ 5 \ 1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 10 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[398] \ 1^{112}31^{14141}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 14 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 15 & 5 \\ 7 & 14 & 1 & 15 & 55 & 17 \\ 2 & 4 & 0 & 5 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{123}31 \ 5 \ 1321^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 44 & 14 \\ 6 & 11 & 11 & 44 & 53 & 17 \\ 2 & 3 & 3 & 14 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{13}31 \ 5 \ 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{111}31 \ 5 \ 1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[402] \ 1^{111}31 \ 5 \ 1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{113}31 \ 5 \ 1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{112}31 \ 5 \ 1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{13}31 \ 5 \ 1321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{111} 31 \ 5 \ 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{112} 31 \ 5 \ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 9 & 5 & 21 & 23 & 11 \\ 5 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{123} 31 \ 5 \ 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 8 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{113} 23151^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 3 & 1 & 5 & 20 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 20 & 5 \\ 7 & 20 & 0 & 20 & 88 & 22 \\ 2 & 5 & -1 & 5 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{112} 23151^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 15 & 4 \\ 2 & 0 & 5 & 3 & 15 & 5 \\ 7 & 0 & 15 & 15 & 53 & 17 \\ 2 & 0 & 4 & 5 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{123} 31 \ 5 \ 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 9 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{143} 31 \ 5 \ 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 16 & 0 & 106 & 24 \\ 2 & 3 & 3 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{112} 31 \ 5 \ 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 8 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{113} 31 \ 5 \ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 2 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[416] $1^{114}31^15^{13}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 61, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[417] $1^{123}31^15^{13}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 8 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

[418] $1^{112}2321^15131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 17 & 4 \\ 2 & 0 & 5 & 3 & 17 & 5 \\ 8 & 0 & 17 & 17 & 67 & 19 \\ 2 & 0 & 4 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

[420] $1^{112}31^15^{13}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[419] $1^{112}31^15^{13}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[421] $1^{112}31^15^{13}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 7 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[422] $1^{112}31^15^{11}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 5 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[423] $1^{111}31^15^{12}1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 6 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$$

[424] $1^{111}31^15^{11}1321^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 9 & 5 & 21 & 23 & 11 \\ 4 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$$

[425] $1^{114}31\bar{5}1321^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 2 & 4 & 5 & 7 & 36 & 7 \\ 12 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

[426] $1^{122}31\bar{5}1321^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 5 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[427] $1^{113}31\bar{5}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 8 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[428] $1^{124}31\bar{5}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 9 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$$

[429] $1^{112}31\bar{5}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 88, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[430] $1^{113}31\bar{5}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 88, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[431] $1^{124}31\bar{5}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 10 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$$

[432] $1^{112}31\bar{5}1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 93, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 7 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

[434] $1^{123}31\ 5\ 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 7 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

[433] $1^{123}31\ 5\ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 1 & 10 & 15 & 3 \\ 5 & 8 & 10 & 34 & 54 & 14 \\ 9 & 12 & 15 & 54 & 79 & 21 \\ 2 & 3 & 3 & 14 & 21 & 5 \end{pmatrix}_{b_0}$$

[435] $1^{122}31\ 5\ 1321^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 2 & 2 & 14 & 14 & 4 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 14 & 9 & 58 & 60 & 18 \\ 7 & 14 & 9 & 60 & 58 & 18 \\ 2 & 4 & 2 & 18 & 18 & 5 \end{pmatrix}_{b_0}$$

[436] $1^{112}31\ 5\ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 6 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

[437] $1^{124}31\ 5\ 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 115, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 9 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$$

[438] $1^{114}23151^2231^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 30 & 4 \\ 2 & 0 & 5 & 3 & 30 & 5 \\ 14 & 0 & 30 & 30 & 218 & 34 \\ 2 & 0 & 4 & 5 & 34 & 5 \end{pmatrix}_{b_0}$$

[439] $1^{114}31\ 5\ 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 121, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[440] $1^{133}31\ 5\ 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 125, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 1 & 11 & 11 & 3 \\ 6 & 11 & 11 & 53 & 55 & 17 \\ 7 & 11 & 11 & 55 & 53 & 17 \\ 2 & 3 & 3 & 17 & 17 & 5 \end{pmatrix}_{b_0}$$

[441] $1^{122}31\ 5\ 1321^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 2 & 2 & 14 & 18 & 4 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 14 & 9 & 58 & 74 & 18 \\ 8 & 18 & 10 & 74 & 90 & 22 \\ 2 & 4 & 2 & 18 & 22 & 5 \end{pmatrix}_{b_0}$$

[442] $1^{113}31^151321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 150, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 8 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[443] $1^{112}31^151321^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 158, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 9 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[444] $1^{114}31^151321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[445] $1^{113}31^151321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 168, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[446] $1^{112}31^151321^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 8 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[447] $1^{114}31^151321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[448] $1^{114}31^151321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 262, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 5 & 7 & 36 & 7 \\ 11 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [449] $1^{111}41414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$

[450] $1^{112}31^1414141^131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.5$

[460] $15 \overset{1}{1} 131 \overset{1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[461] $1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 321 \overset{1}{6} 1^1 2321^{\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 0 & 3 & 5 & 2 \\ 2 & 2 & 3 & 3 & 9 & 5 \\ 6 & 4 & 5 & 9 & 19 & 11 \\ 2 & 2 & 2 & 5 & 11 & 5 \end{pmatrix}_{b_0}$

[462] $1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 321^1 \overset{1}{6} 12321^{\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 0 & 3 & 7 & 2 \\ 2 & 2 & 3 & 3 & 13 & 5 \\ 7 & 6 & 7 & 13 & 39 & 15 \\ 2 & 2 & 2 & 5 & 15 & 5 \end{pmatrix}_{b_0}$

[463] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{2}} 321^1 \overset{1}{6} 12321^{\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 11 & 3 \\ 2 & 2 & 5 & 14 & 4 \\ 4 & 5 & 7 & 22 & 7 \\ 11 & 14 & 22 & 62 & 20 \\ 3 & 4 & 7 & 20 & 6 \end{pmatrix}_{b_0}$

[464] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{4}} 321^1 \overset{1}{6} 1^1 2321^{\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 19 & 3 \\ 2 & 2 & 5 & 24 & 4 \\ 4 & 5 & 7 & 36 & 7 \\ 19 & 24 & 36 & 178 & 34 \\ 3 & 4 & 7 & 34 & 6 \end{pmatrix}_{b_0}$

[465] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 321^1 \overset{1}{6} 1^1 2321^{\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -54, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 18 & 3 \\ 2 & 2 & 5 & 21 & 4 \\ 4 & 5 & 7 & 33 & 7 \\ 18 & 21 & 33 & 145 & 31 \\ 3 & 4 & 7 & 31 & 6 \end{pmatrix}_{b_0}$

[466] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{3}} 321 \overset{1}{6} 1^1 2321^{\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 12 & 3 \\ 2 & 2 & 5 & 15 & 4 \\ 4 & 5 & 7 & 23 & 7 \\ 12 & 15 & 23 & 67 & 21 \\ 3 & 4 & 7 & 21 & 6 \end{pmatrix}_{b_0}$

[467] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{4}} 321^1 \overset{1}{6} 1^1 2321^{\textcolor{red}{1}\textcolor{blue}{4}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -196, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 22 & 3 \\ 2 & 2 & 5 & 26 & 4 \\ 4 & 5 & 7 & 42 & 7 \\ 22 & 26 & 42 & 220 & 38 \\ 3 & 4 & 7 & 38 & 6 \end{pmatrix}_{b_0}$

$T = 14$ [468] $1^1 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 5.0$

[472] $1^{\textcolor{red}{1}\textcolor{blue}{1}} \overset{1}{7} 12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[473] $1 \overset{1}{7} 1231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[474] $1 \overset{1}{7} 1231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$

[475] $1 \overset{1}{7} 1231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[476] $1 \overset{1}{7} 1231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[477] $21 \overset{1}{7} 1231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$

[478] $1 \overset{1}{7} 1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -41, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[479] $1 \overset{1}{5} 1321 \overset{1}{7} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[481] $2 \overset{1}{3} 21712321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 5 & 7 & 36 & 7 \\ 12 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$

[486] $2 \overset{1}{3} 21712321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$

[485] $2 \overset{1}{3} 21712321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[484] $2 \ 3 \ 21712321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[483] $2 \ 3 \ 21712321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[482] $2 \ 3 \ 21712321^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 27 & 54 & 21 \\ 6 & 8 & 10 & 27 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0}$$

[480] $2 \ 3 \ 21712321^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 12 & 2 \\ 2 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 1 & 10 & 20 & 3 \\ 6 & 8 & 10 & 34 & 72 & 14 \\ 12 & 16 & 20 & 72 & 142 & 28 \\ 2 & 3 & 3 & 14 & 28 & 5 \end{pmatrix}_{b_0}$$

$T = 15$ [487] $1^{131} 6^{11} 1316^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[494] $221^{11} 8^{11} 1231^{11} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 3 & 0 & 12 & 8 \\ 3 & 2 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[493] $21^{11} 8^{11} 1231^{11} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 4 & 2 & 2 \\ 4 & 3 & 4 & 11 & 5 & 7 \\ 1 & 1 & 2 & 5 & 1 & 3 \\ 2 & 2 & 2 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[492] $21^{11} 8^{11} 1231^{11} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[491] $1^{11} 8^{11} 12231^{11} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[490] $1^{\frac{1}{1}} 8^{11} 12231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[489] $21^{\frac{1}{1}} 8^{11} 12231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) =$
 $(1, 15, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

[488] $21^{\frac{1}{1}} 8^{11} 1231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 5 & 6 & 5 \\ 2 & 2 & 3 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

[498] $1^{\frac{1}{1}} 8^{11} 1231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[497] $21^{\frac{1}{1}} 8^{11} 1231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 3 & 2 & 3 & 5 & 2 & 5 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[496] $1^{\frac{1}{1}} 8^{11} 12231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 3 & 2 \\ 1 & 1 & 2 & 1 & 4 & 3 \\ 3 & 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[495] $1^{\frac{1}{1}} 8^{11} 12231^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[499] $1^{\frac{1}{1}} 5^{11} 13221^{\frac{1}{1}} 8^{11} 1^{\frac{1}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -38, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} =$

5.0
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [500] \quad 1^{\overline{1}} 1^{\overline{1}} 5^{1321} 8^{1321} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \quad T_{\max} = 16, \quad \Delta = -20, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 16, 1), \quad c_{\text{ext}} = 5.0$$

$$[504] \quad 1^{\overline{1}} 1^{\overline{1}} 6^{1316} 1316^{13151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \quad T_{\max} = 17, \quad \Delta = -39, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 16, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 3 & 5 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[505] \quad 1^{\overline{1}} 1^{\overline{1}} 6^{1316} 1316^{13151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \quad T_{\max} = 17, \quad \Delta = -39, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 16, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[506] \quad 31^{\overline{1}} 6^{11231} 1316131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, \quad T_{\max} = 16, \quad \Delta = -3, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 16, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 6 & 1 & 8 & 16 & 6 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 8 & 1 & 7 & 17 & 7 \\ 5 & 16 & 1 & 17 & 31 & 13 \\ 2 & 6 & 0 & 7 & 13 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [507] \quad 1^{\overline{1}} 1^{\overline{1}} 6^{12321} 8^{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, \quad T_{\max} = 17, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 17, 1), \quad c_{\text{ext}} = 5.0$$

$$[509] \quad 1^{\overline{1}} 1^{\overline{1}} 8^{123151} 13151^{13151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, \quad T_{\max} = 18, \quad \Delta = -40, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [514] \quad 1^{\overline{1}} 1^{\overline{1}} 5^{13161316131} 5^{13161316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -12, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 5.0$$

$$[513] \quad 1^{\overline{1}} 1^{\overline{1}} 8^{1321} 12321^{1321} 8^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -23, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 5.0$$

$$[512] \quad 1^{\overline{1}} 1^{\overline{1}} 8^{1321} 12321^{1321} 8^{12321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -25, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 2), \quad c_{\text{ext}} = 5.0$$

$$[511] \quad 31^{\overline{1}} 6^{13161316131} 131^{13161316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 5.0$$

$$[516] \quad 1^{\overline{1}} 1^{\overline{1}} 7^{12321} 8^{123151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -38, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [519] \quad 1^{1^{11}} 5^{13} 21812321812315^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 23, \quad T_{\max} = 23, \quad \Delta = -11, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 23, 1), \quad c_{\text{ext}} = 5.0$$

$$T = 24 \quad [520] \quad 1^{1^{11}} 6^{12} 321812321812315^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 24, \quad T_{\max} = 24, \quad \Delta = -10, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 24, 1), \quad c_{\text{ext}} = 5.0$$

$$T = 25 \quad [522] \quad 1^{1^{11}} 8^{12} 321812321812315^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -36, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 2 & 5 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [524] \quad 1^{1^{11}} 5^{13} 221(12) \quad 12231513221(12)122315^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -10, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 33, 1), \quad c_{\text{ext}} = 5.0$$

$$T = 36 \quad [525] \quad 1^{1^{11}} 5^{13} 221(12) \quad 12231513221(12)12231513^{11} 6^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 36, 1), \quad c_{\text{ext}} = 5.0$$

$$T = 44 \quad [526] \quad 1^{1^{11}} 5^{13} 221(12)12231513221(12)122315^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 44, 1), \quad c_{\text{ext}} = 5.0$$

3.2.14 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{13} 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 1^{13} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} 1^{13} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.50 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13} 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{13}1^{13}1^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{13}1^{13}1^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 6 & 9 & 0 & 22 & 10 & 10 \\ 3 & 4 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[8] $1^{11}1^{13}1^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 3 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[9] $1^{11}1^{13}1^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{11}1^{13}1^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11}1^{13}1^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [12] $1^{11}2^31^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{12}2^31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{11}2^31^31^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 & 4 \\ 3 & 0 & 5 & 3 & 5 & 5 \\ 3 & 0 & 4 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[15] $1^{11}2^31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [16] \quad 1^{11}22313^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 5.0$$

$$[18] \quad 1^{11}413^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}23^11321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 3 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 5 & -1 & 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[20] \quad 31^{11}3^1151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}3^114131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 0 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}5131^{11}3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}3^1141^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 5 & 3 & 5 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 31^{21}3^1151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12}5131^{21}3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 42 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

[26] $31^3 3^1 15^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 18 & 21 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[27] $1^{13} 5^1 13^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 18 & 21 & 97 & 0 & 21 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 0 & 4 \end{pmatrix}_{b_0}$

[28] $31^1 5^1 13^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 2 & 4 \\ 3 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[29] $31^1 5^1 13^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 2 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 14 & 8 \\ 3 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$

[30] $31^1 5^1 13^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 2 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 34 & 12 \\ 3 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$

[31] $231^{11} 3^1 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 275, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 1 & 5 & 3 & 2 \\ 5 & 33 & 1 & 35 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 20 & 11 \\ 3 & 20 & 0 & 20 & 10 & 6 \\ 2 & 11 & -1 & 11 & 6 & 3 \end{pmatrix}_{b_0}$

$T = 11$ [32] $1^{11} 23^1 141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 5.0$

[34] $1^{12} 31^1 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 6 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$

[35] $1^{12} 3151^1 23^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 74 & 20 \\ 3 & 5 & -1 & 20 & 5 \end{pmatrix}_{b_0}$

[36] $1^{13} 31^1 5^1 13^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 1 & 2 \\ 8 & 0 & 13 & 97 & 11 & 21 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 31^1 513^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11} 31^1 513^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 3 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 1^1 513^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^1 513^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 2 \\ 6 & 0 & 6 & 34 & 7 & 12 \\ 1 & 0 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 31^1 513^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 3 & 6 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 31^1 513^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 12 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 18 \\ 1 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & -1 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 2^3 31^1 513^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{123} 31^1 513^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 4 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{14} 31^1 513^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 8 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{11}3^{11}151^{12}231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 6 & 0 & 9 & 27 & 8 & 11 \\ 1 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14}31^{11}513^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 24 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 3 & -1 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{12}31^{11}513^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 8 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 12 \\ 1 & 1 & 0 & 6 & 0 & 2 \\ 2 & 3 & -1 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11}31^{11}513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 6 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[47] \ 2^3 3^{12} 1513^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 2 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12}3^{11}151^{12}231^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 12 & 0 & 18 & 114 & 16 & 22 \\ 1 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{11}3^{11}15131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 4 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{12}31^{11}513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 5 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{1\bar{1}3}1^{1\bar{1}}5131^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[55] $1^{1\bar{1}3}1413^{21\bar{1}1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 3 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 5 \end{pmatrix}_{b_0}$

[57] $1^{1\bar{1}3}1^{\bar{1}}513^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 2 \\ 4 & 0 & 5 & 17 & 5 & 9 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$

[56] $1^{1\bar{1}3}1^{\bar{1}}513^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 2 \\ 6 & 0 & 9 & 49 & 8 & 15 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$

[58] $1^{1\bar{1}3}1^{\bar{1}}31^{\bar{1}}513^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 9 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$

[59] $1^{1\bar{1}4}1^{\bar{1}}31^{\bar{1}}513^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 10 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$

[60] $1^{1\bar{1}3}1^{\bar{1}}151^{\bar{1}}231^{\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 2 & 1 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 8 & 11 \\ 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$

[61] $1^{1\bar{1}3}1^{\bar{1}}5123^{1\bar{1}1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[62] $1^{1\bar{1}3}1^{\bar{1}}513^{1\bar{1}1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$

[63] $1^{11}3^1151231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{12}3^1513^11^4 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 2 \\ 8 & 0 & 10 & 74 & 10 & 18 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^115131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{12}3^115131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 6 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{12}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 7 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{12}3^115131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 5 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{12}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 5 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{11}3151^{12}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 13 & 4 \\ 9 & 0 & 13 & 43 & 15 \\ 3 & 0 & 4 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12}3^115131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{12}3^1151^{11}231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 2 & 1 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 13 & 0 & 49 & 11 & 15 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & -1 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13}3^115131^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 10 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 9 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{13}3^115131^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{13}3^115131^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 8 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 8 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{13}3^115131^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -97, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{12}3151^{12}231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 18 & 4 \\ 12 & 0 & 18 & 74 & 20 \\ 3 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{13}3^1151^2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -115, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 2 & 1 & 22 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 22 & 0 & 154 & 19 & 26 \\ 1 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & -1 & 26 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{13}3^1151^1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -115, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 2 & 1 & 17 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 17 & 0 & 79 & 14 & 19 \\ 1 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & -1 & 19 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{14}3^115131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -138, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 12 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{14}3^115131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -138, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 10 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{14}3^115131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 9 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{14}3^115131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 11 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{14}3^115131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -158, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 11 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{14}3^115131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -162, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 9 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{14}3^115131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -167, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{14}3^1151^2231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 2 & 1 & 26 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 26 & 0 & 202 & 22 & 30 \\ 1 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & -1 & 30 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [92] \ 1^{43}1^15 \ 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[91] \ 1^{11}3141413^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[107] \ 1^{11}3^{121}6 \ 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{11}3^{121}6 \ 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{33}1^15 \ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 11 & 1 & 3 \\ 6 & 0 & 11 & 53 & 8 & 17 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 3 & 17 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{11}3^115 \ 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{12}3^115 \ 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 6 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{13}3^115 \ 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 9 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{14}3^{11}5^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 12 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [123] \ 1^{12}3^{16}1^{23}15^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -4, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

3.2.15 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix

$$T = 8 \quad [0] \ 1^{11}3^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 5 & 4 & 4 \\ 2 & 3 & -1 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 \\ 3 & 5 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12}3^{11}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 \\ 3 & 5 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}3^{11}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 \\ 3 & 5 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12}3^{11}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{13}3^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{14}3^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{13}3^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{14}3^121^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{14}3^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{14}3^121^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [35] $1^{11}4^{12}13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 7 & 6 & 5 & 18 & 10 \\ 3 & 3 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

[34] $1^{11}4^{11}13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 6 & 5 & 3 & 10 & 8 \\ 3 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$$

[36] $1^{13}4^{13}13^121^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 11 & 9 & 7 & 48 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[37] $1^{11}5^{11}13^113^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 3 & 3 \\ 5 & 5 & 8 & 3 & 5 \\ 6 & 8 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 5 & 8 & 7 & 5 \\ 6 & 8 & 9 & 10 & 7 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 5 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{12}} 4^{\overline{12}} 13^{\overline{12}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 10 & 6 & 46 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{13}} 4^{\overline{13}} 13^{\overline{13}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 12 & 11 & 8 & 60 & 18 \\ 3 & 3 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{12}} 5^{\overline{12}} 13^{\overline{12}} 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 9 & 3 \\ 5 & 5 & 8 & 11 & 5 \\ 6 & 8 & 9 & 16 & 7 \\ 9 & 11 & 16 & 22 & 10 \\ 3 & 5 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{13}} 4^{\overline{13}} 13^{\overline{13}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 15 & 13 & 8 & 90 & 22 \\ 3 & 3 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{\overline{13}} 4^{\overline{13}} 13^{\overline{13}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 13 & 13 & 9 & 76 & 20 \\ 3 & 3 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{\overline{14}} 4^{\overline{14}} 13^{\overline{14}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 16 & 14 & 10 & 108 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{\overline{12}} 5^{\overline{12}} 13^{\overline{12}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 12 & 3 & 3 \\ 5 & 5 & 16 & 3 & 5 \\ 12 & 16 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \ 31^{\overline{12}} 3^{\overline{11}} 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 12 & 5 & 3 \\ 5 & 5 & 16 & 7 & 5 \\ 12 & 16 & 42 & 20 & 14 \\ 5 & 7 & 20 & 8 & 6 \\ 3 & 5 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[47] \ 31^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 11 & 3 \\ 5 & 5 & 8 & 15 & 5 \\ 6 & 8 & 9 & 21 & 7 \\ 11 & 15 & 21 & 39 & 13 \\ 3 & 5 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

- [48] $1^{13} 4^{13} 13^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 102$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 16 & 15 & 9 & 106 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$
- [49] $1^{13} 4^{14} 13^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 110$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 17 & 16 & 11 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$
- [50] $1^{14} 4^{13} 13^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 128$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 17 & 17 & 10 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$
- [51] $1^{13} 4^{14} 13^{13} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 133$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 20 & 18 & 11 & 168 & 30 \\ 3 & 3 & 2 & 30 & 5 \end{pmatrix}_{b_0}$
- [52] $1^{14} 4^{13} 13^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 18 & 18 & 12 & 148 & 28 \\ 3 & 3 & 2 & 28 & 5 \end{pmatrix}_{b_0}$
- [53] $1^{13} 5^{13} 13^{13} 3^{13} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 3 & 3 \\ 5 & 5 & 24 & 3 & 5 \\ 18 & 24 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$
- [54] $31^{13} 3^{13} 1^{13} 5^{13} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 5 & 3 \\ 5 & 5 & 24 & 7 & 5 \\ 18 & 24 & 97 & 30 & 21 \\ 5 & 7 & 30 & 8 & 6 \\ 3 & 5 & 21 & 6 & 4 \end{pmatrix}_{b_0}$
- [55] $1^{14} 4^{13} 13^{13} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 21 & 20 & 12 & 190 & 32 \\ 3 & 3 & 2 & 32 & 5 \end{pmatrix}_{b_0}$
- [56] $1^{13} 3^{13} 1^{13} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$
- [57] $1^{11} 3^{13} 1^{13} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 2 \\ 2 & 0 & 4 & 6 & 7 & 6 \\ 2 & 0 & 3 & 7 & 5 & 5 \\ 2 & 0 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{111}} 3^1 1^{\overline{12}} 5^1 13^1 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 6 & 2 \\ 2 & 0 & 4 & 6 & 15 & 6 \\ 5 & 0 & 6 & 15 & 29 & 11 \\ 2 & 0 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{113}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{111}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{\overline{112}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 & \\ 2 & 0 & 4 & 6 & 23 & 6 & \\ 7 & 0 & 10 & 23 & 67 & 17 & \\ 2 & 0 & 2 & 6 & 17 & 4 & \end{pmatrix}_{b_0}$$

$$[62] \ 1^{\overline{111}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{\overline{13}} 1^{\overline{11}} 5^1 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 4 & 3 \\ 2 & 0 & 3 & 4 & 5 & 5 \\ 2 & 0 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{\overline{111}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 8 & 2 \\ 2 & 0 & 4 & 6 & 19 & 6 \\ 6 & 0 & 8 & 19 & 46 & 14 \\ 2 & 0 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{\overline{112}} 3^1 1^{\overline{11}} 5^1 13^1 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 11 & 2 \\ 2 & 0 & 4 & 6 & 27 & 6 \\ 9 & 0 & 11 & 27 & 96 & 20 \\ 2 & 0 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[86] $1^{12}3^1 1^1 5 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 2 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^1 1^1 5 131^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 46 & 12 \\ 6 & 11 & 0 & 46 & 65 & 17 \\ 2 & 3 & -1 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

[89] $1^{114}3^1 5 13^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 6 \\ 9 & 17 & 0 & 35 & 152 & 26 \\ 2 & 3 & -1 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{14}3^1 1^1 5 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 2 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{134}3^1 5 13^1 1^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 105 & 18 \\ 9 & 17 & 0 & 105 & 152 & 26 \\ 2 & 3 & -1 & 18 & 26 & 4 \end{pmatrix}_{b_0}$$

[92] $1^{142}3^1 5 13^1 1^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & 1 & 16 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 76 & 24 \\ 5 & 9 & 0 & 76 & 44 & 14 \\ 2 & 3 & -1 & 24 & 14 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{124}3^1 5 13^1 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 1 & 8 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 12 \\ 9 & 17 & 0 & 70 & 152 & 26 \\ 2 & 3 & -1 & 12 & 26 & 4 \end{pmatrix}_{b_0}$$

[94] $1^{122}3^1 5 13^1 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 12 \\ 5 & 9 & 0 & 38 & 44 & 14 \\ 2 & 3 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{144}3^1 5 13^1 1^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 1 & 16 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 140 & 24 \\ 9 & 17 & 0 & 140 & 152 & 26 \\ 2 & 3 & -1 & 24 & 26 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{143}3^11^15131^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 124 & 24 \\ 8 & 15 & 0 & 124 & 119 & 23 \\ 2 & 3 & -1 & 24 & 23 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{142}3^11^15131^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 92 & 24 \\ 6 & 11 & 0 & 92 & 65 & 17 \\ 2 & 3 & -1 & 24 & 17 & 4 \end{pmatrix}_{b_0}$$

[98] $1^{121}3^11^15131^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 12 \\ 4 & 7 & 0 & 30 & 27 & 11 \\ 2 & 3 & -1 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 6 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[100] $1^{112}31^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 6 \\ 5 & 9 & 0 & 19 & 44 & 14 \\ 2 & 3 & -1 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[101] $1^{113}31^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 6 \\ 7 & 13 & 0 & 27 & 90 & 20 \\ 2 & 3 & -1 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[102] $1^{133}31^1513^11^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 81 & 18 \\ 7 & 13 & 0 & 81 & 90 & 20 \\ 2 & 3 & -1 & 18 & 20 & 4 \end{pmatrix}_{b_0}$$

[103] $1^{143}3^11^15131^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 24 \\ 2 & 3 & 0 & 28 & 5 & 5 \\ 2 & 3 & -1 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{123}1^15131^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 6 & 0 & 9 & 49 & 18 & 15 \\ 2 & 0 & 3 & 18 & 5 & 5 \\ 2 & 0 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[105] $1^{123}1^15131^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 12 \\ 2 & 3 & 0 & 14 & 5 & 5 \\ 2 & 3 & -1 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{123}1^15131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 2 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

[83] $1^{121}31^15131^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 8 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 12 \\ 3 & 5 & 0 & 22 & 14 & 8 \\ 2 & 3 & -1 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[82] $1^{123}31^15131^{24} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 62 & 12 \\ 8 & 15 & 0 & 62 & 119 & 23 \\ 2 & 3 & -1 & 12 & 23 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{132}31^15131^{32} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 1 & 12 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 18 \\ 5 & 9 & 0 & 57 & 44 & 14 \\ 2 & 3 & -1 & 18 & 14 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{131}31^15131^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 1 & 12 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 18 \\ 3 & 5 & 0 & 33 & 14 & 8 \\ 2 & 3 & -1 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[79] $1^{133}31^15131^{34} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 93 & 18 \\ 8 & 15 & 0 & 93 & 119 & 23 \\ 2 & 3 & -1 & 18 & 23 & 4 \end{pmatrix}_{b_0}$$

[88] $1^{143}31^15131^{43} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 1 & 1 & 16 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 108 & 24 \\ 7 & 13 & 0 & 108 & 90 & 20 \\ 2 & 3 & -1 & 24 & 20 & 4 \end{pmatrix}_{b_0}$$

[78] $1^{132}3^11^15131^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 69 & 18 \\ 6 & 11 & 0 & 69 & 65 & 17 \\ 2 & 3 & -1 & 18 & 17 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{132}3^11^15131^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 45 & 18 \\ 4 & 7 & 0 & 45 & 27 & 11 \\ 2 & 3 & -1 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{113}3^11^15131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 6 \\ 8 & 15 & 0 & 31 & 119 & 23 \\ 2 & 3 & -1 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{141}31^15131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 1 & 16 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 24 \\ 3 & 5 & 0 & 44 & 14 & 8 \\ 2 & 3 & -1 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{112}3^11^15131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 6 \\ 6 & 11 & 0 & 23 & 65 & 17 \\ 2 & 3 & -1 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{141}3^11^15131^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 24 \\ 4 & 7 & 0 & 60 & 27 & 11 \\ 2 & 3 & -1 & 24 & 11 & 4 \end{pmatrix}_{b_0}$$

[87] $1^{113}3^11^15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 2 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{123}31^15131^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 12 \\ 7 & 13 & 0 & 54 & 90 & 20 \\ 2 & 3 & -1 & 12 & 20 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{11}3^{11}5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 6 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{13}3^{11}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 2 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{13}3^{11}5^{13}131^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 18 \\ 2 & 3 & 0 & 21 & 5 & 5 \\ 2 & 3 & -1 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{11}3^{11}5^{12}131^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 0 & 6 & 34 & 15 & 12 \\ 2 & 0 & 3 & 15 & 5 & 5 \\ 2 & 0 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^{11}5^{11}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 6 \\ 4 & 7 & 0 & 15 & 27 & 11 \\ 2 & 3 & -1 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{11}3^{11}5^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[111] $1^{11}3^{11}5^{11}131^{11}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 9 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 13 & 11 \\ 2 & 5 & 0 & 13 & 5 & 5 \\ 2 & 4 & -1 & 11 & 5 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{11}3^{11}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[109] $1^{13}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 2 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[108] $1^{114}31^3513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[107] $1^{111}31^1513^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[106] $1^{111}3^11^1513^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 2 \\ 4 & 0 & 5 & 17 & 11 & 9 \\ 2 & 0 & 3 & 11 & 5 & 5 \\ 2 & 0 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[112] $1^{123}3^11^1513^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 8 & 13 & 3 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 5 & 8 & 8 & 30 & 46 & 12 \\ 8 & 13 & 10 & 46 & 67 & 17 \\ 2 & 3 & 2 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

[114] $1^{111}31^1513^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[113] $1^{111}31^1513^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 2 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[115] $1^{13}3^11^1513^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 2 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

[116] $1^{123}3^11^1513^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 2 \\ 8 & 0 & 10 & 74 & 22 & 18 \\ 2 & 0 & 3 & 22 & 5 & 5 \\ 2 & 0 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[117] $1^{111}3^11^1513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{111} 3^1 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{111} 3^1 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{111} 3^1 5^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 5 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{112} 3^1 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 8 & 3 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 2 & 3 & 1 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 27 & 11 \\ 2 & 3 & 2 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{143} 3^1 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 2 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{11} 3^1 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 5 & 0 & 14 & 8 \\ 2 & 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{141} 3^1 5^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 2 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{121} 3^1 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 2 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0} \\
[127] \quad & 1^{11} 3^1 151^1 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 5 & 0 & 9 & 27 & 8 & 11 \\ 2 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{111} 31^1 513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 2 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{131} 31^1 513^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 2 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{12} 3^1 151^2 231^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 10 & 0 & 18 & 114 & 16 & 22 \\ 2 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0} \\
[130] \quad & 1^{13} 3^1 151^3 231^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 & 3 \\ 15 & 0 & 27 & 259 & 24 & 33 \\ 2 & 0 & 3 & 24 & 1 & 3 \\ 2 & 0 & 3 & 33 & 3 & 4 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{111} 31^1 513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 4 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{112} 31^1 513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 2 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0} \\
[133] \quad & 1^{113} 3^1 1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0} \\
[134] \quad & 1^{121} 31^1 513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5
\end{aligned}$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 5 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{121} 3^{11} 5^{131} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{131} 3^{11} 5^{131} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{121} 3^{12} 5^{131} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 2 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{123} 3^{11} 5^{131} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 9 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 6 & 0 & 29 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{123} 3^{11} 5^{131} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 7 & 0 & 27 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{111} 3^{11} 5^{131} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{111} 3^{11} 5^{131} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{131} 3^{11} 5^{131} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 6 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{113}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$

[144] $1^{122}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$

[145] $1^{121}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -46, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$

[146] $1^{111}3^1151^1231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 2 & 0 & 4 & 2 & 12 & 4 \\ 7 & 0 & 12 & 12 & 46 & 14 \\ 2 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[147] $1^{13}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$

[148] $1^{112}3^1151^1231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 7 & 16 & 0 & 16 & 74 & 18 \\ 2 & 4 & -1 & 4 & 18 & 4 \end{pmatrix}_{b_0}$

[149] $1^{13}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$

[150] $1^{121}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 6 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$

[151] $1^{12}3^11^15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 9 & 0 & 44 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

[152] $2 \ 3 \ 1513^{\overline{1^{121}}}1^{\overline{1^{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[153] $2 \ 3 \ 1513^{\overline{1^{141}}}1^{\overline{1^{14}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{\overline{1^{11}}}3^{\overline{1^{12}}}1^{\overline{1^{11}}}5 \ 131^{\overline{1^{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 5 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[155] $2 \ 3 \ 1513^{\overline{1^{111}}}1^{\overline{1^{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[156] $2 \ 3 \ 1513^{\overline{1^{131}}}1^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{\overline{1^{13}}}31^{\overline{1^{13}}}5 \ 131^{\overline{1^{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 2 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[158] $1^{\overline{1^{14}}}31^{\overline{1^{14}}}5 \ 131^{\overline{1^{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 14 & 10 & 74 & 22 & 18 \\ 2 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[161] $1^{\overline{1^{131}}}31^{\overline{1^{21}}}5 \ 131^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 8 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[160] $1^{\overline{1^{131}}}31^{\overline{1^{21}}}5 \ 131^{\overline{1^{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 6 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{121}31^1513^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 7 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{121}31^1513^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{113}3^11^15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 8 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{113}31^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{111}3^11^15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{121}31^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{113}31^1513^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{114}3^11^15131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[169] $1^{131}315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[170] $1^{13}315131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[171] $1^{121}315131^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 5 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[172] $1^{121}315131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[173] $1^{13}315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 13 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 13 & 10 & 0 & 67 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[174] $1^{121}315131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[176] $1^{123}315131^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 8 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

[175] $1^{122}315131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 29 & 9 \\ 7 & 10 & 9 & 29 & 44 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{112}315131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{141} 31^1 5 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 11 & 15 & 13 & 97 & 25 & 21 \\ 2 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{114} 31^1 5 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{113} 3^1 1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{114} 3^1 1^1 5 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 12 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{121} 3^1 1^1 5 131^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{112} 31^1 5 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 5 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{122} 31^1 5 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 5 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{122} 3^1 1^1 5 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 9 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 9 & 6 & 24 & 29 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

[189] $1^{113}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[186] $1^{113}31^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[187] $1^{13}3^11^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 13 & 0 & 90 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[188] $1^{131}3^11^1513^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 6 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

[190] $1^{141}31^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[191] $1^{131}31^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[192] $1^{14}31^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 12 & 0 & 92 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[193] $1^{113}3^11^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[194] $1^{141}31^1513^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 11 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{131}31^1513^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 7 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{141}31^1513^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 9 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[197] \quad 1^{112}31^1513^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{122}31^15131^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 38 & 12 \\ 7 & 10 & 9 & 38 & 44 & 14 \\ 2 & 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[199] \quad 1^{131}31^15131^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -127, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 8 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[200] \quad 1^{143}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 17 & 3 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 17 & 14 & 0 & 121 & 23 \\ 2 & 3 & 2 & 0 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[202] \quad 1^{141}31^15131^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 8 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[201] \quad 1^{131}31^15131^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[204] \quad 1^{131} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -135, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[203] \quad 1^{113} 3^1 1^5 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -135, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{113} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[206] \quad 1^{112} 2 \quad 3 \quad 15 \quad 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{122} 2 \quad 3 \quad 15 \quad 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 6 & 2 \\ 2 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 5 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[208] \quad 1^{113} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -153, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[209] \quad 1^{131} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -155, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[210] \quad 1^{141} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -161, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 9 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[211] \quad 1^{112} 3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -162, \quad (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[212] \quad 1^{13} 3^1 1^5 131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[213] \quad 1^{123} 3^1 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 9 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[214] \quad 1^{141} 3^1 1^5 131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[215] \quad 1^{123} 3^1 1^5 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 7 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[216] \quad 1^{113} 3^1 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -177, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[217] \quad 1^{141} 3^1 1^5 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -182, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 10 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[218] \quad 1^{133} 3^1 1^5 131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -183, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 13 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 59 & 15 \\ 8 & 13 & 10 & 59 & 67 & 17 \\ 2 & 3 & 2 & 15 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[219] \quad 1^{141} 3^1 1^5 131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -195, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 11 & 16 & 16 & 126 & 44 & 24 \\ 4 & 6 & 5 & 44 & 14 & 8 \\ 2 & 3 & 2 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{114}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 11 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[221] \quad 1^{114}3^{11}5^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -207, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 4 & 4 & 6 & 35 & 6 \\ 13 & 18 & 17 & 35 & 152 & 26 \\ 2 & 3 & 2 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{141}3^{11}5^{13}1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -217, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 9 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[223] \quad 1^{123}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 8 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[224] \quad 1^{141}3^{11}5^{13}1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{124}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 16 & 3 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 16 & 11 & 43 & 96 & 20 \\ 2 & 3 & 2 & 9 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{113}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[227] \quad 1^{124}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -325, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 17 & 3 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 7 & 5 & 17 & 48 & 9 \\ 11 & 17 & 14 & 48 & 121 & 23 \\ 2 & 3 & 2 & 9 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{114}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -333, \quad (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 4 & 4 & 6 & 31 & 6 \\ 11 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [230] \quad 1^{11} 3^{11} 1^1 5^{11} 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$$

$$[229] \quad 1^{11} 3^{11} 1^1 5^{11} 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[231] \quad 1^{11} 3^{11} 1^2 5^{12} 13221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[232] \quad 1^{11} 3^{11} 1^3 5^{13} 13221^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[233] \quad 1^{11} 3^{11} 1^3 5^{13} 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 2 & 2 & 13 & 5 & 4 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 13 & 6 & 43 & 18 & 15 \\ 2 & 5 & 3 & 18 & 6 & 6 \\ 2 & 4 & 2 & 15 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[234] \quad 1^{11} 3^{11} 1^1 5^{12} 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 2 & 2 & 13 & 5 & 4 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 6 & 13 & 6 & 43 & 18 & 15 \\ 3 & 5 & 3 & 18 & 6 & 6 \\ 2 & 4 & 2 & 15 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{11} 3^{11} 1^1 5^{12} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 2 & 5 & 4 & 7 & 8 & 7 \\ 3 & 5 & 3 & 8 & 6 & 6 \\ 2 & 4 & 2 & 7 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{12} 3^{11} 1^1 5^{11} 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 2 & 2 & 14 & 5 & 4 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 6 & 14 & 9 & 58 & 21 & 18 \\ 3 & 5 & 3 & 21 & 6 & 6 \\ 2 & 4 & 2 & 18 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{11} 3^{11} 1^1 5^{11} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 5 & 4 & 7 & 8 & 7 \\ 2 & 5 & 3 & 8 & 6 & 6 \\ 2 & 4 & 2 & 7 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{12} 3^{11} 1^3 5^{13} 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 2 & 2 & 14 & 5 & 4 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 14 & 9 & 58 & 21 & 18 \\ 2 & 5 & 3 & 21 & 6 & 6 \\ 2 & 4 & 2 & 18 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{11} 3^{11} 1^1 5^{11} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 4 & 5 & 4 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 4 & 1 & 2 & 5 & 4 \\ 2 & 5 & 3 & 5 & 6 & 6 \\ 2 & 4 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{13} 1^{11} 5^{131} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 4 & 5 & 4 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 4 & 1 & 2 & 5 & 4 \\ 3 & 5 & 3 & 5 & 6 & 6 \\ 2 & 4 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{12} 1^{13} 1^{131} 5^{131} 1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 7 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{12} 1^{13} 1^{131} 5^{131} 1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 5 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{11} 1^{13} 1^{11} 5^{131} 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 6 & 0 & 15 & 9 \\ 2 & 3 & 3 & 0 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{13} 1^{13} 1^{121} 5^{131} 1321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 8 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{11} 1^{13} 1^{11} 5^{131} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{14} 1^{13} 1^{131} 5^{131} 1321^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 1 & 20 & 6 & 3 \\ 11 & 16 & 20 & 142 & 48 & 28 \\ 4 & 6 & 6 & 48 & 15 & 9 \\ 2 & 3 & 3 & 28 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{11} 1^{13} 1^{21} 6^{131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[248] $1^{11}3^{11}1^{121}5^{121}1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 4 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[249] $1^{11}3^{11}1^{12}5^{121}1321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 9 & 5 & 21 & 13 & 11 \\ 2 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

[251] $1^{12}3^{11}1^{14}5^{1321}1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 18 & 10 & 90 & 26 & 22 \\ 2 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

[252] $1^{11}3^{11}1^{11}5^{1321}1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 9 & 5 & 21 & 13 & 11 \\ 3 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

[250] $1^{12}3^{11}1^{121}5^{141}1321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 18 & 10 & 90 & 26 & 22 \\ 3 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

[253] $1^{11}1^{11}3^{121}6^{131}1^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 5 & 3 & 7 & 7 \\ 3 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[259] $1^{122}3^{11}1^{12}5^{1321}1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 42 & 14 \\ 7 & 10 & 11 & 42 & 48 & 16 \\ 2 & 3 & 3 & 14 & 16 & 5 \end{pmatrix}_{b_0}$$

[254] $1^{12}3^{11}1^{12}5^{1321}1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 11 & 0 & 48 & 16 \\ 2 & 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$$

[255] $1^{11}1^{11}3^{11}1^{12}5^{1321}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[256] $1^{112}3^11^151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

[257] $1^{112}3^11^151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 4 & 14 & 4 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 4 & 1 & 2 & 14 & 4 \\ 7 & 14 & 8 & 14 & 55 & 17 \\ 2 & 4 & 2 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[258] $1^{122}3^11^151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 31 & 10 \\ 7 & 10 & 11 & 31 & 48 & 16 \\ 2 & 3 & 3 & 10 & 16 & 5 \end{pmatrix}_{b_0}$$

[260] $1^{111}3^11^151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[261] $1^{113}3^{121}1^16^1131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[262] $1^{111}3^{121}1^16^1131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 8 & 7 & 5 & 21 & 11 \\ 3 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[263] $1^{131}3^11^151321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 9 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

[264] $1^{111}3^11^151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[265] $1^{121}3^11^151321^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) =$

$$(1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 6 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{123}3^{11}5^{13}1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 14 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 60 & 14 \\ 10 & 14 & 16 & 60 & 99 & 23 \\ 2 & 3 & 3 & 14 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{13}3^{11}5^{13}1321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 16 & 0 & 99 & 23 \\ 2 & 3 & 3 & 0 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{113}3^{11}5^{13}1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{11}3^{11}6^{21}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 7 & 2 \\ 9 & 12 & 1 & 7 & 56 & 16 \\ 2 & 3 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{11}3^{11}6^{21}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 11 & 6 & 42 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{114}3^{11}5^{14}1321^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 1 & 5 & 21 & 3 \\ 2 & 4 & 5 & 7 & 39 & 7 \\ 13 & 18 & 21 & 39 & 168 & 30 \\ 2 & 3 & 3 & 7 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{112}3^{11}5^{12}1321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{111}3^{11}5^{12}1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 9 & 5 & 21 & 36 & 11 \\ 7 & 14 & 8 & 36 & 55 & 17 \\ 2 & 4 & 2 & 11 & 17 & 5 \end{pmatrix}_{b_0}$$

[274] $1^{113}3^{11}5^{13}1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 196, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [275] $1^{11}413^{11}151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 3 \\ 2 & 2 & 5 & 4 & 4 \\ 4 & 5 & 7 & 6 & 7 \\ 4 & 4 & 6 & 4 & 6 \\ 3 & 4 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

[276] $1^{14}1^{13}1^{11}151321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 8 & 4 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 8 & 11 & 35 & 14 & 15 \\ 4 & 4 & 14 & 4 & 6 \\ 3 & 4 & 15 & 6 & 6 \end{pmatrix}_{b_0}$$

[277] $1^{111}413^{11}151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 2 & 2 & 3 & 3 & 6 & 5 \\ 4 & 3 & 3 & 6 & 7 & 7 \\ 2 & 2 & 2 & 5 & 7 & 5 \end{pmatrix}_{b_0}$$

[278] $1^{111}41^{13}1^{11}151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 4 & 2 & 3 & 7 & 8 & 7 \\ 4 & 3 & 3 & 8 & 7 & 7 \\ 2 & 2 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[279] $1^{121}41^{13}1^{11}151321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 8 & 6 & 9 & 47 & 20 & 17 \\ 4 & 3 & 3 & 20 & 7 & 7 \\ 2 & 2 & 2 & 17 & 7 & 5 \end{pmatrix}_{b_0}$$

[280] $1^{11}41^{13}1^{11}151321^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 4 & 3 \\ 2 & 2 & 16 & 4 & 4 \\ 12 & 16 & 74 & 20 & 22 \\ 4 & 4 & 20 & 4 & 6 \\ 3 & 4 & 22 & 6 & 6 \end{pmatrix}_{b_0}$$

[281] $1^{111}41^{13}1^{11}151321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 4 & 6 & 22 & 14 & 12 \\ 4 & 3 & 3 & 14 & 7 & 7 \\ 2 & 2 & 2 & 12 & 7 & 5 \end{pmatrix}_{b_0}$$

[282] $1^{112}41^{13}1^{11}151321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [286] \quad 1^{11} 3^{11} 15^{131} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 1 & 5 & 4 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 5 & 3 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{11} 3^{11} 321^{11} 6131513^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 & 3 \\ 2 & 2 & 9 & 4 & 4 \\ 7 & 9 & 27 & 10 & 13 \\ 3 & 4 & 10 & 3 & 5 \\ 3 & 4 & 13 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{12} 3^{11} 3161^{11} 231513^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 9 & 11 & 40 & 12 & 16 \\ 3 & 4 & 12 & 3 & 5 \\ 3 & 4 & 16 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{12} 3^{11} 3161^{11} 231513^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 & 3 \\ 2 & 2 & 13 & 4 & 4 \\ 12 & 13 & 58 & 15 & 20 \\ 3 & 4 & 15 & 3 & 5 \\ 3 & 4 & 20 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{11} 3^{11} 3161^{11} 231513^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [289] \quad 1^{11} 3^{11} 15^{1321} 8^{11} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[290] \quad 1^{11} 3^{11} 15^{1321} 8^{11} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[288] \quad 1^{11} 3^{11} 15^{1321} 8^{11} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

3.2.16 NHC(-2)(-3) + su₂ + su₃

$$T = 7 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 10, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [4] \quad 1^{\overline{1}1} 3 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}1} 3 21^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 5 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [6] \quad 1^{\overline{1}1} 2 3 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1}1} 2 3 21^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{1}1} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{1}1} 4 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 2 \\ 3 & 2 & 2 & 1 & 3 \\ 3 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1}1} 2 3 21^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [11] \quad 1^{\overline{1}1} 5 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{1}1} 5 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^{11} 31^1 5 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^1 5 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 31^1 5 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -37, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [17] \quad 1^{11} 31^1 5 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 6.0$$

$$[19] \quad 1^{11} 6 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 6 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -34, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 31^1 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -9, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 1 & 5 & 0 & 3 \\ 2 & 4 & 5 & 7 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 7 & -1 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [21] \quad 1^{11} 6 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -8, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

- [10] $1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$
- [12] $1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -120$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [13] $1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -150$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [14] $21^{12} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -187$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 4 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$
- [15] $1^{131} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -187$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$
- [16] $1^{13} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -220$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{14} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -294$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{131} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -307$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$
- [19] $21^{131} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -93$, $\det = -400$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$
- $T = 8$ [21] $1^{11} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -65$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[20] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[22] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[24] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[26] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[27] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[29] $1^1 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^{11} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -65$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{11} 23^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -38$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{11} 23^{11} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -38$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{11} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 6 & 4 & 6 \\ 4 & 2 & 4 \\ 6 & 4 & 6 \end{pmatrix}_{b_0}$$

[36] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3^{11} 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -36$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 4 & 0 & 2 & 4 \\ 2 & 2 & 3 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[41] $1^{11} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

- [40] $1^{\overline{1}1} 4^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$
- [42] $1^{\overline{1}1} 3^{\overline{1}1} 2^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -36$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$
- [43] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 3 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$
- [44] $1^{\overline{1}1} 2^{\overline{1}1} 3^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -38$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$
- [45] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 6 & 6 \\ 4 & 3 & 6 & 5 \end{pmatrix}_{b_0}$
- [46] $1^{\overline{1}1} 4^{\overline{1}1} 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 3 & 3 \\ 5 & 3 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [47] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 1 & 4 & 3 \\ 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [48] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 12 & 8 \\ 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$
- [49] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$
- [50] $1^{\overline{1}1} 4^{\overline{1}1} 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$
- [51] $1^{\overline{1}1} 4^{\overline{1}1} 13^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -63$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 14 & 8 \\ 3 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$

[52] $1^{\overline{11}} 1^{\overline{11}} 31^{\overline{1}} 4^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 9 & 7 \\ 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[53] $1^{\overline{1}} 1^{\overline{12}} 31^{\overline{1}} 4^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 1 & 7 & 3 \\ 10 & 7 & 21 & 11 \\ 4 & 3 & 11 & 5 \end{pmatrix}_{b_0}$

[54] $1^{\overline{11}} 1^{\overline{11}} 31^{\overline{1}} 4^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 27 & 11 \\ 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$

$T = 10$ [55] $1^{\overline{11}} 5^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[56] $21^{\overline{1}} 5^{\overline{11}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[57] $1^{\overline{1}} 5^{\overline{1}} 1231^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

[58] $1^{\overline{1}} 5^{\overline{1}} 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[59] $1^{\overline{1}} 5^{\overline{12}} 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[60] $1^{\overline{1}} 5^{\overline{1}} 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[62] $1^{\overline{1}} 5^{\overline{1}} 1231^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$

- [61] $1^1 \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$
- [64] $1^1 \bar{5} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [63] $1^1 \bar{4} 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 6 & 4 \\ 3 & 2 & 5 & 4 \\ 6 & 5 & 10 & 8 \\ 4 & 4 & 8 & 6 \end{pmatrix}_{b_0}$
- [65] $1^1 \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$
- [67] $1^1 \bar{5} 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [68] $21^1 \bar{5} 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [66] $1^1 \bar{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [71] $1^1 \bar{5} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$
- [72] $1^1 \bar{5} 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [73] $1^1 \bar{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$
- [74] $21^1 \bar{5} 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[70] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[69] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1 \overset{1}{\underset{11}{4}} 1321^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[76] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[77] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[78] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[79] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[80] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 7 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 1 \overset{1}{\underset{11}{5}} 1231^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 1 \overset{1}{\underset{11}{5}} 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

- [83] $1^{11}_{11} 5^{11} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [84] $1^{12}_{11} 5^{11} 131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 4 & 8 & 8 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [85] $1^{13}_{11} 5^{11} 131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 15 & 5 \\ 4 & 8 & 8 & 18 & 6 \\ 9 & 15 & 18 & 34 & 12 \\ 3 & 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$
- [86] $1^{14}_{11} 5^{11} 131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 20 & 5 \\ 4 & 8 & 8 & 24 & 6 \\ 12 & 20 & 24 & 62 & 16 \\ 3 & 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$
- [87] $1^{14}_{11} 5^{11} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [88] $1^{11}_{11} 5^{11} 1^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 15 & 0 & 37 & 13 \\ 3 & 5 & -1 & 13 & 4 \end{pmatrix}_{b_0}$
- [89] $1^{13}_{11} 5^{11} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [90] $1^{11}_{11} 5^{11} 1^{11} 231^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 1 & 11 & 3 \\ 8 & 15 & 11 & 37 & 13 \\ 3 & 5 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [91] $1^{11}_{11} 5^{11} 1^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 \\ 1 & 2 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 \\ 4 & 5 & 5 & 0 & 18 \\ 2 & 2 & 2 & -1 & 8 \end{pmatrix}_{b_0}$
- [94] $1^{11}_{11} 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{1^3} 5^{14} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 45, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{1^2} 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 45, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{1^1} 5^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 53, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 7 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{1^4} 5^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 53, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 10 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{1^1} 5^1 1^{231} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 59, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 63 & 17 \\ 3 & 5 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{1^1} 5^1 1^{231} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 61, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 25 & 0 & 97 & 21 \\ 3 & 5 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{121} 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 61, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 11 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 11 & 0 & 49 & 11 & 13 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 3 & -1 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{1^4} 5^{11} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 62, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1^3} 5^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 69, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 9 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{1^2} 5^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 69, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

- [104] $1^{12} \overset{13}{5} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 9 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$
- [103] $1^{12} \overset{11}{5} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$
- [106] $1^{12} \overset{12}{5} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$
- [105] $1^{11} \overset{1}{5} 1^2 231^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 1 & 22 & 3 \\ 16 & 30 & 22 & 154 & 26 \\ 3 & 5 & 3 & 26 & 4 \end{pmatrix}_{b_0}$
- [108] $1^{11} \overset{14}{5} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$
- [107] $1^{11} \overset{13}{5} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 11 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$
- [109] $1^{11} \overset{11}{5} 1^2 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$
- [111] $1^{12} \overset{11}{5} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [110] $1^{11} \overset{11}{5} 1^2 231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 11 & 0 & 50 & 14 \\ 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$
- [112] $1^{11} \overset{1}{5} 1^2 231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 122$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 30 & 0 & 154 & 26 \\ 3 & 5 & -1 & 26 & 4 \end{pmatrix}_{b_0}$

[114] $1^{12} \overset{14}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[113] $1^{12} \overset{12}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[116] $1^{11} \overset{1}{5} 1^3 231^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 5 & 5 & 45 & 5 \\ 3 & 5 & 1 & 33 & 3 \\ 24 & 45 & 33 & 349 & 39 \\ 3 & 5 & 3 & 39 & 4 \end{pmatrix}_{b_0}$

[115] $1^{12} \overset{13}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[117] $1^{11} \overset{1}{5} 1^2 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 9 & 12 & 0 & 81 & 17 \\ 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[118] $1^{12} \overset{12}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$

[119] $1^{11} \overset{1}{5} 1^2 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 35 & 0 & 202 & 30 \\ 3 & 5 & -1 & 30 & 4 \end{pmatrix}_{b_0}$

[120] $1^{12} \overset{13}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$

[122] $1^{13} \overset{12}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$

[121] $1^{11} \overset{1}{5} 1^2 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 10 & 15 & 0 & 108 & 20 \\ 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[123] $1^{14}_5 1^{13}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[124] $1^{11}_5 1^{12}_2 231^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 40 & 0 & 258 & 34 \\ 3 & 5 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

[125] $1^{13}_5 1^{14}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[126] $1^{13}_5 1^{12}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[127] $1^{13}_5 1^{14}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 14 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[128] $1^{12}_5 1^{11}_2 231^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 11 & 13 & 0 & 101 & 19 \\ 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[129] $1^{12}_5 1^{14}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[130] $1^{12}_5 1^{13}_1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[131] $1^{11}_5 1^{13}_2 231^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 5 & 0 & 45 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 349 & 39 \\ 3 & 5 & -1 & 39 & 4 \end{pmatrix}_{b_0}$$

[132] $1^{11}_5 1^{12}_3 231^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 13 & 19 & 0 & 190 & 26 \\ 2 & 2 & 2 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{13} \overset{1^3}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{13} \overset{1^4}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{12} \overset{1^1}{5} 1^{23} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 12 & 16 & 0 & 130 & 22 \\ 2 & 2 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{12} \overset{1^1}{5} 1^3 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 14 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 14 & 17 & 0 & 181 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{13} \overset{1^3}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{11} \overset{1^1}{5} 1^3 231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 5 & 0 & 50 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 419 & 43 \\ 3 & 5 & -1 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{12} \overset{1^1}{5} 1^{23} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 13 & 19 & 0 & 165 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{13} \overset{1^4}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{12} \overset{1^1}{5} 1^3 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 15 & 2 \\ 1 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 15 & 20 & 0 & 218 & 28 \\ 2 & 2 & 2 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[142] $1^{13}5^11^3231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 16 & 18 & 0 & 211 & 27 \\ 2 & 2 & 2 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{12}5^11^3231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 16 & 23 & 0 & 261 & 31 \\ 2 & 2 & 2 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{12}5^11^3231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 549, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 17 & 26 & 0 & 310 & 34 \\ 2 & 2 & 2 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{13}5^11^3231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 17 & 21 & 0 & 250 & 30 \\ 2 & 2 & 2 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{13}5^11^3231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 24 & 0 & 295 & 33 \\ 2 & 2 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{13}5^11^3231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 748, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 0 & 19 & 2 \\ 1 & 2 & 0 & 0 & 27 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 19 & 27 & 0 & 346 & 36 \\ 2 & 2 & 2 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [150] $1^{11}31^15131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[152] $1^15^11^12321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 6 & 0 & 28 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 28 & 0 & 106 & 24 \\ 3 & 6 & -1 & 24 & 5 \end{pmatrix}_{b_0}$$

[151] $1^{11} 31 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[149] $1^{12} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[153] $1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[154] $1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$

[155] $1^{13} 5^{11} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 6 & 3 & 18 & 6 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 18 & 6 & 43 & 15 \\ 3 & 6 & 2 & 15 & 5 \end{pmatrix}_{b_0}$

[156] $1^{12} 5^{11} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 6 & 3 & 12 & 6 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 12 & 4 & 18 & 10 \\ 3 & 6 & 2 & 10 & 5 \end{pmatrix}_{b_0}$

[157] $1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[158] $1^{14} 5^{11} 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 6 & 3 & 24 & 6 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 24 & 8 & 78 & 20 \\ 3 & 6 & 2 & 20 & 5 \end{pmatrix}_{b_0}$

[162] $1^{111} 4^{11} 14131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 2 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[159] $21^{111} 5^{11} 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[160] $1^{11} \overset{1}{5} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[161] $1^{11} \overset{1}{5} 1^{11} 2321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 6 & 6 & 16 & 6 \\ 3 & 6 & 2 & 12 & 4 \\ 8 & 16 & 12 & 38 & 14 \\ 3 & 6 & 4 & 14 & 5 \end{pmatrix}_{b_0}$$

[163] $1^{11} \overset{1}{5} 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[164] $1^{11} \overset{1}{4} 14131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[165] $1^{11} \overset{1}{5} 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[166] $1^{11} \overset{1}{3} 1 \overset{1}{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -37, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[167] $1^{11} \overset{1}{5} 1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[168] $1^{11} \overset{1}{4} 141^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[169] $1^{11} \overset{1}{5} 1^{11} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 4 & 8 & 0 & 22 & 10 \\ 2 & 2 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[173] \quad 1 \overset{1^3}{\underset{1^1}{5}} 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 6 & 9 & 18 & 6 \\ 4 & 9 & 9 & 21 & 7 \\ 9 & 18 & 21 & 43 & 15 \\ 3 & 6 & 7 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[170] \quad 1 \overset{1^4}{\underset{1^1}{5}} 1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 6 & 9 & 24 & 6 \\ 4 & 9 & 9 & 28 & 7 \\ 12 & 24 & 28 & 78 & 20 \\ 3 & 6 & 7 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[171] \quad 1 \overset{1^2}{\underset{1^1}{5}} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 6 & 9 & 12 & 6 \\ 4 & 9 & 9 & 14 & 7 \\ 6 & 12 & 14 & 18 & 10 \\ 3 & 6 & 7 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[172] \quad 1 \overset{1^1}{\underset{1^1}{5}} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 6 & 9 & 6 & 6 \\ 4 & 9 & 9 & 7 & 7 \\ 3 & 6 & 7 & 3 & 5 \\ 3 & 6 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[174] \quad 1 \overset{1^2}{\underset{1^1}{5}} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[175] \quad 1 \overset{1^1}{\underset{1^1}{5}} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[176] \quad 1 \overset{1^3}{\underset{1^1}{5}} 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[177] \quad 1 \overset{1^2}{\underset{1^1}{5}} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -37, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[178] \quad 1 \overset{1^3}{\underset{1^1}{5}} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[179] $1^{11} \overset{14}{5} 1321^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

[180] $1^{11} \overset{11}{4} 141^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -50, (n_+, n_-, n_0) =$
 $(1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 5 & 6 & 1 & 6 & 15 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{11} \overset{11}{5} 1^{11} 2321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 13 & 0 & 54 & 16 \\ 2 & 2 & 3 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{12} \overset{11}{5} 1321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 9 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[183] $1^{12} \overset{12}{5} 1321^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[184] $1^{12} \overset{12}{5} 1321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 10 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[185] $1^{12} \overset{13}{5} 1321^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

[186] $1^{12} \overset{13}{5} 1321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 11 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{12}_{1^1} 5^{13} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -107, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{12}_{1^1} 5^{14} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -118, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{12}_{1^1} 5^{14} 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -118, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{13}_{1^1} 5^{12} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -147, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 14 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{13}_{1^1} 5^{13} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -172, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 15 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{13}_{1^1} 5^{13} 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -182, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 17 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{13}_{1^1} 5^{14} 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -183, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 16 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{13}_{1^1} 5^{14} 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -213, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 18 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [195] \ 1^{11} 31^1 5^{13} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -9, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.0$$

$$[202] \ 1^{131}_{1^1} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -34, \ \det = 2, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[201] & 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[199] & 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[200] & 1^{12} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[204] & 1^{11} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[205] & 1^{11} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0} \\
[203] & 1^{11} 6^{11} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[206] & 1^{12} 5^{11} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 22 & 5 \\ 12 & 12 & 22 & 88 & 24 \\ 3 & 3 & 5 & 24 & 6 \end{pmatrix}_{b_0} \\
[207] & 1^{11} 5^{11} 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0} \\
[208] & 21^{11} 5^{11} 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0} \\
[209] & 1^{12} 5^{11} 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0} \\
[210] & 1^1 \overset{11}{6} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[212] & 21^1 \overset{11}{6} 12321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 4 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 8 & 7 \\ 4 & 4 & 8 & 6 & 6 \\ 3 & 3 & 7 & 6 & 5 \end{pmatrix}_{b_0} \\
[211] & 21^1 \overset{111}{6} \overset{111}{2321}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 6 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 10 & 7 \\ 6 & 4 & 10 & 10 & 8 \\ 3 & 3 & 7 & 8 & 5 \end{pmatrix}_{b_0} \\
[213] & 1^1 \overset{11}{6} \overset{111}{2321}^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 8 & 3 \\ 3 & 2 & 5 & 5 & 12 & 5 \\ 4 & 4 & 8 & 12 & 22 & 10 \\ 2 & 2 & 3 & 5 & 10 & 4 \end{pmatrix}_{b_0} \\
[214] & 1^1 \overset{14}{5} 13221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 3 & 28 & 5 \\ 16 & 14 & 28 & 138 & 30 \\ 3 & 3 & 5 & 30 & 6 \end{pmatrix}_{b_0} \\
[215] & 1^1 \overset{11}{6} 12321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[216] & 1^1 \overset{11}{31} \overset{11}{5} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & -1 & 7 & 5 \end{pmatrix}_{b_0} \\
[217] & 1^1 \overset{11}{6} 12321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0} \\
[221] & 1^1 \overset{14}{5} 13221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 7 & 10 & 28 & 7 \\ 4 & 10 & 10 & 32 & 8 \\ 12 & 28 & 32 & 94 & 24 \\ 3 & 7 & 8 & 24 & 6 \end{pmatrix}_{b_0} \\
[218] & 1^1 \overset{11}{5} 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 7 & 10 & 7 & 7 \\ 4 & 10 & 10 & 8 & 8 \\ 3 & 7 & 8 & 4 & 6 \\ 3 & 7 & 8 & 6 & 6 \end{pmatrix}_{b_0} \\
[219] & 1 \overset{1^2}{\underset{1^1}{5}} 13221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 7 & 10 & 14 & 7 \\ 4 & 10 & 10 & 16 & 8 \\ 6 & 14 & 16 & 22 & 12 \\ 3 & 7 & 8 & 12 & 6 \end{pmatrix}_{b_0} \\
[220] & 1 \overset{1^3}{\underset{1^1}{5}} 13221^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 7 & 10 & 21 & 7 \\ 4 & 10 & 10 & 24 & 8 \\ 9 & 21 & 24 & 52 & 18 \\ 3 & 7 & 8 & 18 & 6 \end{pmatrix}_{b_0} \\
[222] & 1 \overset{1^1}{\underset{1^1}{6}} 12321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0} \\
[223] & 1 \overset{1^{11}2321^1}{\underset{1^1}{6}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0} \\
[224] & 1 \overset{1^{11}321}{\underset{1^1}{6}} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 4 & 1 & 3 & 5 \\ 3 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0} \\
[225] & 1 \overset{1^{11}321}{\underset{1^1}{6}} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 2 & 5 & 4 \\ 4 & 5 & 4 & 6 \\ 4 & 4 & 6 & 6 \end{pmatrix}_{b_0} \\
[226] & 1 \overset{1^{11}31}{\underset{1^1}{6}} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0} \\
[227] & 21 \overset{1^{112}2321^1}{\underset{1^1}{6}} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 10 & 3 \\ 2 & 1 & 4 & 8 & 3 \\ 5 & 4 & 9 & 18 & 7 \\ 10 & 8 & 18 & 34 & 14 \\ 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0} \\
[228] & 1 \overset{1^1}{\underset{1^{112}2321^1}{6}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 1 & 5 & 13 & 3 \\ 3 & 2 & 5 & 5 & 19 & 5 \\ 6 & 7 & 13 & 19 & 54 & 16 \\ 2 & 2 & 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [230] $1^{11} \overset{1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0}$
- [231] $1^{11} \overset{1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$
- [232] $1^{11} \overset{1}{6} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$
- [229] $1^{11} \overset{12}{6} 12321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$
- [233] $1^{11} \overset{1}{6} 12321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$
- [234] $1^{11} \overset{13}{6} 12321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$
- [235] $1^{11} \overset{12}{6} 12321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$
- [237] $1^{11} \overset{13}{6} 12321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$
- [236] $1^{11} \overset{12}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$
- [239] $1^{11} \overset{1}{6} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

[238] $1^{11} \overset{1^4}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

[240] $1^{11} \overset{1^3}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 15 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[241] $1^{11} \overset{1^3}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 13 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

[242] $1^{11} \overset{1^4}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 16 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[243] $1^{11} \overset{1^2}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

[244] $1^{11} \overset{1^{12} 2^{13} 2^{11}}{6} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[245] $1^{11} \overset{1^1}{6} 1^{13} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[246] $1^{11} \overset{1^2}{6} 1^{14} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 18 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[247] $1^{11} \overset{1^1}{6} 1^{12} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[248] $1^{11} \overset{1^2}{6} 1^{12} 2^{13} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[249] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 15 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[250] $1^{11} \overset{1^2}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 117, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[251] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 19 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[252] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 17 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[253] $1^{11} \overset{1^4}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 18 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[254] $1^{11} \overset{1^4}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 155, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 20 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [257] $1^{1^1 31} 6^{11} 13141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[256] $1^{1^1 31} 6^{11} 13141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 4 & 0 & 2 & 2 & 4 \\ 2 & 4 & -1 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[258] $1^{1^1 2321} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{11} \bar{5}^{11} 1315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{11} \bar{6}^{11} 13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 2 & 4 \\ 2 & 4 & -1 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{11} \bar{6}^{11} 13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 2 & 0 & 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{11} \bar{6}^{11} 131^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 5 & 0 & 8 & 6 & 22 & 10 \\ 2 & 0 & 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{11} \bar{6}^{11} 131^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 2 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 5 & 12 & 0 & 8 & 42 & 14 \\ 2 & 4 & -1 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [264] \ 1^{11} \bar{4}^{11} 1414141^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 6.0$$

$$[266] \ 1^{12} \bar{3}^{12} 161^{12} 231 \bar{5}^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 34 & 12 \\ 2 & 3 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{11} \bar{3}^{11} 216131 \bar{5}^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{13} \bar{2}^{13} 1 \bar{6}^{13} 131 \bar{5}^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 5 & 0 & 12 & 8 \\ 3 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[270] $1^{11}316123151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 15$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

[267] $1^{11}612315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 15$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

[271] $1^{112}3161^{1231}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 24$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 6 & 9 & 0 & 9 & 34 & 12 \\ 2 & 3 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[272] $1^{112}3151321^{11}6^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 27$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 13 & 4 \\ 3 & 2 & 10 & 4 \\ 13 & 10 & 43 & 17 \\ 4 & 4 & 17 & 6 \end{pmatrix}_{b_0}$$

[273] $1^{11}3151321^{11}6^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 27$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 2 & 8 & 4 \\ 10 & 8 & 25 & 13 \\ 4 & 4 & 13 & 6 \end{pmatrix}_{b_0}$$

[274] $1^{12}3161^{1231}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 32$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 10 & 0 & 36 & 14 \\ 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$$

[275] $1^{11}3161^{1231}51^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 42$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 0 & 7 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 21 & 11 \\ 3 & 3 & 0 & 11 & 5 \end{pmatrix}_{b_0}$$

[276] $1^{11}321^{16131}51^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 60$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 29 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

[277] $1^{113}3151321^{11}6^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 82$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 18 & 4 \\ 3 & 2 & 15 & 4 \\ 18 & 15 & 88 & 24 \\ 4 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

[278] $1^{13}3161^{12315}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 87$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 63 & 19 \\ 3 & 3 & 0 & 19 & 5 \end{pmatrix}_{b_0}$

[279] $1^{11}3161^{12315}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 87$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 8 & 0 & 27 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[280] $1^{11}3151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 92$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 15 & 4 \\ 3 & 2 & 13 & 4 \\ 15 & 13 & 58 & 20 \\ 4 & 4 & 20 & 6 \end{pmatrix}_{b_0}$

[281] $1^{12}3161^{12315}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 107$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 11 & 0 & 42 & 16 \\ 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$

[282] $1^{12}321^{161315}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 115$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 56 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$

[283] $1^{12}3161^{12315}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 172$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 14.98 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 0 & 12 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 12 & 0 & 52 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$

[284] $1^{13}3161^{12315}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 182$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 14.96 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 15 & 0 & 73 & 21 \\ 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$

$T = 15$ [285] $1^{11}41414141^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 2)$, $c_{\text{ext}} = 6.0$

[286] $21^{11}6131513221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 6.0$

[287] $1^{11}6131513221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 6.0$

[288] $1^{11}2315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -8$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 6.0$

[296] $21^{11}613151321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[297] $1^{11} \overset{1}{6} 13151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

[298] $1^{11} \overset{1}{6}^{13} 23151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

[299] $1^{11} \overset{1}{6} 13151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0}$$

[300] $1^{11} \overset{1}{6}^{31} 15131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 15, T_{\max} = 15, \Delta = -7, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$T = 16$ [301] $1^{11} \overset{1}{6}^{11} 31^{11} 131612321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 16, T_{\max} = 16, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 6.0$

[305] $1^{11} \overset{1}{6}^{11} 31^{11} 1316131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 2 & 4 \\ 2 & 4 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[308] $1^{11} \overset{1}{6}^{11} 31^{11} 13161231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 16, T_{\max} = 16, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 3 & 5 \\ 2 & 4 & 0 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[306] $1^{11} \overset{1}{6}^{11} 31^{11} 1316131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 2 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[307] $1^{11} \overset{1}{6}^{11} 31^{11} 1316131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 3 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[309] $1^{1^{12}31} 6^{1^1} 1316131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 4 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[310] $1^{1^{12}31} 6^{1^2} 1316131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 5 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$T = 17$ [319] $1^{1^{1^1}5^{1^1}1321} 8^{1^1} 12321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 4 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 2 & 5 & 0 & 5 & 6 & 6 \\ 2 & 4 & 0 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

[320] $1^{1^{1^1}5^{1^1}1321} 8^{1^1} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 8 & 8 \\ 3 & 4 & 0 & 8 & 6 \end{pmatrix}_{b_0}$$

[321] $1^{11} 6^{1^1} 1^{1^1} 2321712321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 2 & 16 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 14 & 16 & 9 & 97 & 25 \\ 3 & 4 & 2 & 25 & 6 \end{pmatrix}_{b_0}$$

[322] $1^{13} 232181^{1^1} 223151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 86 & 24 \\ 3 & 4 & 0 & 24 & 6 \end{pmatrix}_{b_0}$$

[323] $1^{1^{11}31} 6^{1^1} 131613141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 3 & 5 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 3 & 5 \\ 2 & 5 & -1 & 3 & 5 & 5 \end{pmatrix}_{b_0}$$

[324] $1^{14} 232181^{1^1} 223151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 23 & 0 & 132 & 30 \\ 3 & 4 & 0 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{113}232181^{12231}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, \quad T_{\max} = 17, \quad \Delta = 0, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 8 & 16 & 0 & 16 & 78 & 20 \\ 2 & 4 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{11231}6^{1316131}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, \quad T_{\max} = 17, \quad \Delta = -3, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 3 & 14 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 10 & 3 \\ 5 & 14 & 0 & 10 & 46 & 16 \\ 2 & 5 & -1 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [327] \quad 1^{11131}6^{13161315131}11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 18, \quad T_{\max} = 18, \quad \Delta = -6, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 18, 1), \quad c_{\text{ext}} = 6.0$$

$$T = 19 \quad [328] \quad 1^{11231}6^{13161316131}11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -3, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 6.5$$

$$[329] \quad 1^{11131}6^{131613161231}11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -4, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 6.0$$

$$[330] \quad 1^{11131}6^{13161316131}11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -3, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 6.0$$

$$T = 20 \quad [332] \quad 1^{11131}6^{1316131613141}11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 20, 1), \quad c_{\text{ext}} = 6.0$$

$$[333] \quad 1^{11231}6^{13161316131}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 20, 1), \quad c_{\text{ext}} = 6.0$$

3.2.18 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \quad 1^{113}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -95, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{113}1^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -68, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{114}13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -65, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{114}13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -65, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 4 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} \overline{4} 13^{\overline{1}} 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1}} \overline{5} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1}} \overline{5} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^1 4 14 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^1 4 14 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 3^1 1^1 5 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^1 4 13^1 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 4 13^1 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^1 4 13^1 1^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [19] \quad 1^1 3^1 6^1 13 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [20] \quad 1^1 5 13 15 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 5 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 6 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[21] \quad & 1^{\overline{1}} \overline{5} 131513^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = \\
& (1, 13, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 14 \quad [25] \quad & 1^{\overline{1}} \overline{6} 131513^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[24] \quad & 1^{\overline{1}} \overline{6} 1231513^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[23] \quad & 1^{\overline{1}} \overline{3}^{\overline{1}} 2161315^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[22] \quad & 1^{\overline{1}} \overline{3}^{\overline{1}} 2161315^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 15 \quad [27] \quad & 1^{\overline{1}} \overline{3}^{\overline{1}} 21712315^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = \\
& (1, 15, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[26] \quad & 1^{\overline{1}} \overline{3}^{\overline{1}} 21712315^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = \\
& (1, 15, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

3.2.19 NHC(-2)(-3) + su₃ + su₂n₃mix

$$\begin{aligned}
T = 10 \quad [0] \quad & 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 14 & 5 \\ 4 & 8 & 8 & 17 & 6 \\ 9 & 14 & 17 & 29 & 11 \\ 3 & 5 & 6 & 11 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

- $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 5 & 8 & 9 & 5 \\ 4 & 8 & 8 & 11 & 6 \\ 6 & 9 & 11 & 11 & 7 \\ 3 & 5 & 6 & 7 & 4 \end{pmatrix}_{b_0}$
- [2] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 4 & 8 & 8 & 5 & 6 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$
- [3] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 19 & 5 \\ 4 & 8 & 8 & 23 & 6 \\ 12 & 19 & 23 & 55 & 15 \\ 3 & 5 & 6 & 15 & 4 \end{pmatrix}_{b_0}$
- [4] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [5] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [6] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$
- [7] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$
- [8] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 5 & 8 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$
- [9] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$
- [10] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$
- [12] $1 \overset{1}{5} 13 \overset{1}{1} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 64, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 70, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1^2}_{1^1} 5^{1^2} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1^4}_{1^1} 5^{1^4} 13^{1^1} 1^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 75, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 13 & 3 \\ 11 & 9 & 13 & 44 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1^4}_{1^1} 5^{1^4} 13^{1^1} 1^{1^4} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 3 & 24 & 5 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 24 & 10 & 89 & 19 \\ 3 & 5 & 2 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1^2}_{1^1} 5^{1^2} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 125, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 10 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 125, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 11 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1^2}_{1^1} 5^{1^2} 13^{1^1} 1^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1^3}_{1^1} 5^{1^3} 13^{1^1} 1^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 189, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 13 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{12} \overset{14}{5} \overset{11}{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 189, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 14 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12} \overset{13}{5} \overset{11}{13} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 229, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 19 & 3 \\ 15 & 14 & 19 & 99 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{12} \overset{14}{5} \overset{11}{13} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 245, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 19 & 3 \\ 16 & 14 & 19 & 99 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{13} \overset{13}{5} \overset{11}{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 272, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 17 & 3 \\ 15 & 15 & 17 & 92 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{13} \overset{14}{5} \overset{11}{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 280, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 17 & 3 \\ 16 & 15 & 17 & 92 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{13} \overset{13}{5} \overset{11}{13} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 340, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 21 & 3 \\ 17 & 17 & 21 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13} \overset{14}{5} \overset{11}{13} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 364, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 21 & 3 \\ 18 & 17 & 21 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{13} \overset{14}{5} \overset{11}{13} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 424, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 19 & 3 \\ 2 & 3 & 1 & 25 & 3 \\ 20 & 19 & 25 & 176 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13} \overset{11}{5} \overset{13}{23} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 629, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.00 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 23 & 0 & 290 & 32 \\ 2 & 2 & 2 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \ 1^{111} 41 \overset{1}{3} 1 \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 20 & 8 \\ 2 & 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11}413^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11}413^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 2 & 0 & 2 & 4 \\ 3 & 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11}41^{11}3^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 0 & 0 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11}41^{11}3^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 4 & 0 & 14 & 8 \\ 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11}413^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12}41^{11}3^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11}41^{11}3^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 6 & 0 & 22 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11}41^{11}3^{11}51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 5 & 0 & 20 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11}41^{11}3^{11}51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -189, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 7 & 0 & 28 & 12 \\ 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 41^{13} 1^{15} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -272, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 8 & 0 & 38 & 14 \\ 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [42] \quad 1^{11} 6^{13} 131513^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 6^{13} 131513^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 7 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11} 6^{13} 131513^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 43, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 8 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11} 6^{13} 131513^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 50, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 16 & 4 \\ 11 & 9 & 16 & 53 & 17 \\ 3 & 3 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 6^{12} 3^{11} 151321^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 98, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 17 & 4 \\ 3 & 2 & 14 & 4 \\ 17 & 14 & 72 & 22 \\ 4 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

3.2.20 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 5^{13} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [1] \quad 1^{11} 5^{13} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

3.2.21 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 2 \quad 3 \quad 21 \quad \begin{smallmatrix} 1 \\ 1 \\ 1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 5 & 3 \\ 3 & 6 & 6 & 5 & 3 & 5 \\ 3 & 6 & 6 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

3.2.22 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 11 \quad [0] \quad 2 \quad 3 \quad 21 \quad \begin{smallmatrix} 1 \\ 1 \\ 1 \end{smallmatrix} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 & 5 \\ 3 & 6 & 6 & 5 & 1 & 3 \\ 3 & 6 & 6 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

3.2.23 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 2 \quad 3 \quad 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 2 \quad 3 \quad 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 5 & 3 \\ 2 & 3 & 0 & 3 & 4 & 5 & 2 \\ 2 & 3 & 3 & 1 & 6 & 5 & 3 \\ 4 & 6 & 4 & 6 & 10 & 10 & 6 \\ 4 & 5 & 5 & 5 & 10 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 2 \quad 3 \quad 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 & 3 \\ 2 & 3 & 0 & 3 & 4 & 2 & 2 \\ 2 & 3 & 3 & 1 & 6 & 3 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 2 \quad 3 \quad 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 2 \quad 3 \quad 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 2 & 5 & 2 \\ 2 & 3 & 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 8 & 6 & 6 \\ 2 & 3 & 2 & 3 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 2 \overset{1^{11}}{\underset{1^{1112}}{3}} \ 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -30, \ \det = -576, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 6 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 8 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 7 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 8 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 8 & 3 & 3 \\ 6 & 8 & 7 & 8 & 8 & 21 & 9 & 9 \\ 2 & 3 & 2 & 3 & 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1 \overset{1^{11}}{\underset{1^{112}}{2}} \ 2 \ 3 \ 21 \overset{1^{11}}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1 \overset{1^{12}}{\underset{1^{114}}{2}} \ 3 \ 21 \overset{1^{12}}{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1 \overset{1^{21}}{\underset{1^{132}}{2}} \ 3 \ 21 \overset{1^{21}}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 & 2 \\ 1 & 1 & 2 & 0 & 10 & 7 & 3 & 3 \\ 1 & 2 & 0 & 1 & 6 & 5 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 26 & 12 & 12 \\ 4 & 7 & 5 & 0 & 26 & 17 & 9 & 9 \\ 2 & 3 & 2 & 0 & 12 & 9 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1 \overset{1^{22}}{\underset{1^{134}}{2}} \ 3 \ 21 \overset{1^{22}}{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 & 2 \\ 1 & 1 & 2 & 0 & 10 & 14 & 3 & 3 \\ 1 & 2 & 0 & 1 & 6 & 10 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 52 & 12 & 12 \\ 8 & 14 & 10 & 0 & 52 & 74 & 18 & 18 \\ 2 & 3 & 2 & 0 & 12 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1 \overset{1^{22}}{\underset{1^{144}}{2}} \ 3 \ 21 \overset{1^{22}}{1^{22}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 8 & 2 & 2 \\ 1 & 1 & 2 & 0 & 14 & 14 & 3 & 3 \\ 1 & 2 & 0 & 1 & 10 & 10 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 76 & 18 & 18 \\ 8 & 14 & 10 & 0 & 76 & 74 & 18 & 18 \\ 2 & 3 & 2 & 0 & 18 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1 \overset{1^{11}}{\underset{1^{122}}{2}} \ 3 \ 21 \overset{1^{11}}{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 & 2 \\ 1 & 1 & 2 & 0 & 7 & 7 & 3 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 19 & 9 & 9 \\ 4 & 7 & 5 & 0 & 19 & 17 & 9 & 9 \\ 2 & 3 & 2 & 0 & 9 & 9 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1 \overset{1^{21}}{\underset{1^{142}}{2}} \ 3 \ 21 \overset{1^{21}}{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 & 2 \\ 1 & 1 & 2 & 0 & 14 & 7 & 3 & 3 \\ 1 & 2 & 0 & 1 & 10 & 5 & 2 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 38 & 18 & 18 \\ 4 & 7 & 5 & 0 & 38 & 17 & 9 & 9 \\ 2 & 3 & 2 & 0 & 18 & 9 & 4 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 11 \quad [36] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 12 & 3 \\ 4 & 6 & 8 & 12 & 24 & 6 \\ 4 & 8 & 6 & 12 & 24 & 6 \\ 6 & 12 & 12 & 18 & 40 & 10 \\ 12 & 24 & 24 & 40 & 78 & 20 \\ 3 & 6 & 6 & 10 & 20 & 5 \end{pmatrix}_{b_0} \\
& [34] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 15 & 3 \\ 4 & 6 & 8 & 24 & 30 & 6 \\ 4 & 8 & 6 & 24 & 30 & 6 \\ 12 & 24 & 24 & 78 & 100 & 20 \\ 15 & 30 & 30 & 100 & 123 & 25 \\ 3 & 6 & 6 & 20 & 25 & 5 \end{pmatrix}_{b_0} \\
& [18] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 21 & 3 \\ 4 & 6 & 8 & 12 & 42 & 6 \\ 4 & 8 & 6 & 12 & 42 & 6 \\ 6 & 12 & 12 & 18 & 70 & 10 \\ 21 & 42 & 42 & 70 & 243 & 35 \\ 3 & 6 & 6 & 10 & 35 & 5 \end{pmatrix}_{b_0} \\
& [17] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 9 & 3 \\ 4 & 6 & 8 & 18 & 18 & 6 \\ 4 & 8 & 6 & 18 & 18 & 6 \\ 9 & 18 & 18 & 43 & 45 & 15 \\ 9 & 18 & 18 & 45 & 43 & 15 \\ 3 & 6 & 6 & 15 & 15 & 5 \end{pmatrix}_{b_0} \\
& [16] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 12 & 3 \\ 4 & 6 & 8 & 18 & 24 & 6 \\ 4 & 8 & 6 & 18 & 24 & 6 \\ 9 & 18 & 18 & 43 & 60 & 15 \\ 12 & 24 & 24 & 60 & 78 & 20 \\ 3 & 6 & 6 & 15 & 20 & 5 \end{pmatrix}_{b_0} \\
& [15] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.9 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 15 & 3 \\ 4 & 6 & 8 & 18 & 30 & 6 \\ 4 & 8 & 6 & 18 & 30 & 6 \\ 9 & 18 & 18 & 43 & 75 & 15 \\ 15 & 30 & 30 & 75 & 123 & 25 \\ 3 & 6 & 6 & 15 & 25 & 5 \end{pmatrix}_{b_0} \\
& [14] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 21 & 3 \\ 4 & 6 & 8 & 24 & 42 & 6 \\ 4 & 8 & 6 & 24 & 42 & 6 \\ 12 & 24 & 24 & 78 & 140 & 20 \\ 21 & 42 & 42 & 140 & 243 & 35 \\ 3 & 6 & 6 & 20 & 35 & 5 \end{pmatrix}_{b_0} \\
& [13] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 18 & 3 \\ 4 & 6 & 8 & 36 & 36 & 6 \\ 4 & 8 & 6 & 36 & 36 & 6 \\ 18 & 36 & 36 & 178 & 180 & 30 \\ 18 & 36 & 36 & 180 & 178 & 30 \\ 3 & 6 & 6 & 30 & 30 & 5 \end{pmatrix}_{b_0} \\
& [20] \quad 2 \quad 3 \quad 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 15 & 3 \\ 4 & 6 & 8 & 12 & 30 & 6 \\ 4 & 8 & 6 & 12 & 30 & 6 \\ 6 & 12 & 12 & 18 & 50 & 10 \\ 15 & 30 & 30 & 50 & 123 & 25 \\ 3 & 6 & 6 & 10 & 25 & 5 \end{pmatrix}_{b_0}$$

[21] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 21 & 3 \\ 4 & 6 & 8 & 36 & 42 & 6 \\ 4 & 8 & 6 & 36 & 42 & 6 \\ 18 & 36 & 36 & 178 & 210 & 30 \\ 21 & 42 & 42 & 210 & 243 & 35 \\ 3 & 6 & 6 & 30 & 35 & 5 \end{pmatrix}_{b_0}$$

[22] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 9 & 3 \\ 4 & 6 & 8 & 12 & 18 & 6 \\ 4 & 8 & 6 & 12 & 18 & 6 \\ 6 & 12 & 12 & 18 & 30 & 10 \\ 9 & 18 & 18 & 30 & 43 & 15 \\ 3 & 6 & 6 & 10 & 15 & 5 \end{pmatrix}_{b_0}$$

[23] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 6 & 3 \\ 4 & 6 & 8 & 12 & 12 & 6 \\ 4 & 8 & 6 & 12 & 12 & 6 \\ 6 & 12 & 12 & 18 & 20 & 10 \\ 6 & 12 & 12 & 20 & 18 & 10 \\ 3 & 6 & 6 & 10 & 10 & 5 \end{pmatrix}_{b_0}$$

[24] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.7$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 21 & 3 \\ 4 & 6 & 8 & 42 & 42 & 6 \\ 4 & 8 & 6 & 42 & 42 & 6 \\ 21 & 42 & 42 & 243 & 245 & 35 \\ 21 & 42 & 42 & 245 & 243 & 35 \\ 3 & 6 & 6 & 35 & 35 & 5 \end{pmatrix}_{b_0}$$

[25] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 21 & 3 \\ 4 & 6 & 8 & 6 & 42 & 6 \\ 4 & 8 & 6 & 6 & 42 & 6 \\ 3 & 6 & 6 & 3 & 35 & 5 \\ 21 & 42 & 42 & 35 & 243 & 35 \\ 3 & 6 & 6 & 5 & 35 & 5 \end{pmatrix}_{b_0}$$

[26] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.2$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 18 & 3 \\ 4 & 6 & 8 & 6 & 36 & 6 \\ 4 & 8 & 6 & 6 & 36 & 6 \\ 3 & 6 & 6 & 3 & 30 & 5 \\ 18 & 36 & 36 & 30 & 178 & 30 \\ 3 & 6 & 6 & 5 & 30 & 5 \end{pmatrix}_{b_0}$$

[27] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.1$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 15 & 3 \\ 4 & 6 & 8 & 6 & 30 & 6 \\ 4 & 8 & 6 & 6 & 30 & 6 \\ 3 & 6 & 6 & 3 & 25 & 5 \\ 15 & 30 & 30 & 25 & 123 & 25 \\ 3 & 6 & 6 & 5 & 25 & 5 \end{pmatrix}_{b_0}$$

[28] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 12 & 3 \\ 4 & 6 & 8 & 6 & 24 & 6 \\ 4 & 8 & 6 & 6 & 24 & 6 \\ 3 & 6 & 6 & 3 & 20 & 5 \\ 12 & 24 & 24 & 20 & 78 & 20 \\ 3 & 6 & 6 & 5 & 20 & 5 \end{pmatrix}_{b_0}$$

[29] $2 \ 3 \ 21^{1^1 41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

6.8

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 9 & 3 \\ 4 & 6 & 8 & 6 & 18 & 6 \\ 4 & 8 & 6 & 6 & 18 & 6 \\ 3 & 6 & 6 & 3 & 15 & 5 \\ 9 & 18 & 18 & 15 & 43 & 15 \\ 3 & 6 & 6 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

[30] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
6.5

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 6 & 3 \\ 4 & 6 & 8 & 6 & 12 & 6 \\ 4 & 8 & 6 & 6 & 12 & 6 \\ 3 & 6 & 6 & 3 & 10 & 5 \\ 6 & 12 & 12 & 10 & 18 & 10 \\ 3 & 6 & 6 & 5 & 10 & 5 \end{pmatrix}_{b_0}$$

[31] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
6.0

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 & 5 \\ 3 & 6 & 6 & 5 & 3 & 5 \\ 3 & 6 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[32] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}37} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.1

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 21 & 3 \\ 4 & 6 & 8 & 18 & 42 & 6 \\ 4 & 8 & 6 & 18 & 42 & 6 \\ 9 & 18 & 18 & 43 & 105 & 15 \\ 21 & 42 & 42 & 105 & 243 & 35 \\ 3 & 6 & 6 & 15 & 35 & 5 \end{pmatrix}_{b_0}$$

[38] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}56} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.4

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 18 & 3 \\ 4 & 6 & 8 & 30 & 36 & 6 \\ 4 & 8 & 6 & 30 & 36 & 6 \\ 15 & 30 & 30 & 123 & 150 & 25 \\ 18 & 36 & 36 & 150 & 178 & 30 \\ 3 & 6 & 6 & 25 & 30 & 5 \end{pmatrix}_{b_0}$$

[39] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}55} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.3

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 15 & 3 \\ 4 & 6 & 8 & 30 & 30 & 6 \\ 4 & 8 & 6 & 30 & 30 & 6 \\ 15 & 30 & 30 & 123 & 125 & 25 \\ 15 & 30 & 30 & 125 & 123 & 25 \\ 3 & 6 & 6 & 25 & 25 & 5 \end{pmatrix}_{b_0}$$

[40] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}36} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.1

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 18 & 3 \\ 4 & 6 & 8 & 18 & 36 & 6 \\ 4 & 8 & 6 & 18 & 36 & 6 \\ 9 & 18 & 18 & 43 & 90 & 15 \\ 18 & 36 & 36 & 90 & 178 & 30 \\ 3 & 6 & 6 & 15 & 30 & 5 \end{pmatrix}_{b_0}$$

[33] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}44} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.0

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 12 & 3 \\ 4 & 6 & 8 & 24 & 24 & 6 \\ 4 & 8 & 6 & 24 & 24 & 6 \\ 12 & 24 & 24 & 78 & 80 & 20 \\ 12 & 24 & 24 & 80 & 78 & 20 \\ 3 & 6 & 6 & 20 & 20 & 5 \end{pmatrix}_{b_0}$$

[37] $2 \overset{1^{141}}{3} 21^{\textcolor{red}{11}57} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$
8.5

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 21 & 3 \\ 4 & 6 & 8 & 30 & 42 & 6 \\ 4 & 8 & 6 & 30 & 42 & 6 \\ 15 & 30 & 30 & 123 & 175 & 25 \\ 21 & 42 & 42 & 175 & 243 & 35 \\ 3 & 6 & 6 & 25 & 35 & 5 \end{pmatrix}_{b_0}$$

[35] $2 \overset{1^{141}}{3} 21^{1146} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 18 & 3 \\ 4 & 6 & 8 & 24 & 36 & 6 \\ 4 & 8 & 6 & 24 & 36 & 6 \\ 12 & 24 & 24 & 78 & 120 & 20 \\ 18 & 36 & 36 & 120 & 178 & 30 \\ 3 & 6 & 6 & 20 & 30 & 5 \end{pmatrix}_{b_0}$$

[19] $2 \overset{1^{141}}{3} 21^{1126} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 18 & 3 \\ 4 & 6 & 8 & 12 & 36 & 6 \\ 4 & 8 & 6 & 12 & 36 & 6 \\ 6 & 12 & 12 & 18 & 60 & 10 \\ 18 & 36 & 36 & 60 & 178 & 30 \\ 3 & 6 & 6 & 10 & 30 & 5 \end{pmatrix}_{b_0}$$

[41] $2 \overset{1^{111}}{3} \overset{1^{11}}{4} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -256, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [42] $1 \overset{1^{11}}{112} 31 \overset{1^{11}}{5} 1321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$

[44] $1 \overset{1^{11}}{112} 31 \overset{1^{11}}{5} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[43] $1 \overset{1^{12}}{114} 31 \overset{1^{12}}{5} 131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$T = 13$ [45] $1 \overset{1^{11}}{111} \overset{1^{11}}{2321} 6^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 6.0$

[46] $1 \overset{1^{112}}{112} \overset{1^{11}}{2321} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 6.8$

3.2.24 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1 \overset{1^1}{131} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{13} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 13 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{13} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 13 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 1 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 15 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{13} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 17 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 17 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 1 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 16 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{14} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^121^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{14}3^121^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 1 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [20] $1^{11}3^11^51321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

$T = 14$ [23] $1^{11}7^{2321}123^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 6.0$

3.2.25 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix}

$T = 9$ [2] $1^{12}3^121^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 11 \\ 2 & 2 & 2 & 0 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^121^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 15 & 3 \\ 2 & 3 & 0 & 4 & 5 & 24 & 5 \\ 8 & 9 & 0 & 15 & 24 & 83 & 17 \\ 2 & 2 & -1 & 3 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[0] $1^{114}3^121^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{13}3^121^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 10 & 10 & 13 & 0 & 13 & 74 & 16 \\ 2 & 2 & 2 & 0 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^121^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 7 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 10 & 2 \\ 10 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141} 1^{121} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 28 & 16 \\ 4 & 4 & 5 & 0 & 28 & 10 & 6 \\ 2 & 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{141} 1^{121} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114} 1^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112} 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 19 & 5 \\ 8 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114} 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{113} 1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 7 & 2 \\ 8 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{122} 1^{122} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{122} 1^{112} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 25 & 8 \\ 7 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{121} 1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{121} 1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113} 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{111} 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{114} 1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 1^{112} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 9 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{113} 1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 10 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{122} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{122} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 23 & 7 \\ 7 & 7 & 9 & 0 & 23 & 35 & 11 \\ 2 & 2 & 2 & 0 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 10 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{114} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 26 & 5 \\ 11 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -192, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{123} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 10 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{14} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 11 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 11 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -215, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 12 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{123} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{121} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{131} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{141} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 12 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{141} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 10 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 58 & 26 & 14 \\ 4 & 4 & 5 & 0 & 26 & 10 & 6 \\ 2 & 2 & 2 & 0 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -235, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{114} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 12 & 2 \\ 12 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[61] $1^{132} 3^1 21^{142} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$

[62] $1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 5 & 13 & 2 \\ 1 & 2 & 0 & 1 & 5 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 43 & 7 \\ 13 & 13 & 17 & 0 & 43 & 127 & 21 \\ 2 & 2 & 2 & 0 & 7 & 21 & 3 \end{pmatrix}_{b_0}$

[63] $1^{132} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$

[64] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}$

[65] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$

[66] $1^{132} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$

[67] $1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$

$$[68] \quad 1^{131} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 8 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 12 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -315, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -337, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 11 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{132} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 39 & 12 \\ 7 & 7 & 9 & 0 & 39 & 35 & 11 \\ 2 & 2 & 2 & 0 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{124} 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -350, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{123} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -363, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{124} 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{122} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -378, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{131} 1^{131} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{123} 1^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{123} 1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -394, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 1 & 5 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 28 & 7 \\ 9 & 9 & 10 & 0 & 28 & 51 & 13 \\ 2 & 2 & 2 & 0 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{141} 1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{141} 1^{131} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{142} 1^{132} 3^1 21^{142} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{124} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -457, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{142} 1^{132} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{141} 1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 31 & 17 \\ 4 & 4 & 5 & 0 & 31 & 10 & 6 \\ 2 & 2 & 2 & 0 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{113} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{124} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 11 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{132} 1^{132} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{124} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{114} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{124} 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 1 & 8 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 52 & 10 \\ 12 & 12 & 14 & 0 & 52 & 96 & 18 \\ 2 & 2 & 2 & 0 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{124} 1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 1 & 5 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 38 & 7 \\ 12 & 12 & 14 & 0 & 38 & 96 & 18 \\ 2 & 2 & 2 & 0 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{113} 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{142} 1^{142} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 1 & 13 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 55 & 17 \\ 7 & 7 & 9 & 0 & 55 & 35 & 11 \\ 2 & 2 & 2 & 0 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{141} 1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 4 & 2 \\ 1 & 2 & 0 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 36 & 20 \\ 4 & 4 & 5 & 0 & 36 & 10 & 6 \\ 2 & 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{123} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 10 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -687, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{123} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -714, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 1 & 8 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 54 & 10 \\ 12 & 11 & 16 & 0 & 54 & 103 & 19 \\ 2 & 2 & 2 & 0 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -762, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 7 & 2 \\ 1 & 0 & 2 & 0 & 12 & 7 & 2 \\ 1 & 2 & 0 & 1 & 16 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 64 & 20 \\ 7 & 7 & 9 & 0 & 64 & 35 & 11 \\ 2 & 2 & 2 & 0 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -842, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -938, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 13 & 2 \\ 1 & 0 & 2 & 0 & 6 & 13 & 2 \\ 1 & 2 & 0 & 1 & 8 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 60 & 10 \\ 13 & 13 & 17 & 0 & 60 & 127 & 21 \\ 2 & 2 & 2 & 0 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

4 Round 4: NHC External 4-ext (1161 bases)

Colored curves in quiver show attachment points. Matching color in external label: ext1 ext2 ext3 ext4 ext5 ext6
 Total: 1161 bases (46 uni, 1115 non-uni).

T	Count
8	56
9	300
10	157
11	251
12	351
13	21
14	5
15	11
16	6
17	1
19	1
20	1

4.1 Unimodular ($|\det| = 1$) — 46 bases

4.1.1 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \quad 1 \overset{1^1}{\underset{1^{13}}{5}} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 11 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 14 & 11 & 0 & 93 & 11 & 19 \\ 1 & 1 & 2 & 1 & 0 & 11 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 19 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[1] \quad 1 \overset{1^1}{\underset{1^{12}}{5}} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 7 & 10 & 11 & 9 & 0 & 68 & 9 & 16 \\ 1 & 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 16 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[2] \quad 1 \overset{1^1}{\underset{1^{11}}{5}} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 3 & 0 & 5 & 3 & 5 \\ 1 & 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[3] \quad 1 \overset{1^1}{\underset{1^{11}}{5}} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 6 & 5 & 0 & 21 & 5 & 9 \\ 1 & 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[4] \quad 1 \overset{1^1}{\underset{1^1}{5}} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 14 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 8 & 10 & 14 & 11 & 0 & 11 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 2 & 19 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.1.3 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [17] \quad 1^{111} 231^1 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 7 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 0 & 7 & 0 & 14 & 7 & 8 \\ 2 & 0 & 4 & 0 & 7 & 2 & 4 \\ 2 & 0 & 3 & 0 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$[18] \quad 1^{1111} 231^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 & 4 \\ 3 & 7 & 0 & 7 & 14 & 7 & 8 \\ 2 & 4 & 0 & 4 & 7 & 2 & 4 \\ 2 & 4 & -1 & 4 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$T = 11 \quad [28] \quad 1^{1111} 31^1 4131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 & 6 & 3 & 3 \\ 4 & 6 & 1 & 6 & 14 & 7 & 8 \\ 2 & 3 & 1 & 3 & 7 & 2 & 4 \\ 2 & 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[27] \quad 1^{1111} 31^1 41^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & 6 & 4 & 3 \\ 4 & 6 & 1 & 6 & 14 & 9 & 8 \\ 3 & 4 & 0 & 4 & 9 & 5 & 5 \\ 2 & 3 & 0 & 3 & 8 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[26] \quad 1^{1111} 3141^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 5 \\ 2 & 3 & 1 & 3 & 4 & 2 & 4 \\ 2 & 3 & 0 & 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$T = 14 \quad [34] \quad 1^{11} 6^{11} 131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 3 & 2 & 3 & 5 & 3 & 0 & 5 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[33] \quad 1^{11} 6^{11} 131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 3 & 0 & 1 & 2 \\ 2 & 2 & 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & 0 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[32] \quad 1^{11} 6^{11} 131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

6.0

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 1 & 2 \\ 2 & 2 & 3 & 5 & 4 & 3 & 5 \\ 2 & 1 & 2 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 5 & 3 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[31] \ 1^{1^1} 6^{1^1} 13^1 1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[30] \ 1^{1^1} 6^{1^1} 13^1 1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [35] \ 1^{1^1} 8^{1^1} 12^1 3^1 1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 16 \quad [36] \ 1^{1^1} 8^{1^1} 12^1 2^1 3^1 1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.1.4 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 12 \quad [54] \ 1^{1^1} 12^1 3^1 1^1 5^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 4 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[55] \ 1^{1^1} 12^1 3^1 1^1 5^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.5 NHC(-2)(-3) + su₂ + su₂→(-3) + su₂→(-3)

$$T = 12 \quad [11] \quad 1^{13} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[10] \quad 1^{12} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 6 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[9] \quad 1^{12} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 4 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[8] \quad 1^{14} 3^{11} 5^{13} 13^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 9 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.6 NHC(-2)(-3) + su₂ + su₂→(-3) + su_{2n3mix}

$$T = 12 \quad [1] \quad 1^{11} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 3 & 2 \\ 2 & 0 & 4 & 6 & 3 & 7 & 6 \\ 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 7 & 3 & 5 & 5 \\ 2 & 0 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[2] \quad 1^{13} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 1 & 2 & 0 & 3 & 2 \\ 2 & 0 & 3 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[3] \quad 1^{12} 3^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 3 & 2 \\ 6 & 0 & 9 & 49 & 8 & 18 & 15 \\ 1 & 0 & 1 & 8 & 0 & 3 & 2 \\ 2 & 0 & 3 & 18 & 3 & 5 & 5 \\ 2 & 0 & 2 & 15 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[0] \quad 1^{11} 3^{11} 5^{13} 13^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 3 & 2 \\ 6 & 0 & 6 & 34 & 7 & 15 & 12 \\ 1 & 0 & 1 & 7 & 0 & 3 & 2 \\ 2 & 0 & 3 & 15 & 3 & 5 & 5 \\ 2 & 0 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.7 NHC(-2)(-3) + su₂ + su_{2n3mix} + su₂

$$T = 12 \quad [209] \quad 1^{1111} 31^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 7 & 5 \\ 2 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[208] \quad 1^{1131} 31^{13} 15131^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 3 & 13 & 4 & 3 \\ 1 & 2 & 0 & 1 & 10 & 4 & 2 \\ 1 & 3 & 1 & 1 & 13 & 4 & 3 \\ 9 & 13 & 10 & 13 & 67 & 23 & 17 \\ 2 & 4 & 4 & 4 & 23 & 6 & 6 \\ 2 & 3 & 2 & 3 & 17 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[207] \quad 1^{1111} 31^{13} 15131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 3 & 6 & 4 & 3 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 3 & 1 & 1 & 6 & 4 & 3 \\ 4 & 6 & 5 & 6 & 14 & 11 & 8 \\ 2 & 4 & 4 & 4 & 11 & 6 & 6 \\ 2 & 3 & 2 & 3 & 8 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.8 NHC(-2)(-3) + su₂ + su_{2n3mix} + su₂ → (-3)

$$T = 12 \quad [0] \quad 1^{1131} 1^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 1 & 2 \\ 6 & 0 & 6 & 34 & 15 & 7 & 12 \\ 2 & 0 & 3 & 15 & 5 & 3 & 5 \\ 1 & 0 & 1 & 7 & 3 & 0 & 2 \\ 2 & 0 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[3] \quad 1^{1131} 1^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 1 & 2 \\ 2 & 0 & 4 & 6 & 7 & 3 & 6 \\ 2 & 0 & 3 & 7 & 5 & 3 & 5 \\ 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[1] \quad 1^{131} 1^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 2 & 0 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 3 & 4 & 5 & 3 & 5 \\ 1 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 0 & 2 & 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[2] \quad 1^{1231} 1^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 1 & 2 \\ 6 & 0 & 9 & 49 & 18 & 8 & 15 \\ 2 & 0 & 3 & 18 & 5 & 3 & 5 \\ 1 & 0 & 1 & 8 & 3 & 0 & 2 \\ 2 & 0 & 2 & 15 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.9 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 12 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{3}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{2}} 5^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 13^{\textcolor{red}{1}\textcolor{blue}{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 13 & 3 \\ 1 & 2 & 0 & 1 & 3 & 10 & 2 \\ 1 & 3 & 1 & 1 & 4 & 13 & 3 \\ 3 & 4 & 3 & 4 & 5 & 21 & 5 \\ 9 & 13 & 10 & 13 & 21 & 67 & 17 \\ 2 & 3 & 2 & 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 5^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 13^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 3 & 11 & 6 & 3 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 3 & 1 & 1 & 11 & 6 & 3 \\ 8 & 11 & 8 & 11 & 46 & 27 & 14 \\ 4 & 6 & 5 & 6 & 27 & 14 & 8 \\ 2 & 3 & 2 & 3 & 14 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.1.10 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 15 \quad [53] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 15^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{2}\textcolor{red}{1}} 6^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 10 & 2 & 4 \\ 3 & 2 & 8 & 1 & 4 \\ 10 & 8 & 25 & 3 & 13 \\ 2 & 1 & 3 & 0 & 2 \\ 4 & 4 & 13 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.1.11 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 15 \quad [63] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 15^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{2}\textcolor{red}{1}} 6^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 2 & 1 & 8 & 4 \\ 2 & 1 & 0 & 3 & 2 \\ 10 & 8 & 3 & 25 & 13 \\ 4 & 4 & 2 & 13 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.1.12 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 15 \quad [68] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{1}} 6^{\textcolor{red}{1}} 13^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{1}} 15^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}\textcolor{blue}{2}\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 4 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.2 Non-unimodular ($|\det| > 1$) — 1115 bases

4.2.1 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

4.2.2 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -34, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 5 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

4.2.3 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [6] \quad 1^{12} 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 8 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 8 & 14 \\ 1 & 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13} 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 12 & 21 \\ 1 & 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 & 7 \\ 1 & 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

4.2.4 NHC(-2)(-3) + NHC(-2)(-3) + su₂→(-3) + su₂

$$T = 12 \quad [8] \quad 1^{11} \overset{1}{5} 13^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 6 & 8 & 10 & 8 & 0 & 8 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13} \overset{1}{5} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 9 & 12 & 15 & 12 & 0 & 12 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

4.2.5 NHC(-2)(-3) + so₈ + su₂ + su₂n3mix

$$T = 13 \quad [0] \quad 1^{13} \overset{1}{5} 13221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 9.0$$

$$[3] \quad 1^{13} \overset{1}{5} 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 8 & 4 \\ 3 & 0 & 5 & 3 & 8 & 5 \\ 5 & 0 & 8 & 8 & 15 & 9 \\ 3 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} \overset{1}{5} 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 9 & 8 & 4 \\ 7 & 0 & 9 & 23 & 20 & 11 \\ 5 & 0 & 8 & 20 & 15 & 9 \\ 3 & 0 & 4 & 11 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} \overset{1}{5} 1321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 19 & 8 & 4 \\ 13 & 0 & 19 & 79 & 36 & 21 \\ 5 & 0 & 8 & 36 & 15 & 9 \\ 3 & 0 & 4 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13} \overset{1}{5} 1321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 14 & 8 & 4 \\ 10 & 0 & 14 & 46 & 28 & 16 \\ 5 & 0 & 8 & 28 & 15 & 9 \\ 3 & 0 & 4 & 16 & 9 & 5 \end{pmatrix}_{b_0}$$

4.2.6 $\mathbf{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2$

$$T = 12 \quad [0] \overset{11}{1^3 3^1 5^1 1^3 1^1} + [\textcolor{red}{s} \textcolor{blue}{u} \textcolor{red}{2} \cdot \textcolor{blue}{g} \textcolor{red}{2}] + \textcolor{green}{s} \textcolor{blue}{0} \textcolor{red}{8} + \textcolor{brown}{s} \textcolor{blue}{u} \textcolor{red}{2} \rightarrow (-3) + \textcolor{red}{s} \textcolor{blue}{u} \textcolor{red}{2} \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 0 & 4 \end{pmatrix} b_0$$

$$[1] \ 1^{11} 3^1 1^1 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 2, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 6 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix} b_C$$

$$[2] \quad 1^{11}3^1 15131^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 3 \\ 6 & 8 & 0 & 7 & 22 & 10 \\ 3 & 4 & -1 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{13}31^513^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{brown}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 11 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix} b_0$$

[4] $1^{12}31^513^{11}1^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{0}\textcolor{green}{g} + \textcolor{brown}{s}\textcolor{red}{u}_2 \rightarrow (-3) + \textcolor{brown}{s}\textcolor{red}{u}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 9 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix} b_0$$

[5] $1^{11}3^{11}15^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 4 & -1 & 3 & 4 & 4 \end{pmatrix} b_0$$

[6] $1^3 1^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 11$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 2 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[7] $1^{12}31^513^{11}1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 10 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^11^15^{13}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 2 & 1 & 3 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 17 & 3 \\ 14 & 20 & 0 & 17 & 130 & 24 \\ 3 & 4 & -1 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 8 \\ 1 & 1 & 0 & 3 & 0 & 3 \\ 2 & 4 & 0 & 8 & 3 & 6 \\ 2 & 3 & -1 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 3^{11} 5^{11} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 13 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} 3^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 2 & 1 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 3 \\ 10 & 16 & 0 & 13 & 70 & 18 \\ 3 & 4 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12} 3^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 2 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 3 \\ 8 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 7 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{14} 3^{11} 5^{11} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 2 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 20 & 3 \\ 16 & 24 & 0 & 20 & 174 & 28 \\ 3 & 4 & -1 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3^{11} 5^{12} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 8 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12} 3^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 91, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

- [17] $1^{13}3^{11}5^{12}13^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 20 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$
- [18] $1^{12}3^{11}5^{12}13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 14 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$
- [19] $1^{12}3^{11}5^{13}13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 15 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$
- [20] $1^{12}3^{11}5^{12}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 154, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [21] $1^{12}3^{11}5^{13}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 187, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$
- [22] $1^{13}3^{11}5^{13}13^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 197, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 21 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$
- [23] $1^{14}3^{11}5^{13}13^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 203, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 32 & 3 \\ 2 & 0 & 3 & 1 & 28 & 3 \\ 27 & 0 & 32 & 28 & 382 & 40 \\ 3 & 0 & 3 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$
- [24] $1^{13}3^{11}5^{12}13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 214, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 16 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$
- [25] $1^{14}3^{11}5^{13}13^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 226, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 22 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{13}3^11^513^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 305, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 17 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^11^513^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 375, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 23 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{14}3^11^513^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 427, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 20 & 3 \\ 19 & 0 & 24 & 20 & 174 & 28 \\ 3 & 0 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

4.2.7 NHC(-2)(-3) + so₈ + su₂→(-3) + su₂→(-3)

$T = 12$ [0] $1^{13}3^11^513^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 3 & 3 & 1 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

4.2.8 NHC(-2)(-3) + so₈ + su₂→(-3) + su₂n3mix

$T = 12$ [0] $1^{12}3^11^513^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 3 & 0 & 3 & 4 & 17 & 4 \\ 12 & 0 & 15 & 17 & 65 & 17 \\ 3 & 0 & 3 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^513^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 5 & 7 & 0 & 6 & 14 & 8 \\ 3 & 4 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{13}3^11^513^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 13 & 19 & 0 & 16 & 110 & 22 \\ 3 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{12}3^11^513^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 2 & 0 & 3 & 1 & 13 & 3 \\ 13 & 0 & 15 & 13 & 76 & 18 \\ 3 & 0 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

4.2.9 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} 31 \bar{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 6 & 0 & 7 & 14 & 7 & 8 \\ 2 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{121} 31 \bar{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 4 & 3 \\ 12 & 0 & 15 & 76 & 15 & 18 \\ 2 & 0 & 4 & 15 & 2 & 4 \\ 3 & 0 & 3 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1 \bar{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 4 & 3 \\ 9 & 0 & 11 & 39 & 11 & 13 \\ 2 & 0 & 4 & 11 & 2 & 4 \\ 3 & 0 & 3 & 13 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1 \bar{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 8 & 3 \\ 9 & 0 & 11 & 39 & 31 & 13 \\ 6 & 0 & 8 & 31 & 22 & 10 \\ 3 & 0 & 3 & 13 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31 \bar{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 5 & 0 & 7 & 14 & 7 & 8 \\ 3 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{131} 31 \bar{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 19 & 4 & 3 \\ 16 & 0 & 19 & 110 & 19 & 22 \\ 2 & 0 & 4 & 19 & 2 & 4 \\ 3 & 0 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 31 \bar{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 3 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{113} 31 \bar{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 8 \\ 10 & 16 & 0 & 33 & 70 & 18 \\ 3 & 4 & -1 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 23, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 2 & 1 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 8 \\ 6 & 8 & 0 & 19 & 22 & 10 \\ 3 & 4 & -1 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121} 3^1 1 5 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 23, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 6 & 3 \\ 2 & 2 & 1 & 15 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 39 & 17 \\ 6 & 8 & 0 & 39 & 22 & 10 \\ 3 & 4 & -1 & 17 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{131} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 35, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 22 \\ 2 & 4 & 0 & 19 & 2 & 4 \\ 3 & 4 & -1 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121} 3^1 1 5 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 35, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 2 & 1 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 15 & 17 \\ 2 & 4 & 0 & 15 & 2 & 4 \\ 3 & 4 & -1 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 5 & 0 & 7 & 14 & 33 & 8 \\ 13 & 0 & 16 & 33 & 70 & 18 \\ 3 & 0 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 8 & 3 \\ 5 & 0 & 7 & 14 & 19 & 8 \\ 7 & 0 & 8 & 19 & 22 & 10 \\ 3 & 0 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 50, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 2 & 1 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 8 \\ 8 & 12 & 0 & 26 & 42 & 14 \\ 3 & 4 & -1 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{122} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 54, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 11 & 0 & 32 & 38 & 12 \\ 8 & 12 & 0 & 38 & 42 & 14 \\ 3 & 4 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^115131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 11 & 0 & 39 & 42 & 13 \\ 8 & 12 & 0 & 42 & 42 & 14 \\ 3 & 4 & -1 & 13 & 14 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{123}3^115131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 2 & 1 & 15 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 69 & 17 \\ 10 & 16 & 0 & 69 & 70 & 18 \\ 3 & 4 & -1 & 17 & 18 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{112}315131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 5 & 0 & 7 & 14 & 26 & 8 \\ 10 & 0 & 12 & 26 & 42 & 14 \\ 3 & 0 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{141}3^115131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 2 & 1 & 27 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 27 & 0 & 217 & 27 & 31 \\ 2 & 4 & 0 & 27 & 2 & 4 \\ 3 & 4 & -1 & 31 & 4 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^115131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 2 & 1 & 15 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 54 & 17 \\ 8 & 12 & 0 & 54 & 42 & 14 \\ 3 & 4 & -1 & 17 & 14 & 4 \end{pmatrix}_{b_0}$$

$T = 13$ [21] $1^{113}3^1151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 19 & 4 \\ 5 & 0 & 8 & 15 & 36 & 9 \\ 13 & 0 & 19 & 36 & 79 & 21 \\ 3 & 0 & 4 & 9 & 21 & 5 \end{pmatrix}_{b_0}$$

[22] $1^{113}3^1151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 9 & 4 \\ 5 & 0 & 8 & 15 & 20 & 9 \\ 7 & 0 & 9 & 20 & 23 & 11 \\ 3 & 0 & 4 & 9 & 11 & 5 \end{pmatrix}_{b_0}$$

[23] $1^{13}3^1151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 5 & 4 \\ 5 & 0 & 8 & 15 & 8 & 9 \\ 3 & 0 & 5 & 8 & 3 & 5 \\ 3 & 0 & 4 & 9 & 5 & 5 \end{pmatrix}_{b_0}$$

[24] $1^{113}3^1151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0\textcolor{red}{g} + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 14 & 4 \\ 5 & 0 & 8 & 15 & 28 & 9 \\ 10 & 0 & 14 & 28 & 46 & 16 \\ 3 & 0 & 4 & 9 & 16 & 5 \end{pmatrix}_{b_0}$$

4.2.10 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 12 & 0 & 15 & 65 & 17 & 17 \\ 3 & 0 & 3 & 17 & 4 & 4 \\ 3 & 0 & 3 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & -1 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 162, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 13 & 0 & 15 & 76 & 13 & 18 \\ 2 & 0 & 3 & 13 & 1 & 3 \\ 3 & 0 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

4.2.11 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix

$$T = 12 \quad [0] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 11 & 3 \\ 5 & 0 & 7 & 14 & 25 & 8 \\ 10 & 0 & 11 & 25 & 39 & 13 \\ 3 & 0 & 3 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -31, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 8 \\ 10 & 15 & 0 & 32 & 65 & 17 \\ 3 & 4 & -1 & 8 & 17 & 4 \end{pmatrix}_{b_0}$$

4.2.12 NHC(-2)(-3) + $\mathfrak{su}_2$ + NHC(-2)(-3) + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -34, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

4.2.13 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{12} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 9 & 0 & 0 & 28 & 12 & 10 \\ 2 & 3 & 0 & 0 & 12 & 4 & 4 \\ 2 & 3 & -1 & 0 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 1 & 6 & 3 \\ 4 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 3 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -45, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 0 & 3 & 6 & 1 & 3 \\ 2 & 3 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 15 & 7 \\ 3 & 6 & 0 & 6 & 15 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 15 & 7 \\ 4 & 6 & 0 & 6 & 15 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 2^1 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$T = 10 \quad [19] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 3 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 4 & 6 & 8 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 5 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overset{1}{11}, \overset{1}{11}} 5^{\overset{1}{11}} 1^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [29] \quad 1^{\overset{1}{111}, \overset{1}{111}} 31^{\overset{1}{111}} 41^{\overset{1}{111}} 31^{\overset{1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 1 & 3 & 4 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 & 1 & 0 \\ 2 & 3 & 1 & 2 & 4 & 7 & 4 \\ 3 & 4 & 0 & 4 & 5 & 9 & 5 \\ 4 & 6 & 1 & 7 & 9 & 14 & 8 \\ 2 & 3 & 0 & 4 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

4.2.14 NHC(-2)(-3) + su₂ + su₂ + su₂ → (-3)

$$T = 9 \quad [0] \quad 1^{\overset{1}{11}, \overset{1}{11}} 3^{\overset{1}{11}, \overset{1}{11}} 1^{\overset{1}{11}} 31^{\overset{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & -1 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}313^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 3 & -1 & 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 3 & -1 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 2 & 3 \\ 4 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 3 & -1 & 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \ 1^{11}3^{11}141^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 4 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 2 & 4 & 2 & 4 \\ 3 & 4 & 0 & 4 & 5 & 3 & 5 \\ 1 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 3 & 0 & 4 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
T = 12 \quad [19] \quad 1^{143}3^{11}5^{1321}1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[18] \quad 1^{113}3^{11}5^{1321}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[17] \quad 1^{33}3^{11}5^{1321}1^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.6 \\
[15] \quad 1^{23}3^{11}5^{1321}1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.3 \\
[14] \quad 1^{22}3^{11}5^{1321}1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[13] \quad 1^{13}3^{11}5^{1321}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8 \\
[12] \quad 1^{44}3^{11}5^{1321}1^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[11] \quad 1^{43}3^{11}5^{1321}1^{143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[10] \quad 1^{42}3^{11}5^{1321}1^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[9] \quad 1^{111}3^{11}5^{14141}1^{131}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[16] \quad 1^{12}3^{11}5^{1321}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5 \\
[20] \quad 1^{111}3^{11}5^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[21] \quad 2^1 3^{111} 5^{1513} 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 19 & 5 & 9 \\ 4 & 7 & 5 & 19 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112}3^{11}5^{151}2^{231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 1 & 2 \\ 1 & 2 & 1 & 4 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 3 & 4 \\ 6 & 13 & 0 & 13 & 49 & 11 & 15 \\ 1 & 3 & 0 & 3 & 11 & 1 & 3 \\ 2 & 4 & -1 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

4.2.15 NHC(-2)(-3) + su₂ + su₂ + su₂n3mix

$$T = 12 \quad [9] \quad 1^{123}3^{11}5^{1321}1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$$

$$[14] \quad 1^{141}3^{11}5^{1321}1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[4] \quad 1^{422}3^{11}5^{1321}1^{422} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

[17] $1^{144}3^1151321^{1144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$

[16] $1^{143}3^1151321^{1143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$

[15] $1^{142}3^1151321^{1142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$

[3] $1^{421}3^1151321^{1421} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$

[30] $2^3 3^{1111} 1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 3 \\ 3 & 4 & 4 & 6 & 8 & 7 \\ 3 & 4 & 4 & 8 & 6 & 7 \\ 3 & 4 & 3 & 7 & 7 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

4.2.16 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 8$ [0] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

4.2.17 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$T = 12$ [0] $1^{111}31513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 4 & 4 \\ 2 & 3 & -1 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

4.2.18 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 9$ [0] $1^{13}1^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 3 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 & 3 \\ 2 & 3 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 5 & 0 & 2 & 2 & 4 \\ 3 & 5 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 3 & -1 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[9] $1^{12}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 6 & 10 & 0 & 4 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 8 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 10 & 0 & 5 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{13} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 9 & 15 & 0 & 6 & 34 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 \\ 2 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^1 1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 \\ 4 & 6 & 0 & 6 & 5 & 13 \\ 2 & 3 & -1 & 3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{14} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 12 & 20 & 0 & 8 & 62 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 9 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{12} 1^{11} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 11 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[22] $1^{13} 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$

[21] $1^{11} 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$

[24] $1^{12} 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$

[23] $1^{13} 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$

[25] $1^{12} 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$

[26] $1^{14} 1^{11} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$

[27] $1^{13} 1^{11} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$

[29] $1^{13} 1^{11} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 12 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 14 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 14 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{14} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 15 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{14} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{13} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 15 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{13} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 17 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{12}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{14}3^{14}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 18 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{13}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^{14}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 14 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 14 & 20 & 0 & 106 & 22 \\ 3 & 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{14}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{13} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 17 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [52] \quad 1^{111} 3^1 141^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 3 & 4 & 0 & 4 & 3 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [53] \quad 1^{111} 3^1 14141^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[56] \ 1^{111}3151^{123}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{113}1^1 5 \ 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 0 & 2 & 1 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{12}31^1 5 \ 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 5, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 5 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{14}31^1 5 \ 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 1 & 14 & 3 \\ 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 14 & 10 & 0 & 8 & 74 & 18 \\ 2 & 3 & 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{111}31^1 5 \ 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 5 & 2 \\ 2 & 0 & 4 & 6 & 3 & 12 & 6 \\ 1 & 0 & 1 & 3 & 0 & 5 & 2 \\ 4 & 0 & 5 & 12 & 5 & 17 & 9 \\ 2 & 0 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{112}31^1 5 \ 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 9 & 2 \\ 2 & 0 & 4 & 6 & 3 & 20 & 6 \\ 1 & 0 & 1 & 3 & 0 & 8 & 2 \\ 6 & 0 & 9 & 20 & 8 & 49 & 15 \\ 2 & 0 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{113}1^1 5 \ 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 10, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 2 & 4 & 3 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 4 & 0 & 3 & 6 & 6 \\ 2 & 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{124}31^1 5 \ 131^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 12, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 8 & 2 \\ 1 & 1 & 1 & 8 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 64 & 12 \\ 1 & 1 & 0 & 6 & 0 & 12 & 2 \\ 8 & 16 & 0 & 64 & 12 & 126 & 24 \\ 2 & 3 & -1 & 12 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[63] \ 2 \ 3 \ 1513^{11}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 12, \ (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 1 & 5 & 2 \\ 4 & 7 & 5 & 17 & 5 & 19 & 9 \\ 1 & 2 & 1 & 5 & 0 & 5 & 2 \\ 4 & 7 & 5 & 19 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{144} 31 5 13^1 1^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 128 & 24 \\ 1 & 1 & 0 & 12 & 0 & 12 & 2 \\ 8 & 16 & 0 & 128 & 12 & 126 & 24 \\ 2 & 3 & -1 & 24 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{122} 31 5 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 4 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 31 5 13^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 10 & 2 \\ 2 & 0 & 4 & 6 & 3 & 24 & 6 \\ 1 & 0 & 1 & 3 & 0 & 10 & 2 \\ 8 & 0 & 10 & 24 & 10 & 74 & 18 \\ 2 & 0 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{131} 31 5 13^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 1 & 1 & 12 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 24 & 18 \\ 1 & 1 & 0 & 9 & 0 & 3 & 2 \\ 2 & 4 & 0 & 24 & 3 & 6 & 6 \\ 2 & 3 & -1 & 18 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{132} 31 5 13^1 1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 4 & 2 \\ 1 & 1 & 1 & 12 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 48 & 18 \\ 1 & 1 & 0 & 9 & 0 & 6 & 2 \\ 4 & 8 & 0 & 48 & 6 & 30 & 12 \\ 2 & 3 & -1 & 18 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{133} 31 5 13^1 1^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 72 & 18 \\ 1 & 1 & 0 & 9 & 0 & 9 & 2 \\ 6 & 12 & 0 & 72 & 9 & 70 & 18 \\ 2 & 3 & -1 & 18 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{134} 31 5 13^1 1^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 96 & 18 \\ 1 & 1 & 0 & 9 & 0 & 12 & 2 \\ 8 & 16 & 0 & 96 & 12 & 126 & 24 \\ 2 & 3 & -1 & 18 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{111} 31 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 6 & 6 \\ 2 & 3 & -1 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{112}31513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 1 & 4 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 16 & 6 \\ 1 & 1 & 0 & 3 & 0 & 6 & 2 \\ 4 & 8 & 0 & 16 & 6 & 30 & 12 \\ 2 & 3 & -1 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{114}31513^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 1 & 4 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 32 & 6 \\ 1 & 1 & 0 & 3 & 0 & 12 & 2 \\ 8 & 16 & 0 & 32 & 12 & 126 & 24 \\ 2 & 3 & -1 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{122}31513^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 32 & 12 \\ 1 & 1 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 0 & 32 & 6 & 30 & 12 \\ 2 & 3 & -1 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{122}31513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 5 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{111}31513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 4 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{113}31513^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 7 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{123}11513^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 2 & 8 & 3 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 6 & 8 & 8 & 0 & 6 & 30 & 12 \\ 2 & 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{123}11513^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 2 & 7 & 3 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 5 & 7 & 5 & 0 & 5 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{112}3^11^15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{113}3^11^15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 8 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{113}3^11^15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 8 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{111}3^11^15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{114}3^11^15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 14 & 3 \\ 1 & 2 & 0 & 4 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 10 & 2 \\ 9 & 14 & 10 & 24 & 10 & 74 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{13}3^11^15131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 2 & 12 & 3 \\ 1 & 2 & 0 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 12 & 12 & 0 & 9 & 70 & 18 \\ 2 & 3 & 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{112}3^11^15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{13}3^11^15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 2 & 11 & 3 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 8 & 11 & 9 & 0 & 8 & 49 & 15 \\ 2 & 3 & 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{112}3^{11}5131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{112}3^{11}5131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{111}3^{11}5131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 52, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 3 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{112}3^{11}51^{1231^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 2 & 1 & 4 & 3 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 13 & 4 \\ 1 & 3 & 0 & 3 & 1 & 11 & 3 \\ 6 & 13 & 0 & 13 & 11 & 49 & 15 \\ 2 & 4 & -1 & 4 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{112}3^{11}5131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 7 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 7 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{112}3^{11}5131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{113}3^{11}5131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 12 & 3 \\ 1 & 2 & 0 & 4 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 9 & 2 \\ 9 & 12 & 12 & 24 & 9 & 70 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{114}3^{11}5131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 69, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 2 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 11 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{123}3^1 1^1 5 131^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 90, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 1 & 9 & 2 \\ 5 & 8 & 8 & 30 & 6 & 40 & 12 \\ 1 & 2 & 1 & 6 & 0 & 8 & 2 \\ 7 & 11 & 9 & 40 & 8 & 49 & 15 \\ 2 & 3 & 2 & 12 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{112}3^1 1^1 5 131^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 90, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 5 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{114}3^1 1^1 5 131^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 94, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 12 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 2 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 12 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{123}3^1 1^1 5 131^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 8 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 8 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{14}3^1 1^1 5 131^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 2 & 15 & 3 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 10 & 15 & 13 & 0 & 11 & 97 & 21 \\ 2 & 3 & 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{112}3^1 1^1 5 131^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{122}3^1 1^1 5 131^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 1 & 8 & 2 \\ 6 & 8 & 8 & 30 & 6 & 32 & 12 \\ 1 & 2 & 1 & 6 & 0 & 6 & 2 \\ 5 & 8 & 8 & 32 & 6 & 30 & 12 \\ 2 & 3 & 2 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{122}3^1 1^1 5 131^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 7 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 7 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{114}3^{11}5^{13}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 297, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 10 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 3 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 10 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{114}3^{11}5^{13}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 298, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 3 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 11 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

4.2.19 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{11}3^{11}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{11}5^{11}13^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 7 & 0 & 0 & 4 \\ 6 & 7 & 9 & 0 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}5^{11}13^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 6 & 7 & 9 & 5 & 0 & 7 \\ 3 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12}5^{11}13^{11}2^33^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 12 & 14 & 42 & 10 & 0 & 14 \\ 3 & 3 & 10 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}5^{11}13^{11}3^33^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 18 & 21 & 97 & 15 & 0 & 21 \\ 3 & 3 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 3^{11}5^{11}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 11, 0)$$

$$(1, 11, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 2 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 6 & 8 & 6 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 8 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 3^1 1^1 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 2 & 4 & 12 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 9 & 12 & 9 & 34 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 3^1 1^1 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -96, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 2 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \ 1^{14} 3^1 1^1 5^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det =$$

$$4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 10 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{12} 3^1 1^1 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 5 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

4.2.20 NHC(-2)(-3) + su₂ + su₂ → (-3) + su_{2n3mix}

$$T = 12 \quad [4] \ 1^{12} 3^1 1^1 5^{13} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 3 & 2 \\ 8 & 0 & 10 & 74 & 10 & 22 & 18 \\ 1 & 0 & 1 & 10 & 0 & 3 & 2 \\ 2 & 0 & 3 & 22 & 3 & 5 & 5 \\ 2 & 0 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 5^{13} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 3 & 2 \\ 4 & 0 & 5 & 17 & 5 & 11 & 9 \\ 1 & 0 & 1 & 5 & 0 & 3 & 2 \\ 2 & 0 & 3 & 11 & 3 & 5 & 5 \\ 2 & 0 & 2 & 9 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^1 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 2 & 8 & 3 \\ 1 & 2 & 0 & 5 & 1 & 7 & 2 \\ 4 & 7 & 5 & 17 & 5 & 23 & 9 \\ 1 & 2 & 1 & 5 & 0 & 6 & 2 \\ 6 & 8 & 7 & 23 & 6 & 27 & 11 \\ 2 & 3 & 2 & 9 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112}3^{11}5^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121}3^{11}5^{13}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 6 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}3^{11}5^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 3 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{131}3^{11}5^{13}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 3 & 2 \\ 1 & 1 & 2 & 12 & 2 & 4 & 3 \\ 1 & 2 & 0 & 12 & 1 & 3 & 2 \\ 9 & 12 & 12 & 70 & 9 & 21 & 18 \\ 1 & 2 & 1 & 9 & 0 & 3 & 2 \\ 3 & 4 & 3 & 21 & 3 & 5 & 5 \\ 2 & 3 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{133}1^{11}5^{13}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 2 & 12 & 3 \\ 1 & 2 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 12 & 11 & 0 & 9 & 65 & 17 \\ 2 & 3 & 2 & 0 & 2 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112}3^{11}5^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 3 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

4.2.21 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [2] \ 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 & 6 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{111}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 8 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

6.5

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 16 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 4 & 6 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{132} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

6.5

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 33 & 16 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 10 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 5 & 6 \\ 2 & 1 & 3 & 0 & 5 & 1 & 3 \\ 2 & 2 & 2 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 5 & 6 \\ 2 & 1 & 3 & 0 & 5 & 1 & 3 \\ 2 & 2 & 2 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

6.0

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 14 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 19 & 10 \\ 6 & 10 & 0 & 19 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 & 6 \\ 3 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 4 & 6 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{121} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 & 2 \\ 1 & 2 & 0 & 0 & 12 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 8 & 14 \\ 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & 0 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{123} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 15 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 11 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{112} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 4 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{111} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 3 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{141} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 & 2 \\ 1 & 2 & 0 & 0 & 13 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 10 & 16 \\ 2 & 2 & 1 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{112} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 34 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{131} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 7 & 11 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{13} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 10 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 17 & 6 \\ 6 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{123} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 17 & 6 \\ 7 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{121} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{114} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 5 & 0 & 13 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 & 10 \\ 3 & 5 & 0 & 10 & 2 & 4 \\ 3 & 5 & 0 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{121} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 25 & 11 \\ 5 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & 0 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 6 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{122} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{12} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 9 & 11 \\ 2 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & 0 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 8 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 9 & 11 \\ 2 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & 0 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{141} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 6 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{121} 1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 4 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{123} 1^{124} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 46 & 11 \\ 9 & 10 & 10 & 0 & 46 & 58 & 14 \\ 2 & 2 & 2 & 0 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{122} 1^1 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 20 & 10 \\ 6 & 10 & 0 & 20 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{132} 1^{11} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 34 & 18 \\ 6 & 10 & 0 & 34 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{113} 1^{11} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{112} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{141} 1^{121} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 7 & 11 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{121} 1^{11} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 13 & 14 \\ 3 & 5 & 0 & 13 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{121} 3^{112} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^{114} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 28 & 6 \\ 10 & 8 & 14 & 0 & 28 & 70 & 16 \\ 2 & 2 & 2 & 0 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{123} 3^{112} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 30 & 10 \\ 9 & 15 & 0 & 30 & 34 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^{112} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{123} 3^{112} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 14 & 6 \\ 6 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{122} 3^{112} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 9 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{131} 3^{112} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 9 & 11 \\ 2 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & 0 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3^{112} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 40 & 10 \\ 12 & 20 & 0 & 40 & 62 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{141} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 14 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 12 & 18 \\ 2 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 7 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 13 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 & 14 \\ 3 & 5 & 0 & 14 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 9 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 25 & 11 \\ 5 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & 0 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{123} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 10 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{123} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 5 & 0 & 18 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 28 & 14 \\ 6 & 10 & 0 & 28 & 14 & 8 \\ 3 & 5 & 0 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[82] $1^{132} 3^1 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{14} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 & 2 \\ 1 & 2 & 0 & 0 & 13 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 10 & 16 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{113} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{113} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 17 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[86] $1^{121} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -123$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[87] $1^{131} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -123$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 13 & 16 \\ 2 & 1 & 3 & 0 & 13 & 1 & 3 \\ 2 & 2 & 2 & 0 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{132} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{114} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 16 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{111} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 29 & 9 \\ 7 & 5 & 10 & 0 & 29 & 33 & 11 \\ 2 & 2 & 2 & 0 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{131} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 9 & 13 \\ 2 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{131} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 9 & 13 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{113} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 10 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{11} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 39 & 10 \\ 12 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 5 & 0 & 13 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 58 & 10 \\ 18 & 30 & 0 & 58 & 134 & 24 \\ 3 & 5 & 0 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[99] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 13 & 9 \\ 3 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{13} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 & 2 \\ 1 & 2 & 0 & 0 & 13 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 10 & 16 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 23 & 11 \\ 4 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{132} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 13 & 9 \\ 4 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[104] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[105] $1^{131} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 36 & 16 \\ 5 & 4 & 7 & 0 & 36 & 16 & 8 \\ 2 & 2 & 2 & 0 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{114} 3^1 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{124} 3^1 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{141} 3^1 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 13 & 16 \\ 2 & 1 & 3 & 0 & 13 & 1 & 3 \\ 2 & 2 & 2 & 0 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{141} 3^1 21^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 18 & 18 \\ 3 & 5 & 0 & 18 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[112] $1^{131} 3^1 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{132} 3^1 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{114} 3^1 21^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 24 & 4 \\ 12 & 12 & 16 & 0 & 24 & 110 & 20 \\ 2 & 2 & 2 & 0 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[110] $1^{131} 3^1 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{131} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 36 & 16 \\ 5 & 4 & 7 & 0 & 36 & 16 & 8 \\ 2 & 2 & 2 & 0 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 36 & 18 \\ 6 & 10 & 0 & 36 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{132} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 38 & 13 \\ 6 & 6 & 8 & 0 & 38 & 26 & 10 \\ 2 & 2 & 2 & 0 & 13 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 26 & 9 \\ 6 & 6 & 8 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & 0 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -167, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{13} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 10 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{132} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 4 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{12} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -179, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 8 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{111} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -183, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 21 & 9 \\ 5 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 6 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 8 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{122} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -188, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 23 & 11 \\ 5 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -189, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 15 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -192, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 5 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 10 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{131} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 & 2 \\ 1 & 0 & 2 & 0 & 13 & 1 & 2 \\ 1 & 2 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 17 & 21 \\ 2 & 1 & 3 & 0 & 17 & 1 & 3 \\ 2 & 2 & 2 & 0 & 21 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{122} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 41 & 11 \\ 8 & 7 & 11 & 0 & 41 & 45 & 13 \\ 2 & 2 & 2 & 0 & 11 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{112} 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 7 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{123} 1^{113} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 39 & 9 \\ 9 & 9 & 12 & 0 & 39 & 61 & 15 \\ 2 & 2 & 2 & 0 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{131} 1^{132} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 33 & 16 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{114} 1^{12} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 18 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{141} 1^{11} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{131} 1^{12} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -222, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 19 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{113} 1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 9 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1^{14} 1^{141} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 14 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 12 & 18 \\ 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{141} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -227$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 & 2 \\ 1 & 2 & 0 & 0 & 17 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 13 & 21 \\ 1 & 2 & 1 & 0 & 13 & 0 & 2 \\ 2 & 2 & 2 & 0 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 5 & 2 \\ 1 & 0 & 2 & 0 & 13 & 4 & 2 \\ 1 & 2 & 0 & 0 & 17 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 47 & 21 \\ 5 & 4 & 7 & 0 & 47 & 16 & 8 \\ 2 & 2 & 2 & 0 & 21 & 8 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{124} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{124} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 20 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{142} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10.00 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 8 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 11 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 1 & 2 \\ 1 & 2 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 15 & 17 \\ 2 & 1 & 3 & 0 & 15 & 1 & 3 \\ 2 & 2 & 2 & 0 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{142} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 4 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 9 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 16 & 19 \\ 2 & 1 & 3 & 0 & 16 & 1 & 3 \\ 2 & 2 & 2 & 0 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 7 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{124} 3^1 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -260, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 52 & 9 \\ 12 & 12 & 16 & 0 & 52 & 110 & 20 \\ 2 & 2 & 2 & 0 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{124} 3^1 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -263, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 9 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{142} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -263, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 33 & 16 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{142} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{131} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{142} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 4 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{123} 3^1 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -299, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 8 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{141} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 4 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[170] \quad 1^{132} 3^1 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[171] \quad 1^{141} 3^1 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 4 & 2 \\ 1 & 0 & 2 & 0 & 13 & 5 & 2 \\ 1 & 2 & 0 & 0 & 17 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 43 & 21 \\ 4 & 5 & 5 & 0 & 43 & 13 & 7 \\ 2 & 2 & 2 & 0 & 21 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[172] \quad 1^{131} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{122} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -316, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{113} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -319, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 9 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{114} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{132} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{142} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 42 & 9 \\ 10 & 8 & 14 & 0 & 42 & 70 & 16 \\ 2 & 2 & 2 & 0 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 11 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -339$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 24 & 17 \\ 3 & 3 & 4 & 0 & 24 & 5 & 5 \\ 2 & 2 & 2 & 0 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -355$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 11 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -371$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 10 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{124} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 10 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{142} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 12 & 6 & 2 \\ 1 & 2 & 0 & 0 & 14 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 52 & 18 \\ 6 & 6 & 8 & 0 & 52 & 26 & 10 \\ 2 & 2 & 2 & 0 & 18 & 10 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -419$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{114} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -431$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 11 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -431$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 11 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 10 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{142} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 5 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{131} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -447$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 43 & 19 \\ 5 & 4 & 7 & 0 & 43 & 16 & 8 \\ 2 & 2 & 2 & 0 & 19 & 8 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -451$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 48 & 11 \\ 9 & 9 & 12 & 0 & 48 & 61 & 15 \\ 2 & 2 & 2 & 0 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -451$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 0 & 15 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 39 & 17 \\ 5 & 4 & 7 & 0 & 39 & 16 & 8 \\ 2 & 2 & 2 & 0 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 11 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{142} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 36 & 6 \\ 12 & 12 & 16 & 0 & 36 & 110 & 20 \\ 2 & 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{132} 3^1 21^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 26 & 18 \\ 3 & 3 & 4 & 0 & 26 & 5 & 5 \\ 2 & 2 & 2 & 0 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -556$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 27 & 19 \\ 3 & 3 & 4 & 0 & 27 & 5 & 5 \\ 2 & 2 & 2 & 0 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[202] \quad 1^{132} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -580, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 11 & 6 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 54 & 19 \\ 6 & 6 & 8 & 0 & 54 & 26 & 10 \\ 2 & 2 & 2 & 0 & 19 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[203] \quad 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 12 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -715, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 3 & 2 \\ 1 & 2 & 0 & 0 & 17 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 30 & 21 \\ 3 & 3 & 4 & 0 & 30 & 5 & 5 \\ 2 & 2 & 2 & 0 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -796, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 6 & 2 \\ 1 & 0 & 2 & 0 & 13 & 6 & 2 \\ 1 & 2 & 0 & 0 & 17 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 60 & 21 \\ 6 & 6 & 8 & 0 & 60 & 26 & 10 \\ 2 & 2 & 2 & 0 & 21 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [206] \quad 1^{1122} 3^1 1^5 1321^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 6.5$$

$$[210] \quad 1^{1111} 3^1 1^5 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 4 & 3 & 4 & 5 & 7 & 5 \\ 3 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[213] \quad 1^{1114} 3^1 1^5 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 3 & 6 & 15 & 3 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 3 & 1 & 1 & 6 & 15 & 3 \\ 4 & 6 & 5 & 6 & 14 & 39 & 8 \\ 9 & 15 & 13 & 15 & 39 & 97 & 21 \\ 2 & 3 & 2 & 3 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[214] \quad 1^{1112} 3^1 1^5 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 7 & 3 \\ 4 & 6 & 5 & 6 & 14 & 17 & 8 \\ 5 & 7 & 5 & 7 & 17 & 17 & 9 \\ 2 & 3 & 2 & 3 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[212] \quad 1^{112}31^1513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 1 & 3 & 1 & 1 & 4 & 7 & 3 \\ 3 & 4 & 3 & 4 & 5 & 11 & 5 \\ 5 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[211] \quad 1^{111}3^11^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 5 & 2 \\ 2 & 0 & 4 & 6 & 7 & 12 & 6 \\ 2 & 0 & 3 & 7 & 5 & 11 & 5 \\ 4 & 0 & 5 & 12 & 11 & 17 & 9 \\ 2 & 0 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[215] \quad 1^{1114}31^1513^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 15 & 3 \\ 1 & 2 & 0 & 1 & 3 & 13 & 2 \\ 1 & 3 & 1 & 1 & 4 & 15 & 3 \\ 3 & 4 & 3 & 4 & 5 & 25 & 5 \\ 9 & 15 & 13 & 15 & 25 & 97 & 21 \\ 2 & 3 & 2 & 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[216] \quad 1^{1141}3^11^1513^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 17 & 4 & 3 \\ 1 & 2 & 0 & 1 & 1 & 14 & 4 & 2 \\ 1 & 3 & 1 & 1 & 1 & 17 & 4 & 3 \\ 11 & 17 & 14 & 17 & 121 & 31 & 23 \\ 2 & 4 & 4 & 4 & 31 & 6 & 6 \\ 2 & 3 & 2 & 3 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[217] \quad 1^{1113}3^11^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 3 & 6 & 11 & 3 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 3 & 1 & 1 & 6 & 11 & 3 \\ 4 & 6 & 5 & 6 & 14 & 28 & 8 \\ 7 & 11 & 9 & 11 & 28 & 49 & 15 \\ 2 & 3 & 2 & 3 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[218] \quad 1^{1113}31^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 3 & 4 & 11 & 3 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 3 & 1 & 1 & 4 & 11 & 3 \\ 3 & 4 & 3 & 4 & 5 & 18 & 5 \\ 7 & 11 & 9 & 11 & 18 & 49 & 15 \\ 2 & 3 & 2 & 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{1414}31^1513^{1414} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 8 & 2 \\ 1 & 1 & 1 & 16 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 128 & 24 \\ 3 & 5 & 0 & 44 & 14 & 44 & 8 \\ 8 & 16 & 0 & 128 & 44 & 126 & 24 \\ 2 & 3 & -1 & 24 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{1221}31^1513^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 & 2 \\ 1 & 1 & 1 & 8 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 16 & 12 \\ 5 & 9 & 0 & 38 & 44 & 19 & 14 \\ 2 & 4 & 0 & 16 & 19 & 6 & 6 \\ 2 & 3 & -1 & 12 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[259] $1^{134}3^{11}5^{131}3^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 96 & 18 \\ 2 & 3 & 0 & 21 & 5 & 28 & 5 \\ 8 & 16 & 0 & 96 & 28 & 126 & 24 \\ 2 & 3 & -1 & 18 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[258] $1^{1411}3^{11}5^{131}1^{411} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 & 2 \\ 1 & 1 & 1 & 16 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 32 & 24 \\ 3 & 5 & 0 & 44 & 14 & 11 & 8 \\ 2 & 4 & 0 & 32 & 11 & 6 & 6 \\ 2 & 3 & -1 & 24 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[257] $1^{133}3^{11}5^{131}3^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 72 & 18 \\ 2 & 3 & 0 & 21 & 5 & 21 & 5 \\ 6 & 12 & 0 & 72 & 21 & 70 & 18 \\ 2 & 3 & -1 & 18 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[256] $1^{1211}3^{11}5^{131}1^{211} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 8 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 16 & 12 \\ 3 & 5 & 0 & 22 & 14 & 11 & 8 \\ 2 & 4 & 0 & 16 & 11 & 6 & 6 \\ 2 & 3 & -1 & 12 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[255] $1^{1113}3^{11}5^{131}1^{23} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 1 & 1 & 4 & 7 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 24 & 6 \\ 4 & 7 & 0 & 15 & 27 & 45 & 11 \\ 6 & 12 & 0 & 24 & 45 & 70 & 18 \\ 2 & 3 & -1 & 6 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[254] $1^{132}3^{11}5^{131}3^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 4 & 2 \\ 1 & 1 & 1 & 12 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 48 & 18 \\ 2 & 3 & 0 & 21 & 5 & 14 & 5 \\ 4 & 8 & 0 & 48 & 14 & 30 & 12 \\ 2 & 3 & -1 & 18 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[253] $1^{114}3^{11}5^{131}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 32 & 6 \\ 2 & 3 & 0 & 7 & 5 & 28 & 5 \\ 8 & 16 & 0 & 32 & 28 & 126 & 24 \\ 2 & 3 & -1 & 6 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[252] $1^{1221}3^{11}5^{131}2^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 & 2 \\ 1 & 1 & 1 & 8 & 11 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 46 & 16 & 12 \\ 6 & 11 & 0 & 46 & 65 & 23 & 17 \\ 2 & 4 & 0 & 16 & 23 & 6 & 6 \\ 2 & 3 & -1 & 12 & 17 & 6 & 4 \end{pmatrix}_{b_0}$$

[251] $1^{1133}1^{115131^{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 24 & 6 \\ 2 & 3 & 0 & 7 & 5 & 21 & 5 \\ 6 & 12 & 0 & 24 & 21 & 70 & 18 \\ 2 & 3 & -1 & 6 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[250] $1^{1223}1^{115131^{212}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 32 & 12 \\ 2 & 3 & 0 & 14 & 5 & 14 & 5 \\ 4 & 8 & 0 & 32 & 14 & 30 & 12 \\ 2 & 3 & -1 & 12 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[249] $1^{1141}31^{115131^{141}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 1 & 1 & 4 & 17 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 8 & 6 \\ 9 & 17 & 0 & 35 & 152 & 35 & 26 \\ 2 & 4 & 0 & 8 & 35 & 6 & 6 \\ 2 & 3 & -1 & 6 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

[248] $1^{1443}1^{115131^{414}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 128 & 24 \\ 2 & 3 & 0 & 28 & 5 & 28 & 5 \\ 8 & 16 & 0 & 128 & 28 & 126 & 24 \\ 2 & 3 & -1 & 24 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[247] $1^{11213}1^{115131^{131}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 1 & 1 & 4 & 11 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 8 & 6 \\ 6 & 11 & 0 & 23 & 65 & 23 & 17 \\ 2 & 4 & 0 & 8 & 23 & 6 & 6 \\ 2 & 3 & -1 & 6 & 17 & 6 & 4 \end{pmatrix}_{b_0}$$

[246] $1^{1123}1^{115131^{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 16 & 6 \\ 2 & 3 & 0 & 7 & 5 & 14 & 5 \\ 4 & 8 & 0 & 16 & 14 & 30 & 12 \\ 2 & 3 & -1 & 6 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{1423}1^{115131^{412}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 4 & 2 \\ 1 & 1 & 1 & 16 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 64 & 24 \\ 2 & 3 & 0 & 28 & 5 & 14 & 5 \\ 4 & 8 & 0 & 64 & 14 & 30 & 12 \\ 2 & 3 & -1 & 24 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{1133}31^{115131^{133}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 6 & 2 \\ 1 & 1 & 1 & 4 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 24 & 6 \\ 7 & 13 & 0 & 27 & 90 & 81 & 20 \\ 6 & 12 & 0 & 24 & 81 & 70 & 18 \\ 2 & 3 & -1 & 6 & 20 & 18 & 4 \end{pmatrix}_{b_0}$$

[243] $1^{1131}31513^11^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 1 & 1 & 4 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 8 & 6 \\ 7 & 13 & 0 & 27 & 90 & 27 & 20 \\ 2 & 4 & 0 & 8 & 27 & 6 & 6 \\ 2 & 3 & -1 & 6 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[242] $1^{1131}31513^11^{311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 & 2 \\ 1 & 1 & 1 & 12 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 24 & 18 \\ 3 & 5 & 0 & 33 & 14 & 11 & 8 \\ 2 & 4 & 0 & 24 & 11 & 6 & 6 \\ 2 & 3 & -1 & 18 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[261] $1^{1413}31513^11^{413} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 6 & 2 \\ 1 & 1 & 1 & 16 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 96 & 24 \\ 3 & 5 & 0 & 44 & 14 & 33 & 8 \\ 6 & 12 & 0 & 96 & 33 & 70 & 18 \\ 2 & 3 & -1 & 24 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[240] $1^{1121}31513^11^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & 1 & 4 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 8 & 6 \\ 5 & 9 & 0 & 19 & 44 & 19 & 14 \\ 2 & 4 & 0 & 8 & 19 & 6 & 6 \\ 2 & 3 & -1 & 6 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[239] $1^{1411}3^115131^{421} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 & 2 \\ 1 & 1 & 1 & 16 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 32 & 24 \\ 4 & 7 & 0 & 60 & 27 & 15 & 11 \\ 2 & 4 & 0 & 32 & 15 & 6 & 6 \\ 2 & 3 & -1 & 24 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

[238] $1^{1111}31513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & 1 & 4 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 8 & 6 \\ 3 & 5 & 0 & 11 & 14 & 11 & 8 \\ 2 & 4 & 0 & 8 & 11 & 6 & 6 \\ 2 & 3 & -1 & 6 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{1131}3^115131^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 1 & 1 & 4 & 15 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 8 & 6 \\ 8 & 15 & 0 & 31 & 119 & 31 & 23 \\ 2 & 4 & 0 & 8 & 31 & 6 & 6 \\ 2 & 3 & -1 & 6 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

[236] $1^{1312}31513^11^{312} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 4 & 2 \\ 1 & 1 & 1 & 12 & 5 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 48 & 18 \\ 3 & 5 & 0 & 33 & 14 & 22 & 8 \\ 4 & 8 & 0 & 48 & 22 & 30 & 12 \\ 2 & 3 & -1 & 18 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

[235] $1^{1313}31^1513^11^{313} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 6 & 2 \\ 1 & 1 & 1 & 12 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 72 & 18 \\ 3 & 5 & 0 & 33 & 14 & 33 & 8 \\ 6 & 12 & 0 & 72 & 33 & 70 & 18 \\ 2 & 3 & -1 & 18 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[234] $1^{1212}31^115131^{222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 32 & 12 \\ 4 & 7 & 0 & 30 & 27 & 30 & 11 \\ 4 & 8 & 0 & 32 & 30 & 30 & 12 \\ 2 & 3 & -1 & 12 & 11 & 12 & 4 \end{pmatrix}_{b_0}$$

[233] $1^{1321}31^1513^11^{321} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 & 2 \\ 1 & 1 & 1 & 12 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 24 & 18 \\ 5 & 9 & 0 & 57 & 44 & 19 & 14 \\ 2 & 4 & 0 & 24 & 19 & 6 & 6 \\ 2 & 3 & -1 & 18 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{1222}31^1513^11^{222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 4 & 2 \\ 1 & 1 & 1 & 8 & 9 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 32 & 12 \\ 5 & 9 & 0 & 38 & 44 & 38 & 14 \\ 4 & 8 & 0 & 32 & 38 & 30 & 12 \\ 2 & 3 & -1 & 12 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{1242}31^1513^11^{242} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 4 & 2 \\ 1 & 1 & 1 & 8 & 17 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 32 & 12 \\ 9 & 17 & 0 & 70 & 152 & 70 & 26 \\ 4 & 8 & 0 & 32 & 70 & 30 & 12 \\ 2 & 3 & -1 & 12 & 26 & 12 & 4 \end{pmatrix}_{b_0}$$

[228] $1^{1421}31^1513^11^{421} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 & 2 \\ 1 & 1 & 1 & 16 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 76 & 32 & 24 \\ 5 & 9 & 0 & 76 & 44 & 19 & 14 \\ 2 & 4 & 0 & 32 & 19 & 6 & 6 \\ 2 & 3 & -1 & 24 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{1144}31^1513^11^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 8 & 2 \\ 1 & 1 & 1 & 4 & 17 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 32 & 6 \\ 9 & 17 & 0 & 35 & 152 & 140 & 26 \\ 8 & 16 & 0 & 32 & 140 & 126 & 24 \\ 2 & 3 & -1 & 6 & 26 & 24 & 4 \end{pmatrix}_{b_0}$$

[226] $1^{1143}31^1513^11^{143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 6 & 2 \\ 1 & 1 & 1 & 4 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 24 & 6 \\ 9 & 17 & 0 & 35 & 152 & 105 & 26 \\ 6 & 12 & 0 & 24 & 105 & 70 & 18 \\ 2 & 3 & -1 & 6 & 26 & 18 & 4 \end{pmatrix}_{b_0}$$

[225] $1^{1142}31513^11^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 1 & 1 & 4 & 17 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 16 & 6 \\ 9 & 17 & 0 & 35 & 152 & 70 & 26 \\ 4 & 8 & 0 & 16 & 70 & 30 & 12 \\ 2 & 3 & -1 & 6 & 26 & 12 & 4 \end{pmatrix}_{b_0}$

[224] $1^{1231}31513^11^{231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 & 2 \\ 1 & 1 & 1 & 8 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 16 & 12 \\ 7 & 13 & 0 & 54 & 90 & 27 & 20 \\ 2 & 4 & 0 & 16 & 27 & 6 & 6 \\ 2 & 3 & -1 & 12 & 20 & 6 & 4 \end{pmatrix}_{b_0}$

[223] $1^{1322}31513^11^{322} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 4 & 2 \\ 1 & 1 & 1 & 12 & 9 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 48 & 18 \\ 5 & 9 & 0 & 57 & 44 & 38 & 14 \\ 4 & 8 & 0 & 48 & 38 & 30 & 12 \\ 2 & 3 & -1 & 18 & 14 & 12 & 4 \end{pmatrix}_{b_0}$

[222] $1^{1232}31513^11^{232} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 4 & 2 \\ 1 & 1 & 1 & 8 & 13 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 32 & 12 \\ 7 & 13 & 0 & 54 & 90 & 54 & 20 \\ 4 & 8 & 0 & 32 & 54 & 30 & 12 \\ 2 & 3 & -1 & 12 & 20 & 12 & 4 \end{pmatrix}_{b_0}$

[221] $1^{1212}31513^11^{212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 4 & 2 \\ 1 & 1 & 1 & 8 & 5 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 32 & 12 \\ 3 & 5 & 0 & 22 & 14 & 22 & 8 \\ 4 & 8 & 0 & 32 & 22 & 30 & 12 \\ 2 & 3 & -1 & 12 & 8 & 12 & 4 \end{pmatrix}_{b_0}$

[220] $1^{1112}315131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 16 & 6 \\ 4 & 7 & 0 & 15 & 27 & 30 & 11 \\ 4 & 8 & 0 & 16 & 30 & 30 & 12 \\ 2 & 3 & -1 & 6 & 11 & 12 & 4 \end{pmatrix}_{b_0}$

[219] $1^{1113}15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 8 & 6 \\ 2 & 3 & 0 & 7 & 5 & 7 & 5 \\ 2 & 4 & 0 & 8 & 7 & 6 & 6 \\ 2 & 3 & -1 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[262] $1^{1111}315131^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 1 & 1 & 4 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 8 & 6 \\ 4 & 7 & 0 & 15 & 27 & 15 & 11 \\ 2 & 4 & 0 & 8 & 15 & 6 & 6 \\ 2 & 3 & -1 & 6 & 11 & 6 & 4 \end{pmatrix}_{b_0}$

[263] $1^{1431}31513^{1431} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 12$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 & 2 \\ 1 & 1 & 1 & 16 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 108 & 32 & 24 \\ 7 & 13 & 0 & 108 & 90 & 27 & 20 \\ 2 & 4 & 0 & 32 & 27 & 6 & 6 \\ 2 & 3 & -1 & 24 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[229] $1^{1422}31513^{1422} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 12$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 4 & 2 \\ 1 & 1 & 1 & 16 & 9 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 76 & 64 & 24 \\ 5 & 9 & 0 & 76 & 44 & 38 & 14 \\ 4 & 8 & 0 & 64 & 38 & 30 & 12 \\ 2 & 3 & -1 & 24 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[260] $1^{1412}31513^{1412} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 12$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 4 & 2 \\ 1 & 1 & 1 & 16 & 5 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 64 & 24 \\ 3 & 5 & 0 & 44 & 14 & 22 & 8 \\ 4 & 8 & 0 & 64 & 22 & 30 & 12 \\ 2 & 3 & -1 & 24 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

[264] $1^{1112}31513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 20$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 2 & 3 & 1 & 1 & 4 & 7 & 3 \\ 2 & 4 & 3 & 4 & 5 & 11 & 5 \\ 5 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[265] $1^{1111}31513^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 2 & 4 & 4 & 6 & 7 & 8 & 6 \\ 3 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[266] $1^{1112}31513^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 12 & 6 \\ 3 & 4 & 3 & 7 & 5 & 11 & 5 \\ 4 & 7 & 5 & 12 & 11 & 17 & 9 \\ 2 & 3 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[267] $1^{1113}31513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 32$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 4 & 11 & 3 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 2 & 3 & 1 & 1 & 4 & 11 & 3 \\ 2 & 4 & 3 & 4 & 5 & 18 & 5 \\ 8 & 11 & 9 & 11 & 18 & 49 & 15 \\ 2 & 3 & 2 & 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[268] $1^{1111}31513^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 7 & 8 & 6 \\ 2 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{112}3^11^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 52, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 5 & 2 \\ 1 & 1 & 2 & 4 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 3 & 4 & 4 & 6 & 7 & 12 & 6 \\ 2 & 4 & 3 & 7 & 5 & 11 & 5 \\ 5 & 7 & 5 & 12 & 11 & 17 & 9 \\ 2 & 3 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{111}3^11^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 82, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 6 & 4 & 3 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 2 & 4 & 4 & 6 & 11 & 8 & 6 \\ 4 & 6 & 5 & 11 & 14 & 11 & 8 \\ 3 & 4 & 4 & 8 & 11 & 6 & 6 \\ 2 & 3 & 2 & 6 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{112}3^11^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 87, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 2 & 4 & 4 & 6 & 11 & 12 & 6 \\ 4 & 6 & 5 & 11 & 14 & 17 & 8 \\ 4 & 7 & 5 & 12 & 17 & 17 & 9 \\ 2 & 3 & 2 & 6 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[272] \ 2^13^11513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 7 & 8 & 6 \\ 3 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

4.2.22 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [5] \ 1^{11}3^11^1513^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 1 & 2 \\ 4 & 0 & 5 & 17 & 11 & 5 & 9 \\ 2 & 0 & 3 & 11 & 5 & 3 & 5 \\ 1 & 0 & 1 & 5 & 3 & 0 & 2 \\ 2 & 0 & 2 & 9 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12}3^11^1513^11^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 1 & 2 \\ 8 & 0 & 10 & 74 & 22 & 10 & 18 \\ 2 & 0 & 3 & 22 & 5 & 3 & 5 \\ 1 & 0 & 1 & 10 & 3 & 0 & 2 \\ 2 & 0 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}3^11^1513^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 4 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121}3^11^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 5 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{131}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 1 & 2 \\ 1 & 1 & 2 & 12 & 4 & 2 & 3 \\ 1 & 2 & 0 & 12 & 3 & 1 & 2 \\ 8 & 12 & 12 & 70 & 21 & 9 & 18 \\ 3 & 4 & 3 & 21 & 5 & 3 & 5 \\ 1 & 2 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 7 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{141}3^{11}5^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 1 & 2 \\ 1 & 1 & 2 & 16 & 4 & 2 & 3 \\ 1 & 2 & 0 & 16 & 3 & 1 & 2 \\ 11 & 16 & 16 & 126 & 28 & 12 & 24 \\ 3 & 4 & 3 & 28 & 5 & 3 & 5 \\ 1 & 2 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & 2 & 24 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 & 3 \\ 1 & 2 & 0 & 0 & 3 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 3 & 0 & 5 & 3 & 5 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 0 & 8 & 2 & 3 \\ 1 & 2 & 0 & 0 & 7 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 7 & 0 & 27 & 6 & 11 \\ 1 & 2 & 1 & 0 & 6 & 0 & 2 \\ 2 & 3 & 2 & 0 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{122}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 8 & 2 & 3 \\ 1 & 2 & 0 & 5 & 7 & 1 & 2 \\ 4 & 7 & 5 & 17 & 23 & 5 & 9 \\ 6 & 8 & 7 & 23 & 27 & 6 & 11 \\ 1 & 2 & 1 & 5 & 6 & 0 & 2 \\ 2 & 3 & 2 & 9 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 2 & 4 & 8 & 2 & 3 \\ 1 & 2 & 0 & 4 & 7 & 1 & 2 \\ 2 & 4 & 4 & 6 & 15 & 3 & 6 \\ 6 & 8 & 7 & 15 & 27 & 6 & 11 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^11^1513^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{131}3^11^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 1 & 2 \\ 1 & 1 & 2 & 12 & 4 & 2 & 3 \\ 1 & 2 & 0 & 12 & 3 & 1 & 2 \\ 9 & 12 & 12 & 70 & 21 & 9 & 18 \\ 3 & 4 & 3 & 21 & 5 & 3 & 5 \\ 1 & 2 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{121}3^11^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 6 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{131}3^11^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 0 & 12 & 2 & 3 \\ 1 & 2 & 0 & 0 & 11 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 9 & 17 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & 2 & 0 & 17 & 2 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{112}3^11^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 1 & 2 \\ 1 & 1 & 2 & 4 & 8 & 2 & 3 \\ 1 & 2 & 0 & 4 & 7 & 1 & 2 \\ 3 & 4 & 4 & 6 & 15 & 3 & 6 \\ 6 & 8 & 7 & 15 & 27 & 6 & 11 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

4.2.23 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix

$T = 12$ [2] $1^{1114}3^11^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 3 & 4 & 17 & 3 \\ 1 & 2 & 0 & 1 & 3 & 14 & 2 \\ 1 & 3 & 1 & 1 & 4 & 17 & 3 \\ 3 & 4 & 3 & 4 & 5 & 28 & 5 \\ 11 & 17 & 14 & 17 & 28 & 121 & 23 \\ 2 & 3 & 2 & 3 & 5 & 23 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^11^1513^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 11 & 6 \\ 3 & 4 & 3 & 7 & 5 & 10 & 5 \\ 4 & 6 & 5 & 11 & 10 & 14 & 8 \\ 2 & 3 & 2 & 6 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

4.2.24 NHC(-2)(-3) + su₂ + su₃ + su₂

$$T = 8 \quad [0] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & 0 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & 0 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} \ 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & 0 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \ 3 \ 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & 0 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [10] \quad 1^{11} 3^{11} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \quad 1^{11} 4^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 4^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 4^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [15] \quad 1^{11} 5^{11} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 1 & 1 & 0 & 11 & 3 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 11 & 7 & 0 & 40 & 14 \\ 2 & 2 & 3 & 2 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 6 & 4 & 0 & 12 & 8 \\ 2 & 2 & 3 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 5^{11} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 12 & 9 & 0 & 56 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 5^{12} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 42, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 12 & 8 & 0 & 54 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12}_{1^{13}} 5^{1^1} 1321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 0 & 17 & 3 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 17 & 11 & 0 & 102 & 22 \\ 2 & 2 & 3 & 2 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}_{1^{13}} 5^{1^1} 1321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 0 & 18 & 3 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 18 & 12 & 0 & 124 & 24 \\ 2 & 2 & 3 & 2 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [20] \quad 1^{11}_{1^{131}} 6^{1^1} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 2 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}_{1^{131}} 6^{1^1} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 5 & 4 & 0 & 12 & 8 \\ 2 & 3 & 2 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}_{1^{131}} 6^{1^1} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 6 & 5 & 0 & 25 & 11 \\ 2 & 3 & 2 & 2 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [23] \quad 1^{11}_{1^{1231}} 6^{1^1} 1316^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.0$$

4.2.25 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{11}_{1^1} 3^{1^1} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{\overline{11}} 4 13^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.2.26 NHC(-2)(-3) + su₂ + su₃ + su_{2n3mix}

$$T = 11 \quad [0] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 0 & 12 & 5 \\ 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 7 & 0 & 22 & 10 \\ 3 & 5 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 3 & 0 & 20 & 5 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 65 & 17 \\ 3 & 5 & 3 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 2 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 5 13^{\overline{11}} 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 3 & 0 & 28 & 5 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 17 & 0 & 130 & 24 \\ 3 & 5 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} \overset{1}{5} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 7 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} \overset{1}{5} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 8 & 12 & 10 & 0 & 70 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 9 & 8 & 0 & 47 & 13 \\ 2 & 2 & 2 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} \overset{1}{5} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 10 & 10 & 0 & 65 & 15 \\ 2 & 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} \overset{1}{5} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 13 & 11 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \overset{1}{5} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 14 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

4.2.27 NHC(-2)(-3) + su₃ + su₂ + su₂

$$T = 8 \quad [0] \quad 1^{11} \overset{1}{3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 5 & 3 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [4] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [6] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [10] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -1 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3 \quad 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 28, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} \ 3 \ 1^{\overline{11}} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{red}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{su}}_3 + \textcolor{blue}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{teal}{\mathfrak{su}_3} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

[13] $1^{11} 3^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[15] $1^{12} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[16] $1^{11}1^13^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{11} 3 1^4 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 52, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11} 3 1^{11} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{red}{\mathbf{g}_2}] + \textcolor{teal}{\mathbf{su}_3} + \textcolor{teal}{\mathbf{su}_2} + \textcolor{teal}{\mathbf{su}_2} \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 59, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[19] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

$$[21] \quad 1^{12} 3 \quad 1^{11} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_3 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

- [20] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [26] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [27] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [28] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [30] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 16 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{112} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [36] $1^{11} 3 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 7 & 5 \\ 2 & 3 & 0 & 4 & 7 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

7.0
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 3 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 3 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [47] $1^{11} 4^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{11} 4^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 2 & 3 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 3 & 6 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 1 & 3 \\ 6 & 4 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{\overline{11}} 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 11 & 6 \\ 4 & 2 & 8 & 4 \\ 11 & 8 & 19 & 11 \\ 6 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{\overline{11}} 1^{\overline{11}} 3 \ 221^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 4 & 0 & 2 & 5 & 4 \\ 2 & 3 & 5 & 0 & 5 & 6 & 6 \\ 2 & 2 & 3 & -1 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 5 & 4 \\ 3 & 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 3 & 3 \\ 5 & 4 & 3 & 6 & 6 \\ 4 & 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 6 & 1 & 5 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 & 5 & 5 \\ 3 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{\overline{11}} 1^{\overline{11}} 31^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 1 & 3 \\ 1 & 1 & 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{\overline{11}} 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 3 & 1 & 5 & 4 & 5 \\ 3 & 3 & 1 & 4 & 2 & 4 \\ 3 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{11} 4^{111} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 4 & 2 & 6 & 6 \\ 4 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

[58] $1^{11} 4^{111} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 5 & 3 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[59] $1^{11} 4^{111} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 4 & 4 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[60] $1^{11} 4^{111} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 4 & 4 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[61] $1^{111} 4^{111} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 1 & 5 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 5 & 5 & 0 & 8 & 11 & 6 \\ 6 & 6 & 1 & 11 & 14 & 8 \\ 3 & 3 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{11} 4^{111} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 3 & 4 & 3 \\ 5 & 3 & 3 & 4 & 5 \\ 4 & 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[63] $1^{11} 4^{112} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 5 & 3 & 3 & 7 & 5 \\ 7 & 5 & 7 & 12 & 8 \\ 4 & 3 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

[64] $1^{111} 4^{111} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 5 & 4 \\ 5 & 5 & 0 & 5 & 8 & 6 \\ 3 & 3 & 0 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{11} 4^{111} 31^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 3 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[66] $1^{111} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{111} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 1 & 5 & 9 & 7 \\ 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{111} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 6 & 3 \\ 2 & 1 & 1 & 6 & 5 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 6 & 6 & 1 & 14 & 12 & 8 \\ 6 & 5 & 1 & 12 & 9 & 7 \\ 3 & 3 & 0 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{111} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 6 & 6 & 1 & 6 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{111} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 8 & 1 & 8 & 27 & 11 \\ 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{112} 31^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 1 & 1 & 7 & 3 \\ 2 & 1 & 0 & 4 & 2 \\ 10 & 7 & 4 & 21 & 11 \\ 4 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[72] $1^{111} 4^{11} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 9 & 4 \\ 9 & 8 & 1 & 9 & 27 & 11 \\ 3 & 3 & 0 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{111} 4^{11} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 7 & 4 \\ 6 & 6 & 1 & 7 & 14 & 8 \\ 3 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [74] $1^{11} 4^{11} 13221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -5, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 7.0$

$T = 11$ [75] $1^{11} 4^{11} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 6 & 5 & 4 \\ 3 & 2 & 5 & 5 & 4 \\ 6 & 5 & 10 & 9 & 8 \\ 5 & 5 & 9 & 7 & 7 \\ 4 & 4 & 8 & 7 & 6 \end{pmatrix}_{b_0}$$

$T = 11$ [76] $1^{11} 5^{11} 1^{11} 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 8 & 15 & 0 & 11 & 37 & 13 \\ 3 & 5 & -1 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [77] $1^{11} 5^{11} 1^{11} 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 \\ 1 & 2 & 0 & 0 & 3 & 8 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 8 \\ 5 & 6 & 8 & 0 & 8 & 31 \\ 2 & 2 & 2 & -1 & 3 & 11 \end{pmatrix}_{b_0}$$

$T = 11$ [78] $1^{11} 5^{11} 1^{11} 231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 15 & 4 \\ 10 & 20 & 0 & 15 & 63 & 17 \\ 3 & 5 & -1 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [79] $1^{11} 5^{11} 2^{11} 231^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 26 & 4 \\ 18 & 35 & 0 & 26 & 202 & 30 \\ 3 & 5 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [80] $1^{11} 5^{11} 3^{11} 231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 26 & 3 \\ 3 & 5 & 0 & 5 & 50 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 37 & 4 \\ 26 & 50 & 0 & 37 & 419 & 43 \\ 3 & 5 & -1 & 4 & 43 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [81] $1^{11} 5^{11} 2^{11} 231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 30 & 4 \\ 20 & 40 & 0 & 30 & 258 & 34 \\ 3 & 5 & -1 & 4 & 34 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [82] $1^{11} 4^{11} 141321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$

$T = 12$ [83] $1^{11} 5^{11} 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 5 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^{11} 4 14131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 2 & 1 & 3 & 2 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{11} 1^{11} 5 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 4 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

[86] $1^{111} 4 1411^{131} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 3 & 4 & 0 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[89] $1^{121} 1^{11} 5 1321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 6 & 3 & 12 & 6 & 6 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 12 & 4 & 18 & 10 & 10 \\ 3 & 6 & 2 & 10 & 3 & 5 \\ 3 & 6 & 2 & 10 & 5 & 5 \end{pmatrix}_{b_0}$$

[88] $1^{131} 1^{11} 5 1321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 6 & 3 & 18 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 18 & 6 & 43 & 15 & 15 \\ 3 & 6 & 2 & 15 & 3 & 5 \\ 3 & 6 & 2 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

[87] $1^{111} 1^{11} 5 1321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 & 5 \\ 3 & 6 & 2 & 5 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[91] $1^{111} 1^{11} 5 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 7 & 0 & 10 & 8 \\ 3 & 3 & 4 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[90] $1^{112} 1^{11} 5 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 2 & 0 & 12 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 12 & 0 & 29 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \end{pmatrix}_{b_0}$$

- [92] $1^{11}_{1^1} 5^{12} 1321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 5 & 12 & 4 \\ 3 & 2 & 5 & 3 & 12 & 5 \\ 8 & 7 & 12 & 12 & 29 & 13 \\ 3 & 3 & 4 & 5 & 13 & 5 \end{pmatrix}_{b_0}$
- [93] $1^{11}_{1^1} 5^{13} 1321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 5 & 12 & 4 \\ 3 & 2 & 5 & 3 & 12 & 5 \\ 9 & 7 & 12 & 12 & 29 & 13 \\ 3 & 3 & 4 & 5 & 13 & 5 \end{pmatrix}_{b_0}$
- [94] $1^{111}_{1^1} 4^{11} 141^{131} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -3$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 1 & 2 & 2 & 7 & 4 \\ 5 & 6 & 1 & 6 & 7 & 15 & 9 \\ 2 & 3 & 0 & 2 & 4 & 9 & 4 \end{pmatrix}_{b_0}$
- [95] $1^{11}_{1^1} 5^{13} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 2 & 0 & 12 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 12 & 0 & 29 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \end{pmatrix}_{b_0}$
- [96] $1^{12}_{1^1} 5^{12} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 14 & 0 & 46 & 16 \\ 3 & 3 & 4 & 0 & 16 & 5 \end{pmatrix}_{b_0}$
- [97] $1^{12}_{1^1} 5^{12} 1321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 2 & 5 & 19 & 4 \\ 3 & 2 & 5 & 3 & 19 & 5 \\ 12 & 12 & 19 & 19 & 79 & 21 \\ 3 & 3 & 4 & 5 & 21 & 5 \end{pmatrix}_{b_0}$
- [98] $1^{11}_{1^1} 5^{13} 1321^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 6 & 9 & 6 & 18 & 6 \\ 4 & 9 & 9 & 7 & 21 & 7 \\ 3 & 6 & 7 & 3 & 15 & 5 \\ 9 & 18 & 21 & 15 & 43 & 15 \\ 3 & 6 & 7 & 5 & 15 & 5 \end{pmatrix}_{b_0}$
- [99] $1^{12}_{1^1} 5^{13} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 19 & 0 & 79 & 21 \\ 3 & 3 & 4 & 0 & 21 & 5 \end{pmatrix}_{b_0}$
- [100] $1^{12}_{1^1} 5^{13} 1321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} =$

7.5

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 2 & 5 & 19 & 4 \\ 3 & 2 & 5 & 3 & 19 & 5 \\ 13 & 12 & 19 & 19 & 79 & 21 \\ 3 & 3 & 4 & 5 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{1^{21}}_1 5 \ 1321^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 6 & 9 & 12 & 6 & 6 \\ 4 & 9 & 9 & 14 & 7 & 7 \\ 6 & 12 & 14 & 18 & 10 & 10 \\ 3 & 6 & 7 & 10 & 3 & 5 \\ 3 & 6 & 7 & 10 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1^{11}}_1 5 \ 1321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 6 & 9 & 6 & 6 & 6 \\ 4 & 9 & 9 & 7 & 7 & 7 \\ 3 & 6 & 7 & 3 & 5 & 5 \\ 3 & 6 & 7 & 5 & 3 & 5 \\ 3 & 6 & 7 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{1^{13}}_1 5 \ 1321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$$

7.8

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 2 & 5 & 26 & 4 \\ 3 & 2 & 5 & 3 & 26 & 5 \\ 17 & 17 & 26 & 26 & 153 & 29 \\ 3 & 3 & 4 & 5 & 29 & 5 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1^{13}}_1 5 \ 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$$

7.8

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 21 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 21 & 0 & 106 & 24 \\ 3 & 3 & 4 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [105] \ 1^{1^{11}}_1 4 \ 14141^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$$

$$[106] \ 1^{1^{11}}_1 4 \ 1414131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$$

$$[107] \ 1^{1^{11}}_1 4 \ 14141^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 7.0$$

$$[108] \ 1^{1^{11}}_1 6 \ 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 2 & 3 & 4 & 6 & 4 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 2 & 2 & 6 & 4 \\ 5 & 6 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{1^{11}}_1 6 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 2 & 4 \\ 3 & 4 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1^{11}}_1 6 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 2 & 3 & 0 & 6 & 4 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 6 & 6 & 0 & 10 & 8 \\ 3 & 4 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overset{11}{1}} \overset{11}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 2 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 8 & 0 & 26 & 12 \\ 3 & 4 & 3 & 0 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overset{11}{1}} \overset{11}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -36, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 2 & 3 & 4 & 10 & 4 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 2 & 2 & 10 & 4 \\ 7 & 10 & 8 & 10 & 26 & 12 \\ 3 & 4 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overset{11}{1}} \overset{11}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -64, (n_+, n_-, n_0) =$$

$$(1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 2 & 3 & 6 & 10 & 4 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 6 & 6 & 10 & 18 & 8 \\ 7 & 10 & 8 & 18 & 26 & 12 \\ 3 & 4 & 3 & 8 & 12 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [117] \ 1^{\overset{112}{1}} \overset{112}{6} \ 131^{\overset{12}{12}} 6 \ 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 16, 1), c_{\text{ext}} = 7.5$$

$$[116] \ 1^{\overset{11}{1}} \overset{11}{6} \ 131^{\overset{11}{11}} 6 \ 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 16, 1), c_{\text{ext}} = 7.0$$

$$[115] \ 1^{\overset{11}{1}} \overset{11}{6} \ 131^{\overset{12}{12}} 6 \ 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 16, 1), c_{\text{ext}} = 7.0$$

$$[114] \ 1^{\overset{112}{1}} \overset{112}{6} \ 131^{\overset{11}{11}} 6 \ 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 16, 1), c_{\text{ext}} = 7.5$$

$$T = 17 \quad [118] \ 1^{\overset{112}{1}} \overset{112}{6} \ 131^{\overset{112}{12}} 6 \ 131^{\overset{11}{11}} 41^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -2, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 17, 1), c_{\text{ext}} = 7.0$$

$$T = 19 \quad [119] \ 1^{\overset{121}{1}} \overset{121}{6} \ 131^{\overset{121}{12}} 6 \ 131^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -2, \det =$$

$$0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 7.5$$

$$T = 20 \quad [120] \ 1^{\overset{112}{1}} \overset{112}{6} \ 131^{\overset{112}{12}} 6 \ 131^{\overset{11}{11}} 41^{\overset{11}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -1, \det =$$

$$0, (n_+, n_-, n_0) = (1, 20, 2), c_{\text{ext}} = 7.0$$

$$\mathbf{4.2.28} \quad \text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$$

$$T = 8 \quad [0] \ 1^{\overset{11}{1}} \overset{11}{3} \ 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11}4^{11}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{11}4^{11}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[6] $1^{11}4^{11}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 4 \\ 3 & 1 & 1 & 0 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

[7] $1^{11}4^{11}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$T = 11$ [10] $1^{11}5^{12}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 4 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[8] $1^{11}5^{12}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 21 \overset{1}{1} \overset{11}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1 \overset{1}{1} \overset{11}{5} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 7 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1 \overset{1}{1} \overset{11}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1 \overset{1}{1} \overset{12}{5} 13 \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1 \overset{1}{1} \overset{13}{5} 13 \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1 \overset{1}{1} \overset{11}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1 \overset{1}{1} \overset{11}{5} 123 \overset{1}{1} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1 \overset{1}{1} \overset{11}{5} 13 \overset{1}{1} \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^1 5^{11} 123^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 5 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 5^{14} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 10 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 21^1 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 6 & 5 & 6 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^1 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^1 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 6 & 5 & 6 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 3 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^1 5^{11} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1^2} 5^{1^2} 13^{1^2} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1^4} 5^{1^4} 13^{1^4} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 10 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^2} 5^{1^2} 13^{1^2} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1^3} 5^{1^3} 13^{1^3} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^2} 5^{1^2} 13^{1^2} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^3} 5^{1^3} 13^{1^3} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^1} 5^{1^1} 1^{23} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 4 & 5 & 5 & 0 & 18 & 5 \\ 1 & 1 & 2 & 0 & 5 & 0 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{14} \overset{1}{5} 13^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 12 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} \overset{1}{5} 1^{12} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 15 & 0 & 37 & 10 & 13 \\ 2 & 4 & 0 & 10 & 1 & 3 \\ 3 & 5 & -1 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} \overset{1}{5} 1^{12} 23^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 2 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 2 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{13}_{1^1} 5^{14} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -305, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 16 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{13}_{1^1} 5^{13} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{13}_{1^1} 5^{14} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 2 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 18 & 17 & 22 & 137 & 18 & 25 \\ 2 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

4.2.29 NHC(-2)(-3) + su₃ + su₂ + su_{2n3mix}

$T = 11 \quad [0] \quad 1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{11}_{1^1} 5^{12} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^{21}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 4 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1^{31}} 5^{13} 1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1^{11}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 3 & 0 & 9 & 5 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 4 & 0 & 11 & 7 \\ 3 & 5 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1^{11}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1^{12}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1^{11}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{21}} 5^{13} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 3 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1^{11}} 5^{13} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 4 & 8 & 8 & 6 & 5 & 6 \\ 3 & 5 & 6 & 2 & 3 & 4 \\ 3 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1^{12}}_1 5^{13} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 5 & 9 & 5 \\ 4 & 8 & 8 & 6 & 11 & 6 \\ 3 & 5 & 6 & 2 & 7 & 4 \\ 6 & 9 & 11 & 7 & 11 & 7 \\ 3 & 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1^{21}}_1 5^{13} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 10 & 4 & 5 \\ 4 & 8 & 8 & 12 & 5 & 6 \\ 6 & 10 & 12 & 14 & 6 & 8 \\ 3 & 4 & 5 & 6 & 1 & 3 \\ 3 & 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1^{31}}_1 5^{13} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 & 3 \\ 3 & 5 & 8 & 15 & 4 & 5 \\ 4 & 8 & 8 & 18 & 5 & 6 \\ 9 & 15 & 18 & 34 & 9 & 12 \\ 3 & 4 & 5 & 9 & 1 & 3 \\ 3 & 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1^{31}}_1 5^{13} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 3 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1^{11}}_1 5^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1^{12}}_1 5^{13} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 3 & 0 & 14 & 5 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 14 & 6 & 0 & 29 & 11 \\ 3 & 5 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1^{11}}_1 5^{13} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1^{11}}_1 5^{13} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} \overset{11}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 7 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} \overset{11}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -35, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \overset{11}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{113} \overset{11}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 3 & 0 & 19 & 5 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 19 & 8 & 0 & 55 & 15 \\ 3 & 5 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} \overset{11}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{121} \overset{11}{5} 13^{11} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{112} \overset{11}{5} 13^{11} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 10 & 14 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 10 & 4 & 14 & 22 & 8 \\ 8 & 14 & 6 & 22 & 29 & 11 \\ 3 & 5 & 2 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} \overset{11}{5} 13^{11} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -53, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{1^{11}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{1^{12}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 6 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{1^{21}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{1^{21}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{1^{21}} 5^{13} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{1^{31}} 5^{13} 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 20 & 9 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 20 & 8 & 62 & 28 & 16 \\ 5 & 9 & 4 & 28 & 11 & 7 \\ 3 & 5 & 2 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{1^{13}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 9 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{1^{11}} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 7 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12}_{1^1} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13}_{1^1} 5^{13} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 10 & 19 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 10 & 4 & 14 & 30 & 8 \\ 11 & 19 & 8 & 30 & 55 & 15 \\ 3 & 5 & 2 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}_{1^1} 5^{13} 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -70, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 9 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11}_{1^1} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 9 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{14}_{1^1} 5^{13} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -75, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 3 & 5 & 24 & 5 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 14 & 24 & 10 & 19 & 89 & 19 \\ 3 & 5 & 2 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{12}_{1^1} 5^{13} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11}_{1^1} 5^{13} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{13}_{1^1} 5^{13} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 12 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11}_{1^1} 5^{1^{31}} 13^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1^{12}}_{1^1} 5^{1^{12}} 13^{1^{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1^{13}}_{1^1} 5^{1^{13}} 13^{1^{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 9 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1^{21}}_{1^1} 5^{1^{21}} 13^{1^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1^{21}}_{1^1} 5^{1^{21}} 13^{1^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 12 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1^{13}}_{1^1} 5^{1^{13}} 13^{1^{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 13 & 0 & 44 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1^{21}}_{1^1} 5^{1^{21}} 13^{1^{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1^{12}}_{1^1} 5^{1^{12}} 13^{1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{12}_{1^1} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{12}_{1^1} 5^{13} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -142, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{12}_{1^1} 5^{31} 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{13}_{1^1} 5^{31} 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 13 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{12}_{1^1} 5^{31} 13^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -151, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 14 & 2 & 3 \\ 2 & 3 & 1 & 20 & 3 & 3 \\ 15 & 14 & 20 & 106 & 16 & 22 \\ 3 & 2 & 3 & 16 & 1 & 3 \\ 3 & 3 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{12}_{1^1} 5^{31} 13^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{12}_{1^1} 5^{13} 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 15 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{13}_{1^1} 5^{31} 13^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{13} 5^{31} 13^{11} 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 17 & 17 & 22 & 137 & 18 & 25 \\ 3 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{13} 5^{13} 13^{11} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 17 & 0 & 92 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

4.2.30 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 61, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 10 \quad [4] \quad 1^{11} 4^{11} 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$

$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[6] $1^{11} 4^{11} 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 4 \\ 3 & 1 & 0 & 1 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 4 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 4 & 3 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 4 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 21^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{13} 5^{13} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 5^{11} 123^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 5^{11} 123^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 1 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 5^{11} 123^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 3 & 4 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 5^{11} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1^4} 5^{1^1} 13^{1^1} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 10 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 21^{1^1} 5^{1^1} 13^{1^1} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1^3} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 3 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^4} 5^{1^1} 13^{1^1} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 10 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1^2} 5^{1^1} 13^{1^1} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 8 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1^2} 5^{1^1} 13^{1^1} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^3} 5^{1^1} 13^{1^1} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^2} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^1} 5^{1^1} 1^{1^1} 23^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 5 & 5 & 0 & 5 & 18 & 8 \\ 2 & 2 & 2 & -1 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^3} 5^{1^1} 13^{1^1} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 11 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1^2} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^4} 5^{1^1} 13^{1^1} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 12 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[38] $1^1 5^1 1^1 2^3 1^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 8 & 15 & 0 & 10 & 37 & 13 \\ 3 & 5 & -1 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{12} 5^1 1^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{11} 5^1 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{11} 5^1 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[42] $1^1 5^1 1^1 2^3 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 4 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 10 & 20 & 0 & 13 & 63 & 17 \\ 3 & 5 & -1 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{12} 5^1 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{12} 5^1 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{13} 5^1 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 2 & 2 & 3 & 1 & 18 & 3 \\ 18 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [55] \quad 1^{11}413^1 1^1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11}413^1 1^1 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 3 & 3 & 0 & 1 & 2 & 4 \\ 3 & 2 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11}41^1 3^1 1^1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 4 & 2 \\ 4 & 4 & 0 & 0 & 4 & 10 & 6 \\ 2 & 2 & 0 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111}41^1 3^1 1^1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 4 & 2 \\ 2 & 1 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 4 & 4 & 10 & 6 \\ 2 & 2 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^1 41^1 3^1 1^1 5 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 7 & 4 & 0 & 3 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^1 41^1 3^1 1^1 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 5 & 3 & 0 & 3 & 6 & 6 \\ 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11}41^1 3^1 1^1 5 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 6 & 5 & 0 & 4 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11}41^1 3^1 1^1 5 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 10 & 7 & 0 & 4 & 28 & 12 \\ 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{11}41^{13}1^{11}51^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 8 & 6 & 0 & 4 & 18 & 10 \\ 3 & 2 & 0 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$T = 15$ [64] $1^{11}6^{13}131513^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[65] $1^{11}6^{13}131513^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 2 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

4.2.31 NHC(-2)(-3) + su₃ + su₂n3mix + su₂

$T = 11$ [1] $1^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[0] $1^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 2 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 3 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 4 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 4 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 9 & 0 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 9 & 4 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 7 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 9 & 5 & 5 \\ 4 & 8 & 8 & 11 & 6 & 6 \\ 6 & 9 & 11 & 11 & 7 & 7 \\ 3 & 5 & 6 & 7 & 2 & 4 \\ 3 & 5 & 6 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 4 & 8 & 8 & 5 & 6 & 6 \\ 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 4 & 10 & 5 \\ 4 & 8 & 8 & 5 & 12 & 6 \\ 3 & 4 & 5 & 1 & 6 & 3 \\ 6 & 10 & 12 & 6 & 14 & 8 \\ 3 & 5 & 6 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 5 & 8 & 4 & 15 & 5 \\ 4 & 8 & 8 & 5 & 18 & 6 \\ 3 & 4 & 5 & 1 & 9 & 3 \\ 9 & 15 & 18 & 9 & 34 & 12 \\ 3 & 5 & 6 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 5 & 3 & 14 & 0 & 5 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 14 & 6 & 29 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 11 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 7 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 13^{\frac{1}{1}} 1^{\frac{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 9 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11} 5^{11} 13^{11} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -31$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{12} 5^{11} 13^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -35$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -36$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 5 & 0 & 3 \\ 5 & 5 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{131} 5^{11} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -43$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 5 & 3 & 19 & 0 & 5 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 11 & 19 & 8 & 55 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -45$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{112} 5^{11} 13^{11} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -45$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 8 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{121} 5^{11} 13^{11} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -47$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 14 & 10 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 14 & 6 & 29 & 22 & 11 \\ 5 & 10 & 4 & 22 & 14 & 8 \\ 3 & 5 & 2 & 11 & 8 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{13} 5^{11} 13^{11} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -53$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{1^{21}} 5^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{1^{11}} 5^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{1^{12}} 5^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{1^{12}} 5^{11} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{1^{12}} 5^{11} 13^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{1^{13}} 5^{11} 13^{11} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 9 & 20 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 9 & 4 & 11 & 28 & 7 \\ 11 & 20 & 8 & 28 & 62 & 16 \\ 3 & 5 & 2 & 7 & 16 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{1^{31}} 5^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{1^{11}} 5^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{12}_{1^1} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{31}_{1^1} 5^{13} 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 19 & 10 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 19 & 8 & 55 & 30 & 15 \\ 5 & 10 & 4 & 30 & 14 & 8 \\ 3 & 5 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}_{1^1} 5^{13} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -70, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 9 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{21}_{1^1} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 1 & 3 & 7 & 0 & 3 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 8 & 7 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{14}_{1^1} 5^{13} 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -75, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 5 & 3 & 24 & 5 & 5 \\ 2 & 3 & 0 & 10 & 2 & 2 \\ 14 & 24 & 10 & 89 & 19 & 19 \\ 2 & 5 & 2 & 19 & 2 & 4 \\ 3 & 5 & 2 & 19 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{21}_{1^1} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{12}_{1^1} 5^{13} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{12}_{1^1} 5^{13} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 12 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{11}_{1^1} 5^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{121}_{1^1} 5^{13} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{131}_{1^1} 5^{13} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{12}_{1^1} 5^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 12 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{13}_{1^1} 5^{13} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 14 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{11}_{1^1} 5^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 9 & 0 & 3 \\ 2 & 3 & 1 & 13 & 0 & 3 \\ 11 & 9 & 13 & 44 & 0 & 14 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{12}_{1^1} 5^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{12}_{1^1} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 10 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{12}_{1^1} 5^{131} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -123, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 11 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{13}_{1^1} 5^{131} 13^1 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -142, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 3 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{13}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 13 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[56] $1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{12}_{1^1} 5^{13} 13^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -151, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 3 & 20 & 3 \\ 3 & 2 & 3 & 1 & 16 & 3 \\ 15 & 14 & 20 & 16 & 106 & 22 \\ 3 & 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{12}_{1^1} 5^{13} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 13 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 1 & 3 \\ 2 & 1 & 3 & 12 & 0 & 3 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 13 & 12 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{13}_{1^1} 5^{13} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{13} 5^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 3 & 2 & 3 & 1 & 18 & 3 \\ 17 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{13} 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 1 & 3 \\ 2 & 1 & 3 & 15 & 0 & 3 \\ 2 & 3 & 1 & 17 & 0 & 3 \\ 15 & 15 & 17 & 92 & 0 & 20 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 20 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [63] $1^{11} 41^{13} 1^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 4 \\ 3 & 2 & 0 & 0 & 5 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 41^{13} 1^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 21, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{11} 41^{13} 1^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 11 & 4 \\ 5 & 6 & 0 & 0 & 11 & 20 & 8 \\ 2 & 2 & 0 & 0 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{12} 41^{13} 1^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 7 & 10 \\ 1 & 2 & 0 & 0 & 7 & 0 & 2 \\ 2 & 2 & 0 & 0 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{11} 41^{13} 1^{15} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 0 & 3 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 0 & 8 & 10 & 6 \\ 2 & 2 & 0 & 0 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$T = 15$ [70] $1^{11} 6^{13} 15^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 2 & 4 & 0 & 4 \\ 3 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{11} 6^{11} 131513^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 2 & 5 & 4 & 3 & 5 \\ 3 & 3 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 6^{11} 131513^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 7 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{11} 6^{11} 131513^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 2 & 11 & 5 & 4 \\ 7 & 7 & 11 & 26 & 11 & 12 \\ 3 & 2 & 5 & 11 & 3 & 5 \\ 3 & 3 & 4 & 12 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 6^{11} 131513^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -23, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

4.2.32 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \quad 1^{11} 2^3 1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

4.2.33 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [3] \quad 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 1 & 4 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 12 & 4 \\ 10 & 10 & 12 & 0 & 12 & 42 & 14 \\ 3 & 3 & 3 & 0 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 1 & 4 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 6 & 4 \\ 5 & 5 & 6 & 0 & 6 & 9 & 7 \\ 3 & 3 & 3 & 0 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{14} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 3 & 0 & 2 & 20 & 3 \\ 2 & 3 & 1 & 1 & 4 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 24 & 4 \\ 20 & 20 & 24 & 0 & 24 & 174 & 28 \\ 3 & 3 & 3 & 0 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 3 & 0 & 2 & 15 & 3 \\ 2 & 3 & 1 & 1 & 4 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 18 & 4 \\ 15 & 15 & 18 & 0 & 18 & 97 & 21 \\ 3 & 3 & 3 & 0 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{122} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 14 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{122} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 27 & 7 \\ 12 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 4 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{114} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 19 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 7 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 16 & 7 \\ 7 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 5 & 3 \\ 2 & 1 & 3 & 0 & 7 & 5 & 3 \\ 2 & 3 & 1 & 1 & 10 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 17 & 11 \\ 5 & 5 & 6 & 0 & 17 & 9 & 7 \\ 3 & 3 & 3 & 0 & 11 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 10 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 34 & 11 \\ 10 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 11 & 7 \\ 5 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 8 & 15 & 3 \\ 2 & 3 & 1 & 1 & 8 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 48 & 10 \\ 15 & 15 & 18 & 0 & 48 & 97 & 21 \\ 3 & 3 & 3 & 0 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 3 & 0 & 8 & 10 & 3 \\ 2 & 3 & 1 & 1 & 8 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 32 & 10 \\ 10 & 10 & 12 & 0 & 32 & 42 & 14 \\ 3 & 3 & 3 & 0 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 16 & 7 \\ 8 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 8 & 20 & 3 \\ 2 & 3 & 1 & 1 & 8 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 64 & 10 \\ 20 & 20 & 24 & 0 & 64 & 174 & 28 \\ 3 & 3 & 3 & 0 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 12 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 12 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 326, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 11 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 11 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 16 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 11 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 13 & 3 \\ 2 & 3 & 1 & 1 & 12 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 54 & 14 \\ 13 & 13 & 14 & 0 & 54 & 65 & 17 \\ 3 & 3 & 3 & 0 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 27 & 7 \\ 13 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 518, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 13 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 564, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 10 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 17 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 17 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 17 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 758, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 16 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 16 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 806, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 16 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 16 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 870, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 884, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 15 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 15 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 18 & 3 \\ 2 & 3 & 1 & 1 & 12 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 76 & 14 \\ 18 & 18 & 20 & 0 & 76 & 130 & 24 \\ 3 & 3 & 3 & 0 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 38 & 7 \\ 18 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 918, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 18 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 1140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 1332, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 20 & 3 \\ 2 & 1 & 3 & 0 & 10 & 20 & 3 \\ 2 & 3 & 1 & 1 & 12 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 88 & 14 \\ 20 & 20 & 24 & 0 & 88 & 174 & 28 \\ 3 & 3 & 3 & 0 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [77] \ 1^{112} 3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

4.2.34 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂

$$T = 10 \quad [0] \ 1^{121} 3^1 21^{1121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 9 & 6 \\ 2 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 13 & 5 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 5 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1112} 3^1 21^{1112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 0 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 16 & 3 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 3 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 5 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{122} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 11 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 11 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 5 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 18 & 6 \\ 5 & 6 & 8 & 0 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1112} 3^1 21^{1111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{1211}} 3^{\overline{1}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 13 & 10 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{1111}} 3^{\overline{1}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{1112}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 0 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1121}} 3^{\overline{1}} 21^{\overline{1121}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

5 Round 5: NHC External 5-ext (269 bases)

Colored curves in quiver show attachment points. Matching color in external label: ext1 ext2 ext3 ext4 ext5 ext6
 Total: 269 bases (12 uni, 257 non-uni).

T	Count
9	77
10	17
11	15
12	128
13	30
14	2

5.1 Unimodular ($|\det| = 1$) — 12 bases

5.1.1 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 13 \quad [0] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 & 0 & 1 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 11 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[2] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 9 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

5.1.2 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 13 \quad [2] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[0] \quad 1^{13} 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \\ 2 & 0 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

5.1.3 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [2] \quad 1^{11} 4 \quad 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

5.1.4 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [2] \quad 1^{11} 4 \quad 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

5.1.5 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [42] \quad 1^{1111} 4 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 5 & 3 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

5.1.6 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

5.1.7 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

5.1.8 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

5.2 Non-unimodular ($|\det| > 1$) — 257 bases

5.2.1 NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 13 \quad [3] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 10 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

5.2.2 NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 13 \quad [0] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 5 & 3 \\ 2 & 2 & 1 & 3 & 8 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 6 & 3 \\ 6 & 8 & 0 & 7 & 22 & 19 & 10 \\ 5 & 7 & 0 & 6 & 19 & 14 & 8 \\ 3 & 4 & -1 & 3 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 2 & 4 & 0 & 3 & 7 & 4 \\ 5 & 7 & 0 & 6 & 14 & 8 \\ 3 & 4 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 5 & 3 \\ 2 & 2 & 1 & 3 & 16 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 6 & 3 \\ 10 & 16 & 0 & 13 & 70 & 33 & 18 \\ 5 & 7 & 0 & 6 & 33 & 14 & 8 \\ 3 & 4 & -1 & 3 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 4 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 16 & 3 \\ 2 & 4 & 0 & 3 & 2 & 19 & 4 \\ 13 & 19 & 0 & 16 & 19 & 110 & 22 \\ 3 & 4 & -1 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 5 & 3 \\ 2 & 2 & 1 & 3 & 12 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 6 & 3 \\ 8 & 12 & 0 & 10 & 42 & 26 & 14 \\ 5 & 7 & 0 & 6 & 26 & 14 & 8 \\ 3 & 4 & -1 & 3 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

5.2.3 NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 13 \quad [3] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

5.2.4 NHC(-2)(-3) + so₈ + su₂→(-3) + su_{2n3mix} + su₂

$$T = 13 \quad [0] \quad 1^{11} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 6 & 3 \\ 2 & 2 & 1 & 3 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 7 & 3 \\ 5 & 7 & 0 & 6 & 14 & 19 & 8 \\ 6 & 8 & 0 & 7 & 19 & 22 & 10 \\ 3 & 4 & -1 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 10 & 3 \\ 2 & 2 & 1 & 3 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 13 & 3 \\ 5 & 7 & 0 & 6 & 14 & 33 & 8 \\ 10 & 16 & 0 & 13 & 33 & 70 & 18 \\ 3 & 4 & -1 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 7 & 0 & 6 & 14 & 7 & 8 \\ 2 & 4 & 0 & 3 & 7 & 2 & 4 \\ 3 & 4 & -1 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 2 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 & 3 \\ 13 & 19 & 0 & 16 & 110 & 19 & 22 \\ 2 & 4 & 0 & 3 & 19 & 2 & 4 \\ 3 & 4 & -1 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 1^1 2^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 8 & 3 \\ 2 & 2 & 1 & 3 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 10 & 3 \\ 5 & 7 & 0 & 6 & 14 & 26 & 8 \\ 8 & 12 & 0 & 10 & 26 & 42 & 14 \\ 3 & 4 & -1 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

5.2.5 NHC(-2)(-3) + so₈ + su_{2n3mix} + su₂ + su₂→(-3)

$$T = 13 \quad [2] \quad 1^{11} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 16 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 6 & 8 \\ 10 & 16 & 0 & 33 & 70 & 13 & 18 \\ 2 & 3 & 0 & 6 & 13 & 1 & 3 \\ 3 & 4 & -1 & 8 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^1 3^1 1^1 5 13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 2 & 3 \\ 2 & 2 & 1 & 7 & 8 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 6 & 8 \\ 6 & 8 & 0 & 19 & 22 & 7 & 10 \\ 2 & 3 & 0 & 6 & 7 & 1 & 3 \\ 3 & 4 & -1 & 8 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{111}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 6 & 8 \\ 2 & 4 & 0 & 7 & 2 & 3 & 4 \\ 2 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -1 & 8 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 16 & 22 \\ 2 & 4 & 0 & 19 & 2 & 3 & 4 \\ 2 & 3 & 0 & 16 & 3 & 1 & 3 \\ 3 & 4 & -1 & 22 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 2 & 3 \\ 2 & 2 & 1 & 7 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 6 & 8 \\ 8 & 12 & 0 & 26 & 42 & 10 & 14 \\ 2 & 3 & 0 & 6 & 10 & 1 & 3 \\ 3 & 4 & -1 & 8 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

5.2.6 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su}2\text{n}3\text{mix} + \mathfrak{su}_2 + \text{su}2\text{n}3\text{mix}$

$$T = 13 \quad [0] \ 1^{1112}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 4 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 32 & 8 \\ 2 & 4 & 0 & 7 & 2 & 15 & 4 \\ 10 & 15 & 0 & 32 & 15 & 65 & 17 \\ 3 & 4 & -1 & 8 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

5.2.7 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su}2\text{n}3\text{mix} + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 13 \quad [2] \ 1^{113}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 1 & 13 & 3 \\ 10 & 16 & 0 & 33 & 13 & 70 & 18 \\ 3 & 4 & -1 & 8 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 7 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 7 & 3 & 2 & 4 \\ 3 & 4 & -1 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{111}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 6 & 3 \\ 2 & 2 & 1 & 7 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 19 & 8 \\ 2 & 3 & 0 & 6 & 1 & 7 & 3 \\ 6 & 8 & 0 & 19 & 7 & 22 & 10 \\ 3 & 4 & -1 & 8 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^1 3^1 1^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 19 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 & 3 \\ 2 & 4 & 0 & 19 & 3 & 2 & 4 \\ 3 & 4 & -1 & 22 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 8 & 3 \\ 2 & 2 & 1 & 7 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 26 & 8 \\ 2 & 3 & 0 & 6 & 1 & 10 & 3 \\ 8 & 12 & 0 & 26 & 10 & 42 & 14 \\ 3 & 4 & -1 & 8 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

5.2.8 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix + $\mathfrak{su}_2$

$$T = 13 \quad [0] \ 1^{112} 3^1 1^1 5^1 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 7 & 8 \\ 10 & 15 & 0 & 32 & 65 & 15 & 17 \\ 2 & 4 & 0 & 7 & 15 & 2 & 4 \\ 3 & 4 & -1 & 8 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.9 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1112} 3^1 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 6 & 3 & 3 \\ 4 & 6 & 0 & 6 & 13 & 15 & 6 & 7 \\ 3 & 6 & 0 & 6 & 15 & 13 & 6 & 7 \\ 2 & 3 & 0 & 3 & 6 & 6 & 1 & 3 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1112} 3^1 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 9 & 12 & 4 \\ 3 & 6 & 0 & 9 & 13 & 15 & 21 & 7 \\ 4 & 6 & 0 & 9 & 15 & 13 & 21 & 7 \\ 5 & 9 & 0 & 12 & 21 & 21 & 28 & 10 \\ 2 & 3 & -1 & 4 & 7 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{1111} 3^1 1^1 4^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 7.0$$

$$[3] \ 1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \\ 5 & 6 & 8 & 5 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 2 & 0 & 3 \\ 2 & 3 & 5 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [6] \quad 1^{11} 6^{11} 13^1 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

$$[5] \quad 1^{11} 6^{11} 13^1 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

5.2.10 NHC(-2)(-3) + su₂ + su₂ + su₂ + su₂ → (-3)

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.11 NHC(-2)(-3) + su₂ + su₂ + su₂ → (-3) + su₂

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.12 NHC(-2)(-3) + su₂ + su₂ + su₃ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^{1111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

5.2.13 NHC(-2)(-3) + su₂ + su₂ → (-3) + su₂ + su₂

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 2 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.14 NHC(-2)(-3) + su₂ + su₂ → (-3) + su₂ + su_{2n3mix}

$$T = 12 \quad [9] \quad 1^{131} 3^1 1^1 5^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8$$

$$[4] \quad 1^{114} 3^1 1^1 5^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0$$

$$[0] \ 1^{1212} 3^1 1^1 5 13^1 1^{2121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0$$

$$\mathbf{5.2.17} \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$$

$$T = 12 \quad [0] \ 1^{1111} 3^1 1^1 5 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$\mathbf{5.2.18} \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 0 & 12 & 13 & 7 \\ 3 & 4 & 0 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 9 & 13 & 7 \\ 3 & 4 & 0 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 10 \\ 3 & 4 & 0 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 5 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 0 & 12 & 28 & 10 \\ 3 & 4 & 0 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 0 & 9 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 0 & 21 & 28 & 10 \\ 3 & 4 & 0 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 0 & 15 & 13 & 7 \\ 3 & 4 & 0 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

5.2.19 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \begin{matrix} 1^{11} \\ 1^1 3^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

5.2.20 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \begin{matrix} 1^{11} \\ 1^1 3^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \begin{matrix} 1^{11} \\ 1^1 3^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [3] \begin{matrix} 1^{11} \\ 1^1 5^1 13^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \begin{matrix} 1^{11} \\ 1^1 5^1 13^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 8 & 5 \\ 2 & 3 & 1 & 0 & 2 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 4 & 8 & 5 & 0 & 5 & 9 & 7 \\ 3 & 5 & 3 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[5] \begin{matrix} 1^{11} \\ 1^1 5^1 13^1 1^{11} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 2 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[6] \begin{matrix} 1^{11} \\ 1^1 5^1 13^1 1^{12} \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 5 & 3 & 0 & 4 & 16 & 5 \\ 2 & 3 & 1 & 0 & 2 & 10 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 10 & 3 \\ 8 & 16 & 10 & 0 & 10 & 42 & 14 \\ 3 & 5 & 3 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

- [7] $1^{11} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [8] $1^{11} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 3 & 0 & 4 & 24 & 5 \\ 2 & 3 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 15 & 3 \\ 12 & 24 & 15 & 0 & 15 & 97 & 21 \\ 3 & 5 & 3 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$
- [9] $1^{12} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 5 & 7 & 7 & 7 & 0 & 6 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [10] $1^{11} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 6 & 7 & 10 & 9 & 0 & 8 & 52 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [11] $1^{11} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$
- [12] $1^{12} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 11 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 7 & 9 & 11 & 10 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$
- [13] $1^{13} \overset{1}{5} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 8 & 11 & 12 & 11 & 0 & 10 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

5.2.21 NHC(-2)(-3) + su₃ + su₂ + su₂ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 4 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 3 & 4 \\ 8 & 13 & 0 & 12 & 28 & 9 & 10 \\ 3 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -1 & 4 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 6 & 0 & 4 \\ 4 & 5 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 15 & 7 \\ 5 & 8 & 0 & 12 & 10 & 12 & 6 \\ 6 & 9 & 0 & 15 & 12 & 13 & 7 \\ 3 & 4 & -1 & 7 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{121} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 15 & 7 \\ 6 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 0 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 9 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -1 & 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{1111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 9 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{1112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{1111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{1112} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 9 & 3 \\ 6 & 9 & 0 & 6 & 13 & 21 & 7 \\ 8 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -1 & 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [41] $1^{11} 3^{1111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 & 7 & 4 \\ 3 & 5 & 0 & 4 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -1 & 4 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [45] $1^{11} 4^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 4 & 4 & 2 & 3 & 4 & 5 \\ 5 & 3 & 2 & 4 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[44] $1^{11} 4^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 & 4 \\ 3 & 1 & 4 & 3 & 4 & 3 \\ 5 & 4 & 6 & 5 & 5 & 6 \\ 5 & 3 & 5 & 3 & 4 & 5 \\ 4 & 4 & 5 & 4 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

[43] $1^{11} 4^{11} 131^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 & 4 \\ 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 4 & 4 & 2 & 5 & 5 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

5.2.22 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -1 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -1 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 1 & 4 \\ 3 & 1 & 1 & 4 & 0 & 3 \\ 2 & 1 & 0 & 2 & 0 & 2 \\ 5 & 4 & 2 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5^1 1^{11} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 5 & 0 & 5 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 3 & 4 \\ 8 & 15 & 0 & 11 & 37 & 10 & 13 \\ 2 & 4 & 0 & 3 & 10 & 1 & 3 \\ 3 & 5 & -1 & 4 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

5.2.23 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 5 & 4 \\ 3 & 1 & 1 & 0 & 4 & 3 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 0 & 6 & 6 \\ 4 & 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 2 & 3 & 0 & 3 \\ 2 & 3 & 2 & 1 & 2 & 0 & 3 \\ 2 & 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 5^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 2 & 0 & 3 & 2 & 4 \\ 3 & 5 & 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[10] \quad & 1^{1^{21}} 5^{11} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{1^{11}} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{1^1} 5^{11} 13^{11} 23^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 11 & 4 \\ 2 & 4 & 0 & 3 & 1 & 10 & 3 \\ 8 & 15 & 0 & 11 & 10 & 37 & 13 \\ 3 & 5 & -1 & 4 & 3 & 13 & 4 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{1^{12}} 5^{11} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 5 & 3 & 0 & 4 & 10 & 5 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 6 & 3 \\ 6 & 10 & 4 & 0 & 6 & 14 & 8 \\ 3 & 5 & 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{1^{31}} 5^{11} 13^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{1^{12}} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{1^{13}} 5^{11} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 5 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 9 & 3 \\ 9 & 15 & 6 & 0 & 9 & 34 & 12 \\ 3 & 5 & 2 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{1^1} 5^{11} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1^2}_{1^1} 5^{12} 13^{11} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1^3}_{1^1} 5^{12} 13^{11} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1^{12}}_{1^1} 5^{12} 13^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1^{12}}_{1^1} 5^{12} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 3 & 0 & 2 & 8 & 3 \\ 2 & 3 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 7 & 3 \\ 8 & 8 & 8 & 0 & 7 & 22 & 10 \\ 3 & 3 & 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1^4}_{1^1} 5^{12} 13^{11} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 12 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1^{11}}_{1^1} 5^{12} 13^{11} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1^{12}}_{1^1} 5^{12} 13^{11} 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 10 & 4 & 14 & 6 & 16 & 8 \\ 2 & 4 & 2 & 6 & 1 & 6 & 3 \\ 6 & 10 & 4 & 16 & 6 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1^{13}}_{1^1} 5^{12} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 3 & 0 & 2 & 7 & 3 \\ 2 & 3 & 1 & 0 & 3 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 7 & 10 & 0 & 8 & 25 & 11 \\ 3 & 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\frac{1}{1}} \overset{13}{5} \overset{1}{13} \overset{123}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 10 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 10 & 4 & 14 & 6 & 24 & 8 \\ 2 & 4 & 2 & 6 & 1 & 9 & 3 \\ 9 & 15 & 6 & 24 & 9 & 34 & 12 \\ 3 & 5 & 2 & 8 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{\frac{1}{1}} \overset{11}{5} \overset{13}{13} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\frac{1}{1}} \overset{11}{5} \overset{13}{13} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\frac{1}{1}} \overset{11}{5} \overset{13}{13} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\frac{1}{1}} \overset{12}{5} \overset{13}{13} \overset{113}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 8 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{\frac{1}{1}} \overset{12}{5} \overset{13}{13} \overset{113}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\frac{1}{1}} \overset{13}{5} \overset{13}{13} \overset{113}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[33] \quad & 1^{1^{13}}_1 5 13^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 11 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[34] \quad & 1^{1^{13}}_1 5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 10 & 3 \\ 11 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[35] \quad & 1^{1^{13}}_1 5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 3 & 0 & 2 & 13 & 3 \\ 2 & 3 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 12 & 3 \\ 13 & 13 & 14 & 0 & 12 & 65 & 17 \\ 3 & 3 & 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

5.2.24 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$\begin{aligned}
T = 12 \quad [0] \quad & 1^{1^{11}}_1 5 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 2 & 2 & 4 & 2 & 3 & 6 & 4 \\ 3 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1] \quad & 1^{1^{11}}_1 5 13^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 & 7 \\ 3 & 5 & 2 & 4 & 7 & 2 & 4 \\ 3 & 5 & 2 & 4 & 7 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[2] \quad & 1^{1^{12}}_1 5 13^1 1^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 8 & 4 \\ 5 & 9 & 4 & 7 & 11 & 14 & 7 \\ 6 & 10 & 4 & 8 & 14 & 14 & 8 \\ 3 & 5 & 2 & 4 & 7 & 8 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[3] \quad & 1^{1^{13}}_1 5 13^1 1^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 5 & 9 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 5 & 2 & 2 & 7 & 12 & 4 \\ 5 & 9 & 4 & 7 & 11 & 21 & 7 \\ 9 & 15 & 6 & 12 & 21 & 34 & 12 \\ 3 & 5 & 2 & 4 & 7 & 12 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[4] \quad & 1^{1^{21}}_1 5 13^1 1^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 & 11 \\ 3 & 5 & 2 & 4 & 11 & 2 & 4 \\ 3 & 5 & 2 & 4 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1\bar{1}} 5^{1\bar{1}} 13^{1\bar{1}} 1^{1\bar{1}3\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 & 15 \\ 3 & 5 & 2 & 4 & 15 & 2 & 4 \\ 3 & 5 & 2 & 4 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.25 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 \\ 3 & 4 & -1 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1\bar{1}} 3^{1\bar{1}} 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11} 4^{13} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 5 & 4 \\ 3 & 1 & 0 & 1 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 5 & 4 & 0 & 2 & 6 & 6 \\ 4 & 3 & 0 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 & 3 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 2 & 3 & 0 & 2 & 4 \\ 3 & 5 & 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 5^{13} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 5^{13} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 10 & 3 \\ 2 & 5 & 0 & 3 & 2 & 11 & 4 \\ 8 & 15 & 0 & 10 & 11 & 37 & 13 \\ 3 & 5 & -1 & 3 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 5^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 2 & 0 & 2 & 3 \\ 2 & 3 & 1 & 3 & 0 & 4 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 3 & 0 & 2 & 4 \\ 3 & 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 4 & 0 & 10 & 5 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 2 & 4 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 4 & 6 & 0 & 14 & 8 \\ 3 & 5 & 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{1^1} 5^{1^1} 13^{1^1} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$

[15] $1^{1^{13}} 5^{1^1} 13^{1^1} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 5 & 3 & 4 & 0 & 15 & 5 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 2 & 4 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 6 & 9 & 0 & 34 & 12 \\ 3 & 5 & 2 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$

[16] $1^{1^{31}} 5^{1^1} 13^{1^1} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$

[17] $1^{1^{12}} 5^{1^1} 13^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[18] $1^{1^2} 5^{1^1} 13^{1^1} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 2 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$

[19] $1^{1^3} 5^{1^1} 13^{1^1} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 2 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$

[22] $1^{1^4} 5^{1^1} 13^{1^1} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 2 & 5 & 2 & 3 & 2 & 16 & 4 \\ 12 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$

[21] $1^{1^{12}} 5^{1^1} 13^{1^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 2 & 0 & 8 & 3 \\ 2 & 3 & 1 & 3 & 0 & 8 & 3 \\ 2 & 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 7 & 0 & 22 & 10 \\ 3 & 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}_{1^1} 5^{12} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 6 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}_{1^1} 5^{11} 13^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11}_{1^1} 5^{12} 13^{11} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 16 & 8 \\ 6 & 10 & 4 & 6 & 16 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{113}_{1^1} 5^{13} 13^{11} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 4 & 10 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 4 & 2 & 1 & 6 & 9 & 3 \\ 5 & 10 & 4 & 6 & 14 & 24 & 8 \\ 9 & 15 & 6 & 9 & 24 & 34 & 12 \\ 3 & 5 & 2 & 3 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113}_{1^1} 5^{13} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 2 & 0 & 7 & 3 \\ 2 & 3 & 1 & 3 & 0 & 10 & 3 \\ 2 & 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 8 & 0 & 25 & 11 \\ 3 & 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}_{1^1} 5^{11} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11}_{1^1} 5^{11} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1^{12}}_1 5^{11} 13^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 8 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{12}}_1 5^{11} 13^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 3 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{13}}_1 5^{11} 13^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{13}}_1 5^{11} 13^{11} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 3 & 5 & 2 & 3 & 2 & 16 & 4 \\ 11 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^{13}}_1 5^{12} 13^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 2 & 0 & 10 & 3 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 10 & 0 & 42 & 14 \\ 3 & 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^{13}}_1 5^{13} 13^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 2 & 0 & 13 & 3 \\ 2 & 3 & 1 & 3 & 0 & 14 & 3 \\ 2 & 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 12 & 0 & 65 & 17 \\ 3 & 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

5.2.26 NHC(-2)(-3) + su₃ + su₂n₃mix + su₂ + su₂

$$T = 12 \quad [0] \quad 1^{1^{11}}_1 5^{11} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 3 & 2 & 3 & 1 & 3 & 5 & 3 \\ 2 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \quad 1^{11} \quad 5 \quad 13 \quad 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 43, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 & 4 \\ 3 & 5 & 2 & 7 & 4 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} \quad 1^{11} \quad 5 \quad 13 \quad 1^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 46, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 & 14 & 7 \\ 2 & 5 & 2 & 7 & 2 & 8 & 4 \\ 6 & 10 & 4 & 14 & 8 & 14 & 8 \\ 3 & 5 & 2 & 7 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} \quad 1^{11} \quad 5 \quad 13 \quad 1^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 51, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 9 & 5 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 9 & 4 & 11 & 7 & 21 & 7 \\ 2 & 5 & 2 & 7 & 2 & 12 & 4 \\ 9 & 15 & 6 & 21 & 12 & 34 & 12 \\ 3 & 5 & 2 & 7 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{121} \quad 1^{11} \quad 5 \quad 13 \quad 1^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 71, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 & 4 \\ 3 & 5 & 2 & 11 & 4 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{131} \quad 1^{11} \quad 5 \quad 13 \quad 1^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 107, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 & 4 \\ 3 & 5 & 2 & 15 & 4 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

5.2.27 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} \quad 3^1 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 10, 1), \quad c_{\text{ext}} = 8.0$$

$$[1] \quad 1^{111} \quad 3^1 \quad 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 10, 1), \quad c_{\text{ext}} = 8.0$$

$$[2] \quad 1^{121} \quad 3^1 \quad 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 10, 1), \quad c_{\text{ext}} = 8.5$$

$$[3] \quad 1^{121} \quad 3^1 \quad 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 10, 1), \quad c_{\text{ext}} = 8.5$$

$$[4] \quad 1^{1121} \quad 3^1 \quad 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 10, 1), \quad c_{\text{ext}} = 8.5$$

$$[5] \quad 1^{1131} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.8$$

$$[6] \quad 1^{1141} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

5.2.28 NHC(-2)(-3)(-2) + $\textcolor{brown}{su}_2$ + $\textcolor{brown}{su}_2\textcolor{brown}{n}3\textcolor{brown}{mix}$ + $\textcolor{brown}{su}_2$ + $\textcolor{brown}{su}_2$

$$T = 10 \quad [0] \quad 1^{11121} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5$$

$$[1] \quad 1^{11111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.0$$

$$[2] \quad 1^{1112} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5$$

$$[3] \quad 1^{1121} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.0$$

6 Round 6: NHC External 6-ext (25 bases)

Colored curves in quiver show attachment points. Matching color in external label: **ext1** **ext2** **ext3** **ext4** **ext5** **ext6** **ext7**
 Total: 25 bases (0 uni, 25 non-uni).

T	Count
10	23
11	2

6.1 Non-unimodular ($|\det| > 1$) — 25 bases

6.1.1 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{11}} \ 5 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 8.0$$

6.1.2 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} \ 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1121}} \ 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 9 & 6 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 21 & 10 \\ 6 & 9 & 0 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

6.1.3 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 0 & -1 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

6.1.4 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

6.1.5 NHC(-2)(-3) + su₂ + su₃ + su₂→(-3) + su₂ + su₂

$$T = 10 \quad [0] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

6.1.6 NHC(-2)(-3) + su₃ + su₂ + su₂ + su₂ + su₂

$$T = 10 \quad [0] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1121} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 6 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 21 & 10 \\ 6 & 9 & 0 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11111}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11111}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11111}} \ 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11111}} \ 3 \ 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\overline{11111311}} \ 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 9.0$$

6.1.7 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{1111}} \ 3^{\overline{11}} \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & -1 & 4 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} \ 3^{\overline{11}} \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

6.1.8 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [1] \ 1^{\overline{111}} \ 3^{\overline{11}} \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

6.1.9 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

6.1.10 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

7
Round 7: NHC External 7-ext (1 base)

Colored curves in quiver show attachment points. Matching color in external label: ext1 ext2 ext3 ext4 ext5 ext6 ext7 ext8

Total: 1 bases (0 uni, 1 non-uni).

T	Count
11	1

7.1
Non-unimodular ($|\det| > 1$) — 1 bases

7.1.1
NHC(−2)(−3) + su₂ + su₂ + su₂ + su₂ + su₂ + su₂

$$T = 11 \quad [0] \quad 1^{\textcolor{red}{1}} \quad 5^{\textcolor{blue}{1}\textcolor{red}{1}} \quad 1^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{magenta}{su}_2 + \textcolor{brown}{su}_2 \quad T = 11, \, T_{\max} = 11, \, \Delta = -8, \, \det = 0, \, (n_+, n_-, n_0) = (1, 11, 2), \, c_{\text{ext}} = 9.0$$